

TRANSMEDICS GROUP INC (NASDAQ: TMDX)

Walk Like An Egyptian: A “Mafia-Style” Extortion, Racketeering, And Organ Trafficking Scheme Masquerading As A Medical Device Company – Fueled By Kickbacks, Billing Fraud, Unreported Device Failures, And Off-Label Misuse On A Vast Scale; Precipitating A Customer Exodus And Rapidly Unfolding Death Spiral. Target Price: \$0.

We are shortly filing a Citizen Petition with the FDA requesting suspension of PMA (Premarket Approval) of TransMedics Organ Care System (OCS), in conjunction with a public letter to members of Congress probing abuses in the US organ transplant system. Transplant surgeons, hospital administrators, OPO's, or others wishing to submit FDA comments in support of the Petition may email us at contact@scorpioncapital.com or anonymously via the contact form on our site.

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“It’s like racketeering or something, right? That’s what it feels like...I’m sure other surgeons have used some pretty dramatic words...they’re playing like mafia-style...very deliberate and a thug-style way of getting things done...I would call him Waleed’s enforcer. He’s someone you wouldn’t be surprised to find out is knee-capping people...you see those movies where the guys go around and collect their protection money. That’s what he reminds me of - the guy who’s going around to collect the protection money.” - Transplant surgeon at a high-volume academic center; one of TransMedics largest customers; in process of moving to another device.

“You should see Waleed arrive at conferences. They show up in these blacked-out cars...and you think the president is showing up. It’s like three blacked-out SUVs. They all get out. They all look the same in their suits. Yeah, it’s like Waleed and his secret service, but it’s the TransMedics folks.” – Executive at a leading OPO (Organ Procurement Organization), industry veteran

“There’s so much that’s going to be found out. Do you know everything that’s going on with the OIG and DOJ and the IRS?...it’s deeper than [the Washington Post], and I’ll let you know because I’ve been interviewed for it...here’s all the drama. I recently interviewed with the Department of Justice, the OIG, and the IRS. And with the DOJ out of Washington and out of New York...and then the state attorney general is getting involved in this as well because if there’s Medicare fraud...and probably the most prominent attorney general that’s getting involved is Virginia.” – CEO of a major OPO, longtime industry executive

“If that gets revealed, if that gets exposed in any way, that’s curtains...I have known of some sites in our region that would get these kinds of calls...this kind of hotline for organs...it allowed them consequently to boost their numbers substantially...We scratch our heads wondering how in the heck did these guys get these organs? How did that happen?...There’s a lot of game-playing going on out there...they would come from, say, Tennessee...literally a sort of hotline...asking if they want an organ...it just stopped being part of the formal allocation process. This kind of thing, I suspect, is actually happening much more frequently than we know...I think this is how some centers get ahead in their numbers. It’s a below the radar screen kind of arrangement...it’s kind of like one way of getting the good stuff, so to speak...” – Transplant hepatologist at Massachusetts General Hospital; MGH stated their entire usage of TransMedics is at risk

“...they are intentionally keeping people in these dark silos so that people don’t have a grasp of what’s going on...it’s just like this weird lack of transparency, this weird secretive culture, and it just felt like everyone is intentionally kept in the dark so that any sort of information couldn’t be used against the company in any way. And Waleed was always like if anyone has questions, refer them back to us...the pilots were like, “We were told not to talk to you guys”...everything at this company feels like a big secret, and it’s really weird and sketchy.” - Former employee who left recently

“I cannot stand working with them. It feels like I’m talking to a used car salesman...We had a recent situation, where we had some A/R...they declined a heart transplant for one of our patients in the ICU because of this credit hold. They refused to go out and get the heart for us. I could not believe it...I had to spend probably eight hours of my week on the phone with their rep, trying to beg and plead with them to release the hold because we didn’t want to miss an opportunity to get an organ for one of our patients...I felt like I was groveling...these heart patients, this isn’t like a kidney transplant. This person is in the ICU. They’re going to die. They’re status one. They’re getting this organ offer because they’re the sickest patient in need of one, and TransMedics made that call. It was their COO [sic], Tamer Khayal, who is not a clinician...[Magdy Attia] was part of the people threatening us.” – Administrator at a pre-eminent academic transplant center, which expects to eliminate TransMedics usage starting in early 2025

“There was a liver a month ago...it was an infarction, that part of it had died...one of my colleagues was already at a point of no return in the recipient operation. He had already started before realizing that this was happening...we ended up having to use that liver and, sure enough, post-transplant...the right lobe of the liver just was completely dead...The [TransMedics] procuring surgeon could feel it, but it was not reported...that’s just the first part of the story. Fast forward three weeks later, another colleague, we had another liver on pump for a different patient...and again, this time, the right lobe of the liver felt off...we called TransMedics...we’re nervous because of what happened with our previous patient...the response was ‘Wow, yours is the only center that is having these issues,’ which is an outright lie... it’s just a fucking lie. It’s a lie that we’re the only center that’s having these problems but that was the communication to us.” – Head of transplant surgery at a major center; high-volume TransMedics user, who intends to terminate usage

“This is my take-home message to you - that this is just a matter of time. Back in the ‘90s Blackberry dominated. Now it’s history. TransMedics is going to be like Blackberry. TransMedics is going to die because people are so pissed off with the company and this predatory behavior. People are just going to quit using it. They’re very upset, and it’s going to be, “Screw you, I don’t want to use this product anymore.” People are not happy with the management of the company. Everybody hates the CEO, the CFO, and all the people in the upper level...it’s run by the wrong people...doctors are going to run from this company.” – Transplant surgeon in a leadership role at Vanderbilt, one of the top centers in the US, which has already wiped out TransMedics usage in heart, lung, and liver

Table of contents

Introduction to the real TransMedics: a 6-month investigation with over 30 interviews, including ex-employees, surgeons, leading transplant centers, organ procurement organizations, competitors, and its largest customers.	11
Part 1. TransMedics National OCS Program (NOP): an unsustainable, anti-competitive scheme based on what we believe to be coercive tying, price extortion, and retaliation.	41
Part 2. The NOP service is, in our opinion, a large-scale fraudulent billing racket, predicated on overcharging hospitals for unnecessary flights.	70
Part 3. Revenue is dependent on a handful of transplant surgeons/centers - “prostitutes” who are “completely owned and operated by TransMedics” - who allegedly receive kickbacks and inducements.	98
Part 4. These high-volume users may allegedly be under scrutiny as part of multiple investigations by federal and states authorities and agencies, as well as by prominent news outlets, indicating the precarious nature of TransMedics business.	109
Part 5. We believe TransMedics is operating an organ trafficking scheme, shopping and steering rejected organs to its top users as a quid pro quo for accepting them on its device and via its NOP service.	124
Part 6. TransMedics device usage is, we conclude, almost all off-label, the largest such racket we have encountered, putting it at existential risk: “I’ve never seen a device manufacturer so boldly engage in off-label use themselves...it’s their technician.”	154

Table of contents

Part 7. The NOP organ procurement service is “a ticking time bomb” staffed with imported H1B surgeons unlicensed to practice medicine in the US, from high-risk areas like India, Pakistan, and the Middle East; resulting in butchered and lost organs.	193
Part 8. The NOP service, we believe, traffics in and pushes unsafe, damaged organs rejected by reputable centers, as an unplaced organ means loss of device and aircraft fees; with TransMedics allegedly concealing adverse organ information.	227
Part 9. Organs on TransMedics devices are managed by inexperienced, high-risk technicians called OCS Specialists, who allegedly receive only a week of training prior to engaging in the practice of medicine, putting organs and recipients in jeopardy.	240
Part 10. Customer rage: TransMedics business practices have antagonized the small world of transplant, past the point of no return – a short-tempered CEO; allegedly threatening centers who state that organs have been held hostage from ill patients.	249
Part 11. An accelerating customer exodus, the fastest such stampede we have ever encountered. Large, pre-eminent transplant centers have eliminated or sharply reduced usage; its remaining high-volume customers indicate they are in the process of doing so.	269
Part 12. TransMedics device has no value proposition. Even its largest customers admit it has no clinical benefit to organs, using it off-label strictly for “surgeon lifestyle” and scheduling convenience; or, we believe, in exchange for organs steered improperly.	281

Table of contents

Part 13. Competitive tsunami. TransMedics is being rapidly replaced by superior devices at a fraction of the cost, which began rolling out only recently. Customer interviews indicate delivery of OrganOx, Paragonix, and other devices at a weekly clip.	303
Part 14. Disruption by NRP, spreading like wildfire. OPO's and transplant centers are standardizing around an inexpensive, alternative technique which has long been the standard of care in Europe, with TransMedics 2025 revenue at risk of collapse.	317
Part 15. TransMedics device is at risk of recall, plagued by what we think is a high failure rate, resulting in necrotic livers with dead tissue and other issues. Surgeons report losing a significant percentage of organs, and allege a cover-up by dishonest management.	326

In 20 years of shorting, TransMedics is the most extreme and grotesque healthcare fraud we have encountered, not only for its scale, but because it is predicated on the exploitation of the most vulnerable patients – the terminally ill, desperate for an organ. The “lucky” patients who receive a diseased, damaged organ rejected by reputable surgeons and centers – the type that TransMedics NOP service traffics in and flings off-label onto its rig – or ones with dead, necrotic tissue after rotting on the device, are oblivious to the cesspool of perverse, secret incentives that steered the organ their way. The corruption pervades every aspect of the business model. It is more accurately a racket, the closest we’ve seen in the public markets to a Mafia-style extortion scheme. Tony Soprano took pride in clever schemes that showcased his cunning and business acumen, like his Bust-Out Scheme, Esplanade Project, and Bogus Stock Scam – and stock scam is a fitting segue for us to note that any resemblance between real and fictional characters is purely accidental.

TransMedics CEO Waleed Hassanein



Source: <https://www.youtube.com/watch?v=mmadFroPCpc>

Investors sense something is afoot, after a spate of recent bad news: a Q3 miss and startling drop in sequential revenue; the CFO fired a month later, after a 10-year tenure; a disastrous Dec investor day and 2025 guide – back-half weighted, naturally. As short-sellers, we are partial to books like Spy The Lie, by ex-CIA interrogators on spotting deception. Skim it on your Kindle en route to JPM, if you have a 1x1 next week with TransMedics. Two of the highest-risk behaviors, unsurprisingly, are denial and attack. The OPCO analyst asked a question on the Q3 call, and we note the CEO's replies below: "So Waleed, I'd be remiss...quite a shock...trying to connect all your comments together, I confess I'm still struggling to understand...I'm curious, there seems to have been a break in your trajectory of share gain...."

TransMedics Q3 earnings call, Oct 28, 2024 – CEO Waleed Hassanein

*"...there is no shift in gears. **It's your perception of reality.**"*

*"I caution the Street from **trying to create something out of nothing.**"*

*"There's **absolutely, categorically no impact of any technologies**, methodologies, or competitive dynamics on the decline in Q3...**the facts are the facts.**"*

*"**I want to make it crystal clear.** We have not seen any fundamental or competitive dynamics... **Let me repeat it again**, there has not been or we have not seen any fundamental or competitive dynamics..."*

*"**Guys, I want to be crystal clear**...abdominal NRP does not impact OCS."*

*"**There is no clear reason for these declines** other than normal variability."*

*"**Listen**, we're just coming out of a quarter where transplant volumes declined..."*

*"**I repeat again**, our share in heart and liver...remain unchanged."*

If you own the stock, we have bad news for you: it's already over. You just don't know it yet. Fasten your seatbelt and put on some goggles, because the stuff has yet to hit the fan. The TransMedics story is cliché for a fraud - hubris, overreach, calling to mind the usual genre, Icarus, Narcissus, or perhaps for this one, Shelley's ode to the pharaoh Ozymandias. TransMedics has an unusual and cunning business model, a creature of perverse incentives unique to organ transplantation. It sells a device for the transportation, preservation, and assessment of organs for transplant, called the Organ Care System (OCS) – plus an outsourced end-to-end service called the National OCS Program (NOP), comprised of its own fleet of private jets, organ procurement surgeons, and device technicians who manage the organ. The device comes in three versions for lung, heart, and liver. NOP service revenue comprised 39% in the last quarter; and the company is a one-trick pony for liver. Lung, its first product, is a self-evident failure. Heart growth has been underwhelming, and all three versions declined sequentially in the last quarter. The company's disclosure is opaque, as it fails to break out the key figure – device sales by organ – showing only a product/service total by SKU.

Product vs. service revenue

	Annual					Last 4 quarters			
	2020	2021	2022	2023	LTM	4Q23	1Q24	2Q24	3Q24
Product	NA	29.7	79.2	176.1	250.8	51.9	61.3	71.7	65.9
Service	NA	0.6	14.2	65.6	150.3	29.3	35.5	42.6	42.9
Total	25.6	30.3	93.5	241.6	401.2	81.2	96.9	114.3	108.8
Product %	NA	NA	84.7%	72.9%	62.5%	63.9%	63.3%	62.7%	60.6%
Service %	NA	NA	15.2%	27.2%	37.5%	36.1%	36.6%	37.3%	39.4%

Revenue by organ

	Annual					Last 4 quarters			
	2020	2021	2022	2023	LTM	4Q23	1Q24	2Q24	3Q24
Sales by organ									
OCS Liver	5.2	1.9	46.3	151.8	275.3	54.7	66.9	77.0	76.7
OCS Heart	14.2	17.7	38.3	73	92.7	20.8	20.2	27.2	24.5
OCS Lung	6.2	10.7	8.8	11.8	15.8	3.2	4.7	4.2	3.7
OCS Liver	20%	6%	50%	64%	72%	70%	73%	71%	73%
OCS Heart	55%	58%	41%	31%	24%	26%	22%	25%	23%
OCS Lung	24%	35%	9%	5%	4%	4%	5%	4%	4%

The sell-side was in “shock” at the Q3 miss – the CEO, perhaps not so much. Despite “absolutely, categorically” asserting nothing was wrong, his trades suggest the opposite. He appears to have been in quite the hurry, stepping outside of his open 10b5-1 plan on 8/21 and 8/22 to monetize 45,000 non-expiring options, generating a net pre-tax profit of \$7MM – right in the middle of Q3. He stepped outside of the same plan prior to that in May. Stepping outside of a pre-arranged plan is highly unusual and aggressive, especially when the options are not expiring. It is also high risk. The DOJ recently obtained its first conviction for abuses related to 10b5-1 plans, against an executive accused of misusing his plan and trading based on confidential information.

Press coverage - first conviction for 10b5-1 plan abuses

DOJ Brings First Criminal Charges for Trades Under a Rule 10b5-1 Plan

Jury Convicts Milken Protégé Terren Peizer of Insider Trading

Ex-Ontrak chairman was accused of abusing trading strategy that usually shields executives against suspicions of insider trading

TransMedics is an airline with a medical device attached, at 7X LTM revenue, a lofty multiple even by speculative midcap metrics. The only relevant measure, of course, is EV to product sales, which at 10X – with those sales tanking sequentially – makes it the most expensive midcap medtech in the US, and 3rd priciest overall, after ISRG (\$200B) and GKOS (\$8B). Even at an absurd 5X multiple to product sales, the stock would be in the \$30's. The air charter and service half is worthless – a low multiple of book or EBITDA, typical of low-gross margin/high PP&E plays, with PE comps easily available. TransMedics capex is high at \$158MM LTM, with net debt even if it liquidated its jets; has no free cash flow; ran losses until recently before becoming slightly profitable, which is likely to revert as sales decline. The medical air transport industry is high-risk and prone to billing fraud and kickbacks. Large players regularly succumb to DOJ investigations. Its competitors Trinity Air Medical and The Nationwide Organ Recovery Transport Alliance (NORA) allegedly engage in dubious practices, and are allegedly under investigation, as we shall cover. Air Methods, owned by private equity, settled with the DOJ in 2023, filing for bankruptcy a few months later after they blew up its business model.

Air medical transportation – industry articles

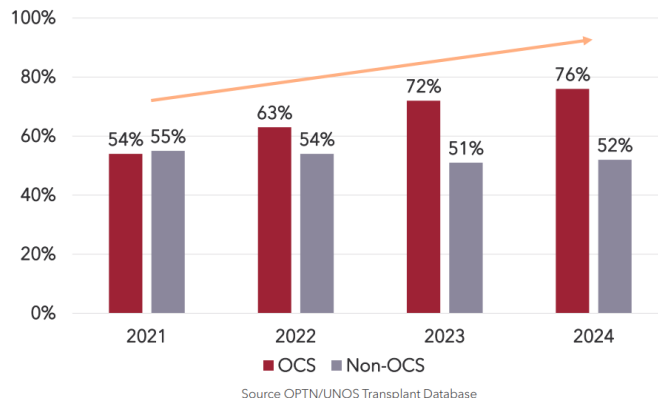
U.S. Air Medical Provider Air Methods Files For Bankruptcy

Air Medical Transport Company Agrees to Pay \$1 Million to Resolve Allegations of False Claims Act Violations

TransMedics was the first company to get FDA approval for its warm perfusion pump, starting with the lung version in 2018, following by heart and liver in 2021. Transplant surgeons flocked to the device. Not because they think that getting gouged for \$70-90K per consumable - plus another six-figures in procurement and aircraft charges - has any clinical benefit for the organ versus static cold storage, but because of a dirty little secret – which they confessed bluntly, one surgeon after another, on our calls. Transplant surgery is a lucrative, prestigious field with one major drawback: the lifestyle sucks. If someone dies at 2AM on Saturday and an organ becomes available, they have to go to work. With TransMedics, they can leave the organ on the pump while they get their “beauty sleep.” Not one surgeon – even its highest-volume users – told us it has any clinical benefit, saying at best, “Don’t know, don’t care.” The wink-wink is visible in TransMedics presentations.

TransMedics Dec 2024 investor day presentation

Day-Time Liver Transplant - **Game Changer**



Impact of OCS NOP on US Transplants

>7,000 US NOP transplants performed to-date & growing rapidly

~75% & ~55% of US DCD transplants for **heart** and **liver** respectively are done using OCS today

12% US national growth in **heart** and **liver** transplant volume primarily driven by OCS and NOP

>76% of liver transplants have shifted to day-time transplants using OCS NOP

Surgeons described the “lifestyle device” as a game-changer, with a disingenuous clinical pretense to badger hospital administrators into letting them use it – that is, Medicare and the US taxpayer footing the bill for hundreds of thousands of dollars per case, just to accommodate a surgeon’s scheduling preferences. They described it like crack cocaine. And as the only FDA-approved pump that you could sling organs on for days, liver degradation be damned if you have golf on Sunday - TransMedics was the only dealer. Predatory pricing ensued, as it must with any good “FDA monopoly” story. The CEO overplayed a weak hand, after a quarter-century of walking the desert with nothing to show for it except a pulled offering in 2007 – only to promptly shat in and thoroughly poison his own well. Transplant medicine is a small world. In 20 years of research, we have never heard anything like the seething, rage, and desire for retribution as on these calls with surgeons and hospitals.

CEO is “so disingenuous” and company is “dishonest”

“I don’t like Waleed or a lot of their upper management...he’s doing the fake it ‘til you make it thing...this goes to show you what a fuck-face he is...he’s so disingenuous... their company, from a corporate culture point of view, is dishonest, to be blunt, and I think their claims are exaggerated.” – Prominent surgeon, Director of a leading academic transplant center

CEO viewed as “an arrogant prick” due to “price gouging”; “people don’t like him”

“Their CEO is pretty arrogant. He’s an arrogant prick. I mean, look at what they did. They bought \$12-13 million aircraft...it works just as well as a \$2 million aircraft or lease the damn thing and have a reasonable logistics expense, but it’s just price gouging. It’s unbelievable. It’s not what transplant is meant to be...people don’t like him... If you ask any questions, they are very aggressive and confrontational...it is frankly putting people’s lives in jeopardy. It’s crazy. Personally, we did not agree with some of their business practices...the CEO was on the call... he just launched on me, threatening that if we didn’t want to work with them, then that’s fine. They could take their device elsewhere. I really don’t know how else to explain it. It was just very aggressive. They don’t like receiving questions...” – CEO of a major OPO (Organ Procurement Organization), longtime industry executive

CEO is “very short-tempered” and “screamed at people a lot”; called customers “idiots”

“Waleed is very short-tempered. He screamed at people a lot and he could be pretty unprofessional...whenever a customer is asking too many questions, the moment they would leave, he’d be like, “Oh my God, such and such idiots.” - Former longtime employee in senior roles including medical safety and clinical affairs

As a preview of a section on enraged customers and the resulting exodus, we quote three large, current customers below. It is self-evident they are not likely to remain so for long. The three below represent a meaningful chunk of the company's liver device usage, the organ that is the bulk of its revenue – the loss of just one, we believe, could easily swing a quarter to a miss. One below uses ~100 liver devices, putting him perhaps in the top 5; another is at one of the most-respected, influential centers in the US and globally: “I hate flying their team all over the fucking country...I hate it...very little transparency...they could have just made it up...it feels like I’m talking to a used car salesman”; “there are a lot of people they pissed off”; “they play hardball...they’ve really alienated a lot of people... a lot of my friends have said, ‘No way, I’m not ever using those guys because of the way they do business’...they’re using OrganOX, and they’re doing fine.”

“I hate flying their team all over the fucking country”; sending extortionate invoices with no transparency

“I hate flying their team all over the fucking country. I hate it. I hate paying for private jets to fly their surgeons all over. I don’t know why we have to pay for all these ridiculous transportation invoices. And I hate when, now that they’ve got their own transportation company, they just started flying their teams and then sending us invoices. there’s very little transparency....we just get a piece of paper, an invoice that says you owe us \$20,000 for this flight going from here to here. It doesn’t say anything about what kind of aircraft it was, how long was it in the air...it’s just a number. They could have just made it up...I cannot stand working with them. It feels like I’m talking to a used car salesman.” - Executive at a pre-eminent academic transplant center, one of the highest volume in the world

“Push down your throat”; “lot of people they pissed off”

“...TransMedics acted as this hard sell, push down your throat approach to the surgical community saying this is our way or the highway, there are a lot of people they pissed off...” – Prominent surgeon, director of a leading academic transplant center

CEO/COO “play hardball” and alienated surgeons who now use other perfusion machines

“Typically, this is not how companies behave with somebody who helped them get FDA approval” - one of the investigators, essentially. It’s very unusual. We’re a very high-volume user but that’s Waleed’s method, he and Tamar Khayal, his COO....that’s how they roll. They play hardball. They’ve really alienated a lot of people and a lot of my friends have said, ‘No way. I’m not ever using those guys because of the way they do business.’ And my friends, of course...they’re using OrganOX, and they’re doing fine, and they’re happy with their volume.” - Transplant surgeon at a high-volume academic center

TransMedics, as we shall get to, is a cesspool of nepotism, an infrequent but reliable red flag. The CEO's sister, Amira Hassanein, runs the failed lung program. Customers described her as "confrontational" as her brother, allegedly coercing surgeons to accept and transplant organs they had already rejected as dangerous to their patient: "Waleed's sister is in charge of the lung program... we had some lungs...where the accepting center declined...the bronch was not good...the lungs were contused...these calls are 100% happening...I saw her on the line with a transplant surgeon...insisting that these lungs would have a good outcome...I was thinking, 'How do you even know that?'"...well beyond her scope to insist, badger a surgeon who made a decision that transplanting this organ could put her patient in jeopardy and to sit here for 15 minutes working to actively convince them to make a different clinical decision...I've never seen anything like that...confrontational, speaking over people...speaking as if she's their boss, their attending surgeon, insisting that the organ will have a good outcome...'you don't understand the machine, you don't understand the machine'...."

CEO's sister allegedly badgering people to transplant defective lungs

Q: "One surgeon told me that there are organs sometimes that he doesn't want to take, and he then gets badgered by Magdy Attia pressuring him to take them. Have you seen them try to force people to take marginal organs?"

A: **"Waleed's sister is in charge of the lung program....she is the one that I have witnessed do this. We had some lungs in that situation where the accepting center declined;** the bronch was not good. The lungs were contused when they were recovered and I was able to witness some of those conversations myself....these calls are 100% happening...**I saw her on the line with a transplant surgeon who's trying to weigh the clinical outcome for her recipient. And Amina would be on the line with her, just insisting that these lungs would have a good outcome after they were transplanted. I was thinking, 'How do you even know that?' This is confusing.** I'm also not a surgeon, but it just seemed well beyond her scope to insist, badger a surgeon who made a decision that transplanting this organ could put her patient in jeopardy and to sit here for 15 minutes working to actively convince them to make a different clinical decision, **I've never seen anything like that...confrontational, speaking over people, really speaking as if she's their boss, their attending surgeon, insisting that organ will have a good outcome** and that they're jeopardizing the patient's safety if they don't transplant it, and it's all because **"you don't understand the machine. You don't understand the machine."** – Executive at a leading OPO, longtime industry veteran

The problem for TransMedics is that its monopoly position was fleeting. Cheaper, better pumps and preservation devices have been approved, with a second wave in the FDA queue. The transplant field is ready to pounce – its “Qaddafi moment,” if we may, to keep our analogies consistent. The speed of the meltdown may be one for the ages as its largest users stampede. The most influential centers are the canary in the coal mine. Vanderbilt is a transplant pioneer and perhaps the highest-volume center globally. We spoke with a transplant surgeon who played a role in TransMedics pivotal liver PROTECT trial, as well as a transplant cardiologist. They thrashed TransMedics device, its management team, and have already eliminated its usage in lung, heart, and liver in favor of NRP, for a mere \$3K to \$10K per case. Roadkill and obsolescence shorts rarely play out at such lightning speed: “I don’t like the TransMedics people that much...a lot of people are very pissed off...because they are acting too greedy and predatory....they really hate it.”

People are very pissed off”; bait and switch

A: **“I don’t like the Transmedics people that much** - I was a [redacted] when they did the initial trial, the PROTECT trial. And, actually, when they were doing the trial, the way that they were advertising the product is like, well, once we get the trial, once we get approved by the FDA, we are going to sell or we are going to lease the pump and then you only have to buy all the stuff that you have to do for each case. So, that’s the way that they were advertising. But once it got approved by the FDA, again, I don’t know what happened at the upper level of the administration of the company, but they completely changed.”

Q: “They basically did a bait-and-switch with the investigators.

A: “Yes, they did. That’s what they did. And that’s the reason that, to be honest with you, **a lot of people are very pissed off with the company...a lot of people are very pissed off because they are acting too greedy and predatory. I really support that letter from that Congressman**. Again, a lot of people in different transplant centers are very pissed off with the company.... **I can tell you, I hate the CEO.**” - Transplant surgeon at Vanderbilt

Vanderbilt heart program refuses to use TransMedics

“Vanderbilt is the biggest heart transplant program not in the country but in the world. The volume is gigantic. **They do zero TransMedics. Zero because they are so upset**. They were extremely upset. So, they don’t use TransMedics at all. They hate it. They hate the company. They hate is so much. **They really hate it.**” -Transplant surgeon at Vanderbilt

Source: Scorpion Capital consultation calls with experts

UCSF, Massachusetts General, UCSD, and Cedars-Sinai are just a partial set of the large, current customers we spoke with who are on the verge of fleeing - all of it, we believe, yet to reflect in the coming quarters. With Vanderbilt gone and UCSF/MGH next – three of the top centers in the US and globally, by case volumes and influence – it is game, set, match. Per their figures, these four alone - just as a vignette of the field - use 255 OCS Liver units per year, which we peg as ~10% of TMDX total product revenue. UCSF and Cedars-Sinai, like others, used profane language to convey their fury at TransMedics and various executives by name. UCSF intends to wipe out usage as fast as possible in early 2025. Cedars-Sinai said they're actively shopping. UCSD said they use ~100 liver devices per year, and that "to be candid" they just installed an OrganOx device. Two senior individuals at MGH, a transplant hepatologist and hospital administrator, both indicated the relationship is at risk. MGH and Duke were key players in its pivotal liver trial, accounting for ~100% of device usage. Duke recently presented a paper stating the device is uneconomic, which "made Waleed's head explode."

MGH expects to cease OCS liver use in next few months in favor of NRP

Q: "Do you see a wave coming?"

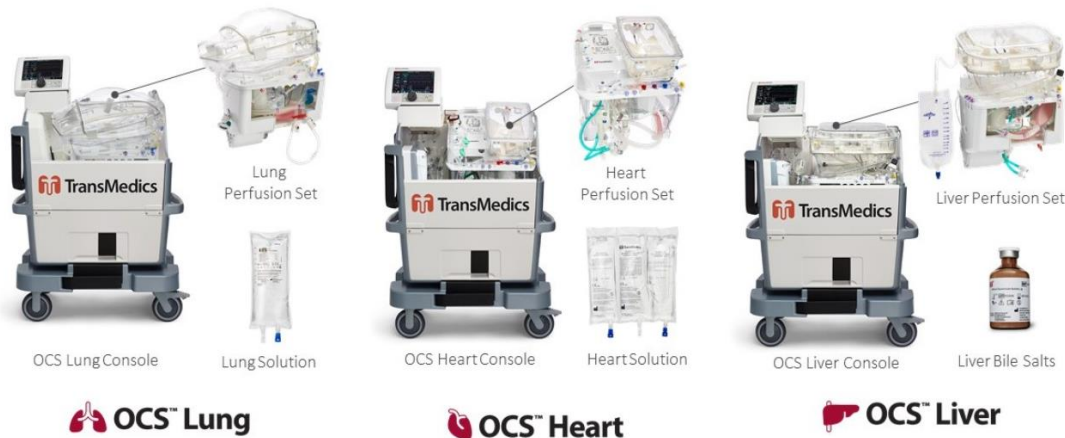
A: "Yeah, that's what's going to happen. It's not new technology, I mean, it's been around for a while. **Like NRP, I think it's going to be a huge game-changer**. And you see some of this initial data that's coming out even from Europe, and you're like, **"Why are we not doing this?" NRP is going to probably be standard. There's going to be no real utility in using an OCS in comparison to other cheaper, more widely available, and more preferable from a business-to-business standpoint—it's a no-brainer."**

Q: "When do you think your volumes will take the next leg down?"

A: **"For livers, probably in the next six months [...] I think we'll probably stop using it. It's gotten to that point [...] the market has shifted...the clinicians who have been very, very concerned about the poor business practices and misalignment of incentives with TransMedics, that there are now going to be other technologies...it's very promising that there's going to be a different option, not only from the actual product standpoint but just from a business standpoint, I would switch right away."** – Veteran transplant administrator/executive in a senior role at Massachusetts General Hospital

The speed of TransMedics meltdown is due to a simple fact: it has no clinical value proposition. Since the 1960's, static cold storage has been the standard of care and gold standard for organ transport and preservation. It's almost free, and surgeons indicate it works fine for the vast majority of organs. Organs are flushed with a cold preservation solution, then stored at 0-4 Celsius to minimize metabolism. TransMedics sells a "normothermic machine perfusion" (NMP) system, which uses a pump and heater to perfuse the organ at near-body temperature, which purports to extend organ preservation time, enabling longer transport distances plus functional assessment of biomarkers like lactate level. The bull case - a multi-organ platform - leaves out a few inconvenient facts: it flopped in Europe; has no kidney product, the largest market by far, as a cheap device has a lock with 90% share; lung, its first FDA-approval in 2018, is a colossal failure; heart, approved in 2021, is crashing and burning; leaving it, best case, as a one-trick pony for a single organ - liver.

TransMedics product line



The concept as well as the “intellectual property” for TransMedics perfusion system have been around since the 1930’s – a plastic bucket, some tubing, a little pump, a heater, and a novel therapeutic solution called “PlasmaLyte perfusate,” which we realized after an exhausting 30 seconds of research is an electrolyte brand sold online by Baxter, the same as Gatorade or Pedialyte, jacked up to \$400 for the IV drip bag. Some transplant surgeons told us their centers had rigged their own versions for a couple of thousand bucks – with duct tape after a Home Depot run, we assume.

TransMedics OCS device in the field



No one bothered to manufacture a warm perfusion pump for 50 years prior to TransMedics, as static cold storage worked fine. But you can't scam Medicare for a \$70-90K consumable if you sell ice, plus another \$50K to \$300K in air transport fees by sending three different private jets for the same organ. TransMedics pump is the kind of obscene, unnecessary contraption that can only exist in the US healthcare system – the Medicare equivalent of the \$10,000 toilet seat, \$400 screw, and other schemes by defense contractors to fleece the federal government. As a preview of a later section on the device's lack of a value proposition, we note scathing comments below – all three surgeons are from prominent centers, one of whom is regarded as perhaps the world's top KOL in liver transplant surgery.

Overpriced and being disrupted by new entrants and NRP

"Why should it cost \$70,000-80,000 for disposables? The disposables are essentially plastic tubings and plastic containers and cartridges, and that's more than the cost of a brand-new luxury car that you're paying for a set of things that you throw in the garbage, essentially. It is, no question, overpriced...there's lots of precedence for this sort of stuff happening in the healthcare industry....you have all these other devices that are coming in, and the price will go down...on top of which...there's no question NRP is picking up." – Transplant surgeon and transplant program director at a leading Northeast academic center

MGH calls perfusion pumps a commodity

"I don't think there's any magic to TransMedics per se. It's got cute bells and whistles; it travels. **Let's face it: there's no secret sauce in this.** Perfusion is in many ways perfusion and, in this case, normothermic perfusion is normothermic perfusion. It would be **very hard to ascribe a special quality to one modality necessarily over another.** If you can get that with one device comparably to another and logistically it's less challenging and certainly a whole lot less costly, then you go that way, **there's no question. So, I think there will be a low threshold to jump over."** – Transplant hepatologist in a leadership role at Harvard/Massachusetts General Hospital; key role in oversight of the liver transplant program

Zero value proposition; no difference in outcomes

"The problem with the marketing is that they've expanded use which doesn't need to be expanded. In other words, they're talking about an \$8 billion industry. That means that every organ would be pumped on the machine at \$100,000; **it's ridiculous. You're replacing a very cheap technology that costs a few thousand dollars with one that costs \$100,000**...one thing that you can take away from all of these studies also is that the **patient and graft survival are no different between the cold storage and machine perfusion...**" – One of the top liver transplant surgeons in the world, based at a leading academic center

Perverse incentives are central to understanding TransMedics business model. It exists solely as a creature of a preposterous Medicare reimbursement loophole unique to transplants – which regulators are racing to kill off, unbeknownst to bulls, taking TransMedics down with it. Private payor coverage is almost non-existent, as they're in on the joke. TransMedics is thus entirely a government pay scam – just like \$10K toilet seats. Medicare reimburses transplant centers for all reasonable and necessary organ acquisition charges, which are rolled up into each center's Medicare Cost Report. The rub: organ acquisition charges – which include TransMedics device and NOP fees - have no cap, as “reasonable” is undefined. An ex-TransMedics reimbursement executive detailed the nuances: “the structure – it's totally crazy...if they for pay for an OCS system on the NOP with the flight, all of these costs get paid at cost plus back to the hospital”; “Waleed will talk about it all the time...his investor calls...he will talk about how Medicare pays the cost...what are you guys worried about?”

Medicare Cost Report structure explanation – “it's totally crazy”

*“In the Gosar letter, it talked about overcharging. And going back to what the CEO would want to hide - reimbursement for organ acquisition, which OCS falls squarely into. **The way the structure – it's totally crazy....all of the costs to procure an organ, as long as they are reasonable and necessary, they go on the organ transplant worksheet or organ procurement worksheet on what's called the Medicare Cost Report**, which is an annual filing and every hospital in the country does whether they're a transplant program or not because that program has a specific worksheet that lists all of the costs of what they incurred to procure the organ. And that is paid at cost plus the overhead rate that varies between 17-25%, depending on the hospital. **So, everything they spent—so if they pay for an OCS system on the NOP with the flight, then all of those costs get paid at cost plus back to the hospital...**”* – Former executive in a key reimbursement role

CEO openly promotes the Medicare loophole

“Waleed will talk about it all the time. If you listen to any of his investor calls, he will talk about how Medicare pays the cost. **What are you guys worried about?..”** – Former executive in a key reimbursement role

With no reimbursement limit, TransMedics has free reign for price gouging – and transplant centers just pass it on to the taxpayer. A surgeon who leads a large academic transplant center: “Medicare because they’re so frickin’ stupid as a government bureaucracy...they have not put a cap...if TransMedics charged \$200,000 for the kit, I would be able to get reimbursed.” An executive at a major Organ Procurement Organization (OPO) – we shortly explain the structure of the US transplant system – called the “organ acquisition charge...a gray area...the same is true for TransMedics...you could exploit any charge...you can exploit the types of planes....you can exploit the surgeon fees...all of those things can be inflated.”

Medicare is “so frickin’ stupid” to reimburse via Medicare Cost Report; no cap on TransMedics cost

“I’ve served on medtech panels. And I’ve been extremely critical. Medicare, because they’re so frickin’ stupid as a government bureaucracy, they’re reimbursing transplant centers through their Medicare Cost Report...at some point, Medicare is going to, once there are other FDA-approved devices, say no, we’re not paying \$100,000 for you guys to provide the surgeon, the aviation company, and the device and the technicians...say my total costs were \$10 million and I did 40% of those transplants were Medicare patients. I would get \$4 million back from Medicare because that’s 40% of the charges that I had. What Medicare has not done so far is they have not put a cap on the amount of money per procedure that you’re doing. Theoretically, if TransMedics charged \$200,000 for the kit, I would be able to get that reimbursed right now because Medicare is so stupid that they’re not restricting it like other companies do like private insurers do. – Prominent surgeon, Director of leading West Coast academic transplant center

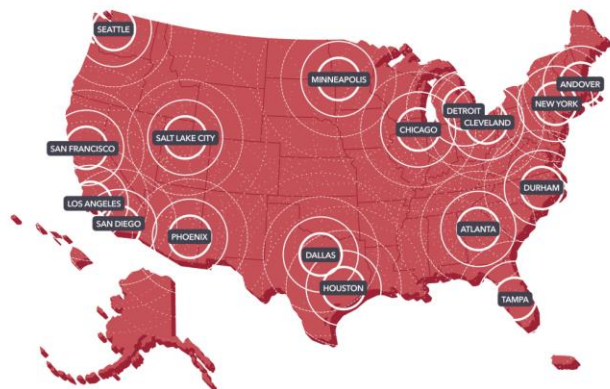
Organ acquisition cost is a “gray area”; ‘you could exploit any charge that they had’

“The organ acquisition charge itself is a gray area...the same is true for TransMedics...there is really a lot of latitude with what they’re charging, and there’s no standardization. So, when TransMedics comes in and says, look, you’ve got to use our guys, use our recovery surgeons - they just set that fee. And the OPO’s have to pay it and pass it through their transplant center...there is the opportunity to take advantage of that...You could exploit any charge that they had. You can exploit the types of planes that you use. You can exploit the surgeon fees. You can expense the pump time, the supply cost, transport cost—all of those things can be inflated. And so, I used to pay the helicopters and the jets for all transplants. And we had an airport that was 12 miles away. By ambulance, it took 10 minutes, and by helicopter, it took 7. And they would require a helicopter that costs \$11,000 versus an ambulance that costs \$1200. An ambulance going a short distance was \$400. If they turned the siren on, it was \$900. Same distance..” – Longtime executive at major OPO (organ procurement organization), and on UNOS committees

If Medicare will reimburse any and all organ acquisition charges, and you're already selling \$70-90K disposable kits for your rig, why not look for other ways to run up the tab? TransMedics thus rolled out its National OCS Program ("NOP") in 2022 – its end-to-end organ procurement and transport service. The company established sixteen "launch points" around the US, staffed with a procurement surgeon and a device operator. The move was a bait-and-switch. As alleged by Congressman Gosar and confirmed by numerous transplant centers we interviewed, TransMedics would no longer train hospital staff to use their device, forcing them to pay for its dedicated device technicians, surgeons, and aircraft. The CEO on a 2023 call: "This is a fact of organ transplant. All aviation and logistics transport are fully reimbursed through the same mechanism of organ acquisition...there's no limit...there is no limit."

TransMedics NOP service – excerpts from investor presentations'

NOP Hubs to Provide Broad Geographical Coverage



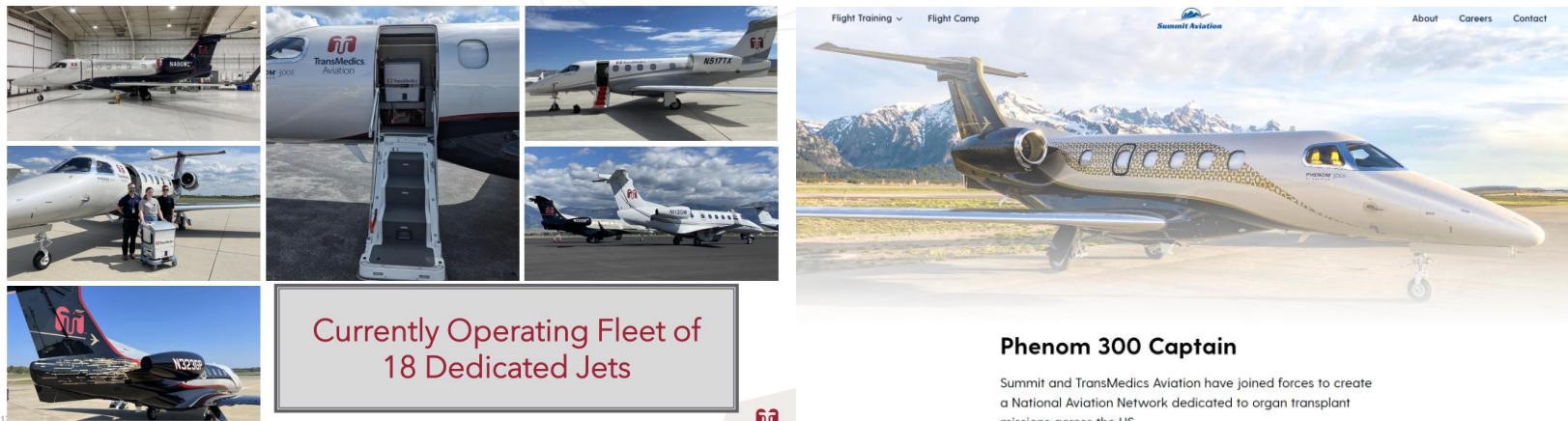
NOP is a Fully Integrated End-to-End Transplant Network, Operating Seamlessly



- OCS™ Product
- Procurement Surgeons
- OCS Clinical Specialists
- 17 Hubs Across the US
- Airplane Fleet
- Pilots
- Maintenance Hub
- Ground Transportation
- 24 / 7 Managed by the Logistics Command Center

TransMedics went full bling with what appear to be 18 Embraer Phenom 300 jets, which seat 11 with a 2,000 mile range – naturally, as one can jam Medicare with a far higher hourly rate with a \$13MM jet than, for example, a PC-12 with similar range at 10-15% of the cost, typical of organ transport services. NOP fees can run \$25K-\$100K per case, plus a \$70-90K disposable, per CEO Q&A on a call. Centers told us they’ve been billed \$300k-\$400K without explanation. The purpose of the jets, at face value, is disingenuous: the company is a one-trick pony in liver, with liver at 73% of total revenue in the most recent quarter. Given the way the national organ allocation system works, livers rarely travel far. Various centers told us that 100% of their liver transplants involve a donor and recipient within a driving radius of 250 miles, as we detail in chapter 6. Thus, no jets are necessary, and neither is a device to enable extended transportation times – making it obvious that TMDX is just a Medicare rig.

TransMedics aircraft fleet -investor presentation, aircraft subsidiary website



Currently Operating Fleet of
18 Dedicated Jets

Phenom 300 Captain

Summit and TransMedics Aviation have joined forces to create a National Aviation Network dedicated to organ transplant missions across the US.

Data as of 10/28/2024

The TransMedics bull case boils down to a simple hope: that the Medicare organ acquisition charge boondoggle continues - into perpetuity. In reality, the abuses it has spawned are too extreme to avoid scrutiny. We spoke with the CEO of major organ procurement organization, who shared color about an unfolding investigation into the industry's practices, and who implied that TransMedics is at risk – “there's so much that's going on to be found out...do you know everything that's going on with the OIG and DOJ and the IRS...it's deeper than [the Washington Post]...here's the drama...I recently interviewed with the Department of Justice, the OIG, and the IRS...and with the DOJ out of Washington and out of New York, in particular...and then the state attorney general...Medicare fraud...the most prominent attorney general that's getting involved is Virginia....”

Multiple regulatory and law enforcement agencies at a federal and state level allegedly investigating industry practices

Q: “A surgeon I just talked to said I'll tell you who their whores are. He mentioned a couple of very high-volume users. It sounds to me like **a bribery, pay-to-play kind of thing.**”

A: **“There's so much that's going to be found out. Do you know everything that's going on with the OIG and DOJ** and the IRS?”

Q: “I read that Washington Post article.”

A: **“It's deeper than that, and I'll let you know because I've been interviewed for it.”**

Q: “Walk me through what you're seeing. And how TransMedics may be a part of this stuff that the Washington Post is talking about as far as kickbacks.”

A: “Here's all the drama. **I recently interviewed with the Department of Justice, the OIG, and the IRS. And with the DOJ out of Washington and out of New York, in particular. And then the state attorney general** is getting involved in this as well because if there's Medicare fraud, then there's Medicaid fraud. And probably **the most prominent attorney general that's getting involved is Virginia**, and the reason why that's important is because UNOS is there, and so is A-OPO, which is the OPO governing body, not a governing body but a trade association, and they also have a terrible OPO based there.”— CEO of a major OPO (Organ Procurement Organization), longtime industry executive

The OPO CEO indicated a parallel investigation by the OIG, separate from the DOJ, focused on the Cost Report, the loophole at the center of TransMedics pricing. He stated that regulators will, for the first time, define “reasonable cost” and close the loophole – perhaps a 2025 event: “they’re going to investigate every single OPO...every single transplant program”; “their investigation has to do with the Cost Report...how we’re reimbursed...overcharging the whole time...there just hasn’t been any oversight on it...things are changing so fast...they’re going to come out and fix things...absolutely, it’s been getting abused...the government is going to come down, and they’re doing to start setting what’s reasonable cost.”

OIG investigating every key transplant center and OPO’s as part of investigation into overcharging via Cost Report

“OIG recently announced that they’re **going to investigate every single OPO, and they’re going to investigate every single transplant program, and it’s going to happen by the end of 2025**. So, that announcement came. Then you’ve got DOJ, OIG, and the IRS investigating some of the OPOs. And **a lot of what their investigation has to do with our Cost Report. It has to do with how we’re reimbursed**. So, we’re a cost-based reimbursement model, one of the few left in the system. If my organization does 200 organ transplants a year and 100 of them are kidneys, 50% of my overhead is pulled toward kidney, so it’s reimbursed on the Medicare Cost Report. At the end of the year, if I say a kidney costs \$60,000 this year and it ends up costing our organization \$65,000, I’d say, federal government, you owe me 5 grand times however many kidneys I transplanted. The same on reverse. If I can do over 55 grand a year, I’d say I owe you 5 grand times however many kidneys I transplanted **because I’ve been overcharging the whole time.**” – CEO of a major OPO (Organ Procurement Organization), longtime industry executive

No oversight of “reasonable cost” cost bucket: “getting abused” and regulators will fix

Q: “You’re saying the origin of the problem is that the way the law and the reimbursement works is that the OPO’s or the transplant centers are able to pass all reasonable cost through to Medicare.”

A: “Yeah, I think that’s part of it. **There just hasn’t been any oversight on it. And then things are changing so fast...they’re going to come out and fix this.**”

Q: “You’re saying that if an OPO does like 100 transplants, then they have another \$5 million in overhead costs, that they just divide that \$5 million in overhead costs by those 100 transplants, and that gets tapped onto the organ acquisition cost and **people are dumping all kinds of crazy stuff into that overhead bucket. Is that right?**”

A: **“Yup, absolutely. Absolutely. It’s been getting abused. I think the government is going to come down, and they’re going to start setting what’s reasonable cost.”** – CEO of a major OPO (Organ Procurement Organization), longtime industry executive

He emphasized the exploitation of “Reasonable Cost” as the heart of TransMedics scam, calling it out as a creature of a perverse reimbursement rule, stating that regulators are locked in and ready to blow up its gravy train - “the glory days of we’re going to charge \$100,000 for a TransMedics device, and you have to use our people and...our expensive jet – I think all that is over...I think the gravy train is coming to an end, honestly”; “they’re going to come in because everything in transplant is based on what’s a reasonable cost...the government is working their way down...this stuff is all new...they’ve never established what reasonable costs are...they’re going to come in and establish ‘reasonable cost’...and now, all of a sudden, the Hawker 800 that costs \$12 million does not make sense anymore because that’s not reasonable cost.”

Regulators about to crack down on the TransMedics gravy train by enforcing “reasonable cost”

“So, I really think the government is going to come in, and they’re really going to shake this up, and I think the glory days of we’re going to charge \$100,000 for a TransMedics device, and you have to use our people, and you’ve got to use our expensive jet - I think all that is over. I think the gravy train is coming to an end, honestly. I think the government is going to really come in.” - CEO of a major OPO (Organ Procurement Organization), longtime industry executive

Regulators about to crack down on the TransMedics gravy train by enforcing “reasonable cost”

Q: “So, you think the government is going to wipe out TransMedics’ gravy train?”

A: “I think so. I think they’re going to come in because everything in transplant is based on what’s a reasonable cost. When you look at this stuff, you look at TransMedics and you look at this logistics stuff and things like that, the expense recovery, it’s very transparent what it costs. It doesn’t cost a ton and it’s used a bunch. And then you work your way down to devices and like TransMedics, and it’s like, okay, well on a medical device, it costs a lot, it’s somewhat transparent, and used kind of a lot. And then you go down to logistics, one step further, there’s no transparency, it’s used all the time, and it’s extremely expensive. I think the government is working their way down. Because what CMS does is like, what’s a reasonable cost? Well, this stuff is all new, so they’ve never established what reasonable costs are. I think what they’re going to do is I think they’re going to come in and they’re going to establish “reasonable cost.” And now, all of a sudden, the Hawker 800 that costs \$12 million does not make sense because that’s not reasonable cost.” - CEO of a major OPO (Organ Procurement Organization), longtime industry executive

In Feb 2024, the Washington Post broke the news of a “wide-ranging federal investigation” involving “US attorneys in various parts of the country” – the one for which, we assume, the OPO executive was interrogated. The authorities raided the homes of ten CEO’s of Organ Procurement Organizations, known as OPO’s. A surgeon at a pre-eminent center told us that the Post may now be working on a story about TransMedics, stating that they have been “reaching out to some transplant surgeons across the country, like a very prominent surgeon...I know that a reporter was reaching out to a surgeon because they were working on a report.” We suspect the surgeon spoke with the reporter, given his stature in the field: “I heard that the Washington Post is working on a report because there is some concern that companies are giving kickbacks to some surgeons or hospitals for using the machine”; “this is something that is not published...I heard that there is a reporter working on TransMedics, specifically.”

Washington Post, Feb 26, 2024

U.S. launches probe into possible fraud by organ collection groups

Washington Post reporter allegedly reached out to surgeons regarding alleged kickbacks from TransMedics

A: “I heard a little rumor, but again, this is rumor, okay? **I heard that the Washington Post is working on a report because there is some concern that companies are giving kickbacks to some surgeons or hospitals for using the machine.”**

Q: “I saw that article. But that was talking about OPO’s. Was that talking about TransMedics—they didn’t mention TransMedics.”

A: “No, no, no. **This is something that is not published. I heard that there is a reporter working on TransMedics, specifically.** This is not about OPO’s”

Q: **“You heard the Washington Post is working on TransMedics getting kickbacks?”**

A: **“Yeah, that’s what I heard. But again, this is just a rumor that I heard from some people, that they are working on a report.”**

Q: “When did you hear this about the “Washington Post?”

A: “I heard that a few months ago. **The Washington Post was reaching out to some transplant surgeons across the country, like a very renowned transplant surgeon; I know that a reporter was reaching out to a surgeon because they were working on a report.”** - Transplant surgeon and division chief of the liver program at a world-renowned academic center

The surgeon indicated the Washington Post inquiries were recent, and while oblique, provided specifics about the reporter's line of questioning that lend credibility: "because everybody is like, how is it possible that some of these hospitals are doing so many TransMedics? Because financially, it's not possible to do too many...usually you tend to be believe that centers that use the device a lot – maybe they have some incentive to use it that much...Houston, maybe...Henry Ford, maybe...." Although he stated several times that this was rumor - "I heard a little rumor, but again, this is rumor, okay?" – we suspect that numerous surgeons, OPO's, and transplant centers in the field have been interviewed by reporters and/or federal/state investigators, as several told us directly that they had been contacted and/or questioned. That something "big" may drop seems to be an open secret – transplant surgery is a small field with only 400 surgeons in the US for liver and ~50-80 responsible for the vast majority, and even fewer for heart and lung.

Washington Post reporter allegedly reached out to surgeons regarding alleged kickbacks from TransMedics

Q: "And they were trying to determine if TransMedics was giving kickbacks to surgeons?"

A: "Surgeons and hospitals. **Because everybody is like how is it possible that some of these hospitals are doing so many TransMedics? Because financially, it's not possible to do too many.**"

Q: "Were there any other names that were thrown around as far as who could be getting kickbacks?"

A: "Those are all rumors, right? **Nobody told me Ghobrial is getting a kickback, or Marwan is getting a kickback or the guy at MGH is getting a kickback. I heard a rumor that some centers are getting a kickback and that's the reason that they are working on a report at the Washington Post. Usually, you tend to believe that the centers that use the device a lot - maybe they have some incentive to use it that much.**"

Q: "If you had to speculate, who do you think those centers or those doctors might be?"

A: **"Houston, maybe; Henry Ford, maybe. Probably, I would say that those are the two centers** that I would be more..." - Transplant surgeon and division chief of the liver program at a world-renowned academic center

As mentioned, we find that nepotism - of both the family and tribal variety - is an infrequent but reliable indicator of fraud. TransMedics bears a striking similarity with Nevro (NVRO), a device company and Scorpion's first report – both are fueled by a clique of high-volume users of Middle East background. We didn't flag it then, given the sensitivity, but have no such hesitation now. Frauds are often based on ethnic networks, such as ponzi schemes – and sunshine is the best disinfectant. Two sell-side TMDX analysts also covered NVRO. The stock has fallen 98% since our 2021 report, from \$6B to \$130MM market cap (0.3x sales). Nevro's ex-CEO was of Middle East background, and we believe its speakers program was a kickback scheme. The top users of its device, per OpenPayments and our research, were "coin-operated" physicians of similar origin. Surgeons who stopped using the device due to safety problems delicately informed us of the dynamic. We include a quote we withheld from the report.

Nevro stock chart, 2020- present



"When Nevro developed their cult, and I don't want to seem...they got a lot of Middle Eastern doctors in there...they got this cult of Middle Easterners, Persians, and Arabs and stuff. And this is not racist – they're all good docs. When we had these think tanks with Nevro, I was the only white guy there. I think that that added to the cult mentality, like this is our thing, this is our bandwagon. It was weird... if you look at the doctors in the forefront...the white-shoe reputable people....that's not who they got....they brought in a bunch of guys – I wouldn't say they were outcasts, but...I am blown away because every doctor I know that does a lot of implant volume doesn't use Nevro except for a small cult." – Surgeon and ex-Nevro user

Egypt punches above its weight when it comes to organ transplants, and seems to export a large number of surgeons to the US and UK, who in the US seem to be rather heavy TransMedics users. We suspect a surgeon named Sir Magdi Yacoub had a lot to do with it. Now 89, he moved to the UK after medical school in Cairo, becoming the UK's most famous transplant surgeon. Yacoub is a flamboyant publicity hound, prone to photo ops with Royals and “daring” surgical stunts that earned him international condemnation. Articles state he put a baboon's heart into a child; did open-heart surgery on a terrorist; and did a heart transplant on a 10-day old infant, sparking global rebuke from surgeons. Predictably, when the press looked into why 29 infants allegedly died at his hospital, they reported that it had the second-worst outcomes in the UK – “the majority of the operations...are believed to have been carried out by the pioneering surgeon Sir Magdi Yacoub.”

“Sir Magdi recognises he was the man for a time when clinical governance was less tight and ethical positions still fluid. Would his pioneering be possible today? Definitely not, he says.” – BMA's The Doctor



Harefield launches death rates inquiry

Nationwide survey finds Sir Magdi Yacoub's hospital scoring better only than the Bristol Royal infirmary in heart operations league

TMDX originally filed to go public in 2007. The S-1 listed Magdi Yacoub as a Board member. A few weeks ago in early December 2024, what we believe to be the “inner circle” at TransMedics appears to have traveled to Cairo for the Egyptian Liver Transplant Society meeting. The group included Yacoub and TransMedics CEO, as well as Magdy Attia, who appears to run its liver program and who we think oversees its surgeons. Ex-employees told us he is unlicensed to practice medicine in the US, yet seems to do exactly that, along with other dubious conduct alleged by customers, that we’ll get to. The entourage included two of TransMedics highest-volume customers, Rafik “Mark” Ghobrial at Houston Methodist and Marwan Abouljoud at Henry Ford in Michigan – both of whom were mentioned by a surgeon in the context of a potential WashPo investigation into kickbacks. Attia was listed on a panel: “How To Approach A Sick Child In Need For Transplant.”

Egyptian Liver Transplant Society, Dec 5-7 2024, Cairo



A number of ex-employees brought the nepotism to our attention. Unusually, we were able to locate zero information on the CEO's sister's educational background or professional qualifications to run their liver product. We located a paper published by Ain Shams university in Egypt that appears to list her as a co-author. Employee color below: “just the toxic culture, this weird secrecy...there's a lot of nepotism...zero experience...there is some substance about favoring a particular culture over another...even down to some of their drivers...it always seems to be Middle Eastern”; “his sister, heads of other divisions, and then technical officer were Egyptian...some others by marriage...they may have some other relatives...they make decisions among themselves and we didn't know what's going on”; “whenever someone's sister is running the lung program...it was just like Larry, Mo, and Curly running things back there....”

“Toxic culture” with “a lot of nepotism”

A: “It's just the toxic culture, this weird secrecy. There's a lot of nepotism. Waleed's sister is the product director for the lung division. What Amira says goes as far as lung cases go. The product director for liver - I want to say that his son is also in the company somewhere. Ahmed is his first name. .”

Q: “I see him on LinkedIn. He's been there a months. He just graduated from college.”

A: “Yeah. So, zero experience...it's hard to know that is it but I think there is some substance about favoring a particular culture over another. Even down to some of their drivers, it always seems to be Middle Eastern...and it's typically all men, never women.” – Former employee

Inner circle allegedly “make decisions among themselves.”

“The CEO, his sister, heads of other divisions, and then the technical officer were Egyptian. Some others by marriage; they may have some other relatives. They make decisions among themselves and we didn't know what's going on.” – Former senior employee in a regulatory affairs role

Sister running the lung program allegedly “has no formal education”

“Whenever someone's sister is running the lung program - things like that. They just built out without knowing what they were doing. And if my team calls their 1-800 number, it was just like Larry, Mo, and Curly running things back there, trying to get a case done.” - Transplant surgeon at a high-volume CA academic center

The three global hotspots for organ trafficking are China, India, and Egypt – the latter being “the largest hub of organ transplant in North Africa” and a top transplant tourism destination, per Wikipedia. In 2016, Egypt busted the “biggest illegal organ trafficking ring in the country’s history.” The 25 individuals arrested included professors at Cairo and Ain Shams universities – Egypt’s two main academic institutions. We allege no link to their alumni, but as short-sellers we take note - with a small asterisk - of origin stories from high-risk jurisdictions. TMDX’s entire “inner circle” appears to have done all of part of their medical training at these two schools: the CEO, the Chief Commercial Officer; the head of its liver program and “Medical Director” who oversees its surgeons; and others.

Press coverage of Egyptian organ trafficking abuses

Medical professors in alleged organ trafficking racket

Egyptian court sentences 41 organ traffickers to up to 15 years in prison

Egypt struggles to stem organ trafficking

Organ trafficking in Egypt: ‘They locked me in and took my kidney’

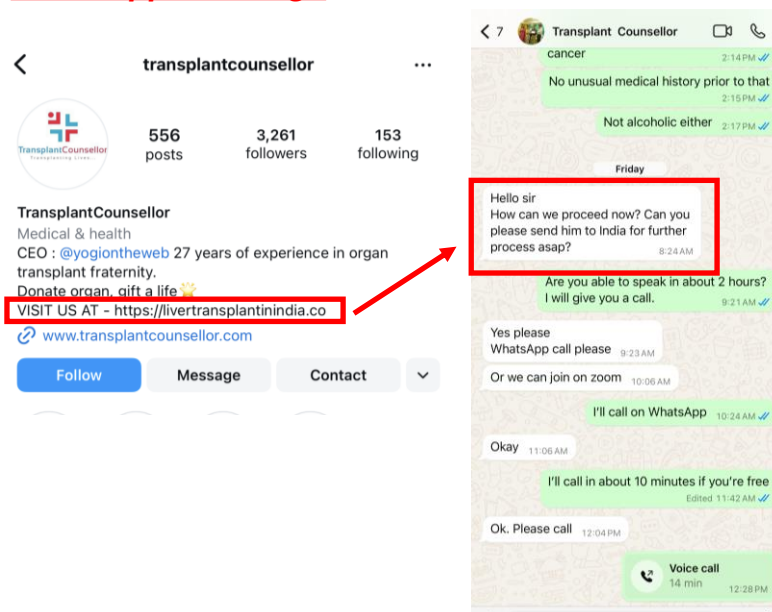
Egypt’s organ traffickers: ‘I woke up screaming’

Given the high-risk nature of the field, we took note that CEO Waleed Hassanein, “super-user” Mark Ghobrial at Houston Methodist, and ex-Board member Magdi Yacoub were on a panel with Dr. Arvinder Soin, a prolific liver transplant surgeon in Delhi, India. Soin is affiliated with a transplant middleman who has an aggressive social media presence promoting transplant tourism. As we detail in later section, a TransMedics surgeon recently imported from India appears on this middleman’s website. We traded WhatsApp messages and spoke with the middleman, who offered right away to facilitate a dual liver and kidney transplant, and asked us to send the patient immediately to India. That broker’s Instagram profile links directly to Dr. Arvinder Soin’s website: www.livertransplantindia.com.

Egyptian Liver Transplant Society, Dec 2024, Cairo



Transplant middleman Instagram profile; our WhatsApp exchange



A former employee in a regulatory role alleged a number of troubling interactions with the CEO, such as allegedly being pressured to doctor adverse event reports to downplay any link to TransMedics device, as we cover in the last chapter. The ex-employee further alleged being pushed to find ways to sell the device in countries where it is not approved; and shared grave concerns about a team of “six or seven” transplant physicians who allegedly visited from China to train on the device – “the leader was a German doctor...he was not able to continue medical practice in Germany and then went to China...he was saying he was trying to be a pioneer of transplants at this Chinese hospital....I mean, the way he spoke...somewhere between Wuhan and Beijing...when the Chinese doctors disappeared, we felt kind of easy because we didn’t want to do anything illegal...exporting to any country illegally could shut down the business.”

Employees allegedly asked to find ways to sell the device “countries without approval”

Q: “Were you ever asked to do anything from a regulatory standpoint that you thought was in a gray area or improper?”

A: **“They tried to sell products to countries without approval - how we can do it?”**

Q: “They asked you to look into how you could do that?”

A: “[Laughs] That’s funny, isn’t it? **I was asked what are the kinds of ways they can sell without approval. We got some Chinese doctors who came over for training, and [redacted] was involved with that.**”

Q: “Who asked you to look at countries you could sell to without approval? Waleed?”

A: **“Yeah, from the top...Chinese doctors came and went through the training. This group of Chinese doctors, I think, it was six or seven people who came, and the leader was a German doctor. He was not able to continue medical practice in Germany and then went to China and where was working for several years. He was saying he was trying to be a pioneer of transplants at this Chinese hospital.”**

Q: “This German doctor, he was not allowed to practice in Germany? It’s kind of weird that he went to China.”

A: **“We didn’t know but that was our suspicion – that was the suspicion among those who attended the Chinese training session.”**

Q: “You guys were worried that this guy was not allowed to practice in Germany?”

A: **“Right, right. I mean, the way he spoke...he said that he was helping them to establish a transplant center in the southern area of China somewhere between Wuhan and Beijing...when the Chinese doctors disappeared, we felt kind of easy because we don’t want to do anything illegal...exporting to any country illegally could shut down the business.”** - Former employee in a regulatory affairs role

Part 1. TransMedics National OCS Program (NOP): an unsustainable, anti-competitive scheme based on what we believe to be coercive tying, price extortion, and retaliation.

On Feb 21, 2024, Congressman Paul Gosar wrote CEO Waleed Hassanein a detailed letter alleging “anti-competitive business practices deployed by TransMedics” tantamount to its medical devices “being held hostage.” The letter was obtained and published by a news site. The stock fell ~20% from \$89 to \$72 in days, hovering near the lows for several weeks before spiking again.

[Congressman Paul Gosar letter to TransMedics CEO, Feb 21, 2024](#)

[Introduction from the letter, link to full text below](#)

EXCLUSIVE: Rep. Gosar Accuses TransMedics Of Holding Lifesaving Medical Devices ‘Hostage’ — Vows Oversight



Photo by Drew Angerer/Getty Images

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COMMITTEE ON
OVERSIGHT AND ACCOUNTABILITY
SUBCOMMITTEES
HEALTH CARE AND FINANCIAL SERVICES
NATIONAL SECURITY, THE BORDER AND
FOREIGN AFFAIRS

February 21, 2024

Dr. Waleed Hassanein
Chief Executive Officer
TransMedics, Inc.
200 Minuteman Road STE 302
Andover, MA 01810

Dear Dr. Hassanein,

As a member of the United States Congress serving on the House Committee on Oversight and Accountability, I am writing to you regarding anti-competitive business practices deployed by TransMedics. Furthermore, as a doctor who served patients for a vast majority of my career, I have questions about lifesaving technology being held hostage by a major corporation.

Gosar accused the CEO of a bait and switch after FDA approval when TransMedics “began to change the entirety of its business model.” He detailed a number of specific practices tantamount to extortion, such as a spike in the cost of disposable cartridges from \$7K to \$60K. The company also ceased training centers on the device and forced them to pay for a TransMedics technician who would now solely operate the device and “labeled this the National OCS Program, or NOP.” Thus, hospitals could no longer purchase OCS as they had previously done, nor receive training on it. He further stated that centers “were mandated by TransMedics to utilize the OCS no less than three to five times per month” – “if volumes were not maintained, they would not offer more cassettes for purchase.”

Congressman Paul Gosar letter to TransMedics CEO, Feb 21, 2024

Once FDA approval was achieved, TransMedics began to change the entirety of its business model. Transplant centers that participated in the trial and previously purchased the equipment were then mandated by TransMedics to utilize the OCS no less than three to five times per month, or centers were forced to return the OCS machine without reimbursement of their upfront purchase capital. TransMedics informed these initial centers that if volumes were not maintained, they would not offer more cassettes for purchase. Almost immediately, the cost of the one-time, disposable cassette utilized to encompass the organ during transportation and perfusion increased from the initial \$7,000 to greater than \$60,000 per disposable cassette. Additionally, transplant centers would no longer receive training for their medical teams to utilize the device at their discretion. Instead, TransMedics created its own team of individuals as the sole source for any initiation of the OCS device and labeled this the National OCS Program, or NOP. Transplant centers could no longer purchase the medical device, rather lease the device and request the necessary TransMedics personnel for any OCS heart, liver, or lung organ recovery. Costs for TransMedics surgical recovery are approximately \$20,000 per request.

A key allegation stated that TransMedics coerces hospitals to use its recently-acquired expensive aircraft as part of its new NOP “full service” business model: “in addition to the significantly increased costs borne by transplant centers to utilize the device itself, TransMedics is pressuring hospitals to use their more expensive aircraft, not only for transportation of organs, but also for the convenient of their staff and equipment.” Gosar indicated that “transplant centers have long relied on private air transportation” using inexpensive “no-frills older-model aircraft based near their hospital” to “keep transportation costs low.” However, TransMedics made “significant investments in new model aircraft similar to the types utilized by luxury air charter companies such as Netjets.”

Congressman Paul Gosar letter to TransMedics CEO, Feb 21, 2024

Given the very short time a human organ can remain viable in transport, transplant centers have long relied on private air transportation to ensure that organs make it to their recipients in time. Historically, transplant centers have sourced their own air transport, typically utilizing well-maintained, but "no-frills," older model aircraft based near their hospital. This enabled hospitals to minimize aircraft repositioning and keep transportation costs low. Recently, TransMedics has made significant investments in new model aircraft similar to the types utilized by luxury air charter companies, such as NetJets. In addition to the significantly increased costs borne by transplant centers to utilize the device itself, TransMedics is pressuring hospitals to utilize their more expensive aircraft, not only for transportation of organs, but also for the convenient movement of their staff and equipment. In some cases, TransMedics has been reported to hold its

Gosar then detailed a startling allegation: “TransMedics has been reported to hold its device hostage, withholding access to lifesaving technology unless hospitals agree to utilize TransMedics aviation’s costly planes” - a “continuous forced bundling of services” with centers “held hostage by public company that has lost its true north.” In response to “protests” by hospitals alarmed at “significantly increased costs” of the NOP service, TransMedics CEO guided them to aggressively exploit a loophole in Medicare reimbursement for organ transplants, which we shall cover in more detail. Gosar cited the CEO’s comments on an earnings call: “not to worry, the American taxpayer will foot the bill through Medicare”; “this is a fact of organ transplant, all aviation and logistics costs are fully reimbursed...there is no limit.”

Congressman Paul Gosar letter to TransMedics CEO, Feb 21, 2024

movement of their staff and equipment. In some cases, TransMedics has been reported to hold its device hostage, withholding access to their lifesaving technology unless hospitals agree to utilize TransMedics aviation's costly planes. Protests related to the significantly increased costs are often met by the same response from TransMedics: “not to worry, the American taxpayer will foot the bill through Medicare.” In fact, on your recent Q3 earnings call, when pressed about the ability of hospitals to absorb the significantly increased costs associated with TransMedics OCS and TransMedics aviation you stated: “This is a fact of organ transplant, all aviation and logistics costs are fully reimbursed...there is no limit” (emphasis added).

Gosar’s letter ended with four specific questions for the CEO. In particular, he asked “Has TransMedics ever used private aircraft to transport staff and equipment without the purpose of transporting organs”; and “Has Transmedics ever denied a transplant center access to your life saving device unless they use TransMedics aircraft and pilots?”

Congressman Paul Gosar letter to TransMedics CEO, Feb 21, 2024

1. Although it is well understood that the use of private aircraft is necessary to ensure that human organs reach their recipient in time, my staff has received allegations that TransMedics uses private aircraft for convenient transportation of their staff and equipment, where no such urgency exists. Has TransMedics ever use private aircraft to transport staff and equipment without the purpose of transporting organs?

4. We understand that TransMedics has made significant investments in new model aircraft and has placed pressure on their hospital customers to utilize their aircraft, despite protests from hospitals that TransMedics Aviation carries significantly higher costs than their current providers. Some transplant centers have reported being pressured to use TransMedics' captive aircraft, at nearly double the cost, or risk losing access to TransMedics' life saving device. Has TransMedics ever denied a transplant center access to your life saving devices unless they use TransMedics’ aircraft and pilots?

Five days after Gosar’s letter, TransMedics CEO Hassanein foamed at the mouth in a fiery reply blasting the “extremely serious accusations” as “hostile and accusatory”; as “misinformation and false statements from...a competitor”; and as “inaccurate third-hand reports and unidentified allegations” – lest one confuse them for mere second-hand ones. The CEO’s denial was absolute and unambiguous: “the allegations of anticompetitive business practices are unfounded...none of these actions are practiced by TransMedics.”

TransMedics CEO Waleed Hassanein reply to Congressman Gosar’s letter Feb 26, 2024

We are writing in response to your letter dated February 21, 2024, related to TransMedics, Inc. Your letter contained extremely serious accusations and we are responding to these accusations with the same level of seriousness and with factual evidence. Respectfully, we are very disappointed that you have been provided with inaccurate information about TransMedics and the transplantation system, which appears to have led to inaccurate conclusions.

While we appreciate the opportunity to address the issues raised in your letter, we cannot do so without noting that the hostile and accusatory statements in the letter suggest that you reached inaccurate conclusions based on misinformation and false statements from what appears to be a transplant logistics competitor of TransMedics. We are very disappointed

Similarly, the allegations of anticompetitive business practices are unfounded. Anticompetitive conduct, according to the antitrust laws, refers to practices that unfairly impede competition. None of these actions are practiced by TransMedics. We have always

Hassanein vehemently denied each specific allegation in turn, thereby underscoring the gravity of the charges Gosar leveled: that he “prohibited transplant centers from training their personnel on the OCS procedure” – “this statement is not correct”; stated that “transplant centers had the option to either use NOP or continue to operate the OCS on their own”; and that “it couldn’t be further from the truth” regarding “pressuring hospitals to utilize our more expensive logistics.”

TransMedics CEO Waleed Hassanein reply to Congressman Gosar’s letter Feb 26, 2024

- Your letter asserted that TransMedics prohibited transplant centers from training their personnel on the OCS procedure. This statement is not correct. To-date, we have trained more than 400 transplant retrieval surgeons for heart, lung, and liver transplants across the U.S. Importantly, given our clinically focused mission, we are offering to train transplant fellows at academic transplant centers by our senior retrieval surgeons on how to perform OCS donor procurement surgery.
- Your letter accused TransMedics of pressuring hospitals to utilize our more expensive logistics. This couldn’t be further from the truth. TransMedics presents the logistics quotation to transplant centers using our NOP service. The transplant centers then have the final say on whether they want to use our services or not. We are confident,

Our research indicates that not only are the CEO's denials utterly false, but that the conduct detailed in Gosar's letter is far more serious than he is aware. The allegations and key points in the letter were confirmed by numerous former employees, as well as by major transplant centers that we spoke to across 30+ interviews. We start with an ex-logistics manager who played a key role in managing TMDX's flight operations as part of the NOP, which he told us is "essentially like Amazon for organs." He began by saying that "in August 2023, we didn't have a single plane" and "wouldn't touch logistics" – "we would contract out" and "hire third parties," but that it all changed in a key moment when "a decision was made: we need to bring all the logistics in house." In a revealing moment, he stated TransMedics is not even really a medical device company: "instead of selling devices, we started selling a service."

Not really selling devices but "selling a service"

"Instead of selling devices, we started selling a service, and we established what we call a National OCS Program, NOP, where we are essentially like Amazon for organs. We have 17 hubs around the country where we have our devices; we have our specialists and our surgeons. Then we bought our own planes. We have a fleet of 14 planes; the fleet is growing. This has been built since last year. **In August 2023, we didn't have a single plane.** It was managed by third-party providers, and now we have brought all that in-house." – Ex-logistics employee at Transmedics

Sudden pivot from refusing to touch logistics to basing the model around moving aircraft and organ transport

"We would contract out all of the logistics and hire third parties for the aviation. We would hire third parties for the ground transportation and so on. **We wouldn't touch logistics; somebody else would take care of it for us. A decision was made: we need to bring all the logistics in-house."** – Ex-logistics employee at Transmedics

Centers are now forced to use the organ procurement service and TMDX aircraft, according to the ex-employee, contradicting the CEO's denials: "they have to use our clinical service." He stated they no longer sell the devices in the US, but still do so abroad – **a telling admission** given the CEO's claims that TransMedics must operate the device for quality control: "we sell devices in other countries...in the US we **no longer sell devices because it just doesn't make sense.**" **Two former staff provided the same color.** A reimbursement executive bluntly stated "yes, you do" when we asked if centers must use the NOP, and noted the switch was so heavy-handed that even centers who already bought and had devices on the shelf could no longer use them. An ex-organ procurement surgeon confirmed that **"centers now are obliged to use their transportation service."**

US transplant centers must use the NOP service, while customers abroad can still operate the device themselves

A: **"They have to use our clinical service.** It's our device and our specialist who's going to operate it...when you call us, we're going to send the device with a clinical specialist to operate it...**we would like them to use our plane because that's part of our business."**

Q: "So, you guys don't sell the device anymore? You just use it as part of your service? It's a very interesting business model."

A: "Correct, yes. "

Q: "Do customers have the option of buying the station with a cartridge, or you don't sell the device, you bundle it with a service now?"

A: "Historically, we did sell them. **We sell devices in other countries** because our devices are available in Europe, the Middle East, and Asia. So, we sell devices there and we sell consumables to them. **In the U.S., we no longer sell devices because it just doesn't make sense...**" – Ex-logistics employee at Transmedics

Hospitals now "obliged" to use TransMedics air transport

"I've been told that the **centers now are obliged to use their transportation service.**" – Transplant procurement surgeon previously working with TransMedics

Centers were no longer allowed to operate devices that they already had on the shelf

Q: "What was correct and incorrect in that Gosar letter? **Do you have to use their NOP to use OCS?"**

A: **"Yes, you do.** Centers that had bought OCS systems were not allowed to buy cartridges unless they used the national program. They implemented that policy in November of 2021. **[Redacted surgeon name] was one of the centers that had OCS on the shelf, and they wanted to use it and they were told that they were not going to be able to use it any longer and that it had to go through the national program."** – Former executive in a key reimbursement role

Transplant surgeons at centers across the US corroborated that they are forced to use the TransMedics NOP service in order to access the device. One of the highest volume users of the device in the US indicated he has no other option, stating that he uses the device 45-50 times a year and that **“we use their national NOP program for all of them.”** Although the answer was self-evident in his 100% percent utilization of the service, for the avoidance of doubt we asked if using the service is a requirement: **“to be completely candid, this is a very small world and I don’t know of any program that uses them that doesn’t use the NOP.”** A second surgeon, based in Utah, and also one of the company’s highest volume users, corroborated that **“you don’t have access to their pump” unless you use their NOP service.**

High volume user indicated NOP is utilized 100% of the time

A: “We are a medium-sized transplant program, This year, we’re slated to do over 100 transplants, which we’ve never done in the almost 35-year history of the program. Our involvement with TransMedics began less than three years ago. I predict **TransMedics OCS will approach 45 organs, maybe even 50 for the year.** I would say that without the use of OCS, we would not be doing these extra transplants. **We use their national NOP program for all of them.**”

Q: “Is that a requirement? I’ve heard if you want to use the device, you basically have to use their NOP. Is that correct?”

A: **“Well, I don’t know if it’s a requirement. To be completely candid, this is a very small world and I don’t know of any program that uses them that doesn’t use the NOP.”** – Transplant hepatologist and director of the liver program at a Midwest academic center; one of TransMedics highest volume users

Another surgeon indicated no access to the device without using NOP service

Q: “You’re saying if you want to use the device, you have to use their procurement service. The only tweak in the procurement service is you can save 5 or 8 grand by sending your own transplant surgeon but you’ve got to use their perfusionist, their specialist, their team, and their aircraft. Is that how you understand it?”

A: “Yes.”

Q: “The organ procurement service. Do they mandate that as part of the deal - that you have to use their pump?”

A: **“Yeah, you don’t have access to their pump.”**

Q: **“Unless you use their service?”**

A: **“Yeah.”** – Transplant surgeon and surgical director at a high volume academic center in the western US; also medical director at a major OPO (organ procurement organization); one of TransMedics highest volume users

Vanderbilt University is one of the world's pre-eminent and highest volume transplant centers. Two physicians - a liver transplant surgeon in a leadership role and a transplant cardiologist – detailed their dubious view of TransMedics and why their usage of the device has fallen off a cliff. The surgeon emphatically agreed with Congressman Gosar's allegations, stating that his letter is “completely honest...it's a very honest letter” and that “you don't have another option” besides using TMDX's technicians and air transport: “You cannot do it. They have to do it. That's for sure.” The cardiologist stated they now rarely use the device since they're forced to use the NOP: “the other thing that's really driven use away is they were essentially mandating, requiring you to use their surgeons and their procurement teams for these hearts.”

No other option but to use the NOP service

Q: “Do you have to use their perfusionist and transportation?”

A: “Oh yeah, that's for sure. Yeah, yeah, yeah. Because **they include as transportation, even if you send your own surgeon.**”

Q: “They're not forcing you to use their surgeon, but they are forcing you to use their technician and their transportation?”

A: “That's absolutely right, yes. **You don't have another option.**”

Q: “That is what the congressman was saying. They're putting a gun to people's heads. You have to use the NOP to use their product.

A: “Yup. Oh yeah, the perfusionist, you have to do it. You cannot send your own perfusionist to go and to pump the liver, no. **You cannot do it. They have to do it. That's for sure. That's the reason I'm telling you, I believe that that letter is completely honest. It's a very honest letter.**” - Transplant surgeon in a leadership role at Vanderbilt

“Mandating” usage of NOP

A: “**The other thing that's really driven us away is they were essentially mandating, requiring you to use their surgeons and their procurement teams for these hearts.** I trust my surgeons. My surgeons trust our surgeons when they go and visualize something and say that it'll be fine. **It's harder to do this when a team that you don't work with on a regular basis is just sending you cell phone videos of this is what it looks like, etc., and you're making that critical decision as to, yes, I'm going to accept his heart.**”

Q: “That's how it works with their team? They're sending cell phone videos to the team on the other end to see if they like it?”

A: “**It's like phone calls and cell phone videos and that's it, yeah.** Because none of our team members are there, so that's the only way **if there are any concerns whatsoever. It just makes it very challenging if there's some injury** that happened during the donor chest opening and dissecting out that heart. Whereas, if it's one of our surgeons...” - Transplant cardiologist at one of the top centers in the US/world

The Vanderbilt transplant cardiologist indicated that “everything related to surgical procurement, use of the OCS device, air transport etc.” was “mandated” and wryly noted the company’s promotion of its brand new jets on quarterly calls. His surgical colleague in the liver program described a key aspect of how the scheme works. If a transplant center refuses to use TransMedics aircraft, they allegedly retaliate by the forcing the center to absorb the cost of two planes – one with the TransMedics device technician, and a second that the center hires for its surgical team: “if I send my own surgeon, it’s going to cost me more money because I have to pay for two transportations...basically, they are forcing you to do it...if I send my own surgeon, it’s going to cost me more money.”

TransMedics forces centers to use the NOP by spiking the cost if they use their own transportation

Q: “When you want to use TransMedics, do you have to use their NOP, their procurement service or can the OPO send their own person?”

A: **“The way that they set it up, it is cheaper for me to ask a TransMedics surgeon to go and recover the liver than sending my own surgeon. If I send my own surgeon, it’s going to cost me more money because I have to pay for two transportations. I have to pay the transportation for my surgeon to go to the hospital, and I also have to pay the transportation of the people from TransMedics that they have to go to the hospital and then they have to come back to my hospital. They created a very, very smart business model.”**

Q: “Did you see that letter that the congressman wrote?”

A: “Oh yeah. I did. Right now, the way that the business model is developed, yes. **Basically, they are forcing you to do it.** If you don’t do it, I mean, they’re going to tell you, oh no, you can send your own surgeon. Yeah, you can, but it’s more expensive. You don’t want to do that, right? You are trying to reduce costs. **If I send my own surgeon, it’s going to cost me more money, so I don’t want to do it. So then, I just let them do it.**” – Transplant surgeon in a leadership role at Vanderbilt

“Mandated” to use the NOP

Q: “You said they essentially mandate the use of their NOP. Is it mandated or essentially mandated?”

A: **“My understanding is that it’s mandated. The way our surgeons describe it is that it’s mandated,** and in the last 7-8 months, for example, whenever we use the OCS system, it’s always been the TransMedics procurement team.”

Q: “So, they’re forcing you to use it. They won’t let you use it with your own surgeon, your own air transport?”

A: “Exactly. Everything relating to surgical procurement, **use of the OCS device, aviation, etc., they’re trying to put it on their end to the point where I think in one of their recent quarterly reports they talked about how they purchased x number of new airplanes.**” - Transplant cardiologist at one of the top centers in the US/world

The Vanderbilt surgeon stated he played a role in TransMedics' pivotal PROTECT trial but soured after they did a bait-and-switch – allegedly misleading centers into thinking they could continue to purchase and operate the device only to be duped after FDA approval: “that’s what they did...people are very pissed off with the company...they are acting too greedy and kind of predatory...I really support that letter from the Congressman...a lot of people in different transplant centers are very pissed off with the company...I can tell you, I hate the CEO.” His cardiac colleague stated that Vanderbilt hadn’t used the device in months due to “their rigidity and needing to use their team...to spend so much more money procuring hearts compared to what we’re comfortable with.”

Duped trial centers; alienated transplant centers across the country

A: “I don’t like the Transmedics people that much - I was a [redacted] when they did the initial trial, the PROTECT trial. When they were doing the trial, the way that they were advertising the product is like, well, once we get the trial, once we get approved by the FDA, we are going to sell or we are going to lease the pump and then you only have to buy all the stuff that you have to do for each case. So, that’s the way that they were advertising. But once it got approved by the FDA, they completely changed.”

Q: “They basically did a bait-and-switch with the investigators.”

A: “Yes, they did. That’s what they did.” And that’s the reason that, to be honest with you, a lot of people are very pissed off with the company...a lot of people are very pissed off because they are acting too greedy and kind of predatory. I really support that letter from the Congressman. Again, a lot of the people in different transplant centers are very pissed off with the company.... I can tell you, I hate the CEO.” Transplant surgeon in a leadership role at Vanderbilt

Vanderbilt slashed usage of the device after being forced to use the NOP services

A: “For us, it’s been their rigidity and needing to use their team and needing to spend so much more money procuring hearts compared to what we are comfortable with that’s driven down our intent to utilize this company.” The other part, if you want more details, is that there are other competing companies that may be cheaper but may also offer similar distance-wise benefits.”

Q: “Have you said we want to use your device, but we want to use our own surgeon and air transport, and they said no?”

A: “From what I’ve been told by our surgeons, the last time we had requested that was many months ago and at that time, we were told no. And that’s part of the reason that we only used them six times in the last six months, and then every single time it’s been their own team.” It’s been during situations when we didn’t have enough surgeon bandwidth to send one of our own team.”

Q: “It was just last, last resort was when you used them?”

A: “Yeah. Exactly.” - Transplant cardiologist at one of the top centers in the US/world

In sharp contrast to the CEO's denials, every major transplant center we spoke with described what sounds to us like a racket, where they have no choice but to use TransMedics high-cost jets for air transport. UCSF stated "I don't think there's another option. That's what you have to do." USC provided the same color. A transplant surgeon complained that "we've pushed back saying we don't need your surgeon and we don't need your aviation company" but were bulldozed to the point that "in 2023, almost every liver was through the NOP." He stated TransMedics recent growth has come from "pushing every liver program not to use their OCS system as an a la carte item but as part of their NOP program."

UCSF indicated that NOP usage is required

Q: "I've heard that even if transplant centers have the option to not use the NOP, one way or another, they're essentially forced to.

A: "I think that's **all they do know is NOP. That was my understanding.**"

Q: "Do you use the NOP for every liver transplant essentially?"

A: "Yes.

Q: "That's because you're forced to?"

A: "Yeah, **I don't think there's another option. That's what you have to do.**" – UCSF administrator, one of the highest volume and pre-eminent academic transplant centers in the world

USC tried pushing back but steamrolled into using the NOP service

"They **increased their revenue in liver in 2023 by pushing every liver program not to use their OCS system as an a la carte item, but as part as part of their NOP program.** They provide the device, the disposable, the technician, the surgeon, and then they added the aviation company... **We've pushed back saying we don't need your surgeon and we don't need your aviation company.** We're in LA. We've got all these rock stars and C-suite people who are chartered jet flying everywhere, and so we've already got a contract with a really good transport logistics company. We don't need to pay your aviation company when we've already got that... **In 2023, almost every liver was through NOP.**" – USC transplant surgeon

As part of the aggressive conduct, centers stated that TransMedics no longer allows transplant physicians or technicians to even train on the device, in contrast to its previous practice. The CEO vehemently denied this in his reply to Gosar's letter, but his statements appear to be false – excerpted again below. One of the top liver transplant surgeons in the world, based at a Chicago academic center, stated that “originally you could operate the device yourself...more recently, they don't allow you to do it...they won't train you at all...it is a little disconcerting.” He confirmed widespread anger across transplant centers: “it is gimmicky...I just don't like the way they keep ratcheting down the screws...that's really unethical...that was a topic of discussion” at the recent national meeting.

TransMedics CEO Waleed Hassanein reply to Congressman Gosar's letter Feb 26, 2024

- Your letter asserted that TransMedics prohibited transplant centers from training their personnel on the OCS procedure. This statement is not correct. To-date, we have trained more than 400 transplant retrieval surgeons for heart, lung, and liver transplants across the U.S. Importantly, given our clinically focused mission, we are

TransMedics allegedly no longer trains centers on the device, in contrast to denials in the letter

A: **“Originally, you could do operate the device yourself or use their service. I've heard this now, that more recently, they don't allow you to do it...they won't train you to do it...it is a little disconcerting. I don't know why they would do it that way.”**

Q: “You said the won't even train you anymore if you want to learn on the device?”

A: “That's what I've heard. I think they're **trying to change their whole business model to just fly around and procure.**” – One of the top liver transplant surgeons in the world, based at a leading Midwest academic center

Alienating the field, “gimmicky” device

Q: “It looks like this is just trying to find a device that they could sell for \$100k versus something that really does a lot.”

A: “Yeah, I think **it is gimmicky...I don't like the way they keep ratcheting down the screws, and as I've heard, if they're not training programs to operate the machine and they're forcing everybody to use the NOP program, that's really unethical.**”

Q: “Where did you hear they're not training people?”

A: “I was at a meeting, probably at the last American Transplant Congress. **That was a topic of discussion about the machines.** I heard that during the discussion, a few months ago.” – One of the top liver transplant surgeons in the world, based at a leading Midwest academic center

In the context of anticompetitive behavior as alleged in Congressman Gosar's letter, extortion occurs when customers are coerced to use a more expensive product/service through the unlawful practice of product bundling. TransMedics strikes us as textbook case whereby it appears that customers are 1) forced to pay for a NOP service they do not want, requiring them to use the company's device technicians, procurement surgeon, and air transport, when they would rather purchase the device and operate it themselves as they did previously; and 2) to do so at a higher price than if they purchased each service themselves versus a bundle. Thus, a large portion of CEO Hassanein's 7-page reply to Gosar attempted to cast the NOP as a cost-saving service.

TransMedics CEO Waleed Hassanein reply to Congressman Gosar's letter Feb 26, 2024 - excerpts

Your letter characterized our NOP program as a means to reduce access to the OCS technology and increase prices and profitability. This is not correct. Contrary to the

Your letter accused TransMedics of pressuring hospitals to utilize our more expensive logistics. This couldn't be further from the truth. TransMedics presents the logistics quotation to transplant centers using our NOP service. The transplant centers then have the final say on whether they want to use our services or not. We are confident, given the above efficient way of operating our fleet, and our cost-sharing model with transplant centers that our quotes are lower than competitors who use the outdated brokerage model.

Your letter painted an inaccurate picture of the charter flight logistics for organ transplants in the US and accused TransMedics of creating a logistical network with newer aircrafts to unjustly increase cost on our clinical users. Again, this is not correct. Here are the facts.

Transplant centers had the option to either use NOP or continue to operate the OCS on their own by sending their own teams. Over the past 24 months, more and more centers elected to transition to NOP due to its more efficient cost profile, better clinical outcomes, and the ability to increase their transplant volumes without surgical fatigue of their staff.

The statutory framework is well-established by the Sherman Act, Clayton Act, and the Federal Trade Commission Act; as well as in key case law that defines illegal tying and bundling such as *International Salt, Jefferson Parish, and Eastman Kodak*. Extortion occurs through the practice of “coercive tying” where the buyer is compelled to accept the tied service to accept the tying product, leading to higher costs and less choice. Thus, coercion and pricing are both central elements in anticompetition law. As a basic fact to gauge whether the pricing is extortion, we note that devices for kidneys cost \$2,000 vs. TransMedics pumps at \$70,000-\$90,000 per consumable. Centers indicated that \$110,000- \$120,000 is the floor when the consumable is bundled with the NOP service, prior to including the Organ Procurement Organization’s charge of \$50,000 to \$70,000.

Kidney perfusion pumps are a small fraction of TransMedics price

“We have a cold **perfusion pump for kidneys. They cost less than \$2,000 each** so that just gets absorbed into the OPO cost, but it’s \$2,000.” – Transplant surgeon and transplant program director at a major academic center in Pennsylvania

Floor is \$110,000 to \$120,000 prior to OPO organ acquisition fee

“The **bills from TransMedics could be \$110,000-120,000, and that’s in addition to the transplant center’s organ acquisition fee that they’re getting from the OPO, which in itself is \$50,000 to \$70,000.** So, if you include the OPO’s fee for the organ, think of it like buying a car, and TransMedics is the transport company for that car, the total cost out the door could be **approaching \$150,000-200,000** and the profit margin for a lot of the payors that we deal with is only about \$300,000. So, you could be **eroding a significant amount of your profit at the transplant center if you were using their NOP program soup-to-nuts.**” – Prominent surgeon, leadership role at a major West Coast academic transplant center

Over \$100,000 with the NOP

- A: “The NOP, when it first came out, was described as \$30,000 for the surgeon and the technician, and then you had to pay for the transportation of the organ. That was on top of the device cost. **We were looking at the cost of an organ on the NOP of over \$100,000 - that’s not including the organ. That’s the cost of just procuring the organ through the TransMedics program.**”
- Q: “You said \$100,000 for the NOP plus the cost of the disposable, the device?”
- A: “Correct. And then you added up the transportation and that’s variable, but it’s at least \$10,000.” - One of the top liver transplant surgeons in the world, based at a leading Midwest academic center

However, customers told us that \$100,000+ is merely a floor and that the price can explode to \$300,000 to \$400,000 per transplant run when the device is bundled with the NOP and if TransMedics sends multiple jets to run up the charge, an outrageous practice. according to centers who stated they have received such invoices. Moreover, as Congressman Gosar alleged, users stated that TransMedics spiked the price after FDA approval. A transplant cardiologist at an Ivy League center stated that the pre-approval price per consumable went up by 3 or 4X from \$24 to \$25,000 but to now \$80 to \$90,000 – before including the NOP service.

Price per device consumable spiked after FDA approval

A: "I actually looked into this a few months ago; it's like the price increased significantly. **When we were doing it ourselves, I think it was around \$20 or \$30,000 per disposable. And now, I think bundled together, it's getting closer to \$80 to 90,000.**"

Q: "The disposable was how much during the trial?"

A: **"It was mid-20s, I think, 24-25."**

Q: "Which period was the trial?"

A: "2020, and then it continued after as protocol, which was 2021. **The cost was nowhere near what it is now.**" –
Transplant cardiologist at a major Ivy League academic center

Surgeons and transplant administrators we spoke with – including TransMedics highest volume users – assailed the pricing as extortionate and predatory. They disputed the CEO’s claim that the bundled NOP service was cheaper than traditional air transport or resulted in any cost saving, and stated that the charges were exponentially higher due to the forced tying of the device to the service – classic anticompetitive conduct. A cardiac surgeon at a major Midwest center stated they “very strongly want you to use their procurement service”; that “their new contracts, they are mandating that if you are going to use our product, you have to use our planes and procurement”; that “they’re persistent in pushing” their aircraft and their “price is so egregious that a lot of programs, including ours can’t or won’t do it” – “the additional cost is an annoyance when it’s unnecessary to be spent.”

“Persistent” in forcing their procurement service and “egregious” pricing

*“I’m also a transplant director, so I wear a couple of different hats...you need to use their clinical specialist, but you don’t have to use their procurement service, although they **very strongly want you to use their procurement service, and in some of their new contracts, they are mandating** that if you are going to use our product, you have to use our planes and procurement...I am aware that they’re pushing people to use their procurement service...as a company and as the salesforce or the clinical specialist, they’re **persistent in pushing their model of using their flight services...their price is so egregious that a lot of programs, including ours, can’t or won’t do it.**”* – Cardiac transplant surgeon at a major Midwest academic center

“Weird predilection” in pushing their aircraft; cost is “unnecessary”

*“What’s been a little bit questionable from my own personal experience is there is this **weird predilection that they have to push you towards using their modes of transportation. They’ve been a little pushy** at times on that. When given the option of booking through our OPO versus using their own modes of transportation, they would always prefer and then use their own transportation services...the additional **cost is an annoyance when it’s unnecessary to be spent.**”* – Cardiac transplant surgeon at a major Ivy League academic center

TransMedics bundled pricing scheme is, we believe, inherently coercive. The hourly flight charges are about double, according to one of the top liver transplant surgeons in the world - a highly published and widely-followed KOL – who stated that the charges escalated after TransMedics established its own transportation network: “their prices are higher than what we pay for our own planes that we contract privately.” He estimated TransMedics charges about \$8,000 per hour while his institution pays \$4,500 for private charter; this underestimates the price discrepancy as TransMedics often sends multiple planes. As another data point, the surgeon pointed to other third-party organ procurement services that do not tie their service to a device. As a result, their price is \$20,000 to \$35,000 including the jet versus \$130-140,000 via TransMedics – “a much more reasonable price because it’s not associated” with a bundled device; “they’re not buying the jets...they’re using charters just like we use currently.”

NOP air transport cost is about double versus what centers pay when they use a la carte charter services

“Their transportation charge has gone up because TransMedics has their transportation network, and their prices are higher than what we pay for our own planes that we contract privately. That reason is because their planes are bigger. They manage all the expenses through this modified NOP program that incorporates transportation into it. For the smaller jets, we were paying about \$4500 an hour. And for the larger planes, that’s about \$7,000. It’s sort of all bundled in when we get a bill for TransMedics for their NOP, and that’s usually over \$110,000. I estimate it’s around \$7500 to \$8500 per hour.” - One of the top liver transplant surgeons in the world, based at a leading Midwest academic center

Other organ procurement services are at more reasonable cost; \$20-35,000 versus \$130-140,000

“There’s a program on the East Coast, there’s a company on the West Coast, they hire retired heart surgeons that know how to harvest heart and lungs and send them out and they’ll do it. But they do it at a much more reasonable price because it’s not associated with OCS. It’s also I think that a lot of those places, they’re not buying the jets; they’re using charters just like we use currently. when Transmedics talked with us about it, it’s \$130,000 to \$140,000. To put that in perspective, what we currently pay is done through—we’re in Region 8, so it’s run through our NPM and UNOS. They make all those arrangements, and for us and for us to charter a jet and fly, it’s \$20,000 to \$35,000.” – Cardiac transplant surgeon at a major Midwest academic center

We cannot recall physicians thrashing the pricing of a medical product as harshly as they do with respect to TransMedics. A surgeon and director of a leading transplant center in the Northeast asked “why should it cost \$70,000 to \$80,000 for disposables” when they are just “plastic tubings and plastic containers...it is, no question, overpriced.” A transplant hepatologist who is one of the highest-volume users and who believe to be on the dole via kickbacks was nonetheless unvarnished when he spoke with us, calling it “ridiculously expensive...I could argue predatory”; “outrageously overpriced”; “completely unreasonable, to be very, very clear”; “yeah, it is pretty polarizing”. Regarding the CEO Waleed Hassanein, he noted “a level of arrogance...I think they are generally known to be fairly assertive and aggressive.”

“No question, overpriced” for disposables that are just plastic tubing

“They’re first on the market, so they have a monopoly, and they priced it like a monopoly because **why should it cost \$70,000-80,000 for disposables? The disposables are essentially plastic tubings and plastic containers** and cartridges, and that’s more than the cost of a brand-new luxury car that you’re paying for a set of things that you throw in the garbage, essentially. **It is, no question, overpriced.**” – Transplant surgeon and transplant program director at a leading Northeast academic center

High volume user slammed the pricing as “outrageous”

“It is a **ridiculously expensive solution. I could argue predatory**, perhaps...I think my biggest issue with it is that it is **outrageously overpriced**...should there be **more transparency in contracting, etc.? Probably, yeah... Yeah, it is pretty polarizing...I don’t like the price. I think the price is completely unreasonable, to be very, very clear**...I have attended national presentations where folks from TransMedics, Waleed, have presented data...there is **a level of arrogance...I think they are generally known to be fairly assertive and aggressive.**” -Transplant hepatologist and director of the liver program at a Midwest academic center; one of TransMedics highest volume users

Case study 1: Massachusetts General Hospital. MGH is one of the leading medical centers and transplant programs in the world, and may have a longer and **deeper history with TransMedics than any other program** given their physical proximity. An MGH surgeon was the lead author on the company's pivotal PROTECT trial. **We spoke with a transplant administrator, as well as a transplant hepatologist in a leadership role,** who confirmed key aspects of the scheme and the dubious business practices that have led them to **sour on the relationship**. They stated that they are forced to use the NOP service and that the pricing is much more expensive as a result of the bundling: “that’s the only way you can use it”; “we’re kind of **forced to...we don’t really have an option**”; “two years ago TransMedics said...we’re going to dictate the terms of the utilization of our pump...we are not going to let transplant centers buy our pump”; “they have to use our service...so **that’s why it’s so much more expensive.**”

MGH corroborated key aspects of TransMedics bundling and extortion scheme

Q: “For those liver cases that you use TransMedics, are you also using their NOP service?”

A: **“That’s the only way you can use it** - there is a model where, our center has three surgeons trained on putting livers on pump...so, we do have three surgeons trained for a TransMedics pump. **Of the 38 OCS pumps that we’ve done, our surgeons have done only two of them, to give you an idea.”**

Q: “One thing that I’ve heard from a lot of people is that they’re not even allowed to train on the pump. That if you want to use the pump, you have to use their NOP service. So, you’re saying that you use it because you actually don’t even have any other option even if you wanted to?”

A: **“Yeah, financially, we’re kind of forced in that way. We don’t really have an option** because the pumps reside with TransMedics. So, they have to send a team anyway...That there’s no what we call back-to-base model, where we own the pump, we can bring a liver back here.. So, **two years ago, TransMedics said we’re going all in, and we’re going to not only set the pricing for our pump but we’re going to dictate the terms of the utilization of our pump,** which includes we’re going to use our own surgeon, we’re going to use our own perfusionists, we are not going to let transplant centers buy our pump. **Not only do they have to use our pumps and the cartridges and the equipment but they have to use our service line. So that’s why it’s so much more expensive than the other pump options.”** – Veteran transplant administrator/executive at Massachusetts General Hospital

Case study 1: Massachusetts General Hospital (cont'd). MGH stated the NOP service is about control, not cost savings or efficiencies – “as an institution, you have to start asking...is this viable?” Shockingly, the transplant administrator indicated that TransMedics logistics are a circus where aircraft or staff may be unavailable, jeopardizing MGH’s ability to procure the organ. However, TransMedics allegedly remains inflexible even in such emergencies and forces the center to sit on its hands until their plane is available, as a condition for accessing the device. In those situations, MGH has to find its own plane for its procurement team but still pay for a second plane TransMedics sends with its device operator – “it’s about control...if you can control the surgeon, the team, the airplane – everything – as a consumer, I’m stuck”; “sometimes, we can’t even get a plane because of the delay...where we have to then field another jet because we’ve got 90 minutes...that adds to the additional cost.”

Bundling is to lock in and control customers; NOP service and logistics are not particularly efficient and increase cost

“I think it’s about control. So, if you can control the surgeon, the team, the airplane - everything - as a consumer, I’m stuck. I might know what the cost would be, but then there are all these delays. You mentioned the two-hour additional warm ischemic time—that eats into my cost. Sometimes, we can’t even get a plane because of the delay in the procuring arena where we have to then field another jet because we’ve got 90 minutes. So, that adds to the additional cost, too. But I would say when it comes down to why their team needs to be the one and only to make the decision, I think that’s a control thing. I think that’s them making the call. And I have no control over that.” – Veteran transplant administrator/executive at Massachusetts General Hospital

MGH asking “is this viable?”

“...so there may be a feeling of being kind of forced or leveraged into taking on that service. At some point, the cost, the economic analysis—I think they’re bordering on eating into all of the margin or most of the margin that would be gotten from doing another case of liver transplantation. I think that when you start pushing up against that, at least as an institution, you have to start asking the question, is this viable?” – Transplant hepatologist in a leadership role at Massachusetts General Hospital

Case study 1: Massachusetts General Hospital (cont'd). MGH provided anecdotes which we think the FTC and/or DOJ would view as textbook coercion – “they’re holding [the device] hostage” - when its procurement team was “mobilized and everything ready to go” but TransMedics would say “well, we can’t get the module to you in time.” Given the requirement that only TransMedics staff can operate the device, the company would allegedly say the only alternative is to fly a separate TransMedics plane just for the device. MGH would push back and offer to send an Uber to their facility to get the device, only to be told “oh no, that’s not how we do things” – “it’s primarily related to locking us in to use the OCS team, flights, their modules”; “I’m stuck...literally, I’m stuck...I literally don’t have the device and they’re holding it hostage.”

MGH anecdotes describe TransMedics holding the device “hostage” to force them to use the NOP

Q: “Are saying TransMedics can’t activate and get everybody there in time, or that their surgeons are unskilled and incompetent? Or both?”

A: “It’s both. Let me just separate. The first part wasn’t related to delays based on the skillset and experience of the procuring team. **It’s primarily related to locking us in in order to use the OCS team, flights, their modules, and everything, versus using our own.** For instance, if we were to work on trying to get our team all mobilized and everything ready to go and they’re saying, well, we can’t get the module to you in time. So, we’re just going to fly the person out from our facility over to wherever the hospital is, but you’re going to have to eat that cost. And I’m like, why would we ever do that? **I will literally have one of my surgeons take an Uber over to your facility, bring it back, meet you at Logan, and then they’ll go on the flight. “Oh no, that’s not how we do things.”** Well, how about you get your tech to then take the Uber with the module over to either my hospital, where the team is waiting, or to get to the airport? “Yeah, that’s going to cost \$7000.” And I’m like that right there is going to cost \$7000? “Yes. You have one hour to give us your answer.” **That’s how the conversation goes. So, I’m stuck. Literally, I’m stuck. I literally don’t have the device and they’re holding it hostage, so I have to then say I’m going to open it up, \$7000, come on over.”** – Veteran transplant administrator/executive at Massachusetts General Hospital

Case study 1: Massachusetts General Hospital (cont'd). MGH's experience exposes the TransMedics NOP as a charade to pressure centers into paying for bundled services and air transport at inflated prices, not as a logistics "hub" that makes organ procurement faster or cheaper. MGH painted the logistics as amateurish and marred by delays and bottlenecks, in contrast to MGH's in-house ability to manage the complex "orchestration" of staff and resources to procure an organ. The "delays" appear intentional to force the center to pay for a second TransMedics plane versus traveling with the procurement team, as well as driven by incompetence that may result in stranded and lost organs – "...and they're dragging their feet...and here we are scrambling...to get the orchestration done, but we ended up finding out TransMedics has been 'working on certain things' for hours...potentially the organ will not be used...."

Centers indicate that the TransMedics NOP is inefficient and marred by delays that jeopardize organ procurement

Q: "You made a comment that's consistent with what I've heard from others about delays, the patient may or may not get the organ, we're the only option if you don't take it, the organ doesn't get used, even a local center can't use it. I've heard from several centers that they're basically holding organs hostage, somebody used the word 'extortion.' Walk me through what you were referring to."

A: "We all know that as you're coordinating a procurement, time is of the essence. So, everyone knows what the planned time is but also everyone knows that there could be delays: the patient may not expire, which would cause a delay. The team didn't get there on time, so the procurement gets delayed. Sometimes, even when you're getting on procurement and the liver team, wherever they're from, they request another biopsy, and then the rest of the team has to stop and not cross-clamp, so you just have to wait, and we get that. That happens. That's a procurement. **It's an orchestration, and there's an art to it. When it comes to TransMedics...and we're going to need the team...and they're dragging their feet. And here we are scrambling to try to get to that timeline of can we get the team? Not just the procuring team who work for us but also the air charter and making sure the pilots don't time out if we do a round trip, making sure that we have all the logistics set up** for later to getting the team and if we have to use the device or not or a different technology or not or instead of equipment, the donor could be in way north Maine or they could be in an urban setting. We have to factor all of those things in. And so, here we are **scrambling to get the orchestration done, but we ended up finding out TransMedics has been "working on certain things" for hours, and then we get notified, and so the only last-ditch effort before we forego the organ and potentially the organ will not be used because it might be up in eastern Maine, as I mentioned, then there's no other option other than eat the cost.**" – Veteran transplant administrator/executive at Massachusetts General Hospital

Case study 2: University of California, San Diego. UCSD is one of TransMedics highest volume users and described coercion similar to but even more overt than at Massachusetts General. A transplant surgeon stated that “when we started this off...we had our own device and we could use them as we pleased...then they took our machine away and said ‘No, you can’t.’” UCSD stated that TransMedics forces them to use the NOP by escalating the cost if they use their own air transport versus the company’s jets – “they make it so onerous to do your own travel...they find ways to increase your costs...giving you no option but to...use their transportation”; “lack of control over the device and now the lack of control over travel...they’ve really boxed everyone into a corner to use their transportation...”

TransMedics recently took away control of the device and transport and “boxed everyone into a corner...to use their transportation services.”

*“When we started this off, it used to be we had our own device. We bought our own cartridges, and we could use them as we pleased. And **then they took our machine away and said, “No, you can’t.” It’s the lack of control over the device and now the lack of control over travel. So, they’ve really boxed everyone into a corner to use their transportation services.**”* - Transplant surgeon at a high volume West Coast academic center

TransMedics “find ways to increase your costs” if you refuse to use their air travel

*“While I think they’ve backed off from saying you can only use our transportation company; **they make it so onerous to do your own travel. They find ways to increase your costs - they’re kind of like giving you no option but to sign with the company and use their transportation.**”* - Transplant surgeon at a high volume West Coast academic center

Case study 2: University of California, San Diego. UCSD shared detailed examples to illustrate how TransMedics allegedly puts a gun to their head to run up the cost and allegedly coerce them to use its jets. Previously, if a DCD organ donor was available in Seattle, TransMedics would send a procurement team available locally at their Seattle hub. If the donor didn't expire, UCSD didn't incur the cost to fly from San Diego to Seattle. Now, the only option they allegedly offer is to fly a TransMedics device and technician *from San Diego to Seattle* even though they have both locally in Seattle. This forces UCSD to eat the \$50,000 transport cost to Seattle if the donor doesn't expire and no organ is available – pure extortion to punish TransMedics if it doesn't use their jets: “if I say I want to use the NOP to get that liver in Seattle, but I don't want to use their transportation company, the only option they give me is traveling from San Diego...that's a round-trip ticket that's...\$50,000...rather than just using their team in Seattle to recover the liver and fly it down to me.”

Detailed example of what appears to be extortion if UCSD doesn't use TransMedics air transport

A: “I'll give you a real-world example. I'm in San Diego. They have hubs all over the country with specialists, machines, surgeons. **If I wanted to get a liver from Seattle, where they have a hub, the only way they would do it because we haven't signed with their transportation company was to fly the specialist and the machine from [major California city] to Seattle to do the case, even though they already have people in Seattle.** Now, it's a DCD case, so it may not expire, and I've got to pay the \$50,000 just to get the team up there, which defeats a large part of the purpose of having hubs everywhere. If I can't use the hub, I have to travel there. If I use their company, they'll do the case from their hub rather than fly in from San Diego. So, it's a \$50,000 price difference in travel for me.”

Q: “Just so I understand - they have a team locally in Seattle, and there's a donor in Seattle. If you want to send your own team, they're saying you have to do what?”

A: “It's not even sending our own team. **If I say I want to use the NOP to get that liver in Seattle, but I don't want to use their transportation company, the only option they give me is traveling from San Diego.** So, they'll send a separate team that has nothing to do with me, their team from San Diego, which is also a hub, all the way to Seattle. **That's a round-trip ticket that's going to cost me \$50,000 to get there, rather than just using their team in Seattle to recover the liver and fly it down to me.”**

Q: “So, if you said I want to use my own air travel, but they're like you have to use our team. They'll fly their team from [major California city] to Seattle and then back if you want to use your own air travel. But if you use theirs, they'll just send the team from Seattle?”

A: “Right.” -Transplant surgeon at a high volume West Coast academic center

Source: Scorpion Capital consultation calls with experts

Case study 2: University of California, San Diego. UCSD described the tactics as **retaliation and as “something new because our contract negotiations with them have been slow”** regarding use of the air transport service. The surgeon stated that TransMedics offers no explanation for refusing to utilize teams available locally at its hubs and for forcing UCSD to incur an unnecessary round-trip expense from San Diego and back - “they will talk themselves in circles trying to explain their rationale”; “they’re making an offer I can’t refuse...**it’s like racketeering...mafia-style...a thug-style way of getting things done.**”

No explanation for running up the cost by refusing to use local NOP teams

Q: “This just sounds highly unethical because you’re adding a lot of hours when there’s a life in play, an organ in play.”

A: “I agree and **they will talk themselves in circles trying to explain their rationale.** We’ve been using TransMedics from the beginning and we were early heavy users. When they first opened up their hubs - and we had a DCD in that city - they would send a team from their hub, and if the donor didn’t expire, there was no cost to us. So, I had nothing to lose by going after that liver, and I had everything to gain if the liver was good. I got a liver, and then I’ll pay for the flight to down here. But now. I’m turning down organs because it’s now going to cost me \$50,000 if that donor doesn’t expire, and I get nothing to show for it.”

Q: “So, they’re saying that if you don’t use the NOP, you’ve got to take the risk and send your whole team up there.”

A: “Yes.”

Q: **“So, they’re just trying to make it as expensive as possible for you to not use their service.”**

A: **“Yes, exactly. Exactly. They’re making an offer I can’t refuse.”** -Transplant surgeon at a high volume West Coast academic center

“It’s like racketeering”; “mafia-style”; “thug-style”

Q: “And if you want to use your own air travel, they should just send the team from Seattle is what they should do.”

A: **“Yeah, which is what they’ve done in the past. This is something new because our contract negotiations with them have been slow. We have not signed a contract with their aircraft company.** Things at our academic center move at a glacial pace, and they’re **trying to force us to move that contract faster so that we sign to use their air travel company.** The regulations at the [academic center name redacted] tell me I can’t use their air company if we don’t have an agreement with them.”

Q: “This sounds like a business model that’s one lawsuit away from being blown up.”

A: **“It’s like racketeering** or something, right? That’s what it feels like. Those are some of the aspects of the business model I have a problem with. They’ve approached several things that way. I’m sure other surgeons have used some pretty dramatic words - but it does feel kind of like they’re playing like **mafia-style.** It does feel **very deliberate and a thug-style way of getting things done.”**

Q: “Surgeons have used the word **predatory** on almost every call I’ve done.”

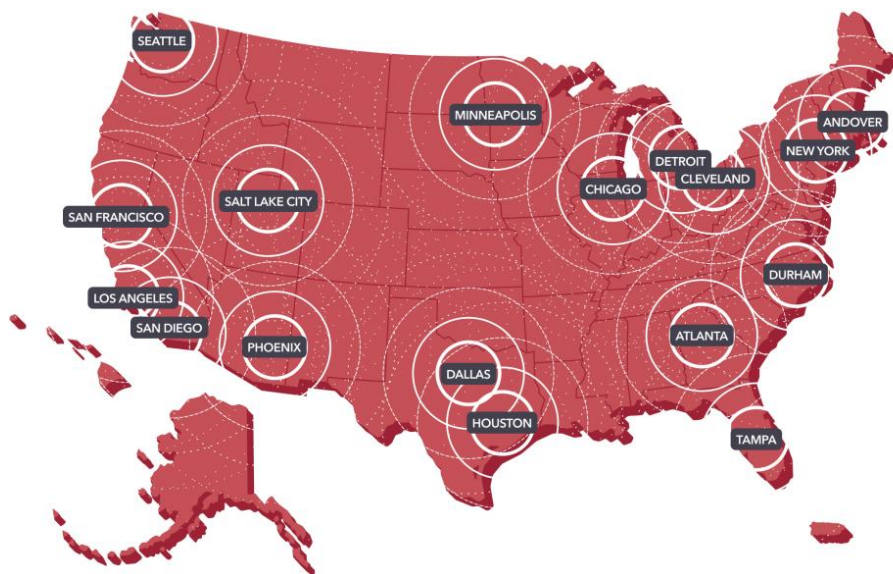
A: **“[Chuckles] Yeah. I think that’s fair.”** -Transplant surgeon at a high volume West Coast academic center

Part 2. The NOP service is, in our opinion, a large-scale fraudulent billing racket, predicated on overcharging hospitals for unnecessary flights.

TransMedics National OCS Program (NOP) is, we believe, a large-scale fraudulent billing conspiracy whereby customers - transplant centers and organ procurement organizations (OPO's) - are overcharged for its air transport service. Our investigation uncovered the details of how the scheme operates and how TransMedics allegedly tries to cover up its tracks, based upon interviews with former employees based at these hubs as well as transplant centers who conveyed their exasperation and outrage. As background, its investor presentation indicates 17 hubs in major cities across the US where it stations devices, OCS specialists who are dispatched to operate the device, surgical procurement teams, and aircraft and flight crews.

National OCS Program (NOP) hubs around the US – investor presentation, Oct 2024

NOP Hubs to Provide Broad Geographical Coverage



Ex-employees and hospitals described two key mechanisms of systematic billing fraud: 1) flying in non-local procurement teams by jet when a local team is already available at its hub and could be driven, indicating the sole purpose was to exploit the customer via unnecessary air transport charges; 2) sending staff on multiple jets to the same location to further inflate the charge. We begin with a former “OCS Specialist” who operated the device and worked in the Seattle hub, who we cloak as “Specialist #1.” The specialist left recently because the practice was “just entirely unethical,” beginning when the NOP was established: “it’s a big reason why I decided to get out of the company”; “since they purchased aircraft, they were flying in nonlocal teams versus driving the local team.” The specialist stated that over half of organ procurements were within driving distance – for example, the donor and recipient were both within Seattle - but TransMedics would still fly in nonlocal aircraft to run up the charge. In addition, the specialist indicated that “I would typically fly independently,” meaning hospitals were billed for multiple aircraft for a single procurement with the device operator on one plane and surgical procurement staff arriving in others.

Flying in nonlocal teams versus driving the local team

*“If it stayed local, it would be driven. If it would be taken within a 4-hour radius, roughly, it would be driven. And then basically anything over 4 hours, we would fly. **Since they purchased aircraft, they were flying in nonlocal teams versus driving the local team.**”* – Former OCS Specialist who left recently

Over half of procurements were within driving distance; specialist would typically fly “independently”

A: “I would assist the surgeon in putting the organ on the OCS device. And then, **I would typically fly independently**, fly or be driven independently, and manage the organ while en route, and then until that organ was implanted at the recipient center. .”

Q: “How far would they send you?”

A: **“Mostly locally, so Seattle to Seattle, Spokane to Seattle. I would say 50% of the time, we were staying within Seattle, and then often we would be going to San Diego, Phoenix, Portland.”**

Q: “What percentage of it would you say was local, like within driving range?”

A: “I would say **probably over 50%.**” – Former OCS Specialist who left recently

The OCS Specialist provided numerous examples of TransMedics flying in nonlocal procurement teams to Seattle when teams were already present at the local hub and with nothing to do. The specialist stated the company was secretive and that Seattle staff were not “given a reason” – “why are you flying a team from Texas or California when we have a full team of 10 people up here waiting to do cases?”; “it’s cheaper to send a local team...they are utilizing their aircraft to bill for the aircraft.” TransMedics flew in planes even when both the donor and recipient were within driving distance, describing specific incidents such as when planes were flown in from Texas “which makes zero sense...even one of the cardiac surgeons was like, why are you sending this team when we’re here?”

Flying in teams via jets just to bill for the aircraft, when local teams are available

Q: “Why would they fly in non-local teams? You’d think it’s more efficient to send the local team.”

A: “You are correct. I could only speculate.”

Q: “What’s your speculation about why they’re flying in teams - I’m guessing it’s overbilling.”

A: **“I think that’s exactly what it is.** We’ve not been given a reason, and we’ve had a lot of confusion within the local teams because we’re up here in Seattle. **Why are you flying a team from Texas or from California when we have a full team of 10 people up here waiting to do cases? We’re not given a great reason. All we’re told is that’s where the planes are, in Texas and in California, and there’s a plane coming. It’s cheaper to send a local team, and I think that they are utilizing their aircraft to bill for the aircraft.**”

Q: “That sounds completely screwed up.”

A: **“Yeah. It absolutely is. And it’s a big reason why I decided to get out of the company. It’s just entirely unethical.”** – Former OCS Specialist who left recently

Flying in nonlocal teams from places like Texas when it “makes zero sense”

Q: “You’re saying it would have been more efficient for them to just use the local team to procure it versus flying in a team?”

A: **“Yes. We had one instance where a heart was going from Montana to Seattle, and they sent out the Texas team to Montana to take the heart back to Seattle, which makes zero sense. Even one of the cardiac surgeons was like, why are you sending this team when we’re here? But Texas to Montana to Seattle.”**

Q: “Because you guys are right there.”

A: “Yeah.”

Q: **“Were there cases where they flew in a team where it could have been done under four hours within driving distance?”**

A: **“If it started local, yes.”** - Former OCS Specialist who left recently

The OCS Specialist indicated the practice of flying in nonlocal teams is now the norm, not an outlier, suggesting fraudulent air billing on a vast scale; and that it began only after TransMedics purchased its own jets, providing further evidence that the real purpose of the NOP's air transport operation is to inflate charges to transplant centers, not to drive efficiency. The specialist described aircraft flying in regularly from San Diego to Seattle, for example, even though local teams were available at the Seattle hub: “the last couple of months prior to me leaving, it was pretty consistent...since they purchased the planes, they basically were like, we’re not going to do this model anymore [of using the closest team to the donor]”; “it seemed to match up with” when TransMedics first bought the aircraft; “that seems to be when it started happening.”

Regularly flying in nonlocal teams; allegedly began after TransMedics purchased the aircraft

Q: “Obviously, it’s cheaper to send the local team. What percentage of local cases would they fly in somebody for?”

A: “It was variable, but I would say **the last couple of months prior to me leaving, it was pretty consistent. If the organ was going outside, like let’s say it was Seattle to California, Seattle to San Diego, they would fly up a San Diego team or a San Ramon team even if the case was a brain death donation...** It used to be how they that if it was a brain death donor, they would fly in the team closest to the recipient center because they were quite certain that the organ was viable and it would be atypical for that organ to be turned down once the team showed up. They would fly the team from closest to the recipient center because they would have to go back there anyway. If it was a DCD, where the donor has to still expire within a certain amount of time for the organ to be viable, they would send the local team because if the donor didn’t expire, then you haven’t invested a significant amount of money to get people from out of state here. So you only have the front-end cost to just be basically the transport from the local hub. That’s how it used to be. **But since they purchased their planes, they basically were like, we’re not going by this model anymore.”**

Q: “This business of flying in people from elsewhere started after they got the planes. They were trying to use the planes?”

A: “**Yes, that’s what it seemed to match up with. That seems to be when it started happening.”**

Q: “And before that, obviously, they would not have sent a charter service from across the country; it could have been done locally. They only did that once they had planes they could bill?”

A: “Unless there just wasn’t a local team available because they were on cases.” – Former OCS Specialist who left recently

As further evidence of the scale and regularity of inflated air transport charges, the OCS Specialist indicated that teams at the Seattle hub were now rarely utilized in favor of flying in nonlocal teams instead – and indicated the transport charge was likely \$30,000 to \$100,000 versus “just a few grand” if driving a local team. Prior to the aircraft being purchased, the ex-employee “was doing almost a case a day to the point...this is overwhelming” but “before I left, I hadn’t done a case in almost two month because cases that were scheduled for Seattle were being assigned elsewhere.” The Seattle had between 9 to 11 OCS Specialists to operate the device, per the ex-employee, but they just sat idle after TransMedics began flying in teams: “it wasn’t unusual for them to go a month or more without doing a case...I would say more so after they got the planes.”

Specialists sat idle for months when TransMedics started flying in planes instead

Q: “How many cases were you doing every month or every quarter? How many organs were you procuring?”

A: “It’s interesting because I was pretty busy. When I was in [redacted], I was doing almost a case a day to the point where I was like this is overwhelming. Like any day I was on call, I was getting called on a case. But out of seven days on call, I was probably doing three to four cases or two to four cases a week. **Before I left, I hadn’t done a case in almost two months because cases that were scheduled for Seattle were being assigned elsewhere.**”

Q: “Do you know what they charged for a local team versus using a plane?”

A: “They would use a third-party service to get us from our local hub to the hospitals, and if it was within a three-hour radius, I assume it was billed based on time or mileage. As far as transportation goes, that’s it. It would just be a few grand, I’m assuming, versus upwards of - depending on where you’re going - \$30,000 to \$100,000.” – Former OCS Specialist who left recently

9 to 11 OCS Specialists in Seattle rarely utilized after TransMedics bought planes

Q: “How many people did they have in Seattle? What was the composition of the team?”

A: **“I want to say between 9 and 11. All OCS specialists.”**

Q: “Were they all sitting around, or were some of them being utilized and you weren’t being utilized?”

A: “Yeah, when I left, it was kind of similar...**it wasn’t unusual for them to go a month or more without doing a case.**”

Q: “Was it always those kinds of gaps or did those gaps start after they got the planes?”

A: **“I would say more so after they got the planes.”** – Former OCS Specialist who left recently

The OCS Specialist stated that the practice occurred nationally, not just at the Seattle hub, and was cloaked in secrecy, given “upwards of 100 OCS Specialists across the country” who could observe questionable aircraft movements on an internal phone app for case assignment and who “probably have the same questions” – “it just doesn’t make any sense...we’ve just not been told...I could see on our app planes being assigned but not understanding why...I could see the whole country.”

OCS specialists at hubs around the entire country could see questionable usage of aircraft

A: “I want to say there were **upwards of 100 OCS Specialists across the country**. We all had company phones, so we would keep in touch using company or personal phones.”

Q: “What would the other OCS specialists say? Did they confirm some of this stuff, like why are they flying in teams when we have local staff?”

A: “I would say that they probably have the same questions. We just don’t know, and **it just doesn’t make sense, and we’ve just not really been told...I could see on our app planes being assigned to what, but not understanding why. It was an internal app for case assignment. I could see the whole country**, anyone who’s on call or not on call; you could go into the app and see who’s being assigned for what...I could see exactly what specialist is being assigned to a particular case. If it’s not a local person that’s being assigned to a local case, it makes me wonder why.” - Former OCS Specialist who left recently

The OCS Specialist resigned because the company was “entirely unethical,” and described a “weird secretive culture” with CEO conduct suggestive of a racket and cover-up, when we asked what specifically was unethical - “weird way that they are utilizing their staff without an explanation and a complete lack of transparency”; “one of the TransMedics pilots...was told by Waleed to not talk to the specialists...they are intentionally keeping people in these dark silos so that they don’t have a grasp of what’s going on”; “everyone is intentionally kept in the dark so that any sort of information couldn’t be used against the company in any way...Waleed was always like, if anyone has questions, refer them back to us....the pilots were like, ‘we were told not to talk to you guys.’”

Secrecy and lack of transparency around usage of aircraft

Q: “You said part of the reason why you left is you didn’t think it was an ethical company. Walk me through the different things that make you think that.”

A: “The biggest one that stands out to me is just this **weird way that they are utilizing their staff without an explanation and a complete lack of transparency** as to what is happening on many levels. One of my close colleagues was told by one of the **TransMedics pilots that he was told by Waleed to not talk to the specialists...they are intentionally keeping people in these dark silos so that they don’t have a grasp of what’s going on.**”

Q: “What were your colleagues’ theories?”

A: “It’s just like this weird lack of transparency, this **weird secretive culture, and it just felt like everyone is intentionally kept in the dark so that any sort of information couldn’t be used against the company in any way. And Waleed was always like if anyone has questions, refer them back to us...the pilots were like, “We were told not to talk to you guys.”** - Former OCS Specialist who left recently

A second OCS Specialist confirmed the air transport billing scheme and provided additional evidence suggesting active, ongoing attempts to manipulate the company's internal systems as a cover-up: "everything at this company feels like a big secret, and it's really weird and sketchy." The ex-employee described shock among staff stationed at hubs when they would see TransMedics fly in teams from around the country, when the local team already at the hub could have simply driven to retrieve the organ – "we have multiple people on call in Seattle...and then all of a sudden...they would fly someone in from Southern California and have them do the case in a hospital in Seattle...no one could ever make sense of why they were doing the things they were doing...people sitting in Seattle doing nothing and saying, why am I not doing this case? And they'd be like, oh, that's logistics, don't worry about it."

Flying in teams when local staff at the hub could drive to hospital instead

Q: "I heard is they have this local team in Seattle, and they're just flying in teams from other parts of the country. I heard sometimes they're even flying in two planes with people from different parts of the country. They're not telling the team why they aren't using them. The suspicion is they want to bill the flight to make more money."

A: **"[Laughs] You're not the only one with that suspicion."**

Q: "Walk me through what you observed there because two or three people have mentioned that to me."

A: "It was all very - **everything at this company feels like a big secret, and it's really weird and sketchy. There would be times when we would have multiple people on call in Seattle. They would have a specialist available. And then you would probably have a surgeon available, too. And then all of a sudden - and it would be at some different part of Washington, like Spokane, which is like a four-hour drive So, instead of sending me from Seattle and a Seattle surgeon, they would fly someone in from Southern California and have them do the case in a hospital in Seattle or somewhere else, And no one could ever make sense of why they were doing the things they were doing.** All of a sudden, these other cases are happening, and nobody knows how they're figuring out who's doing these cases or where they're coming from, and you have **people sitting in Seattle doing nothing and saying, why am I not doing this case? And they'd be like, oh, that's logistics; don't worry about it.**" - Former TransMedics OCS specialist

The second OCS Specialist indicated the practice occurred nationally – “it was happening kind of frequently...I know other people complained about it a lot...it seemed really weird to be flying people in from somewhere else to do cases that were here in the state...nobody knows the answer to why.” For the avoidance of doubt as to the level of egregiousness, we asked if the specialist witnessed cases where both the donor and recipient were in Washington and the local team could have simply driven with the device and surgeon, but where TransMedics still flew in a team from another state: “I believe so [laughs]...this is so ridiculous.”

Flying in nonlocal teams “frequently” from other states, even when the donor and recipient are both in the same state and the entire process could be handled entirely by the local team

Q: “How often would this happen? Did this happen a few times, or was it a very regular occurrence?”

A: **“It was happening kind of frequently. I know other people complained about it a lot. It seemed really weird to be flying people in from somewhere else to do cases that were here in the state.”**

Q: “You said other people were complaining about it in Seattle or somewhere else?”

A: **“In Seattle. I know there were people in Texas complaining because they were getting sent on cases that they thought people in Seattle should be doing. Nobody knows the answer to why.** It was just like, the answer is it’s logistics, and it’s their decision, and that’s all they got.”

Q: “So, people were complaining. Was there ever a case where the procurement was in Washington, and the recipient was in Washington, and they still flew in somebody?”

A: **“I believe so [laughs]...this is so ridiculous.”** - Former TransMedics OCS specialist

The most startling evidence provided by the second OCS Specialist suggests that TransMedics is doctoring records in its internal scheduling system to make it look like a local team is unavailable at a hub, or lacks the credentials for a particular organ - when in fact they are, and are certified - to create a pretext and cover-up for flying in nonlocal teams on its aircraft. The ex-employee described catching TransMedics in the act – “they kind of fucked up one day, and they didn’t realize it”: “another thing that really disturbed me and was absolutely insane and blew my mind was how they assign people...they’re manipulating the system...taking those certifications off people...that makes it look like they’re not on call, so it makes it look like there are less people able to work in this region...they were manipulating it by taking the liver certifications from everyone else....”

Allegedly manipulating internal systems to cover up their tracks to make it appear that no local specialists are available

A: “Another thing that really disturbed me and was absolutely insane and blew my mind was how they assign people. Management was manipulating the system to pick and choose who was having to work...stuff like that was happening pretty regularly. But they kind of fucked up one day, and they didn’t realize it, I think, because they had taken everybody off the board. So, logistics was like, there’s nobody in Seattle to do this case - I don’t think it was Seattle to Seattle, but I think it was Tacoma to Seattle. And so, they flew someone in from LA to do the case. And then management was like, “Why are you there?” And she’s like, “You tell me.”

Q: “You’re saying that they were supposed to pick people based on who’s on call?”

A: “Yeah. It’s supposed to be based on what organ you do, and if you’re next up in the algorithm. And you have this certification; they can see the certifications - essentially, that’s how it was supposed to work. They’re manipulating the system to be like, we just want these people to work basically, so they were taking those certifications off of people. And that makes it look like they’re not on call, so it makes it look like there are less people able to work in this region.”

Q: “You’re saying they were manipulating the system to make it seem like no one was available locally so they could fly in somebody.”

A: “I just know that it looks like there was no one available here and then someone else had to fly here.”

Q: “I see. So, they were making it seem like there was nobody available locally.”

A: “They would have me, who is only doing liver, and then you’d have three other people who do liver and heart. So, they were manipulating it by taking the liver certifications from everyone else and only leaving me. And then I think they forgot that they were doing this, because then it would come to the point where if I was off or I was busy and they’re like, “We have no one else to do liver now.” And then all of a sudden, they would sometimes find someone else and bring their certification back.” - Former TransMedics OCS specialist

UCSF is a pre-eminent transplant center and one of the highest-volume in the world. They are a significant TransMedics customer, but told us that they hope to eliminate their usage of the device in early 2025. A transplant administrator corroborated the air transport billing scheme detailed by the two ex-OCS Specialists. UCSF expressed outrage at the business practices, portraying an out-of-control company pushing any angle to deceive, manipulate, and defraud them for >\$100,000 to \$200,000 in inexplicable air transport charges per case – on top of almost \$100,000 for the device itself – “I hate flying their team all over the fucking country. I hate it. I hate paying for private jets to fly their surgeons all over. I don’t know why we have to pay for all these ridiculous transportation invoices...they just started flying their teams and sending us these invoices”; “very little transparency...they couldn’t tell us what the rates were...we just get a piece of paper that says you owe us for this flight...it doesn’t say anything about what kind of aircraft it was, how long it was in the air...it’s just a number...they could have just made it up...very frustrating.”

UCSF loathes TransMedics “ridiculous transportation invoices”

“Now, I don’t even bother. **I would never even attempt to negotiate anything with them because they won’t negotiate.** The transportation is a major problem for us. Part of the reason why our PO took so long to execute is because they just created their own transportation company and started flying - **I hate flying their team all over the fucking country. I hate it. I hate paying for private jets to fly their surgeons all over. I don’t know why we have to pay for all these ridiculous transportation invoices. And I hate when, now that they’ve got their own transportation company, they just started flying their teams and sending us invoices.** We can’t just pay a random—we can’t pay for flights from a company that we’re not contracted with. They have all these outstanding flight invoices because we don’t have a contract with...there’s **very little transparency.** Contracting with them for the transportation felt ridiculous because they **couldn’t tell us what the rates were. Basically, we just get a piece of paper, an invoice that says you owe us for this flight going from here to here. It doesn’t say anything about what kind of aircraft it was, how long was it in the air—all this stuff. It’s just a number. They could have just made it up. I find that to be very frustrating.**” – UCSF transplant administrator

UCSF stated that the out-of-control air travel invoices started recently, concurrent with the NOP business model and new jets, a sign that TransMedics is resorting to extreme measure to show growth; and that they were given no choice but to use the bundled air transport in order to utilize the device: “now that they’re using their own planes, these are the invoices that we’re getting...that was recent, in the last few months”; “it was messaged to me...that we didn’t have a choice...they’re just telling us what it’s going to cost.” The administrator bristled that TransMedics circumvents their attempts to control the cost, bypassing the administrators and pushing surgeons directly in the middle of the night to approve air transportation, an aggressive tactic alleged by other centers such as Massachusetts General – “there’s no oversight over how they manage logistics”

Invoices suddenly spiked after TransMedics bought their planes; UCSF told “we didn’t have a choice”

A: “Now that they’re using their own planes, these are the invoices that we’re getting.”

Q: “So, was there a recent change where you didn’t have the option of using another transportation vendor, and they forced you to use theirs? Is that a recent change?”

A: “Yeah, that was **recent, in the last few months. The first way that it was messaged to me was that we didn’t have a choice**...the way that it currently works is no one’s getting a quote from another vendor - **they’re just telling us what it’s going to cost.**” - UCSF transplant administrator

Circumventing UCSF administration/oversight by going to surgeons directly for last minute approval; no oversight

“In the middle of the night, there’s an organ offer, and they come up with a transportation quote, and they send it to the surgeon for approval. Now, the surgeon isn’t going to stop what he’s doing and go and get other quotes from other transportation companies or from our preferred vendor. He’s just going to text back and say approved. Currently, I’m trying to get myself in the mix so that they come to me for approval so I can at least monitor it a little more closely, but that means I’m going to be keeping my phone on 24/7, waiting for TransMedics to call me with a flight quote. **There’s no oversight over how they manage the logistics.**” – UCSF transplant administrator

UCSF's billing experience lines up perfectly with the mechanism as described by former TransMedics employees – allegedly flying in teams to run up aircraft charges instead of using a local team that's already available. UCSF recently came to realize that they are being duped, with TransMedics providing no explanation and allegedly flying in multiple planes when none were necessary: “There's no transparency...this just started happening...they're supposed to be invoicing us with flight times and stuff like that...there's zero information...flying their staff from multiple locations to a case...I naively thought that they had a team here...then I realized we are flying in people from other locations...it really is ridiculous.” We note that UCSF has still not grasped the extend of the fraud – that local teams are in fact available but are bypassed to bill unnecessary flights.

Flying in teams across the country when local teams available; no transparency

Q: “So they have a **team available locally and then bill a whole team coming from the other end of the country?**”

A: “Yeah, and there's **no transparency.**”

Q: “Have you asked them for details on flights and logistics, and they've refused?”

A: “**This just started happening...they're supposed to be invoicing us with flight times and stuff like that, and they're not...there's zero information. It doesn't even say what city it went to or where it came from** or anything like that. So, I need to request that. **I'm not sure if they'll do it.**” - UCSF transplant administrator

Flying in *multiple* planes when unnecessary

Q: “Have you seen cases where they are from the Bay Area or **within driving distance, and TransMedics is sending their procurement service, and they're jamming you with an air travel fee?**”

A: “Yes. There have been instances where we are flying people, **flying their staff from multiple locations to a case. I naively thought that they had a team here that would always just be doing the cases here on the West Coast but then I realized we are flying people from other locations.**”

Q: “So, they're flying in two planes to bring a team there when they have a local team?”

A: “That has happened before.”

Q: “**So, they have a local team and they're jamming you with the cost of like two planes.**”

A: “**Yeah. It really is ridiculous.**” - UCSF transplant administrator

UCSF provided multiple examples where several planes were dispatched for the same organ recovery, with air transport charges of \$160,000, and expressed skepticism that TransMedics explanations were truthful: “Now, do I have any way of validating whether that was true...do they have an incentive to lie about it...yes, they do...you just have to trust what they’re telling you...it’s ridiculous...it’s infuriating.” We note again that UCSF does not appreciate the extent of the scheme, as they are led to think no local team was available, as the reason for TransMedics billing air transport costs: “TransMedics didn’t have people locally, so they had to pull people from another location – at least that’s what they said.”

Example of \$160,000 in air transport charges with no visibility or ability to verify

Q: “Explain to me how you ended up having two planes needed?”

A: “We had a donor in Alaska and we needed to take a larger jet so it had enough fuel to get there. Based on the availability of aircraft at that time, we had to take two planes. And then another plane came to bring the team back. And **the first flight was \$100,000. The second flight, I think, was closer to \$60,000. Now, do I have any way of validating whether that was true? Does the transportation vendor have any incentive to tell me the truth? Do they have an incentive to lie about it? Yes, they do. You have to just trust what they’re telling you.**”

Q: “How was it that two planes ended up there - was it that TransMedics kind of just jumped the gun and they just tried to send a plane?”

A: “We realized that these **two planes had been activated; there was really nothing we could do to stop it.** Once you have the plane, the crew, and the fuel, it’s too late to cancel without any exorbitant fees. But I think it was just the fact that **TransMedics didn’t have people locally, so they had to pull people from another location - at least that’s what they said.**” – UCSF transplant administrator

Air travel invoices can be \$40,000 to over \$100,000 per case; \$200K total with device

A: “It’s about \$65,000 for the disposable and then \$20,000 on top of that for the surgical recovery and the perfusionist that accompanies it. **A combined cost of \$89,000, and that doesn’t include the transportation, which can be from \$40,000 to over \$100,000...sometimes there’s a donor in Texas, and to get to Texas and back, we might need to take two planes because of a timeout issue, so that’s going to cost us 80 grand.**”

Q: “So, \$40-100k purely air travel on top of the \$90,000, so **it could cost up to \$200,000?**”

A: “**Yeah, it’s ridiculous.**”

Q: “**And they’re not even telling you where it’s coming and going from?**”

A: “**Yeah, it’s infuriating.**” - UCSF transplant administrator

A West Coast organ procurement organization (OPO) confirmed and elaborated upon UCSF's description of conduct that we interpret as fraud, replying "yeah, oh yeah" when we used the label. We interviewed an industry veteran and executive at the OPO, who stated that the Gosar letter is correct and that centers are coerced into paying for multiple aircraft. TransMedics has a hub in San Ramon, CA "across the street" from the OPO. Previously, if an organ was available in Nevada, for example, the OPO and UCSF would charter and share a single plane and bring back three organs - two kidneys and liver - from a single donor. However, TransMedics now allegedly forces the OPO and UCSF to pay for a separate plane for their device and technician, necessitating two planes by refusing to share one: "TransMedics has now essentially forced the transplant center...if you want access to this technology for their patient, then the transplant center is not able to share a plane with us...they have to go and use the TransMedics-owned and operated aircraft."

OPO and UCSF forced to fly multiple planes even though right next to a TransMedics depot; threatening access to the technology

A: "We're based in the San Francisco Bay area. Let's say that we have an organ donor at a hospital in Reno, and the organ is going to go to UCSF for transplant. You probably saw this mentioned in that letter. Historically, we would charter one plane, and we would all share it together. The UCSF team would join, they would get on a plane. Our team would get on there. We'd go to the OR and procure two kidneys and a liver, and we're able to bring the three organs to the Bay area on that one plane. **TransMedics has now essentially forced the transplant center; if you want access to this technology for their patient, then the transplant center is not able to share a plane with us. They have to go and use the TransMedics-owned and operated aircraft. Their depot is right across the street from our office, our clinical office space.**"

Q: **"This sound like extortion."**

A: **"Yeah, oh yeah."** – Executive at a West Coast OPO, longtime industry veteran

TransMedics conduct makes a mockery of the CEO's reply to Congressman Gosar, cloaking itself in altruistic garb as trying to save lives. Google Maps shows that TransMedics' hub in San Ramon, CA is a 3-minute drive or short walk from the OPO, per a hub address in a job posting. Rather than simply sharing one plane with UCSF and the OPO, the executive described a tragicomical scene where "you can see our team walking out and getting in the car and their team walking out and getting into their car, and we drive to the airport together, and everybody gets onto their own aircraft, and we charter two aircraft, and that's how we get to Reno...and then two planes come right back side by side, but TransMedics is in their plane and they pass all that cost to the transplant center."

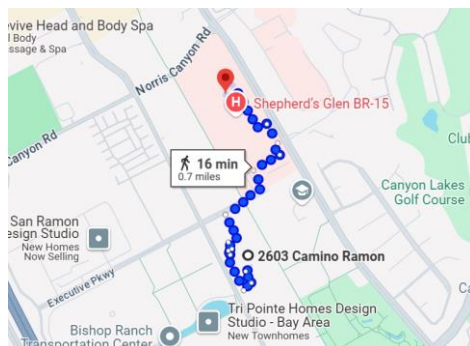
OPO and UCSF forced to fly multiple planes even though right next to a TransMedics depot; threatening access to the technology

Q: "You're saying TransMedics has a depot across the street?"

A: "They have these hubs all over the country, and **they have a hub right across the street in San Ramon**, right across the street from our clinical depot. So, today, we'll have that same organ donor in Reno, and you can see our team walking out and getting in the car and then their team walking out and getting into their car, and we drive to the airport together, and **everybody gets onto their own aircraft, and we charter two planes now, and that's how we get to Reno. And then the two planes come right back side by side, but TransMedics is in their plane and they pass all that cost over to the transplant center.**"

Q: "You are doing a procurement for UCSF. You're chartering a plane to Reno, and they won't let your staff fly on their plane in exchange for sharing the cost, nor will they come with you to make it less expensive for UCSF?"

A: "Yeah. That's exactly what I'm saying." – Executive at a leading West Coast OPO, longtime industry veteran



TransMedics hub is a short walk from OPO facility

The OPO executive described TransMedics conduct as “crazy...they’ll deny it.” He provided examples when “three planes land for the TransMedics operation” for a single case and “get passed on to the transplant center” – one bringing the device from their hub in San Ramon, CA; one with a procurement surgeon flown in from Phoenix, AZ; and a third carrying the device technician from San Diego, CA – “it is crazy what they’re doing...I don’t understand.” He indicated that centers have no other option as TransMedics no longer allows them to operate the device themselves – “they’re not training anyone other than their own staff to operate the device...they also won’t sell you the device.”

Three planes charged to transplant center; multiple ways that TransMedics coerces centers

Q: “It sounds like there are DOJ claims and indictments waiting to happen, which is kind of what Gosar’s letter was saying.”

A: **“Yeah, it’s crazy. They’ll deny it.** There are times where—again, we might have that donor in Reno—perhaps they have a machine available here, and there are times when we have an OR and **three planes land for the TransMedics operation.** One brought the machine from San Ramon. The other one, they sent a surgeon in from Phoenix, and the other plane brought in an OCS specialist, their technician, to operate the machine from San Diego. **It is crazy what they’re doing. I don’t understand.”**

Q: “So, they’re charging people for three planes.”

A: “They charge the fee for the disposable, the charge the transplant centers, to my knowledge, a \$10,000 fee for the technician that operates the device, and as far as I know at this point, **they’re not training anyone other than their own staff to operate the device. They also won’t sell you the device.** Then they’ll charge you \$10,000 for the surgeon when they secure a surgeon, and then on top of that, they’ll pass along to the transplant center the cost of any flight that came in. **So, sometimes, three flights get passed on to the transplant center.**” – Executive at a leading West Coast OPO, longtime industry veteran

The OPO expressed fury, stating that TransMedics conduct resulted in wasted organs that were viable and could have saved lives, by refusing access to the device unless they commit to paying for multiple aircraft and “mystery fees” – what a reasonable person may label textbook extortion – “we might reach out to TransMedics and request that we need to use their technology to save this liver...they will just flat-out say no...organs are wasted where we would happily pay for their service...and they will blanketly refuse to move – they won’t move a finger unless somebody commits to paying not just for the disposable but all the extra mystery fees...you don’t really know how many flights you’re going to pay for until they tell you many flights they’ve used.”

Lost organs by refusing OPO’s access to device unless they pay for flights and mystery/hidden fees; no visibility into their fees Q:

A: ***“They come in and say that they’re in it to increase the number of available organs transplanted to save lives - all of that stuff. We might have an organ donor in Reno, and there have been times where we have an organ donor, and we have to start traveling to that OR because there might be some religious time constraint. But we’re waiting on testing to come back to tell us if the donor is HIV positive, hepatitis, all the standard testing. We might reach out to TransMedics and request that we need to use their technology to save this liver because the test result will be ready 30 minutes before the surgery is supposed to start. They will just flat-out say no. Unless there’s a transplant center that’s interested because that’s their client and they don’t want to say no—I don’t know if the nuances come across—but there are organs that are wasted where we would happily pay for their service and their device or use the device, could rescue these organs that could be transplanted to save people’s lives and they will blanketly refuse to move—they won’t move a finger unless somebody commits to paying, not just for the disposable but all the extra mystery fees that you might get because when you request them when you hire them or commission them to go and perfuse an organ, you don’t know where their machine is coming from. You don’t know whether the technician is coming from another city. You don’t know if you’re going to get the cost of their service plus one round-trip flight from the Bay Area or if you’re going to be on a case that just happens to require three charters. There are these hidden—for the transplant centers, it’s all these mystery fees that you don’t really know how many flights you’re going to pay for until they tell you how many flights they’ve used.”***

Q: *“You’re saying that there’s an organ you want to go get from Reno, and you think the device may be useful for it, and they will not commit to going there until all the testing etc. is done and they know there’s a recipient because they don’t want to do a dry run?”*

A: *“That’s right.”*

Q: ***“And they won’t let you take the device because they have to be trained on it.”***

A: ***“Correct.”*** – Executive at a leading West Coast OPO, longtime industry veteran

The OPO confirmed one of the most damning anecdotes we heard, where TransMedics allegedly extorted a center into paying for “\$300,000, \$400,000 worth of invoices” that appear fraudulent, giving them 48 hours to pay or “they are going to shut off access to the machine” – “they show up with three planes...[the center] believed they ordered a \$100,000 machine and service, but then a month later they get \$300,000, \$400,00 worth of invoices...none of this was agreed to...Waleed reaches out to this transplant center...gives them 48 hours...I just found it very disturbing that they come in with these threats...holding them hostage.”

Allegedly held a center hostage after sending \$3-400K in charges that were not agreed to

“Most recently, I got a message from [redacted]. [Redacted] was a heavy user of TransMedics liver machine. They went into a contract, and then TransMedics started operating in this really inefficient way that we discussed earlier where they sometimes will commit to going to a donor maybe in Reno, but **they show up with three planes, and so then [redacted] believed that they just ordered a \$100,000 machine and service, but then a month later, they get \$300,000, \$400,000 worth of invoices because TransMedics is passing along all of these aircraft fees and the ground transportation fees to them. This colleague was explaining to me the situation where when they went into contract, none of this was agreed on.** And so, internally, they were having trouble getting this routed appropriately through procurement at the hospital and needing to renegotiate or get it into the agreement. And so, **it had been several months where they had not yet been able to reimburse TransMedics for the flight charges that they were trying to pass through to them. Well, Waleed reaches out to this transplant center, which is the largest transplant center in [redacted], and essentially gives them 48 hours to pay all of these flight charges that they never agreed to pay. TransMedics went on and incurred the charges on their own and then tried to pass along the charges to [redacted]. And essentially tells them that they are going to shut off access to the machine to them. They never even agreed to these expenses...I just found it very disturbing that they come in with these threats that their patients will no longer have access to these lifesaving grafts because they now want to be reimbursed for these flights retrospectively that they acknowledged was never in the agreement, but the incurred the charges and they want them to be paid anyway. And holding them hostage.**” – Executive at a leading West Coast OPO, longtime industry veteran

Massachusetts General Hospital, a pre-eminent high-volume transplant center, detailed the exact same abuses on the east coast that UCSF and the OPO alleged on the west coast – indicating that TransMedics’ scheme is national in scope. MGH is one of TransMedics earliest and closest customers – “we were up front, first users of the device because we had a backyard relationship” – but stated that the company’s “tune has changed” and it’s now “about utilization at all costs.” We interviewed a longtime transplant administrator, who described – echoing his peer at UCSF – how TransMedics circumvents their controls and he has to “get involved at 2AM on a Saturday” because MGH suddenly gets hit with outrageous charges for a routine organ recovery: “I find out down the line...it’s upwards of \$180,000 to almost a \$200,000 all-in with the procurement team decision that was made without any one of us involved.”

MGH soured on the relationship, bypassing protocols to run up the cost to \$200K; “utilization at all costs”

*“When I first became involved with the discussion of partnering with TransMedics for their liver and their heart OCS device, and then lungs after FDA approval...yes, it does come at a cost, but we were upfront, first users of the device because we **had a backyard relationship with them**, but also, we, at my institution, were trying to be innovators across all fronts, and that’s also one of the reasons why we wanted to partner with them. **As the relationship evolved over the years, the tune has changed**. It has become less about what can we do together and trying to figure out the best way, the best logistical setup, and the best-aligned incentives to achieve our mutual goals to become **more about utilization at all costs**, meaning sometimes there would be situations where I get involved at 2:00AM on a Saturday because so-and-so had said we want to use the OCS device and then now the entire team takes a hold of the entire logistical setup and I find out down the line that something where it’s a typical we fly out the teams, we take the procurement team with us with the modules, and **now it’s upwards of like \$180,000 to almost a \$200,000 all-in with the procurement team decision that was made without any one of us involved.**” – Veteran transplant administrator/executive at Massachusetts General Hospital*

MGH employs a term called “Never Events” for outlier situations that should never occur, which happen with TransMedics “at a higher frequency...sometimes even a couple of times a week” where MGH is hit with extreme charges for an organ recovery. The administrator was careful in using diplomatic language - “a certain threshold that gets crossed as far as costs” – but was clearly referring to the same transportation billing scheme as UCSF, painting TransMedics as a recidivist offender: “I’ve had to have very candid conversations with their leadership...many times...they’ve apologized profusely, so sorry that this has occurred, never our intention, won’t ever happen again...and then again, there will be a situation where someone had made a call independent of our surgeons”; “I would then have to...go straight up to their leadership...this is not what we discussed...we either need to end this or address this...these types of issues continue to sprout....”

“Never events” occur on a regular basis, monthly or “couple of times a week,” where “cost threshold” crossed
“And we’ve had a couple of these situations in the past where I’ve had to have very candid conversations with their leadership. Many times, there have been situations where they’ve apologized profusely, so sorry that this has occurred, never our intention, won’t ever happen again. We won’t have any disagreements at all for a couple of months, and then again, there will be a situation where someone had made a call independent of our surgeons, and it just would continue to occur at a higher frequency, almost once a month, sometimes even a couple of times a week where now I have to get involved whenever there’s a certain threshold that gets crossed as far as costs go where I would then have to bypass the entire team, go straight up to their leadership, and say, this is not what we discussed, and we either need to end this or address this,...these types of issues continue to sprout, and I’ll be honest, part of me also—I’ve been in situations where “never events” should never have occurred.” – Veteran transplant administrator/executive at Massachusetts General Hospital

The “Never Events” which MGH stated occur with weekly regularity refer to the practice of allegedly flying in multiple jets for a single organ recovery, a recurring theme in our transplant center interviews – “a free check to drive up the cost...any time that two planes fly in...I mean, this just happened the other day...indiscriminate use just to drive up utilization.” The executive detailed an organ recovery in Denver, as an example, where TransMedics attempted to route and send multiple planes in a way that “was just weird” – “I literally had to walk through with the coordinator, no, that doesn’t make any sense...try again...I got a phone call in fifteen minutes, and they gave me the actual option that should have happened...that’s a lot more economical.”

“Never events” like flying in multiple planes indiscriminately

Q: “So, what is happening once or twice a week? Are they trying to run circles around you? What are they doing exactly? And then I want to talk about the never events you mentioned and not reporting stuff to the FDA.”

A: “It runs the gamut of just having **a free check to drive up the cost, having indiscriminate financial stewardship of the resources that are being used...it’s really just indiscriminate use just to drive up the utilization,** which happens - now **any time that two planes fly in I mean, this just happened the other day.** Two planes...it was a pair of lungs coming over from Denver, and it had just been a plane that took the team out and came back with the organ or put it on the pump, and it would be an unmanned, one-way organ over to Boston and that would have been it. But instead, they had to take one plane from Dallas, and the other interesting thing is the Dallas plane over to Denver, and there was another procurement team that’s coming from nearby that needed to be flown out. And then the other team had to come back to Dallas. **It was just weird,** hey, we’re going to be doing this...I literally had to walk through with the coordinator; no, **that doesn’t make any sense; try again.** Call me back in fifteen minutes, and you can give me a couple more options. What ended up happening is **I got a phone call in fifteen minutes, and they gave me the actual option that should have happened, which was to take a plane, go to Denver, fly it back, and then we’ll fly the rest of the crew back on a commercial jet. That’s a lot more economical.**” – Veteran transplant administrator/executive at Massachusetts General Hospital

Racketeering strikes us an appropriate term given the conduct MGH described – a label another customer used directly: **“when you ask questions...you get a bunch of roundabout answers...oh sorry, we mixed up that time zone because it was central times versus eastern time.”** MGH naturally expressed incredulity and dismay – **“if you mix up the time zones, I don’t know...it’s their team that racks up the price...all these added fees like a used car salesman...it’s not what we signed up for, nor is it what we need.”**

“Roundabout answers” and acting like a “used car salesman”

Q: *“Do they even give you transparency as to where the planes are coming from, who’s on them, and what the invoices are? I’m hearing that they don’t provide that information even when asked. Somebody used the phrase mystery fees.”*

A: *“Not only that of the mystery, when you ask questions to them, you get the answer then, and then **you get a bunch of roundabout answers** down the line: Oh, sorry, we mixed up that time zone because it was central time versus eastern time. And I’m like, guys, we’re professionals here. This is what you’re being paid to do. **If you mix up the time zones, I don’t know. I’m literally at odds** and on top of that, here I am doing your job for you. Again, it just gets to a point where it’s a big point of frustration...**it’s their team that racks up the price** because depending on the size of the jet, depending on if you’re going to bring the surgeons on or not, if it’s just a module or not, and you have **all these added fees like a used car salesman** saying, oh, by the way, you can this and get it financed at zero percent but then there are all these other things that can just be added on, and **it’s not what we signed up for, nor is it what we need.**”* – Veteran transplant administrator/executive at Massachusetts General Hospital

Every major transplant center we interviewed conveyed what appear to be racketeering-type tactics. A major center, whose experience we already covered and who referred to TransMedics conduct as “mafia-style” and “thug-style,” described the modus operandi on air transport overcharges: “they jack up the prices like crazy...there’s a lot of variability...and they do not commit to a price on their air travel...it is very hard to pin them down.” Cedars-Sinai in Los Angeles, another high volume user, indicated shock after getting charged for multiple flights for a single organ recovery: “we have had extra cost for transportation and we’re like, ‘what is this?...and we hear, ‘oh, the surgeon didn’t come with the perfusion team...I’ve gotten bills like that.”

Allegedly won’t provide air transport pricing ahead of time; “hard to pin them down”

A: “There’s **so much variability in the flights**, it depends on aircraft, it depends on. If the Kentucky Derby is going on, you can’t find a plan anywhere, and **they jack the prices up like crazy. There’s really a lot of variability in air travel and they do not commit to a price on their air travel. It is very hard to pin them down**, and the University of [redacted] is very strict about this. You have to give a price...you can’t say, okay, I know if I’m going to San Francisco, I’m paying \$15,000 because some days it might be \$15,000, and it might be—they’re not going to tell you that amount of detail. They’ll tell you when you’re leaving. So, you book a case and they’re going to go, “Here’s the cost of your case.”

Q: “They don’t want you to be able to comparison shop with air travel.”

A: “That’s right.”

Q: **“So, you’ve tried getting pricing from them for the air travel and they won’t tell you?”**

A: **“Right.”** - Transplant surgeon at a high-volume academic center

Cedars-Sinai: lack of transparency and surprising charges for air transport

“I mean, we have had extra cost for transportation and we’re like, “What is this?” And we hear, “Oh, the surgeon didn’t come with the perfusion team. The surgeon came from Dallas, so we had to send him back to Dallas.” I’ve gotten bills like that.” – Transplant surgeon

The University of Utah is a high-volume user of TransMedics device, and not only are its air transport charges as egregious as UCSF, Mass General, UCSD, and other centers but it illustrates the extreme level to which TransMedics has pushed its scheme – one could accurately describe it as an orgy of billing fraud. A transplant surgeon affiliated with the University of Utah stated they pay a mind-boggling \$120,000 to \$150,000 in air transport charges per case. But here's the rub: he indicated that basically 100% of these cases involved a donor and recipient within a driving radius of 250 miles – in other words, they are paying six-figure transport charges to fly in TransMedics jets when a van would suffice with perhaps a \$3,000 cost.

Charging \$120,000 to \$150,000 for air transport when donor and recipient are within driving distance

A: "Depending on where the flight is and stuff like that - **it's anywhere between \$120,000 and 150,000.**"

Q: "The livers that they're procuring for you, how far away are they coming from typically - within the region?"

A: "Most of the time, they're within the region because that's what we're being offered. The new liver allocation is getting ready to start, and we're going to start seeing national sharing, but **livers are still within the 250 nautical miles**. We don't really get offered anything of significance, say, from New Jersey."

Q: "**The 10 times you've used them recently, it's all been within 250 miles?**"

A: "**Yeah, pretty much.**"

Q: "They're using their air transport to make it quicker? Or are they using a van or something? It's kind of within the four-hour distance, driving distance."

A: "**No, it's all air** because you want to get the organs back quickly." – Transplant surgeon, Utah center

When we asked the surgeon why they pay up to \$150,000 for air transport when the organ recovery is within the standard 4-hour driving range, his answer was telling and betrays the heart of the scheme - “it’s all air because you want to get the organs back quickly.” The main benefit that TransMedics pitches is keeping organs viable for longer on a pump, thus enabling long-haul transport – which means they have no reason to dispatch jets as part of its NOP service when almost all of these Utah cases are less than 4 hours driving range – except, we conclude, to bill jets when none are necessary. The details the Utah surgeon shared are shocking – that TransMedics allegedly bills them for multiple private jets even both the donor and recipient are both “just down the street from us” in Salt Lake City: “they fly for almost everything”; “since they’ve gotten their own jets, everybody just flies in their own jets,” meaning the device operator in one jet and the procurement surgeon in another.

Flying in multiple planes even when the donor and recipient are in the same city

Q: “Do they do flights even if it’s an hour’s drive or something?”

A: **“I feel like they fly for almost everything,** but most of the time, the surgeon is not local, and the perfusionist isn’t local, and that may be something that we’re trying to change for us where if **the donor’s just down the street from us** but you have to bring in a surgeon from California to do the donor and the perfusionist is from wherever, **that’s what seems to happen....since they’ve gotten their own jets, everybody just flies in their own jets.”**

Q: “So, you’re saying even if the donor is nearby, they just fly in a plane from across the country?”

A: “Yeah, because you have two components. You have the perfusionist and the surgeon, and it depends on where they are.”

Q: “Have there been transplants you’ve done that were intracity or really close by, but they still had to fly in the team?”

A: “Yeah....every perfusionist I’ve met is from elsewhere.”

Q: **“When the donor and the recipient were within Salt Lake City, did they still fly in a plane from someplace else?”**

A: **“Yeah.”** – Transplant surgeon, Utah center

As a preview of a later section on what we believe to be an organ trafficking scheme in exchange for device volumes, we note a particularly chilling comment by the surgeon, when we asked how he feels about being charged stratospheric amounts for private jet transport when the organ could easily have been recovered by car: “the metrics I’m held to as a transplant surgeon are the number of liver transplants I do...so, I don’t really care about cost.”

Transplant surgeon doesn’t care about cost, only transplant volumes

Q: “How does it make you feel that to be charged \$50,000 just to fly a perfusionist and get utilization on a \$13 million plane?”

A: “I think it’s highly expensive, but **the metrics that I’m held to, as a transplant surgeon, are the number of liver transplants I do...so, I don’t really care about cost.**” – Transplant surgeon, Utah center

Part 3. Revenue is dependent on a handful of transplant surgeons/centers - “prostitutes” who are “completely owned and operated by TransMedics” - who allegedly receive kickbacks and inducements.

TransMedics revenue - and growth to date - is dependent on a handful of dubious physicians and centers, often of the same Egyptian or Middle East descent as the CEO and members of the leadership team – allegedly “prostitutes” who are “completely owned and operated by TransMedics,” according to other surgeons we interviewed. We believe these high-volume OCS users a) receive what we think are inducements and kickbacks via stock, lavish travel, and other means; and b) we think that they are beneficiaries of high-risk organs that reputable surgeons won’t touch, which we believe to be improperly steered their way as part of a quid pro quo that they arrive on an OCS pump; and c) that they achieve these unusual volumes via vast off-label usage. A surgeon who runs a leading West Coast academic transplant center described a dynamic we see in almost every medtech or biotech fraud we short, when we asked if the CEO has a “little inner circle”: “He does...Waleed’s got people like that, that will stick by him because they’re conflicted...they’re making a lot of money consulting and speaking.”

Dependent on surgeons who are allegedly questionable

Q: “How much customer concentration does he have in liver? Are there centers that are known as Waleed or OCS centers that are just –”

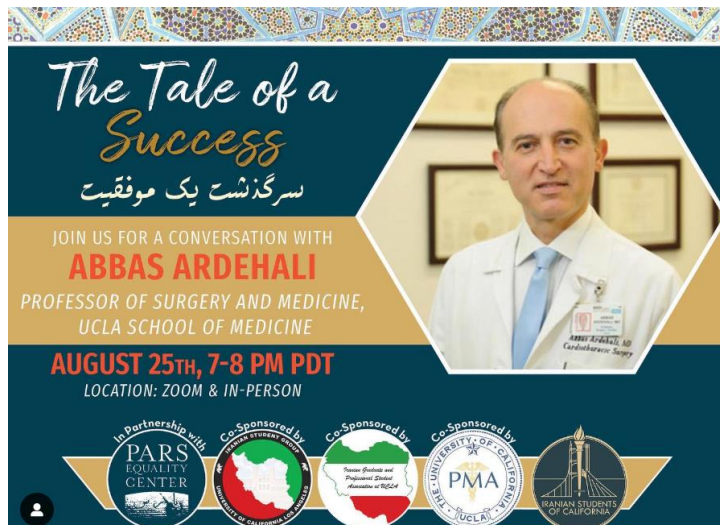
A: **“Prostitutes? [Laughs]”**

Q: “Yeah. He must have his little circle.”

A: **“He does. He has his paid - the biggest whore in my field in cardiac surgery was a guy named Abbas Ardehali at UCLA who was going around telling everybody that every heart should be put on the OCS system.** And he’s even in the trial for doing standard brain-dead donor hearts, putting it on saying that it’s going to have less coronary artery disease. **That’s an example of someone who’s completely owned and operated by TransMedics.** He’s got his box of golden PowerPoint slides, travels around the world, and probably makes \$100,000 per year in consulting fees. Fortunately, there are not that many people like that out there in the field, but there are certainly enough. **Waleed’s got people like that, that will stick by him because they’re conflicted. They’re making a lot of money consulting and speaking. But the vast majority of surgeons in the field are not wedded to him and are in other trials right now.”** – Prominent transplant surgeon, leadership role at a West Coast academic transplant center

The suspect surgeons in transplant surgery appear to be well-known, as it is a small field. In the previous quote, the transplant chief stated that the “biggest whore is my field in cardiac surgery was a guy named Abbas Ardehali” who allegedly “was going around telling everybody that every heart should be put on the OCS system...that’s an example of someone who’s completely owned and operated by TransMedics.” Ardehali is a self-promotional, LA-based “celebrity surgeon” who gives TED talks with claims about the OCS pump that we view as false, reckless, or quackery – “he is developing something called the...”heart in a box” for TransMedics...as the principal investigator, he is quite enthusiastic.” He even sports his own Wikipedia entry, presumably self-written – “a pioneer in the field...constantly pushing the boundaries of what is possible,” who has been interviewed by ABC News, CNN, Fox News, NBC News, Al Jazeera, and others.

Social media posts featuring Dr. Ardehali



Source: Scorpion Capital consultation calls with experts; <https://www.laweekly.com/abbas-ardehali-i-a-transplant/>; <https://www.instagram.com/p/ChFh3k-pORX/?igsh=MWNvNW9iZTBxdDF0eQ==>; <https://images.app.goo.gl/ITHWTHiiGx9k7PQA>; https://www.ted.com/talks/abbas_ardehali_the_miracle_of_organ_donation_and_a_breakthrough_for_the_future/transcript?subtitle=en

In his TED talk, Ardehali suggests one can put pneumonia-infected lungs on the OCS pump which “can actually repair a donor organ” with supernormal doses of antibiotics that would cause “toxicities” in the donor – transplanting infected lungs strikes us a good way to lose one’s medical license and be sued out of business for malpractice. Moreover, we are curious how the TransMedics pump protects lung cells and tissue from doses so high that they would harm the rest of a donor’s body. The claims for machine perfusion become increasingly preposterous: “we can convert different blood types to an O blood type...we can actually make changes in the genetic makeup of the donor organ...we can add genes...silence genes...those foreign proteins that were the cause of rejection.”

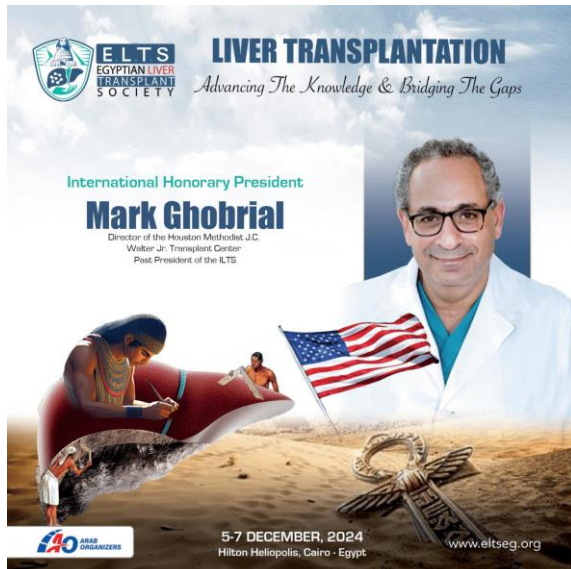
TED talk by Dr. Ardehali, Dec 2023 – video stills



For liver, the highest volume and most suspect users of the device appear to be well-known, as it is a small field. Surgeons indicated the following are among the top users and repeatedly flagged them across our interviews as engaging in questionable practices: 1) Rafik “Mark” Ghobrial at Houston Methodist; 2) Marwan Abouljoud at Henry Ford in Detroit; and 3) Mayo Arizona. Ghobrial attended medical school at Cairo University and seems to be “Honorary President” of the Egyptian Liver Transplant Society, while Abouljoud attended medical school in Lebanon; both then trained in the US and are now feted at galas for being “internationally” or “world renowned.”

Drs. Ghobrial and Abouljoud; headline excerpted from 2021 article about the latter

World-renowned transplant surgeon named Arab American of the Year



One surgeon estimated that “those true religious zealots to Waleed represent 50% of the business.” Another prominent surgeon indicated that Ghobrial and Mayo alone may be 300 devices a year. We think Abouljoud does at least 100, for 400 total across these 3 users – roughly \$34MM in product revenue using an average consumable cost of \$85,000, and \$60MM in total revenue using \$150,000 inclusive of NOP fees, by our math, which implies that just 3 users are 15% of total LTM revenue of \$400MM. The customer concentration is striking when one adds the volumes from a just a few others. whose questionable practices we detail in other sections: UCSD, which we estimate does 100 TransMedics cases; University of Utah, which we suspect will do over 50; and a center run by another Lebanon-trained physician, who told us he’s on track for 50 OCS pumps in 2024; and one in LA using 50 – another 250 devices. Using the same assumptions, these 4 comprise another \$38MM in revenue – that is, an estimated 25% of total TMDX revenue from just 7 high-risk users.

Half of TransMedics business may be dependent on “religious zealots to Waleed”

*“I would say that maybe **those true religious zealots to Waleed represent 50% of the business.**” – Prominent transplant surgeon, leadership role at a West Coast academic transplant center*

Two top users may be ~300 devices per year

***“Mayo in Arizona is their biggest center, and the second-biggest center is Methodist in Houston. Arizona might be doing 180 livers a year, something like that, with a pump. And then, Houston, maybe 100, 110.”** -Transplant surgeon and division chief of the liver program at a world-renowned academic center*

Henry Ford allegedly is one of TransMedics highest-volume users

*“Those are the biggest users that I know. And I think that MGH in Boston, I don’t know now that the chief of MGH is gone. He moved to UPenn, but the chief at MGH was one of the PIs for the initial trial, the PROTECT trial. They were using the pump in a fair number of patients. I also know that **Henry Ford in Detroit, he was also using the machine quite a bit. Marwan Abouljoud.**” - Transplant surgeon and division chief of the liver program at a world-renowned academic center*

Our interview with one of the high-volume users of Middle East origin - we redact his name - was particularly amusing for the number of times he protested his “innocence,” a classic case of *mens rea* – “guilty mind” in Latin, a concept from criminal law - as we never asked about his ties to TransMedics. When we finally noted that he was a top user, he tap danced faster: “I don’t know the CEO at all. I don’t know him personally. Have I actually seen him, met him? Yes, but I don’t have any personal connection. I honestly don’t have any personal connection to anyone at this company. I’ve never had an email from TransMedics...honestly... there’s no conflict of interest...I don’t know what their stock is doing....I am not by any means an insider to that circle...I do not have any involvement...if I was someone that TransMedics in any way engaged with and I was involved with activities that they in any way sponsor...which I absolutely never have...I literally have never received as much as an email or communication from TransMedics...”

One of TransMedics highest volume users played dumb and denied any interaction with TransMedics

A: **“I don’t know the CEO at all. I don’t know him personally. Have I actually seen him, met him? Yes, but I don’t have any personal connection. I honestly don’t have any personal connection to anyone at the company.** They’ve never asked for my opinion about anything. They’ve never asked me to do - nothing, really. **I’ve never had an email from TransMedics.”**

Q: “Based on your volume, you are one of their highest volume users. Usually, companies do a full court press to at least stay in touch.”

A: “Our transplant institute director may have more contact, but I certainly don’t. Honestly, I don’t invest in healthcare on purpose in any healthcare sector so that there’s no conflict of interest with my job. **But I don’t know what their stock is doing. I have no idea what any of that stuff is doing. Because they are a publicly traded company, right?... I am not by any means an insider to that circle**....for kidney, I think we selectively use OCS. I do not have any involvement in kidney, so I don’t know.”

Q: “They don’t have a kidney product.”

A: “They don’t? Okay. Well then, there’s heart and lung, right?”

Q: “And you run the liver transplant program there? You’re in charge of it?”

A: “On the medical side, correct. I run the liver center. If I was someone that TransMedics in any way engaged with and I was involved in activities that they in any way sponsor, **which I absolutely never have. I literally have never received as much as an email or communication from TransMedics** other than updates on the data that they have or anything like that.” – Transplant hepatologist and director of the liver program at a Midwest academic center

Multiple surgeons warned us about Ghobrial at Houston Methodist and Abouljoud at Henry Ford, expressing skepticism at the spike in their volumes. These surgeons indicated they have personal and/or professional relationships with both – “certain things are a little cloudy, especially with Methodist...it is well-known that he owns a lot of stock from...that is a lot of conflict of interest there...it is very well known by everybody...this is not a secret.”

Alleged conflict of interest at Houston Methodist

A: “Mayo in Arizona is their biggest center, and the second-biggest center is Methodist in Houston. Arizona might be doing 180 livers a year, something like that, with a pump. And then, Houston, maybe 100, 110.”

Q: “That’s a big chunk of the company’s revenue right there.”

A: “It is, but let me tell you something [chuckles]. There are **certain things that are a little cloudy**, especially with Methodist. So, **the chief of transplant at Methodist, it is very well known, that he owns a lot of stock from TransMedics. That is a lot of conflict of interest there**, which I don’t know how he managed with the cost, but it is **very well known by everybody that the chief of transplant at Houston Methodist owns a lot of their stock.**”

Q: “This is Mark Ghobrial?”

A: “That is correct. **It’s very well known. This is not a secret.** Everybody knows that Mark Ghobrial owns a lot of stock for TransMedics. Yes, they use a lot of their pumps.”

Q: “The other thing I noticed about this company is Egyptian. The CEO is Egyptian, the chief commercial officer is Egyptian, the CEO’s sister runs the lung product is Egyptian, and Mark Ghobrial is Egyptian.”

A: “I know that Mark Ghobrial is Egyptian..”

Q: **“He’s known as TransMedics’ whore, basically?”**

A: **“That’s correct.”** -Transplant surgeon and division chief of the liver program at a world-renowned academic center

The chief of a liver transplant program, who stated he knows Ghobrial well and appears to be in communication, indicated the relationship with TransMedics is improper and may be seen as a kickback: “Methodist is a huge user...yeah, the conflicts with Mark, you can see the obvious one...the Ghobrial relationship is a different flavor to me...there are similar backgrounds [between the CEO and Ghobrial].” The surgeon stated that he participated in the TransMedics pivotal study but “cut those ties very early on after the trials” – “we’re very, very, intentional about not having any relationship with them or financial conflicts of interest” – in contrast to Ghobrial who aggressively promotes the device: “Mark’s been fairly deliberate about having his junior folks out front in a lot of the talks and whatnot for TransMedics.”

Transplant chief troubled by Ghobrial’s conflicts of interest with TransMedics

A: “Certainly, **Methodist is a huge user...yeah, the conflicts with Mark, you can see the obvious one...the Ghobrial relationship is a different flavor to me...it is a little bit different...there are similar backgrounds.**”

Q: They’re both Egyptian.”

A: **“And the stock ownership - one has to question that for sure. On our end, we’re very, very, very intentional about not having any relationship with them or financial conflicts of interest and we cut those ties very early on after the trial.”**

Q: “Is Ghobrial, essentially a kickback - is that what it is with him?”

A: “That’s what I was going to say....**the stock and consulting and stuff like that could potentially be seen as such** in some manner. Mark’s been fairly deliberate about having his junior folks out front in a lot of the talks and whatnot for TransMedics.”

Q: “So, he’s really promoting it. Is he viewed as a competent, ethical guy? It looks like he just kind of rode this horse, and their transplant volumes went through the roof.”

A: **“Yes. [Chuckles] I’m surprised, probably more than anybody, at their success with him at the helm but they’ve been very successful in growing a very big program. I’m surprised by it...having seen what they started with, I’m shocked that it has blown up into the program that it is.”** - Transplant surgeon at a high-volume academic center

One surgeon after another questioned the transplant practices that led unremarkable centers like Ghobrial at Methodist and Abouljoud at Henry Ford to skyrocket – and their utilization of TransMedics with it: “I’m surprised by it...having seen what they started with...I’m shocked that it has blown up into the program that it is.” As part of what we believe to be an organ steering or “trafficking” scheme orchestrated by TransMedics, surgeons indicated that the growth has come from accepting organs that reputable centers won’t touch, which they sling onto the TransMedics pump to create plausible deniability about the organs’ inferiority – a practice that even some high-volume TransMedics “whores” recoiled at, such as one high volume OCS user, who appears to have an ethical line even he won’t cross: “we’re not in the business of transplanting people with 70-year old livers, like other places do. I know Methodist does a lot of older donors.”

Houston Methodist allegedly implants livers from extremely old patients

A: “There’s a peak probably in the 30s, that’s the MVA group, and then there’s another peak in the 60s.”

Q: “If you had to ballpark what percentage of the OCS liver donors are over 55, what’s a very rough guess?”

A: “30%...the older peak, **we’re not in the business of transplanting people with 70-year-old livers, like other places do. I know Methodist does a lot of older donors.**” – Transplant hepatologist and director of the liver program at a Midwest academic center

Transplant chief surprised by sudden growth at Methodist

Q: “So, he’s really promoting it. Is he viewed as a competent, ethical guy? It looks like he just kind of rode this horse, and their transplant volumes went through the roof.”

A: **“Yes. [Chuckles] I’m surprised, probably more than anybody, at their success with him at the helm but they’ve been very successful in growing a very big program. I’m surprised by it...having seen what they started with, I’m shocked that it has blown up into the program that it is.”** - Transplant surgeon at a high volume academic center

Surgeons singled out both Methodist and Mayo Arizona as growing their transplant and OCS volumes by being “willing to take organs that others haven’t been willing to” – we emphasize that this quote is from a liver transplant surgeon who may be the most respected KOL in the field, based at an academic center in Chicago. If perhaps the top transplant surgeon in the world won’t touch these organs, we wonder what makes Ghobrial, Mayo Arizona, and Abouljoud think they know something he doesn’t. Surgeons alleged that Ghobrial has “more than doubled” the size of his program through such questionable transplants, that “the guy at Mayo Clinic at Scottsdale has more than doubled theirs”; and that “Scottsdale went from being a medium-sized program to two years ago being the largest transplant program by volume in the country...I believe that was mostly through the expanded of organs, specifically OCS.”

Highest-volume TransMedics users have allegedly ramped their transplant programs by using questionable organs

“Mark Ghobrial has definitely grown his program, more than doubled its size. The guy in Mayo Clinic at Scottsdale has more than doubled theirs. Part of it is that they’re **willing to take organs that others haven’t been willing to** - older donors, all that stuff.”
– One of the top liver transplant surgeons in the world, based at a leading Midwest academic center

Mayo Scottsdale allegedly became largest transplant program in the US by aggressive use of OCS

“So, a place well-known for using TransMedics nationally is Mayo Scottsdale. Scottsdale went from being a medium-sized program to two years ago being the largest transplant program by volume in the country. I believe that was mostly through the expanded use of organs, specifically OCS, is my understanding.”– Transplant hepatologist and director of the liver program at a Midwest academic center

Part 4. These high-volume users may allegedly be under scrutiny as part of multiple investigations by federal and states authorities and agencies, as well as by prominent news outlets, indicating the precarious nature of TransMedics business.

We suspect that TransMedics and its highest-volume users may be under scrutiny as part of multiple federal/state regulatory investigations – which we would find unsurprising, irrespective of hearsay, given the flagrant conduct and rapidly unfolding backlash at centers across the US. Furthermore, a surgeon in a leadership role at a world-renowned center told us that the Washington Post has been “reaching out to some transplant surgeons across the country, like a very prominent surgeon...I know that a reporter was reaching out to a surgeon because they were working on a report.” We suspect the surgeon spoke with the reporter, given his stature in the field: “I heard that the Washington Post is working on a report because there is some concern that companies are giving kickbacks to some surgeons or hospitals for using the machine”; “this is something that is not published...I heard that there is a reporter working in TransMedics, specifically.”

Washington Post reporter allegedly reached out to surgeons regarding alleged kickbacks from TransMedics

A: “I heard a little rumor, but again, this is rumor, okay? **I heard that the Washington Post is working on a report because there is some concern that companies are giving kickbacks to some surgeons or hospitals for using the machine.”**

Q: “I saw that article. But that was talking about OPO’s. Was that talking about TransMedics—they didn’t mention TransMedics.”

A: “No, no, no. **This is something that is not published. I heard that there is a reporter working on TransMedics, specifically.** This is not about OPO’s”

Q: **“You heard the Washington Post is working on TransMedics getting kickbacks?”**

A: **“Yeah, that’s what I heard. But again, this is just a rumor that I heard from some people, that they are working on a report.”**

Q: “When did you hear this about the “Washington Post?”

A: “I heard that a few months ago. **The Washington Post was reaching out to some transplant surgeons across the country, like a very renowned transplant surgeon; I know that a reporter was reaching out to a surgeon because they were working on a report.**” - Transplant surgeon and division chief of the liver program at a world-renowned academic center

The surgeon indicated the Washington Post inquiries were recent, and while oblique, provided specifics about the reporter's line of questioning that lend credibility: "because everybody is like, how is it possible that some of these hospitals are doing so many TransMedics? Because financially, it's not possible to do too many...usually you tend to be believe that centers that use the device a lot – maybe they have some incentive to use it that much...Houston, maybe...Henry Ford, maybe...." Although he stated several times that this was rumor - "I heard a little rumor, but again, this is rumor, okay?" – we suspect that numerous surgeons, OPO's, and transplant centers in the field have been interviewed by reporters and/or federal/state investigators, as several told us directly that they had been contacted and/or questioned. That something "big" may drop seems to be an open secret – transplant surgery is a small field where word travels fast, with only 400 surgeons in the US for liver and ~50-80 responsible for the vast majority, and even fewer for heart and lung.

Washington Post reporter allegedly reached out to surgeons regarding alleged kickbacks from TransMedics

Q: "And they were trying to determine if TransMedics was giving kickbacks to surgeons?"

A: "Surgeons and hospitals. **Because everybody is like how is it possible that some of these hospitals are doing so many TransMedics? Because financially, it's not possible to do too many.**"

Q: "Were there any other names that were thrown around as far as who could be getting kickbacks?"

A: "Those are all rumors, right? **Nobody told me Ghobrial is getting a kickback, or Marwan is getting a kickback or the guy at MGH is getting a kickback. I heard a rumor that some centers are getting a kickback and that's the reason that they are working on a report at the Washington Post. Usually, you tend to believe that the centers that use the device a lot - maybe they have some incentive to use it that much.**"

Q: "If you had to speculate, who do you think those centers or those doctors might be?"

A: **"Houston, maybe; Henry Ford, maybe. Probably, I would say that those are the two centers** that I would be more..." - Transplant surgeon and division chief of the liver program at a world-renowned academic center

The CEO of a major OPO (organ procurement organization), a longtime industry veteran, seemed to corroborate the surgeon's color and stated that he had been "interviewed for it," although it was unclear whether he meant the Washington Post or something else – "there's so much that's going on to be found out...do you know everything that's going on with the OIG and DOJ and the IRS...it's deeper than [the Washington Post]...here's the drama...[redacted] recently interviewed with the Department of Justice, the OIG, and the IRS...and with the DOJ out of Washington and out of New York, in particular...and then the state attorney general...Medicare fraud...the most prominent attorney general that's getting involved is Virginia...."

Multiple regulatory and law enforcement agencies at a federal and state level allegedly interviewing industry participants

Q: "A surgeon I just talked to said I'll tell you who their whores are. He mentioned a couple of very high-volume users. It sounds to me like **a bribery, pay-to-play kind of thing.**"

A: "**There's so much that's going to be found out. Do you know everything that's going on with the OIG and DOJ** and the IRS?"

Q: "I read that Washington Post article."

A: "**It's deeper than that, and I'll let you know because I've been interviewed for it.**"

Q: "Walk me through what you're seeing. And how TransMedics may be a part of this stuff that the Washington Post is talking about as far as kickbacks."

A: "Here's all the drama. [Redacted] **recently interviewed with the Department of Justice, the OIG, and the IRS. And with the DOJ out of Washington and out of New York, in particular. And then the state attorney general** is getting involved in this as well because if there's Medicare fraud, then there's Medicaid fraud. And probably **the most prominent attorney general that's getting involved is Virginia**, and the reason why that's important is because UNOS is there, and so is A-OPO, which is the OPO governing body, not a governing body but a trade association, and they also have a terrible OPO based there." – CEO of a major OPO (Organ Procurement Organization), longtime industry executive

The CEO of the OPO implied an extensive, far-reaching OIG investigation into industry practices and suggested that TransMedics is vulnerable, indicating potential action in 2025: “they’re going to investigate every single OPO...every single transplant program.” Regulators crosshairs are focused on the Cost Report, the loophole at the center of TransMedics’ extortionate pricing for consumables and air transport: “a lot of their investigation has to with the Cost Report...how we’re reimbursed...overcharging the whole time...there just hasn’t been any oversight on it...things are changing so fast...they’re going to come out and fix thigs...absolutely, it’s been getting abused...the government is going to come down, and they’re doing to start setting what’s reasonable cost.”

OIG investigating every key transplant center and OPO’s as part of investigation into overcharging via Cost Report

“OIG recently announced that they’re going to investigate every single OPO, and they’re going to investigate every single transplant program, and it’s going to happen by the end of 2025. So, that announcement came. Then you’ve got DOJ, OIG, and the IRS investigating some of the OPOs. And a lot of what their investigation has to our Cost Report. It has to do with how we’re reimbursed. So, we’re a cost-based reimbursement model, one of the few left in the system. If my organization does 200 organ transplants a year and 100 of them are kidneys, 50% of my overhead is pulled toward kidney, so it’s reimbursed on the Medicare Cost Report. At the end of the year, if I say a kidney costs \$60,000 this year and it ends up costing our organization \$65,000, I’d say, federal government, you owe me 5 grand times however many kidneys I transplanted. The same on reverse. If I can do over 55 grand a year, I’d say I owe you 5 grand times however many kidneys I transplanted because I’ve been overcharging the whole time.” – CEO of a major OPO (Organ Procurement Organization), longtime industry executive

No oversight of “reasonable cost” cost bucket; “getting abused” and regulators will fix

Q: “You’re saying the origin of the problem is that the way the law and the reimbursement works is that the OPO’s or the transplant centers are able to pass all reasonable cost through to Medicare.”

A: “Yeah, I think that’s part of it. There just hasn’t been any oversight on it. And then things are changing so fast...they’re going to come out and fix this.”

Q: “You’re saying that if an OPO does like 100 transplants, then they have another \$5 million in overhead costs, that they just divide that \$5 million in overhead costs by those 100 transplants, and that gets tapped onto the organ acquisition cost and people are dumping all kinds of crazy stuff into that overhead bucket. Is that right?”

A: “Yup, absolutely. Absolutely. It’s been getting abused. I think the government is going to come down, and they’re going to start setting what’s reasonable cost.” – CEO of a major OPO (Organ Procurement Organization), longtime industry executive

The OPO executive stated that TransMedics is a creature of this abusive reimbursement loophole and that regulators are locked in and ready to blow up its gravy train. It can only charge hundreds of thousands per case for the consumable and run up the tab by sending multiple private jets by stuffing everything into the Cost Report as “Reasonable Cost” - “the glory days of we’re going to charge \$100,000 for a TransMedics device, and you have to use our people and...our expensive jet – I think all that is over... I think the gravy train is coming to an end, honestly.” The executive described the investigations: “a lot of these investigations, they’re asking me these questions, like in [state redacted], the guy took the private jet...to the CEO’s dad’s funeral...did they put that on the Cost Report....the same thing goes on the transplant center side.”

Regulators about to crack down on the TransMedics gravy train by enforcing “reasonable cost”

“So, I really think the government is going to come in, and they’re really going to shake this up, and I think the glory days of we’re going to charge \$100,000 for a TransMedics device, and you have to use our people, and you’ve got to use our expensive jet - I think all that is over. I think the gravy train is coming to an end, honestly. I think the government is going to really come in.” - CEO of a major OPO (Organ Procurement Organization), longtime industry executive

OIG investigating transplant centers and OPO’s investigating improper expenses like private jet use that are shifted onto Cost Report

“So, there’s a lot that you can do on the OPO side and you can do on the transplant center side in terms of cost shift, even to get it on your Cost Report to get it reimbursed by Medicare. So, a lot of these investigations, they’re asking me these questions, like in [state redacted], the guy took the private jet or the company jet to the CEO’s dad’s funeral. And it’s like, okay, is that illegal? I don’t know. Did they put it on the Cost Report? If they did, then taxpayers pay for you to take a private jet to a funeral. That’s jacked, right? So, the same thing goes on the transplant center side. A lot of these investigations are coming down on that. What they’re going to do is they’re going to have OIG, the DOJ, and the IRS in those investigations on the OPO side and stuff. And then they’re going to have the OIG investigations on all the OPOs and all the transplant centers.” – CEO of a major OPO (Organ Procurement Organization), longtime industry executive

The OPO executive emphasized the exploitation of “Reasonable Cost” as the heart of TransMedics scam – without it, we note that traditional kidney pumps sell for a mere \$2,000 – and its game over now that government is about to define what reasonable cost actually is: “they’re going to come in because everything in transplant is based on what’s a reasonable cost...the government is working their way down...this stuff is all new...they’ve never established what reasonable costs are...they’re going to come in and establish ‘reasonable cost’...and now, all of a sudden, the Hawker 800 that costs \$12 million does not make sense anymore because that’s not reasonable cost.”

Regulators about to crack down on the TransMedics gravy train by enforcing “reasonable cost”

Q: “So, you think the government is going to wipe out TransMedics’ gravy train?”

A: “I think so. **I think they’re going to come in because everything in transplant is based on what’s a reasonable cost.** When you look at this stuff, you look at TransMedics and you look at this logistics stuff and things like that, the expense recovery, it’s very transparent what it costs. It doesn’t cost a ton and it’s used a bunch. And then you work your way down to devices and like TransMedics, and it’s like, okay, well on a medical device, it costs a lot, it’s somewhat transparent, and used kind of a lot. And then you go down to logistics, one step further, there’s no transparency, it’s used all the time, and it’s extremely expensive. **I think the government is working their way down. Because what CMS does is like, what’s a reasonable cost?** Well, this stuff is all new, so they’ve never established what reasonable costs are. I think what they’re going to do is I think **they’re going to come in and they’re going to establish “reasonable cost.” And now, all of a sudden, the Hawker 800 that costs \$12 million does not make sense because that’s not reasonable cost.**” - CEO of a major OPO (Organ Procurement Organization), longtime industry executive

The OPO executive offered real-time intelligence regarding a rapidly developing regulatory tsunami in the organ transplant arena. He detailed the new IOTA model, which despite only being focused on kidney transplants signals momentum to tackle a variety of abuses in the space, such as breaking up the OPTN/UNOS monopoly that has governed the transplant system for decades. TransMedics entire model, as a creature of this abusive regime, appears to be in jeopardy: “at the end of 2025, they’re going to have all these investigations wrapped up...they’re going to pick winners...they’re going to pick losers...OPTN has been fighting this, and HHS and OPTN are going at each other’s throats right now...the government is coming in, and they’re going to start controlling a lot of this....”

New IOTA model is a precursor to government scrutiny of the gravy train

“And then they just released what’s called IOTA. IOTA is this new increased organ transplant act or something like that. And the whole idea is this massive document, and they’re looking to put half the country’s transplant programs on this IOTA system. They’re going to pick a transplant program based on OPO-designated service areas that are supposedly going to be “random.” It’s not. They’re going to know who they’re picking. **Essentially, I think what’s going to happen is at the end of 2025, they’re going to have all these investigations wrapped up, and they’re going to put everything down, and they’re going to pick winners, and they’re going to pick losers.** – CEO of a major OPO (Organ Procurement Organization), longtime industry executive

New OPTN governance model is a preview of crackdown in organ transplant abuses

“The OPTN, which is organ procurement transplant network, which that **OPTN contract that’s held by UNOS to oversee the donation transplant system; it’s been that way for thirty-some years. That’s getting completely broken up.** After the Senate hearing, Congress passed it unanimously that they change that act and then the president signed it into law. **OPTN has been fighting this, and HHS and OPTN are going at each other’s throats right now.** And recently, there was a call and HHS announced - I don’t know if they really announced it - but they said we’re going to have a contract that’s going to come out. We’re going to appoint somebody, a third-party entity to work with the OPTN board to essentially show them how a board should behave, how they should be behaving. And then, I don’t know if she slipped up when she said it, the woman from HHS, or she meant to say it, but she said, “And then eventually, they’re going to take over.” Because **the government is coming in, and they’re going to start controlling a lot of this.** I talked to a surgeon today who talked about ASTS, which is the American Society of Transplant Surgeons, and they were all getting into it with HHS, the same woman. You’re not getting any of our expertise. You guys are just making decisions. And he said her response was, “You guys haven’t been doing a good job anyway. **So, we’re going to take this over and use a third party to assess things based on data.**” – CEO of a major OPO (Organ Procurement Organization), longtime industry executive

Anytime we research a fraudulent biotech or medtech company, we check the CMS OpenPayments database – a treasure trove which can reveal kickbacks to doctors disguised as speaker programs, the names of physicians or institutions with suspicious utilization patterns, the level of prescriber concentration, etc. We immediately noticed that the total payments reported by TransMedics are unusually low, which suggests a company trying to cover-up the amounts it pays. A former executive in a reimbursement role noted the “glaring omission,” telling us bluntly that the company fails to report the payments data, and that it repeatedly blocked the executive’s attempts to do so: “when I say, well, it’s part of the Open Payments requirement, and here are the NCI numbers for all of the doctors that were out to dinner...they say it doesn’t apply to them”; the executive replied affirmatively when we asked if TransMedics is simply not reporting to the system.

Allegedly refusing to report payments to doctors to OpenPayments

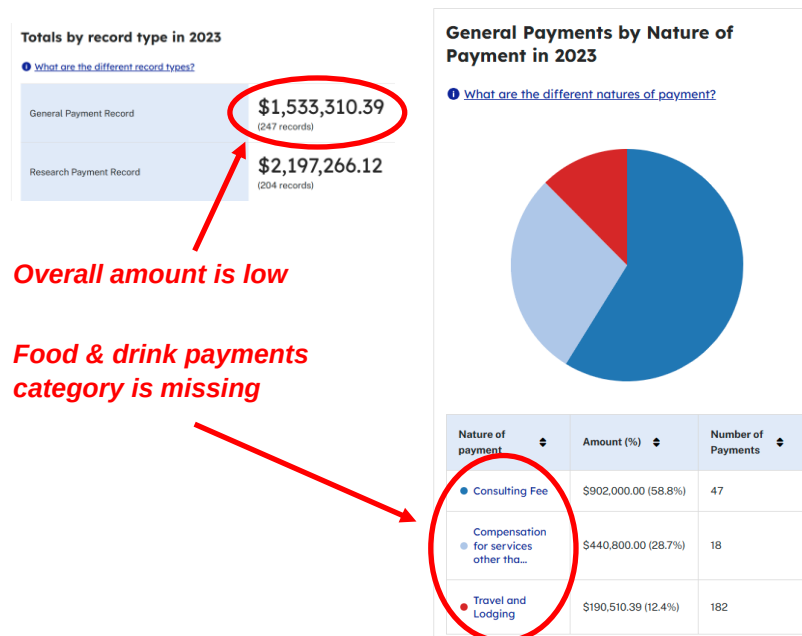
A: “If you open their open payments page and the types of payments that have been paid to providers, **there’s a glaring omission...I said I have noticed that every time I take a doctor to dinner and I send you all of the requirements for open payments requirements, it comes back to me as I don’t need this but thank you. And when I say, well, it’s part of the open payments requirements, and here are the NCI numbers for all of the doctors that were out for dinner** that I took them to, they said, we’re not part of AdvaMed, so that doesn’t apply to us. AdvaMed is the medical device association, and they put forward a statement, like a thought paper, on how you should comply and what’s the process to comply with this regulation. **And they say it doesn’t apply to them.**”

Q: “You’re saying they’re just not reporting a lot of stuff to open payments.”

A: **“Yeah.”** – Former executive in a key reimbursement role

The executive detailed several anomalies in TransMedics entries on Open Payments. We note that the annual amount for “General Payments” was a mere \$1.5MM in 2023, straining credibility. For example, the executive pointed to the absence of payment entries for “Food and Drink” – typically a large sum with the largest number of discrete payments, as wining and dining at expensive restaurants at conferences or speaker events is a common mechanism for inducement. We also note the cluster of fifteen “consulting” payments of exactly \$50,000 each, mostly paid on the same day to various doctors – suggestive of kickbacks – as well as other suspicious clusters in the data.

Overall payment amount reported is suspiciously low, and no data shown for food & drink



Cluster of \$50,000 “consulting” payments made on the same day; other suspicious clusters as well

JAMES MARKMANN	Consulting Fee	04/18/2023	\$50,000.00
Parsia Vageli	Consulting Fee	04/18/2023	\$50,000.00
Rafik Ghobrial	Consulting Fee	04/18/2023	\$50,000.00
MARWAN ABOULJOUD	Consulting Fee	04/18/2023	\$50,000.00
MARYJANE FARR	Consulting Fee	06/01/2023	\$50,000.00
SEAN PINNEY	Consulting Fee	06/01/2023	\$50,000.00
JOSEF STEHLIK	Consulting Fee	06/01/2023	\$50,000.00

Recipient	Nature of payment	Date	Total amount
MARIE BUDEV	Consulting Fee	06/01/2023	\$50,000.00
PUNEET GARCHA	Consulting Fee	06/01/2023	\$50,000.00
GABRIEL LOOR	Consulting Fee	03/17/2023	\$50,000.00
KENNETH MCCURRY	Consulting Fee	06/01/2023	\$50,000.00
MANDEEP MEHRA	Consulting Fee	06/01/2023	\$50,000.00
Matthew Hartwig	Consulting Fee	09/19/2023	\$50,000.00
LAURIE SNYDER	Consulting Fee	09/19/2023	\$50,000.00
CARMELO MILANO	Consulting Fee	09/19/2023	\$50,000.00

The lack of Open Payments reporting is striking given what surgeons detailed as lavish, unusual travel inducements. The head of a major transplant program was astonished at the “eyebrow-raising” red-carpet treatment TransMedics provided to an estimated 150 to 200+ attendees at a “TransMedics-sponsored private event” – as well as to the entourages each doctor could bring: “...one of the nicest trips I’ve ever been invited to...first-class tickets...I was allowed to invite numerous members of my team,” also flown first-class; “the most generous that I’ve ever...doctors are not really accustomed to that in this day and age...I’m not accustomed – we’re not accustomed to that level of attention...it’s a little eyebrow-raising...about 200 surgeons from across the country....everyone was given excellent treatment,” including hotels.

Lavish treatment for a large number of surgeons and their entourages

Q: “How many surgeons who are transplant leaders were at that meeting in the fall?”

A: “They probably had well over 200, maybe 150 attendees. It was at the Hotel Intercontinental in Boston. It was a TransMedics-sponsored private event... **I will say the Boston trip was one of the nicest trips I’ve ever been invited to: first-class tickets, Boston Hotel Intercontinental. I was allowed to invite numerous members of my team, so I brought three additional members of my team, so there were four of us. That was the most generous that I’ve ever.** I get invited with other companies, but this was very much a nicer experience. it was more like first-class tickets and whatnot. But **doctors are not really accustomed to that in this day and age.** I fly economy all the time. **I’m not accustomed - we’re not accustomed to that level of attention.”**

Q: “So, you’re saying they sent first-class tickets for you and four other people on your team to come to this conference?”

A: “Three other people.”

Q: “So, four total were sent first-class tickets to come to this conference?”

A: “Correct.”

Q: “Are there any other medical device or pharmaceutical companies that have offered to fly you first-class or was this unusual?”

A: “For our field, **it’s a little bit eyebrow raising**, because devices are not as regulated as pharma now is. Pharma used to be like that. What was different about this... **it was a really big group - I think there were about 200 surgeons from across the country.”**

Q: “Were they all flown first-class?”

A: “I think everyone was given excellent treatment.”

Q: “Do they pay for hotels also?”

A: “Yeah, they paid for Intercontinental Hotel.” – **Chief of transplant at a major West Coast center; high volume TransMedics user**

Am executive at a pre-eminent academic transplant center, one of the highest volume in the world, described similar invitations from the CEO and persistent attempts to pay for flights, hotels, registration fees – all of which this transplant executive had to decline as it was inappropriate “to accept anything from a vendor”: “...I interacted with Waleed...they often invite us and all transplant programs...they’ll even offer to pay for us to go...I have to always turn it down....they do pay for other people to go...they offered to...pay for my commercial flights and registration...they offered again to pay for the cost of my travel...it never got past that.”

Persistent attempts to pay for travel and other expenses for transplant surgeons and administrators

“Yeah, I interacted with Waleed...they often invite us and all transplant programs that they work with to these meetings near their headquarters or whatever. Or they’ll have meetings at transplant conferences, and **they’ll even offer to pay for us to go. I have to always turn it down because [our center] will not allow me to accept anything from a vendor but they do pay for other people to go**. And so, I think I probably would interact with him more if I attended those meetings. But I don’t...I told them I’m not going to be at that meeting because I can’t travel right now. And **they offered to reimburse me or pay for my commercial flight and registration for the meeting but I declined that**. And then, they had the in-person meeting at their headquarters, and I declined the invitation because of the travel, and **they offered again to pay for the cost of my travel ticket. It never got past that.**” – Transplant administrator

TransMedics aircraft-based business model is inherently high risk and prone to abuse and potential kickbacks, given similar alleged business practices at its competitors in the organ air transport industry. The CEO of a major organ procurement organization (OPO) alleged that Trinity Air Medical, owned by Blade Air Mobility (NASDAQ: BLDE) since 2021, is “under a federal investigation” for “giving kickbacks to different transplant programs...Blade’s getting into trouble...they’re about to get kicked in the teeth.” The CEO alleged practices almost identical to those indicated by ex-TransMedics employees: “they’ll fly planes in from like Arizona and to do stuff in New York because they can charge for all that...they’ll also pay the transplant program or pay somebody, whoever makes the decisions...they’ll give them a kickback to use their transplant logistics services even though they’re excessively more expensive...because whoever makes the decision is getting a piece of that.” We note that the color above re Trinity is merely allegation and that we have not verified its accuracy.

Trinity Air allegedly engaging in abusive business practices similar to TransMedics

Q: “A former employee told me that even if they have a team that’s based in a metro area, they will often fly in a jet from the other end of the country, and the staff is like, “What are you doing? We’re already here.” I presume the reason is because then they get to bill all that?”

A: “Yeah, and if you’ve heard of Trinity Air Medical, which is owned by Blade? Blade bought Trinity for \$22 million and it does \$110 million now. But **that’s what they do. They don’t know it, but they’re under a federal investigation, and Blade’s only doing what it’s doing because of Trinity. But Trinity is under investigation. I don’t know if they even know it. But they’re giving kickbacks to different transplant programs in New York, like, hey, transplant administrator; I’ll kick you x amount of dollars, you just keep calling us, but that’s why the DOJ in New York has tuned in because they’re primarily based on New York. So, Blade’s getting in trouble...they’re about to get kicked in the teeth.**”

Q: “So, Trinity, they’re giving kickbacks to people to use their service even if a plane is not necessary?”

A: **“Yeah, they’ll fly planes in from like Arizona to do stuff in New York because they can charge for all that, but then they’ll also pay the transplant program or pay somebody, whoever makes the decisions, they’ll give them a kickback to use their transplant logistics services even though they’re excessively more expensive than other people, but they’ll keep calling and using Trinity because whoever makes the decision is getting a piece of that.”** - CEO of a major OPO (Organ Procurement Organization), longtime industry executive

The CEO of the OPO alleged that Trinity Air Medical is under investigation by the DOJ, and that it is an open secret in the industry given the large number of industry participants such as other OPO's who have been interviewed as part of an alleged investigation: "it's probably that well-kept of a secret at this point...all those people have been interviewed...." He further suggested that NORA – the Nationwide Organ Recovery Transport Alliance – engages in similar practices to Trinity and TransMedics, portraying the entire organ transport industry as inherently corrupt.

Trinity allegedly under DOJ investigation

Q: "How does Trinity give a piece to the transplant administrator?"

A: "I don't know what the mechanism is, but **the DOJ is investigating it.**"

Q: "Is that public? Is that known that Trinity is under investigation?"

A: "No. I mean, **I don't think it's probably that well-kept of a secret at this point.**"

Q: "And so, who talks about it? Do heads of OPO's all know that Trinity - I mean, they probably subpoenaed people, people have been interviewed—is that how people know about it?"

A: **"Yeah, all those people have been interviewed. There's enough people at this point."**

Q: "Is Trinity part of this Washington Post article, this big investigation that you mentioned?"

A: "They're caught up in all this investigation stuff."

Q: "If Trinity is doing this, my assumption is there's no way that TransMedics NOP is not. It's the exact same business."

A: "That's what I would think. **I would think the same thing, and then NORA is the other major logistics provider, and they've offered me private flights, and I'm like, "No, I'm not doing it."**" - CEO of a major OPO (Organ Procurement Organization), longtime industry executive

Confirming our suspicion about the potential to use TransMedics jets for free private air travel as a kickback or inducement, the CEO of the OPO recounted anecdotes of such offers from Trinity and NORA, allegedly made to him and others as a common practice – “the Trinity thing is well known” - which he rejected as being highly inappropriate: “NORA offered me private flights”; “the rumor is that Trinity does that...I mean, NORA offered me stuff...and I’m like, ‘What?’...and they’re like, ‘We’ll send a plane for you’...and I’m like, ‘No, I’m not doing that’”; “the NORA thinking, I don’t it’s as well-know, but I was at a meeting...and I made the comment...”Yeah, they offered me a private flight”...and the woman sitting across from me was like, ‘Yeah, they did it to me, too’...we both kind of laughed about it.”

Trinity and NORA allegedly offering free private air travel

Q: “You just said something that confirms a major hunch to me. Part of the reason they have all these nice planes is because it’s probably all for the management team and all these surgeons.”

A: **“The rumor is that Trinity does that, a private plane, they give you a helicopter and that kind of stuff. I mean, NORA offered me stuff** because they wanted me to come up and see all their stuff in Chicago. And I was like, “Okay, well, I really don’t want to spend an overnight. So, we’ll kick it down the road to next month.” And they’re like, “Oh better yet, we’ll just come get you.” **And I’m like, “What?” And they’re like, “We’ll send a plane for you.” And I’m like, “No, I’m not doing that.”**

Q: “This is Nationwide Organ Recovery Transport. So, basically, it sounds like this entire corner of the industry, these transport alliance guys are sort of corrupt.”

A: “Yeah. We don’t use any of them.”

Q: “Now I understand where part of the kickback stuff comes from. It comes from private flights - that’s very valuable to people. Is given known whether they offer OPOs or transplant center administrators kickbacks in the form of flights or whatever or inducements?”

A: **“The Trinity thing is well known. The NORA thing, I don’t think it’s as well-known, but I was at a meeting in Boston, and I made the comment, I’m like, “Yeah, they offered me a private flight,” and the woman sitting across from me was like, “Yeah, they did it to me, too.” We both kind of laughed about it.”** – CEO of a major OPO (Organ Procurement Organization), longtime industry executive

Part 5. We believe TransMedics is operating an organ trafficking scheme, shopping and steering rejected organs to its top users as a quid pro quo for accepting them on its device and via its NOP service.

We conclude, based on extensive research, that TransMedics is engaging in a sinister scheme whereby organs are illegally steered to centers under the implicit or explicit quid pro quo that a) they accept the organ on its device and b) that it is transported via its NOP service on its private jets. The organ, we believe, therefore constitutes a kickback under the Anti-Kickback Statute (“Stark Law”) and also, in our opinion, meets the legal definition of organ trafficking per the National Organ Transplant Act which makes it unlawful to “acquire, receive, or transfer any human organ for valuable consideration” – under penalty of fines and/or imprisonment; and per international conventions, such as the Istanbul Declaration and others.

National Organ Transplant Act of 1984 (NOTA)

42 U.S. Code § 274e - Prohibition of organ purchases

(a) PROHIBITION

It shall be unlawful for any person to knowingly acquire, receive, or otherwise transfer any human organ for valuable consideration for use in human transplantation if the transfer affects interstate commerce. The preceding sentence does not apply with respect to human organ paired donation.

(b) PENALTIES

Any person who violates subsection (a) shall be fined not more than \$50,000 or imprisoned not more than five years, or both.

The full extent of the scheme became clear across interviews with ex-employees, transplant surgeons, and OPO's: 1) TransMedics organ procurement team arrives for a retrieval; 2) the first center on the UNOS/OPTN transplant waiting list declines the organ, typically because it is old, defective, or otherwise compromised; 3) the organ is then classified as an "Expedited Offer," a loophole in the OPTN organ allocation system which is now routinely abused to bypass the waiting list and preferentially steer the organ; 4) TransMedics and/or OPO's with whom it conspires allegedly start "dialing for dollars," according to ex-employees and others, to offer the organ to centers willing to accept it with the understanding that it comes on a TransMedics pump and on its NOP aircraft – unsurprisingly, its highest-volume and most corrupt customers seem to get the call. OPTN instituted expedited placement rules in March 2021, with criteria to prevent a "jump ball" when the first center declines an organ – a well-meaning rule that backfired by providing an official pretext for rampant abuses.

OPTN Notice of Policy Change for expedited organ offers, effective March 25, 2021

OPTN

Notice of OPTN Policy Change

Expedited Placement of Livers

Sponsoring Committee: Organ Procurement Organization (OPO)
Policies Affected: 1.2: Definitions
5.3.D: Liver Acceptance Criteria
5.6.B: Time Limit for Review and Acceptance of Organ Offers
9.10.A: Expedited Liver Placement Acceptance Criteria 9.10.B: Expedited

Excerpt of policy change - criteria under which expedited organ offers can be made

9.10.B Expedited Liver Offers

The host OPO or the Organ Center is permitted to make expedited liver offers if *both* of the following conditions are met:

1. The donor has entered the operating room or, in the case of a DCD donor, withdrawal of life sustaining medical support has been initiated, whichever occurs first.
2. The host OPO or Organ Center is notified by the primary transplant hospital that the primary potential transplant recipient will no longer accept the liver.

Transplant centers recognize that expedited placement has devolved into the Wild West, with OPTN now proposing reforms for “Expedited Placement Variance” – a euphemism for rampant out-of-sequence organ placement, favoritism, secrecy, and misuse. In Feb 2024, the American Society of Nephrology detailed its concerns in a 12-page letter to OPTN and HHS. The language is watered-down but spotlights the abuses: “systems violations”; “an increase in out-of-sequence offers, ‘open offers,’ and similar workarounds to the allocation system”; “a patchwork of opaque organ allocation practices”; patients are not “getting a fair opportunity” for an organ if they are at a center that is “not favored as an aggressive center by the participating OPO”; and that not a single OPO or transplant center has been “held accountable for any violation” of match rules.

American Society of Nephrology letter to OPTN/HSS, Feb 2024 - excerpts

The growth in these types of workarounds (system violations, at present) strongly suggests that the current allocation system is not optimized to efficiently match organs with a patient and a

consider expanding expedited placement practices. The current match-run algorithm, in apparently a growing number of cases, does not make match suggestions that the dedicated professionals at OPOs or transplant centers have confidence in.

it takes for an OPO to progress through the match run to place a kidney.”ⁱⁱ Recently, there has been an increase in out-of-sequence offers, “open offers,”² and similar workarounds to the

pilots are made permanent in various regions, a patchwork of opaque organ allocation practices that are not transparent to patients or the community at large could arise, exacerbating existing

are at a transplant center that is not favored as an aggressive center worthy of being offered certain organs by the participating OPO. While aspects of this inequitable dynamic exist in our

We spoke with surgeons who we believe participate in the scheme and unwittingly let us in on the details, as well as surgeons who have long suspected something improper but were unable to form a complete picture of its mechanics. We probed a transplant hepatologist at Massachusetts General, who seemed to leap out of his chair when we shared some of our findings, as the color appeared to confirm his worst suspicions and help crystallize his own observations: “if that get’s revealed, if that gets exposed in any way, that’s curtains...I have known of some sites in our region that would get these kinds of calls...other regions...this kind of hotline for organs that were turned down...this kind of workaround...it allowed them consequently to boost their numbers substantially in a given year with these organs...we scratch our heads wondering how in the heck did these guys get these organs? How did that happen? That’s not what ought to be happening through our new allocation system...there’s a lot of game playing going on out there.”

Suspicious practices and “workarounds” in organ allocation; “lot of game playing”

Q: “I’m hearing that if the recipient center doesn’t want the organ, TransMedics, according to a surgeon at a major center, who said, yeah, there are clinical guys badgering me to take a liver I don’t want to take. And it appears they’re then basically circumventing the OPO and the whole allocation algorithm, and they’re calling around for somebody that’s willing to take the organ, and, of course, they’re only calling people that are going to take it on a TransMedics device. So, they’re calling their highest-volume users.”

A: **“If that gets revealed, if that gets exposed in any way, that’s curtains. It’s really interesting you bring that up because I have known of some sites in our region that would get these kinds of calls from other sites, other regions, and would have this kind of hotline for organs that were turned down in that region...it was this kind of workaround, this shunt that found its way to the center and it allowed them consequently to boost their numbers substantially in a given year with these organs. We scratch our heads wondering how in the heck did these guys get these organs? How did that happen? That’s not what ought to be happening through our new allocation system. Those organs should still be offered to other regions’ highest-priority patients who haven’t yet received offers. There’s a lot of game-playing going on out there.”** – Transplant hepatologist at Massachusetts General Hospital

The physician described how unscrupulous centers are boosting their organ volumes via suspicious “hotlines” coming from certain areas like Tennessee – basically a gray or black market for livers: “it just stopped being part of the formal allocation process...this kind of thing, I suspect, is actually happening much more frequently than we know...this is how some centers get ahead in their numbers...it’s a below the radar screen kind of arrangement.” The physician shared various anecdotes, such as “centers where we heard they got these organs from out of region and they would seem to come from a similar kind of procurement area...Tennessee or something....literally a sort of hotline from the procuring team there to this center in our region asking if they want the organ...the fact was these guys were clearly getting them not in a sequential orderly way.”

Suspicious “hotlines” steering organs inappropriately to centers aggressively increasing their volumes

Q: “So, you’ve heard of centers where you’re like, how did they get organs?”

A: “Oh yeah. These are centers where we heard that they got organs from out of region and they would seem to come from a similar kind of procurement area. Like they would come from, say, Tennessee or something. And there would be literally a sort of hotline from the procuring team there to this center in our region asking if they want an organ...the fact was these guys were clearly getting them not in a sequential orderly way that would have put them next up for that organ, but rather there were local turndowns by local recipient centers and then all of a sudden it became this kind of free for all. And so, it never diverted effectively; it just stopped being part of the formal allocation process. This kind of thing, I suspect, is actually happening much more frequently than we know...I think this is how some centers get ahead in their numbers. It’s a below the radar screen kind of arrangement.” – Transplant surgeon, Utah center

TransMedics, we believe, exploits the fact that smaller, third-tier centers are desperate for organs to get their volumes up, and are therefore willing, we think, to pay exorbitant fees and to accept organs on its pump and its aircraft. The Mass General physician detailed the do-or-die incentive that lead smaller centers to look the other way when TransMedics calls, stating the NOP is mainly for “small programs wanting to get into the game...I think exactly what they’re exploiting to increase the use of TransMedics.” He stated that centers “absolutely would take” an organ on the condition that it’s on their pump – “even if it’s a wash financially, it gets your numbers up...you will, in the eyes of your institution and of payors, take on enhanced status...for being viewed as a higher volume center...you can put yourself in a position to compete for contracts...one way of getting the good stuff, so to speak...larger numbers aided and abetted by perfusion...a really interesting calculus being practiced by...centers that want to get bigger.”

Perverse incentive for smaller centers desperate for volume to accept organs on TransMedics pump

Q: “How much of an incentive is there for a center, if they get a call from TransMedics and they say this other center didn’t want it. You’ve got to take it, but it’s already on a TransMedics device, or we’re going to put it on. That’s the deal.”

A: “Yeah, they would take it. **They absolutely would take it because it represents - even if it’s a wash financially, it gets your numbers up, particularly if you’re not already a high-volume center. You get your numbers up.** You will, in the eyes of your institution and of payors, take on enhanced status, at least for being viewed as a higher volume center and potentially can enhance your margins accordingly. In a way, this is kind of like a loss leader where **you’ve got to do a bunch of these perfusion-supported cases so that you can put yourself in a position to compete for contracts** so that you can get even more of the conventional risk cases that don’t require perfusion. In a way, **it’s kind of like one way of getting the good stuff, so to speak, to take on larger numbers aided and abetted and improved by perfusion It’s a really interesting calculus that’s being practiced by these kinds of centers that want to get bigger.**” – Transplant hepatologist in a leadership role at Harvard/Massachusetts General Hospital; key role in oversight of the liver transplant program

NOP is primarily for smaller programs “wanting to get into the game”

“I understand the NOP service from the vantage point of early **small programs wanting to get into the game**, particularly when it comes to high-risk donors like DCDs. And that’s, I think **exactly what they’re exploiting to increase the use of TransMedics.**” – Transplant hepatologist in a leadership role at Harvard/Massachusetts General Hospital; key role in oversight of the liver transplant program

A senior medical employee of OrganOx, a competitor with a near-identical normothermic pump, provided the same color – that TransMedics makes no sense for large transplant centers, particularly in large metropolitan areas like New York, and it's only useful for a small centers looking to jump the UNOS organ allocation algorithm and access organs via the NOP service: “let’s say I’m a very small center with limited resources and I need an organ...it’s probably advantageous to do a TransMedics case...I’ll be able to probably get a match faster than I would with my current OPO.”

No value for large centers ;just small centers looking to jump UNOS allocation and get organs via NOP

A: “Like I mentioned, not every case needs to be a TransMedics case.”

Q: “So, you think there’s overutilization?”

A: “Yeah, I think there’s **overutilization when there clearly is not a clear clinical indication that it’s going to be beneficial for the patient...Let’s say I’m a very small center with limited resources, and I need an organ...it’s probably advantageous to do a TransMedics case in that regard because I’ll be able to get the organ faster...I’ll be able to probably get a match faster than I would with my current OPO** infrastructure because if I’m a small center, I’m going to have a small utilization from the OPO who’s going to want to give organs to a larger center. So, it helps me in that regard. Conversely, let’s say I’m a very large center, and I’m doing hundreds and hundreds of transplants a year. Do I really need this extra TransMedics case in order to do what exactly what I’m doing on my own? That’s what I’m talking about: **for large centers, perhaps there is less clinical utility than for a small center.**”

Q: “What I’m hearing is that there’s tremendous overutilization, that it’s used for perfectly fine organs that would be fine on ice with no issue.”

A: “Yeah, I agree. If it’s a donor service area that’s very small, like **if you’re near a very large metropolitan area like New York or something, it’s out of the question. Why would you ever do that? You just have to drive across the river or something. It makes no sense.**” – Senior medical employee at OrganOx

The Mass General physician's quip - "one way of getting the good stuff, so to speak" - reminds one of a cartel servicing desperate addicts, and desperate is exactly the tone we sensed from TransMedics highest volume users, albeit for organs not drugs. One such high-volume user, a transplant surgeon at a Utah university, unwittingly led us into the belly of the scheme. We asked why he was willing to lose money by paying >\$150,000 for TransMedics' consumable and NOP service, and he stated his center and OPO need to boost organ volumes at any cost to stay certified and in business: "the transplant administrators...they're addicted to the volume"; "if we can increase our lung volumes by, say 40 lungs...we're a tier one OPO and that's how close the margins are as far as being in compliance." Volumes are also critical as the denominator – "if I do 10 transplants, one or two deaths can shut down the program...but if I do 100, then I have a lot more leeway...the money is funny money."

Desperate for volume to stay certified and in compliance

"My OPO, the only thing they're truly interested in is two metrics...so, if we can increase our lung volumes by, say, 40 lungs ... we're a tier-one OPO, and that's how close the margins are as far as being in compliance. OPO's were never required to have metrics. Then, about two or three years ago, they imposed metrics on them. But those metrics still didn't have consequences. So. And our OPO is a tier one but the way that they are keeping their buffer is by DCDs and lungs. Next year, those metrics will have consequences where the threat is the tier three OPOs will be closed and absorbed by the other OPO's." – Transplant surgeon, Utah

Centers and OPO's "addicted to volume"; denominator important to provide buffer for transplant patient deaths

Q: "What happens to the profitability of a case if you use TransMedics?"

A: **"It drops**, and the transplant administrators are very concerned about that, but **they're addicted to the volume**. You can dramatically increase your volume and **your volume is also important because of your denominator. So, if I do 10 transplants, one or two deaths can shut down the program. But if I do 100, then I have a lot more leeway**....it's a really weird concept because you're so laser-focused in on the money, but the money is **funny money**, especially at a university, and most transplant programs are at universities...As we transplant more, our list isn't as thick because **if I'm transplanting another 40 as opposed to 20, everything's different."** – Transplant surgeon, Utah

The Utah surgeon detailed the inner workings of how TransMedics and corrupt OPO's could conspire to exploit the "expedited offer" loophole – which can be used at "any time": 1) certain OPO's are allegedly induced to store TransMedics devices - a kickback, we believe, since it is at no cost to the OPO; 2) these OPO's and/or TransMedics organ procurement team would then jump the gun and put organs on the device – prior to the organs even being accepted by a center; and 3) TransMedics would then hijack and take over the organ allocation process from the OPO, and bypass the match list to start offering the organ preferentially. Legally, the OPO has to place the organ, but the surgeon stated that "the OPO's are being cut out of the process by TransMedics...dictating where the organ is going or how the list is run." In a slip, he even appeared to mention Mayo Arizona – one of TransMedics heaviest users if not the largest: "...maybe someone in Arizona is upper aggressive...we can basically bypass the list and call that person directly...would you accept this organ only if we put it on Transmedics? That's how you would steer it."

TransMedics allegedly bypassing OPO's and UNOS with expedited offers to force usage of the device

A: "If TransMedics gets into the job of the OPO that would be inappropriate...somehow the OPO's are being cut out of the process by TransMedics. They're just basically another group that comes in to do the actual surgery. Anything beyond that is dictating where the organ is going or how the list is run..."

Q: "So, if TransMedics has a close relationship with an OPO, are there things the OPO could do to steer business to TransMedics?"

A: "It could be perceived that way...there's something called an expedited offer. UNOS has allowed for expedited offers on organs that need timely referrals. So, say we're going to the OR immediately, and we know that all the local centers aren't going to really accept it, but we know maybe someone in Arizona is super aggressive, and they'll accept this organ. So, we can basically bypass the list and call that person directly and say, hey, would you accept this organ? And these are the parameters. And they'll be like, yeah. But then, if they go, well, would you accept this organ only if we put it on TransMedics? That's how you would steer it."

Q: "When are you allowed to do an expedited offer?"

A: "Any time, as long as you can justify it in retrospect."

Q: "An expedited offer is a way to bypass the UNOS allocation system?"

A: "Yes." - Transplant surgeon, Utah

Source: Scorpion Capital consultation calls with experts

The conduct the surgeon described strikes us as nefarious and the type that risks FBI raids and indictments: “...we have a liver on pump...we’re moving quick...if we bypass the list, would you accept the liver,” and stated “yeah, that’s exactly right” when we asked if the center is forced to take it on a TransMedics pump. He confirmed that the devices are in storage at the OPO for use as “expedited offers” and at no charge to the OPO – a flagrant kickback as they run ~\$80-90,000 each. OPO’s rush to sling organs onto “free” devices before the organs are even tested or a center agrees to accept them, with no charge to the OPO if no one takes it: “the agreement is we have a rapid donor that may not have serologies...may not have list run...TransMedics is willing to because they’re using it as storage...put it on pump with no home and then offer it out, and they’ll accept the cost if they don’t because ultimately, most of the time, those organs will be accepted.”

Utah OPO allegedly storing TransMedics pumps to jump the gun on expedited offers

Q: “Is that one of the reasons why those machines could be in storage is to assist in cases of expedited offers? The way that the OPO could steer business to TransMedics is by saying we have an expedited offer, but they only make the expedited offer to a transplant center if they say it’s already on the pump. Is that kind of the goal?”

A: “**Yeah, that’s exactly right. I’ve seen that where it’s like we have a liver on pump - other OPO’s where they’ve called us and said we’re just moving quick, if we bypass the list, would you accept this liver?**”

Q: “**The transplant center said we have an expedited offer and it’s already on TransMedics, that’s how you have to take it?**”

A: “**Yeah, that’s exactly right.**” – Transplant surgeon, Utah

OPO’s allegedly not being charged if organ on pump is not accepted

Q: “**Are they charging the OPO, or is the OPO passing that on to somebody in advance?**”

A: “**No.**”

Q: “Someone’s has to be paying for it, no?”

A: “**The agreement is if we have a rapid donor that may not have serologies, we may not have list run; we may not have certain things, TransMedics is willing to because they’re using it as storage, send out a team, put it on pump with no home and then offer it out, and they’ll accept the cost if they don’t because ultimately, most of the time, those organs will be accepted.**” – Transplant surgeon, Utah

The surgeon stated that TransMedics “approached my CEO” – who we believe to be the CEO of DonorConnect, an OPO based in the Salt Lake City area with responsibility for Utah and parts of Idaho, Wyoming, and Nevada: “my CEO and their CEO have known each other a long time. And this was something that they’re going to try out.” He alleged that TransMedics gives the devices to the OPO at no cost – the very definition of a kickback: “they said if you let us keep our stuff there...if there are organs that are homeless or that are rapid, you can put them on...there won’t be any repercussions if they don’t get places...like the OPO won’t have to pay for it....”

CEO’s of TransMedics and Utah OPO allegedly have a pre-existing personal relationship; OPO allegedly offered devices for free

Q: “TransMedics approached you as the medical director of the OPO or the CEO of the OPO?”

A: “They approached my CEO. My CEO and their CEO have known each other for a long time. And this was something that they’re going to try out.”

Q: “What did TransMedics propose? What did they come and say?”

A: “They said if you let us keep our stuff there, then if there are organs that are homeless or that are rapid, you can put them on and resuscitate them. TransMedics will put them on but there won’t be any repercussions if they don’t get placed. Like the OPO won’t have to pay for it...and then, if the lungs or the liver are functioning a certain way on the pump, it’s predictive of how they’re going to work, and then you can place those organs...what the OPO gets out of it is even more organs placed.” – Transplant surgeon, Utah

The surgeon stated that TransMedics has similar arrangements with other OPO's where devices are stored for free, which we heard directly from other OPO's: "I heard that the company is storing devices at certain places"; "I don't think that they're making them pay for that...it's just sort of a business relationship those two are working out" – referring to the CEO's of his OPO and TransMedics. He stated the practice extends to both livers and lungs, and once again highlighted how centers are then basically forced to accept these organs on the device – with the OPO helping to pressure the center on TransMedics behalf: "I think that's where our OPO is most interested...and so, if a lung center is saying we'll use this if it's resuscitated on a TransMedics device...."

OPOs allegedly not paying for the devices; "just sort of a business relationship"

"I know that they've been talking to our OPO about storing that stuff at a hub in Salt Lake City, but **I don't think that they're making them pay for that** because that's different because the OPO's don't pay TransMedics. It's **just sort of a business relationship those two are working out**...our area is getting more access to TransMedics, and the reason our OPO is open to that is because it's really increasing the use of DCD organs for liver, but also it's increasing the use of lungs. And **I think that's where our OPO is most interested...and so, if a lung center is saying we'll use this if it's resuscitated on a TransMedics device...**" – Transplant surgeon, Utah

Kickbacks, in our opinion, in the form of discounts for utilization and for keeping devices at OPO for expedited offers

"We decided to use them for everything just because I think we got a deal with them on the first dozen livers or something like that...I had heard that the company is **storing devices at certain places and maybe you can get a discount with that.**" – Transplant surgeon, Utah

An executive at a major West Coast OPO – one of the largest in the country – stated that TransMedics similarly tried to induce them to store devices and that it was likely a prevalent practice: “I wouldn’t be surprised...that would not surprise me at all”; “we have been very clear with them...we don’t agree with their business practices...we want nothing to do with them.” The executive alleged that TransMedics fabricated a pretense for why the OPO needed to store the devices: “TransMedics tried to get us to store their devices for them here also...they alleged that without that their service wouldn’t work...we’ve always been very upfront...we’re not going to engage in anything like that.”

TransMedics allegedly tried to induce OPO to store their devices; shady business practices

Q: “It looks like there’s been some DOJ settlements over the last 20 years with some of these air transport companies getting busted for offering free travel as kickbacks. It just seems like an obvious area of abuse waiting to happen. Have they tried to cultivate you or offered stuff like that? Or do you think they’re doing stuff like that with other OPOs that they’re friendly with or other inducements or kickbacks?”

A: **“I wouldn’t be surprised. That would not surprise me at all...I think we have run into - we have been very clear with them. We don’t agree with their business practices. We want nothing to do with them.** When we started, I mentioned that we were directly purchasing their supplies and then passing them through the transplant centers; when we started having conversations about no longer doing that and that we were happy to share our chartered aircraft and try to share the cost to keep costs down for the transplant centers because **TransMedics tried to get us to store their devices for them here also. They alleged that without that their service wouldn’t work. Obviously, they were able to figure it out just fine. I think we’ve always been very upfront. We’re not going to engage in anything like that.**” – Executive at a West Coast OPO, longtime industry veteran

One of TransMedics' highest-volume users, a surgeon at an academic center, provided extensive color on what we interpret as an organ steering scheme and suggested that the company has hijacked the UNOS organ allocation process and national transplant waiting list. He replied "definitely" when we asked if TransMedics is exploiting its relationship with certain OPO's to run up utilization and direct which center receives the organ: "I heard a lot of complaints about how TransMedics is trying to over the case....trying to take over the organ allocation...TransMedics is just pushing, and they're rubbing people the wrong way." He stated that TransMedics relationship with various OPO's has deteriorated with the more ethical OPO's refusing to play along: "the OPO's, the one in Los Angeles, in particular, was starting to refuse to do that...OPO's were looking at a way to retaliate or somewhat to put some pressure onto TransMedics...that relationship is already under a lot of strain."

TransMedics alleged trying to take over the organ allocation process from OPO's; some OPO's pushing back

Q: "Are there any games that they played with your local OPO, where they're trying to press their thumb on the OPO to run up TransMedics device volumes? I heard they're doing that with OPO's, like storing machines at the OPO. For example, the OPO says the patient's already on TransMedics and they're trying to leverage relationships with OPOs to get more utilization. Have you seen anything like that?"

A: **"Definitely...I heard a lot of complaints about how TransMedics is trying to take over the case; TransMedics is trying to take over the organ allocation; TransMedics is just pushing, and they're rubbing people the wrong way.** What you saw is some fallout from that - this putting a liver onto the TransMedics device requires five units of blood. For the heart, it comes from the donor. For the liver, it comes from the blood bank. And the practice is then for TransMedics to ask the OPO or to request the hospital where the donor is located to procure five units of blood so that they can arrive there, and you get the blood from the blood bank to put the organ on the machine. And **the OPOs, the one in Los Angeles, in particular, was starting to refuse to do that. They were starting to say we will not supply blood for TransMedics.** And that became a real big sticking point that would get transplant programs in the middle of it. You've got to call the OPO and tell them to get the blood, or you've got to ship the blood—it was just a big mess. So, **OPOs were looking at a way to retaliate or somewhat to put some pressure onto TransMedics. You can see that relationship is already under a lot of strain.**" - Transplant surgeon

The surgeon provided particularly damning detail which shows, in our opinion, that TransMedics is steering organs as a quid pro quo for device volumes – which would comprise a kickback as well as organ trafficking, per our reading of the National Organ Transplant Act, which prohibits the exchange of organs for “valuable consideration.” The transplant surgeon stated that, prior to a current contracting dispute concerning extortionate air transport charges, his center would “get calls probably a couple of times a week” where TransMedics offered them organs as “expedited offers” but that it’s “definitely died off,” perhaps because “we’re not in good graces with them anymore because of our contracting dispute”: “they would call and say, hey, would you be interested in this?...and then they say to the OPO, oh, so-and-so, the guy over in [redacted] is interested in this liver...so, they’re getting into the organ allocation business rather than just the organ recovery and preservation business...it’s definitely improper.”

Allegedly used to call twice a week to offer organs but now no longer call due to contracting dispute

Q: “If the surgeon on the other end at the center says they don’t want the organ and it’s already on a TransMedics device, they’re actually shopping it around? Doesn’t that violate the law, which has this protocol for how organ allocations are done?”

A: “It can’t go down that way, for sure. They can’t give the organ away, but they would call and say, hey, would you be interested in this? And then someone says, oh yeah, I’d be interested in that, and then they say to the OPO, oh, so-and-so, the guy over in [redacted] is interested in this liver. You should give him a call. So, they’re getting into the organ allocation business rather than just the organ recovery and preservation business. It’s definitely improper.”

Q: “How often do they do that? Have you seen them do that?”

A: “Oh sure. I get phone calls from them. They’ve called me.”

Q: “What do they say? This other surgeon didn’t want the organ; do you want it?”

A: “Yes. Exactly that.”

Q: “How often are they calling you?”

A: “Me and my other partner who had the interactions; we are well known for our aggressive use of organs - for a while, we would get calls probably a couple of times a week, I would say.”

Q: “Are you still getting calls a couple of times a week?”

A: “It’s definitely died off. And we’re not sure whether they realize that’s not perhaps the best way or whether we’re not in good graces with them anymore because of our contracting dispute.” - Transplant surgeon at a high volume academic center

Our interviews with Utah and other high-volume centers were striking, as they detailed participation in what sounds to us like an overt organ trafficking racket while at the same time acknowledging that “it’s definitely improper” – an illustration of the conflicted, corrupt cesspool that the current organ transplant system has become and why various national/state agencies and authorities are preparing to crack down. The center we quoted two pages ago, for example, admitted to boosting its volumes with questionable organs shopped to it by TransMedics – “we are well known for our aggressive use of organs” - while at the same time slamming the practice: “there’s a lot of gray area when it comes to how late declines are handled...that has certainly happened to me where I’ve been inappropriately skipped...that’s against regulations...that’s unethical...you can’t skip me without telling me...these things happen...”

“Gray area” with inappropriate skipping of centers “unethical”

Q: “Is there risk that somebody who would have gotten an organ through the algorithm did not because they just hijacked it and they’re working some relationship and pressuring some OPO? It seems like if this is happening at 2 am –”

A: “There’s a **lot of gray area when it comes to how late declines are offered. That has certainly happened to me where I’ve been inappropriately skipped**, and I had to call an OPO and say you can’t do that. **That’s against regulations. That’s unethical; you can’t skip me without telling me. These things happen. What the recourse is for anybody is a tougher question.**” -

Transplant surgeon at a high volume academic center

TransMedics alleged interference in the organ allocation process, in what appears to be a violation of the UNOS transplant waiting list rules, was corroborated by former employees as well as current staff at transplant centers who are inside operating rooms while the NOP procurement team retrieves organs. A former TransMedics OCS specialist - who operates the device and manages an organ on the pump – described personally witnessing how the team allegedly hijacks the registry: “it’s supposed to go to the OPO to reallocate the liver...but as soon as TransMedics finds out, they have quite the list of backups, they are already calling other surgeons and being like, could you potentially use this liver?...a lot of these conversations [are] happening...I’m not exactly sure what is said...I just know that these phone calls are being made and then it’s...coming back into the OR and being like, this particular surgeon could use this liver. Why don’t you call them?” The ex-employee stated TransMedics bypasses the OPO: “they’re just kind of doing it and saying, oh, this is an aggressive center and they could use it.”

TransMedics allegedly takes over the organ allocation process inside the OR and calls potential recipient centers itself

Q: “What did you observe as far as the interaction between the OCS team, the TransMedics surgeon, and the transplant surgeon waiting on the other end?”

A: “A couple of times, the first center would say we don’t want this liver for whatever reason. And then, **it’s supposed to go to the OPO to reallocate this liver. But as soon as TransMedics finds out, they have quite the list of backups, and they are already calling other surgeons and being like, could you potentially use this liver? So, there are a lot of these conversations happening. I’m not exactly sure what is said or how it goes. I just know that these phone calls are being made, and then it’s like coming back into the OR and being like, this particular surgeon could use this liver. Why don’t you call them?**”

Q: “Are they going through the system and the OPO, or are they just kind of doing it themselves?”

A: “**No, they’re just kind of doing it and saying, oh, this is an aggressive center, and they could use it.** I mean, the OPO does push back. I know that they do unless they go through the process like they’re supposed to. But there are **a lot of incoming phone calls and text messages from the powers that be in Boston trying to have their input.**” - Former TransMedics OCS specialist

The ex-OCS specialist stated that TransMedics executive team is on text messages and internal chats to direct the scheme during organ procurements in the OR: “there are a lot of incoming phone calls and text message from the powers that be in Boston trying to have their input.” The ex-employee singled out Magdy Attia in particular – a fellow Egyptian and confidante of the CEO, the “Chief of OCS Liver Transplant program at TransMedics” per LinkedIn – “ones that I knew who for sure knew what was going on were Magdy and them...they’re always on the internal and external chats...all of the higher-ups can always see what’s going on at all time because they’re part of these threads.” The OCS specialist stated that the calls from the OR were going to TransMedics highest volume customers – particularly UC San Diego and Mayo Arizona – who we believe to be fully aware of the quid pro quo that they get the organ in exchange for taking it on the pump: “the ones that they would call aggressive centers, we were at very frequently....it would be like San Diego or Arizona Mayo.”

Senior TransMedics management allegedly directing organs to “aggressive” centers

Q: “Do you know how that worked? Were they offering organs to surgeons that were heavy TransMedics users?”

A: “**Pretty much, because the aggressive centers are all the ones - or the ones that they would call aggressive centers, we were at very frequently. So, they would say an aggressive center, and it would be like San Diego or Arizona Mayo.**”

Q: “So they basically have a list of people who would take an organ that other people didn’t want?”

A: “Yeah.”

Q: “You said text messages and calls were flying. Who was doing those? Was this the CEO or Magdy Attia?”

A: “The only **ones that I knew who for sure knew what was going on were Magdy and them. They’re always on the internal and external chats. All of the higher-ups can always see what’s going on at all times because they’re part of these threads.**”

Q: “And this chat is like a Slack channel or something like that?”

A: “There’s an app and it’s also text messages.”

Q: “Like regular text? Like not WhatsApp, like regular text?”

A: “**Like regular iPhone text messages.**”

Q: “Wow, is that like HIPAA compliant? Aren’t there rules around this stuff?”

A: “I don’t know the correct answer to that.” - Former TransMedics OCS specialist

Another OCS Specialist providing what we believe is a smoking gun that TransMedics not only seems to steer organs to “aggressive” centers, but that it explicitly appears to skip those unwilling to take it on the device, moving to the next “aggressive center” on their list until they land one – a clear indication, in our view, that organs are trafficked as a kickback for device usage: “there was an instance where I was in the OR...the original center that was supposed to take a liver said we don’t want it...the next place was like, yeah, we’d like the liver.” The center was then allegedly asked if they will use the device: “we have TransMedics in the OR...do you want to use them?” The center stated they wanted it transported on ice, not on the device: “they talked to the people from corporate at TransMedics...they were trying to get them to be like, oh, we can do a contract, and you can take it on the OCS...they were like, never mind, we want it on ice, so it didn’t go to that aggressive center....a lot of the times TransMedics would call after the original hospital...refused [the liver]...they would say, here’s a list of aggressive centers...they were very much campaigning for them...they really did want it to on allegedly the OCS once they were already there.”

Smoking gun – allegedly steered organ away from a high-volume center because wouldn’t take it on the device

Q: “You mentioned that they’re calling around if the receiving center didn’t want the organ - texting and whatever.”

A: “Yeah...there was an instance where I was in the OR and the original center that was supposed to take a liver said we don’t want it. So, they had gone to the next place. And the next place was like, yeah, we’d like the liver. And so, we have TransMedics in the OR. Do you want to use them? And I believe it was a hospital that I think was in Arizona, and they’re like, well, yeah, I think we would love to do that but we don’t know what to do and everything. So, they were thinking about it. I remember they were on the phone, and then they were like, actually, never mind. We just want it on ice. We don’t want - they talked to the people from corporate at TransMedics because they were trying to get them to be like, oh, we can do a contract, and you can take it on OCS, and I think they were like, never mind, we want it on ice, so it didn’t go to that aggressive center...I know a lot of the times TransMedics would call, after the original hospital that was supposed to take this liver refused it. So, they would say, here’s a list of aggressive centers. Make sure you remind them who the aggressive centers are...they were very much campaigning for them. But they were always organs that had already been turned down by the first place...and then the next place would get to decide. But they really did want it to go on OCS once they were already there.” – Former OCS

TransMedics alarming. alleged conduct inside operating rooms appears to be well-known among surgical staff at major transplant centers. An organ procurement technician at a leading center in the Pacific Northwest – who travels frequently to retrieve organs and is often in the operating room with TransMedics procurement teams – **confirmed the modus operandi:** “I was in a recent operating room where TransMedics was” and that its team “**was definitely contacting other centers,** saying if we go past the 120 minutes....” The technician was referring to a DCD donor that didn’t expire in time, meaning the **organ was compromised due to an extended lack of circulation.** The first center on the match list therefore **declined the organ – but TransMedics allegedly contacted centers** willing to take an organ with a warm ischemic time of over 2 hours. In other words, as we detail in a separate chapter, TransMedics is allegedly **pushing questionable organs that reputable centers won’t touch – and these dubious organs appear to be the currency by which TransMedics boosts device volumes** and by which its top users boost their transplant volumes.

Surgical technician: TransMedics contacting centers to shop questionable organs

Q: “I’m hearing that sometimes the receiving center doesn’t want the organ and TransMedics is bypassing the organ allocation system - they’re just texting and calling surgeons that use the device a lot, saying will you take it? And, obviously, they’re never going to call anybody who’s not a TransMedics user because the deal is we can get you this organ, but it’s going to be on a TransMedics device. That sounds like a hardcore kickback.”

A: “Yeah, for sure. **I was in a recent operating room where TransMedics was there.** It was a DCD, and the patient ended up not passing, **but because the time frame expired, they went to another center and TransMedics did contact that center. I’m not 100% sure how the OPO was involved in that, but TransMedics was definitely contacting other centers, saying if we go past 120 minutes...**” – Organ procurement technician at a high volume Northwest academic transplant center

An executive at a major West Coast OPO – who we quoted earlier as refusing to do business with TransMedics due to these and other practices – corroborated the conduct described by ex-employees and other surgical staff. He stated that transplant patients are “routinely” skipped and denied an organ after TransMedics shows up and starts shopping organs to centers participating in its racket: “they are very high pressure...they just want somebody to use it so they get paid...so, they start making calls...they’re going down their Rolodex of their paying customers...if my mom is listed at a transplant center down the road, she doesn’t get a call...patients are being skipped routinely and often...the calls are only going to the centers that are willing to pay for these machines.”

TransMedics “going down their own Rolodex” and “patients are being skipped routinely and often”

Q: “It sounds like they’re very high pressure, like manipulating and pushing the organ allocation process.”

A: **“They are.** In a real-life scenario, what ends up happening is when an organ is declined at the last minute like that, the OPO is really charged and held responsible with the utilization of the organ...but, **they are very high pressure.** I think they’ve got two things on their mind. One is that—I am confident that they’re not doing that work because they want to rescue this organ. **What they want is they want to get paid...they just want somebody to use it so that they get paid...so, they start making their calls...they’re going down their own Rolodex of their paying customers. If my mom is listed at a transplant center down the road, she doesn’t get a call. Patients are being skipped routinely and often. The calls are only going to the centers that are willing to pay for these machines.**”

Q: “That’s 100% a kickback.”

A: “Yeah.” – Executive at a West Coast OPO, longtime industry veteran

A transplant administrator at Massachusetts General stated that bypassing the UNOS waiting list is “not only [violating] anti-kickback, that’s going against the whole OPTN rule...that’s just flat out illegal.” He stated that it likely also constitutes an inducement, a separate problem for TransMedics beyond the Anti-Kickback Statute. The OIG and DOJ actively prosecute violations of the Beneficiary Inducement Statute which prohibits discounts, freebies, and items or services offered below fair market value. When the first center on the match list declines an organ, the administrator questioned if the centers that TransMedics allegedly shops the organ to are paying the full cost of the NOP procurement and transport, leaving aside the device cost – if not, as suspected, the discount or freebie would be an inducement: “they get a dry run 50% discount...what happens with the actual cost related to sending the organ back over to me? If that’s zero, then even at a discount, to be quite frank, that is an inducement.”

Bypassing OPTN allocation is “flat-out illegal”; discounts for organ procurement cost are an inducement

- A: ***“Bypassing that, that’s just not only anti-kickback, that’s going against the whole OPTN rule.*** So, that’s one thing. OPTN. How you allocate organs. It would just be like me turning down the organ and saying hey, by the way, I’m going to call my buddy across the street to see if he wants it...***that’s just flat-out illegal.*** The second question I asked is whoever received that second call, are they not paying for the organ? That would be an inducement to me because here you are, that primary transplant program that was supposed to receive the initial organ that turned down the organ, they get a dry run 50% discount. Whereas either I get the organ for not free because I still have to buy the organ, the organ acquisition cost. But ***what happens with the actual cost related to sending the organ back over to me? If that’s zero, then even at a discount, to be quite frank, that is an inducement.”***
- Q: “I see. So, you’re saying if there’s a scramble to place an organ somewhere, and whoever agrees to take it gets a discount or doesn’t have to pay the cost of organ procurement, they just pay for OCS or whatever, you’re saying that’s an inducement?”
- A: “Yeah. Take OCS out of it. ***Let’s go back on the match run.*** Okay, let’s go all the way down. Okay, I want to take it. I’m fifth in line. And then we the organ gets sent over to us. ***Because I’m not going to send an entire team out, right? But that procurement team is going to come over to my center.*** I still have to pay for that.” – Veteran transplant administrator/executive at Massachusetts General Hospital

Multiple TransMedics “super users” indicated that the company bypasses OPTN rules to allegedly take over the organ allocation process from OPO’s in order to preferentially offer them organs – already on the device, naturally. A transplant chief described a large meeting held by Transmedics – “the leaders of many transplant programs were invited...I was one of them” – where the problem of hijacking the OPTN list was acknowledged – but behind closed doors. We emphasize that this surgeon appears to benefit from the scheme: “calling TransMedics experienced users and saying...’would you be interested?’...[redacted] got up to the podium at that meeting and said...do it under OPTN guidelines, do not bypass the list...the OPO where that liver was procured should be the one to go down the central list...it should not be TransMedics calling UCSD, another super-user, and saying, hey [surgeon name redacted], would you have a patient....”

Open secret among high volume TransMedics users that the organ allocation process is bypassed

“TransMedics held this in-person meeting in Boston right before the American College of Surgeons last year. The leaders of many transplant programs were invited; I was one of them...and at that meeting, this very issue was discussed, this concept of bypassing the list...so, **a center that is not normally accustomed to using the liver on pump, they’re not going to take a liver like that.** And time is important to figure out who you would use the graft for...**calling TransMedics experienced users and saying, the primary center we procured for no longer wants to use this graft. Would you be interested?”** So, [redacted] **got up on the podium at that meeting and said, I understand why you are doing this...make sure that you do it under OPTN guidelines, do not bypass the list.** And everybody in the room was in agreement...**I get the practice that they’re talking about.** What should happen, for example, the liver I just told you about where we decided not to use it, what should happen is the **OPO where that liver was procured should be the one to go down the central list** and say the next patient on the list is a patient from [redacted]. Let’s call [redacted] and explain the situation and see if they are willing to use. **It should not be TransMedics calling [redacted], another super-user, and saying, hey [surgeon name redacted], would you have a patient on your list?** The OPO should be the one to say the next patient on the list is from [redacted], and then reach out to the surgeon and say your patient is next on the list. Would you be willing to use this liver?” – Transplant surgeon; high volume user

A key component of what we believe to be an organ steering scheme involves what we interpret as improper access to DonorNet, the national system used by OPO's and managed by UNOS to generate match runs and the ranked list of potential transplant recipients. Multiple sources told us that TransMedics has essentially infiltrated the system to obtain an "insider view of potential organs for the device." A transplant administrator at a pre-eminent center, one of the highest volume in the world, alleged that TransMedics attempted to "bully" the local OPO into granting them credentials to log into DonorNet, which they have allegedly obtained via another OPO: "I work with our OPO pretty closely...they also really despise TransMedics...they were being bullied...they wanted their unit access to DonorNet so they could see all of the donors...that were part of the [OPO name redacted] service area....they really didn't like that...they were pushing back...it kind of soured the relationship."

TransMedics allegedly "bullied" OPO for Improper access to DonorNet

A: "I work with our OPO pretty closely. I know a lot of the senior leaders there. They also really despise TransMedics, and were not interested in it - they did feel like they were being bullied. They wanted their unit access to Donor Net so they could see all of the donors that were coming that were part of the [OPO name redacted] service area so they could monitor them and have this insider view of potential organs for the device. They really didn't like that, and they were pushing back. It kind of soured the relationship."

Q: "TransMedics is pushing for access to Donor Net—that's this US organ allocation system?"

A: "Yeah, exactly." – Transplant administrator

The UNOS website indicates the broad role DonorNet plays in running the national organ transplant system, allowing OPO's to “launch match runs and make organ offers to hospitals” and transplant hospitals to “review organ imaging, accept or decline organ offers....” The system appears to be property of the US government, which we presume would create far more serious legal jeopardy for TransMedics than infiltrating a private network, if the allegations are true: “this system is a U.S. Government-designated Federal Information System operated by a contractor of an executive agency...unauthorized use of the system is prohibited and subject to civil and criminal penalties.”

DonorConnect overview and login screen

The image displays two screenshots from the UNOS website. The left screenshot shows the main navigation bar with the UNOS logo and links for Organ donors, Patients, Professionals, Media, and About. Below this is a row of five service tiles: Waitlist™ (Transplant candidate data), DonorNet™ (Organ donor data), TransNet™ (Safeguarding organ transport), Data Services (Insights for improvement), and TIEDI™ (Pre-/post-transplant data). A red dashed box highlights the DonorNet section, which contains the UNet DonorNet logo and a list of functions: Organ procurement organizations (OPOs) to manage deceased donor data; OPOs to launch match runs and make organ offers to transplant hospitals; Transplant hospitals to review organ imaging, accept or decline organ offers, and predict the likelihood of future outcomes; and UNOS researchers to study how to increase transplants. To the right of this box is a 'Technology partners' section listing SQL Server, Azure, .NET, apigee, twilio, AMBRA, and ANGULAR, with a note that UNet APIs seamlessly exchange data with DonorNet. The right screenshot shows the DonorNet login interface, featuring a blue header with the UNet DonorNet logo, a 'GIVE NOW' button, and a 'UNET' dropdown. The login form includes fields for 'Username' (Enter username) and 'Password' (Enter password), with a 'Show' eye icon next to the password field. Below the form are links for 'Log in with UNet', 'Forgot password?', and 'View system status'. The footer of the login screen shows the UNOS logo and the copyright notice '© 2024 United Network for Organ Sharing'.

The transplant executive stated that “yeah, it is inappropriate and they weren’t granted access, thankfully” – at least by that OPO. Access to DonorNet would allow TransMedics to view patient medical information and the ability to steer cases onto its device: “you can see at any given times which donors are in the pipe...they wanted to go and review their clinical information to see if they might be a good candidate for the device...to persuade the OPO or try to persuade the surgeon at the transplant center that accepted it...I’m sure they have a code...it’s not something that’s transparent...an OPO could give a vendor access through their own login.”

TransMedics allegedly has a login to DonorNet

Q: “Why did they want access to that system? That seems highly inappropriate.”

A: **“Yeah, it is inappropriate, and they weren’t granted access, thankfully.** But I think what they wanted to see was—because every time a new potential donor becomes available, someone at the OPO is entering that data into this computer system. **You can see at any given time which donors are in the pipe,** which ones are in the cue. And so, they probably, I would imagine **they wanted to go and review their clinical information to see if they might be a good candidate for the device.** I don’t know what they were going to do with that information, **like try to persuade the OPO or try to persuade the surgeon at the transplant center that accepted it;** I don’t know.”

Q: “Do you think they have—they must have access to Donor Net through other OPOs that they’re close to?”

A: “That’s all easy to - yeah, they might. You have to have a business code. You have to be a vendor in the UNOS system to be able to assign access to you. **I’m sure they have a code.** Anyone could look and see who they have access to. It’s not something that’s transparent. Although, **I suppose an OPO could give a vendor access through their own login.”** – Transplant administrator

A TransMedics “super user” and one we consider to be a willing participant in their organ steering scheme – was nonetheless candid about the company’s use of DonorNet to improperly shop organs and bypass the OPO’s ranked list of potential recipients: “they should not be allowed to have access to DonorNet...that really certainly pisses off some OPO’s when...TransMedics has arrived to recover...and people start calling around to other surgeons...saying there’s a liver...do you want to take it...the OPO’s are like, you can’t do that...don’t be making these calls....in DonorNet, you’ll see when you get an organ offer...there’s like a little blurb, the blurb will often say TransMedics is active in the area and can be utilized for organ recovery...so they’ve already inserted themselves.”

Allegedly accessing Donor Net and “calling around” to offer the organ and bypass the organ

Q: “I heard they were trying to badger an OPO to give them access to Donor Net.”

A: **“Yeah. They should not be allowed to have access to Donor Net.** And I know that really certainly pisses off some OPOs when, for example, you get a situation where **a transplant center turns down a liver that TransMedics has arrived to recover. And the TransMedics people start calling around to other surgeons, other programs, saying there’s a liver out there. Do you want to take it? And the OPOs are like, you can’t do that; that’s our job. We’ll allocate the organ; don’t be making these phone calls.”** - Transplant surgeon at a high volume academic center

TransMedics has allegedly already penetrated Donor Net to hijack the organ allocation process

Q: “Are there ways that they could be leveraging their relationship with these OPOs - Nevada, Massachusetts - to do things like look at Donor Net, do things like jumping the line.”

A: “Let’s say we have a scenario where we have a rapid DCD. **We’re just going to put this liver on the machine and then allocate it. And then you have to take it on the machine and you have to pay for it.** In Donor Net, you’ll see when you get an organ offer, and you look at there’s like a little blurb, **the blurb will often say TransMedics is active in this area and can be utilized for organ recovery. So, they’ve already inserted themselves.”** - Transplant surgeon at a high volume academic center

An executive at one of the largest OPO's in the country detailed how TransMedics tried to push them for access to DonorNet: "they tried to do that with us...give me access as if I were your employee...register me in the system, give me a password...we'll go in...to go through the chart, the assessment of the donor that is a good candidate for OCS." The executive speculated that the New England OPO, led by a controversial CEO who we were told is close to TransMedics CEO, provided a login, which would provide the company access to the most sensitive patient data on a national basis: "you have access to the matches of the recipients for all of the patients in the country...we lost trust in their business practices...that is a violation of the data use agreement that UNOS requires us all to commit to."

Allegedly pushing for access into DonorNet to "go through the chart"

Q: "I heard that they have access to Donor Net, and that they're trying to wedge their device in there by trying to see who's doing what and trying to call people. That doesn't seem like that would be allowed, but clearly, they're in bed with some OPO's that let them do it."

A: **"Yeah, they tried to do that with us early on...TransMedics wanted access to that. And so what they were saying to the OPO's, give me access as if I were your employee. Register me in the system, give me a password; we just want to help you. When you have an organ donor that might be a candidate for this technology, we'll go in. You don't have to give me all the information verbally; I'll go into the system. I'll be able to go through the chart; the assessment of the donor is a good candidate for OCS."** –

Executive at a West Coast OPO, longtime industry veteran

Allegedly obtained access via their local New England OPO; "violation of the data use agreement"

"...the New England OPO. I think they're one of the ones that do this. Ms. Glazier over there, what I heard is that she did add some sort of a data use agreement into their contract. The point of that, they believe that covers them. But once you give someone access to that service - if I were to do that for you right now and you logged in, you don't just have access to the organ donor information that I put in there; you have access to the matches of the recipients for all of the patients in the country that have matched. So, if you're a recipient at Mayo Clinic in Florida, TransMedics is going to have access to your name and information in there. We decided not to go that route because we lost trust in their business practices...we believed that is a violation of the data use agreement that UNOS requires us all to commit to." – Executive at a West Coast OPO, longtime industry veteran

The OPO executive explained that a DonorNet login would provide TransMedics with “the broadest level of access...they could be saving information” – “they system is not really designed to let a third-party company into this...once they’re in, they’re in...there’s patient-level information in there...potential recipients, where they are...it could essentially be giving them a competitive advantage...an advantage as a vendor over the competition about which next transplant center you might go out and try to court...”

DonorNet access provides “broadest level of access” and a competitive advantage

Q: “And how could they be abusing this?”

A: “For a while, they were going in with the access that they had from another OPO. It depends on what sorts of permissions that OPO has and how tightly they’re monitoring the privileges that they grant. **But the system is not really designed to let a third-party company into this, and once they’re in, they’re in, and the OPOs typically have the broadest level of access. They could be saving information. There’s patient-level information in there. Potential recipients, where they are. It could essentially be giving them a competitive advantage, if you can go in there and see what center has all the top listed, the sickest liver patients. I imagine that might give them an advantage as a vendor over the competition about which next transplant center you might go out and try to court to be a client of yours.**” – Executive at a West Coast OPO, longtime industry veteran

Part 6. TransMedics device usage is, we conclude, almost all off-label, the largest such racket we have encountered, putting it at existential risk: “I’ve never seen a device manufacturer so boldly engage in off-label use themselves...it’s their technician.”

Upon extensive research, we conclude that the vast majority of TransMedics device usage is off-label, to such an extreme degree that we believe it to be the largest off-label racket we have ever encountered and likely one of the largest in US medical device history. The purported purpose of the pump is clinical – to preserve and assess organs during transportation to increase their utilization and improve outcomes. However, according to dozens of interviews with ex-employees, large OPO's, and the transplant surgeons and centers who are its highest volume users - who were unusually transparent with us – the primary use case is non-clinical, with no clinical benefit: scheduling and convenience for surgeons and OR staff, who park organs on the pump for highly prolonged periods to stack cases and avoid late night work. In our opinion, payors – the government and a few private ones – are defrauded out of \$100,000 to \$400,000 per case with NOP charges, under false pretenses, to improve surgeon lifestyles. The indication on the label is exclusively clinical, using OCS liver as an example.

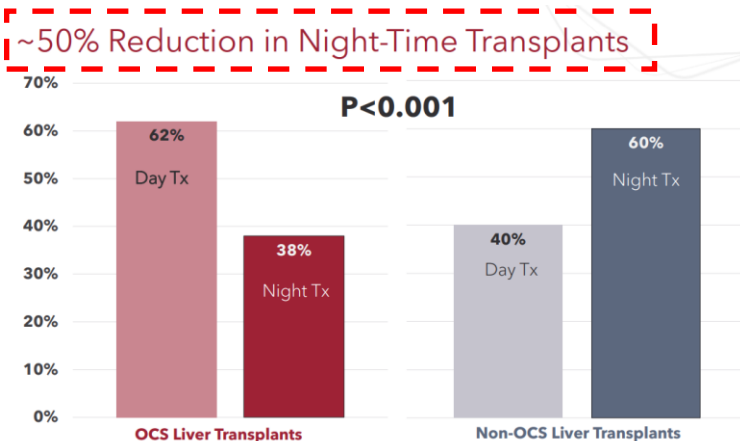
FDA PMA Approval Order Statement, Summary of Safety and Effectiveness Data – Sept 28, 2021

II. INDICATIONS FOR USE

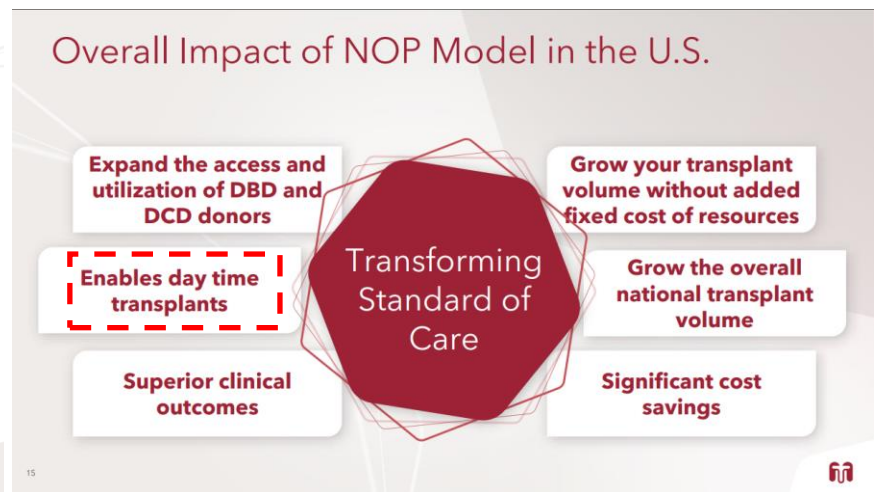
The TransMedics® Organ Care System (OCST™) Liver is a portable extracorporeal liver perfusion and monitoring system indicated for preservation and monitoring of hemodynamics and metabolic function which allows for ex vivo assessment of liver allografts from donors after brain death (DBD) or liver allografts from donors after circulatory death (DCD) ≤55 years old and with ≤30 mins of warm ischemic time, macrosteatosis ≤15%, in a near-physiologic, normothermic and functioning state intended for a potential transplant recipient.

TransMedics entire premise is that its pump and aircraft enable organ transport for longer duration and farther distances than on ice, thus improving their viability and use. No one disputes that a liver with both a donor and recipient in Salt Lake City is perfectly fine on ice and does not require a \$90,000 device and retrieval staff flown in on three separate jets - a real example from a Utah surgeon. TransMedics may claim that putting an organ on pump still enables assessment of biomarkers and viability, but **the claim is specious: every single surgeon we interviewed admitted they have never seen an NOP retrieval surgeon decline an organ,** as we shall detail – naturally, given the obvious incentive to sling every organ possible onto the device no matter how defective and move it via its private jets. Moreover, **even high-volume users told us the “clinical benefit” is just a charade so they can get away with using it for scheduling and convenience.** The ruse is visible in the company’s investor presentation:

TransMedics investor presentation, Oct 2024



11 Source: TransMedics' OCS Liver Perfusion (OLP) Registry & Scientific Registry of Transplant Recipients (SRTR)
Presented at the 2024 ASTS & ILTS Annual Meetings



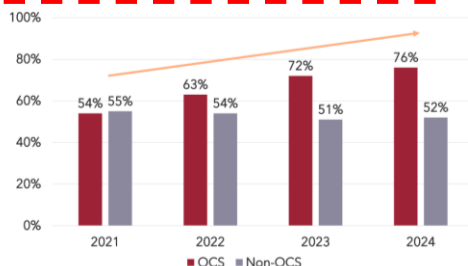
A senior ex-TransMedics employee spilled the beans, telling us what appears to be well-known among employees and surgeons – that the “the whole purpose of why the device was built and why it exists...to offer the flexibility”; “the beauty of the technology is saying there is no rush...when the liver arrives...they will just park it in the corner and wait until the doctor wakes up and is rested...that’s kind of the whole purpose....” He boasted that livers are on the pump for 40 hours, with an average of “at least 10-11 hours”; “while doctors are “getting their beauty sleep, our clinical specialist is sitting right next to the machine with the organ...so, it’s kind of weird.” The recent investor day presentation in Dec was overt in talking up the shift to “Day-Time Liver Transplant” as a “Game Changer.”

Senior Transmedics employee: “whole purpose” of the device is scheduling convenience

*“The **beauty of the technology is saying there is no rush**...it can sit there as long as you need it to. You don’t have to rush. All the hospitals that transplant livers using our devices, they don’t rush at all. If a liver is coming to their patient at 2:00 in the morning or 11:00 at night, they’re not going to rush to schedule the surgery for the middle of the night. They schedule the operating room time for 6:00 in the morning where it’s optimal for everybody. And **when the liver arrives, they will sit on that - they will just park it in the corner and wait until the doctor wakes up and is rested, and so on. That’s the whole purpose of why the device was built and why it exists, to have the flexibility, to offer the flexibility [..]** the doctors are so comfortable with the machine now, they are not in a rush. While they’re **sleeping and getting their beauty sleep**, our clinical specialist is sitting right next to the machine with the organ. So, it’s kind of weird. **So, on some of these missions, our clinical specialists are gone up to 40 hours long because there’s no rush for procurement.**”* - Senior ex-employee at Transmedics

TransMedics investor day presentation, Dec 10, 2024 – slide #25

Day-Time Liver Transplant - Game Changer I



Source: Scorpion Capital consultation calls with experts; <https://investors.transmedics.com/static-files/ae80ac5c-1729-44f6-9c45-4cbb1ff14d7f>

Leaving aside that neither TransMedics surgeons nor transplant centers ever seem to reject an organ irrespective of whatever “data” the pump spits out, the majority of transplants that the device and NOP service are utilized for appear to be ones where both the donor and recipient are nearby – within driving distance. Leaving an organ on a \$90,000 pump for 30 hours and mandating the use of private long-haul aircraft is therefore preposterous and unnecessary – unless you’re a surgeon abusing payors to avoid going to an OR at 2AM. We spoke with an ex-logistics employee at TransMedics, who revealed the audacious nature of the scheme and indicated that half the utilization of the device – as well as of the NOP service – is not only within driving distance, but that a large percentage is where both the donor and recipient are intra-city: “we get a huge volume of share from...within the city, like Boston to Boston, New York to New York...why would they use it when it costs a lot...you’re figuring out your logistics...we’ve had cases where the OCS device was used and our service was used within the same hospital...within a three-hour radius, about 50% of the cases.”

Substantial percentage of device usage is local and intra-city

A: “We get a huge volume of share from local - within the city, like Boston to Boston, New York to New York type of transplant on our device. And you would ask why would they use it when it costs a lot of money. It costs a lot of money...you keep it there for whatever, a few hours while you’re figuring out your logistics between hospital A and hospital B. We’ve had cases where the OCS device was used, and our service was used within the same hospital. So, they take out the organ, they put it on the OCS device, and a couple of hours later.”

Q: “What percentage of transplants are really close, like within a hospital? Within the city, an hour or two?”

A: “Within a three-hour radius, about 50% of the cases, the donors and recipient are within a three-hour driving range. Roughly 50% within a three-hour driving range.” – Ex-logistics employee at Transmedics

A former OCS specialist also stated that most usage of the device is local and/or within the same city: “mostly locally, so Seattle to Seattle, Spokane to Seattle...50% of the time we were staying within Seattle.” One transplant center after another emphasized that the vast majority of liver transplants in the US are within the same OPO region and within driving distance, given how the organ allocation algorithm works, and bluntly rejected the notion that livers – the vast majority of TransMedics business – are traveling from California to Ohio, for example. Vanderbilt, one of the highest volume centers in the world, stated 90% of livers are procured locally – “most of the DCD’s are recovered by their own OPO...DCD livers don’t travel very far.” Massachusetts General told us 80-85%.

Most device usage is allegedly local

A: “I would assist the surgeon in putting the organ on the OCS device. And then, I would fly typically independently, fly or be driven independently, and manage the organ while en route, and then until that organ was implanted at the recipient center.

Q: “How far would they send you?”

A: **“Mostly locally, so Seattle to Seattle, Spokane to Seattle. I would say 50% of the time, we were staying within Seattle, and then often we would be going to San Diego, Phoenix, Portland.”**

Q: “What percentage of it would you say was local, like within driving range?”

A: “I would say **probably over 50%.**” – Former OCS Specialist

Vanderbilt: 90% of livers are locally retrieved

Q: “That means **you’re able to procure 90% of your livers within the OPO region?**”

A: **“For DCD, yes.** Not for brain death, for DCD.”

Q: “Is that the case for most centers, that they’re able to get DCD livers within their OPO region?”

A: **“Most of the DCDs are recovered by their own OPO. DCD livers don’t travel for very far.** Now there is some problem that they are traveling more because of TransMedics.” – Transplant surgeon in a leadership role at Vanderbilt

Massachusetts General: 80-85% procured within driving distance

A: “Most of our procurements usually are in a radius that is achievable for us, mainly New England, basically.”

Q: “What percentage of your transplants are **within driving distance, four hours or less?**”

A: “I’d say that’s still the great majority of it. So, it’s probably **about 80-85%.**” – Transplant hepatologist in a leadership role at Harvard/Massachusetts General Hospital; key role in oversight of the liver transplant program

A transplant program director in the Midwest derided the notion that livers travel far for transplant. One of TransMedics highest-volume users implied that most of his device and NOP utilization is hence unnecessary and off-label: “it’s not like we have a slew of organs in California, Oregon, Hawaii that are getting transplanted here, for sure”; he told us that he uses the NOP aircraft and retrieval service 100% of the time, which is troubling as “75-80%” of his organs are local. A second transplant director, in the Northeast, explained that livers are by definition procured locally under the algorithm, as he would only be offered one from California if every center in between rejected the organ. He stated that the max radius is within 250-500 miles, and that while one could use the device, that four hours “even in the cooler” is “not very detrimental” – “for practical purposes, it’s pretty unusual I would get an organ from that far away.”

High volume user says 75-80% of livers are local

A: “The allocation of organs to you has nothing to do with where you are and everything to do with where the donor hospital is, which has to be typically within 500 miles...to be fair, **it’s not like we have a slew of organs in California, Oregon, and Hawaii that are getting transplanted here, for sure.**”

Q: “So **75-80% are still within the region?**”

A: “Correct. And, of course, any DCD - a DCD is not going to go to California. **That’s an organ that would be transplanted locally.**” - Transplant hepatologist and director of the liver program at a Midwest academic center

Organ allocation algorithm means most livers procured within driving distance

“The **majority of our organs that are going to come are going to be within a 250 to max 500-mile radius**, and that’s because of the way that allocation is structured in the United States. For example, for me to get an organ from California, that means that all centers within that California area and 500 miles around there have to have turned that organ down before it’s going to come to me, even being offered to me. So, the **likelihood that I’m going to get an organ offered from California is actually very low**. For all practical purposes, while we’re willing to go and travel to those because you can travel from California to here now in four hours by jet plane, so **that four hours if an organ is on machine preservation, is nothing. Even in the cooler, if it’s an ideal organ, it’s not very detrimental. But for practical purposes, because of the way that allocation is, it’s pretty unusual that I would get an organ from that far away.**” - Transplant surgeon and transplant program director at a leading Northeast academic center

An executive at one of the largest OPO's stated that "I've never seen a device manufacturer so boldly engage in off-label use themselves" - "It's their technician...they don't permit anyone to operate the device...you have to go through their national program...they are the ones operating it." While physicians have latitude to use devices off-label, it is illegal for medical device companies to market or promote their devices for off-label use. TransMedics is therefore, we believe, extremely reckless, as the off-label use occurs by its own device technicians, known as OCS Specialists – who hospitals must use. Moreover, ex-employees stated that the company's case coordinators direct the NOP retrieval surgeons and clinically decide that an organ is acceptable and to place it on the pump, suggesting that they jump the gun in the case of expedited offers before a center has even accepted the organ, thereby implicating TransMedics directly in off-label usage – where the rubber meets the road in terms of legal jeopardy.

TransMedics allegedly engages "in off-label use themselves"

Q: "The problem is if a company promotes off-label usage, have you heard TransMedics promote the convenience of the pediatric usage?
I mean, to some extent, by definition, they're promoting it because they're the ones using it, their technician, their surgeon."

A: "That's exactly right. **It's their technician. They don't permit anyone to operate the machine. You have to go through their national program. They are the ones operating it. They're the ones doing it. I've never seen a device manufacturer so boldly engage in off-label use themselves.**" – Executive at a West Coast OPO, longtime industry veteran

The same OPO executive – one of the highest-volume in the country as far as organs placed – expressed shock at the off-label practices, using Houston Methodist as example, who we think is a top 2 or 3 “super user”: “I’ve heard they do crazy things...sometimes the organ donor will be in the hospital and the liver is going to a patient next door and they’ll still put it on that machine...that’s just crazy...there’s no clinical benefit to that liver going on that machine to go from one room to another.” We noted that surgeons freely admit that the only use case is convenience, to which the executive replied “100%...they told me first-hand...I’ll sleep overnight...TransMedics can go and get it...but TransMedics won’t let you just hire their surgeon to go...not something they will permit...you’ve got to use the cartridge...”

Highest-volume users allegedly “do crazy things” like using device when donor and recipient are in the same hospital

Q: “There’s a very insular circle of Egyptians at the top: the CEO, Tamer, Magdy, Waleed’s sister, runs the lung business. It seems to be a similar Egyptian circle among their highest-volume users, this guy at Houston Methodist, Mark Ghobrial is Egyptian. There’s a guy at Henry Ford in Michigan; one of their highest-volume users is Lebanese. So, I noticed a lot of their highest-volume implanters share some similarity in background. I’ve heard Houston Methodist is using 170 of their devices. That could be \$20-30 million of revenue to this company once you add in all the airfare and all that stuff. One guy.”

A: **“I’ve heard they do crazy things, too. That’s a big hospital. Sometimes, the organ donor will be in the hospital, and the liver is going to a patient next door, and they’ll still put it on that machine. That’s just crazy. There’s no clinical benefit to that liver going on that machine to go from one room to another.”** – Executive at a West Coast OPO, longtime industry veteran

Surgeons admit the use case is convenience, but can’t only use TMDX surgeon – have to pay for device

Q: “I’m hearing that most usage is off-label - it doesn’t actually need to be used because the organs are perfectly fine; there’s no issue with cold storage. Some surgeons that have been candid - they’re using it for convenience because they don’t want to get up at 2 am.”

A: **“Oh, 100%. I’ve got that first-hand. They told me first-hand.** I’ve been operating all day. I can’t travel to Reno to go do a donor. If TransMedics is available, I’ll take it. **I’ll sleep overnight. TransMedics can go and get it. But TransMedics won’t let you just hire their surgeon to go and get it and bring it back. That is not something that they will permit. You’ve got to use the cartridge.”**

Q: “One West Coast center, they have one OR, even though they’re a high-volume center, so they want to stack the cases, so they just stick them on the machine. If Medicare or the few private payors that cover this that are paying \$100,000-200,000 to help with the logistics and scheduling, I mean, they would hit the roof. Nobody’s paying six figures so you can do it when it’s convenient for you.”

A: **“You’re 100% spot on.”** – Executive at a West Coast OPO, longtime industry veteran

We have never heard surgeons – or any physicians – so openly confess that their usage is for non-clinical, off-label reasons. One of TransMedics highest-volume users implied that surgeons are manipulating administrators with purported benefits: “I will freely admit that most surgeons are using mostly subjective criteria” and that they wrongly attribute any “perceived” benefit to the device” – “the biggest cheerleader for this is a busy surgeon who doesn’t want to come in at 2AM...of course, when you go say [reduced hospitalization] to the hospital system, they love that...the main and perhaps only value proposition was scheduling convenience.” A second surgeon, who leads a major transplant program in Pennsylvania, was crystal clear: “delaying cases for the morning is true...let me real blunt...when we do this, it’s for the operating room team to start at the normal time at 7:30 in the morning...do some centers use it on everyone when they don’t need to? Absolutely correct and absolute fact.”

Surgeons are “real blunt” in admitting the usage is only for scheduling convenience

“Delaying cases for the morning is true - let me be real blunt. You have to beg the nurses to come in and pay them a lot of money and anesthesia, the same way. In fact, some of the anesthesia people will quit after a while because they don’t want to work all night. **So, when we do this, it’s for the operating room team to start at the normal time, at 7:30 in the morning...So, do some centers use it on everyone when they don’t need to? Absolutely correct and absolute fact..”** – Transplant surgeon and transplant program director at a major academic center in Pennsylvania

Only benefit is “scheduling convenience”; surgeons misleading hospitals into “perceiving” reduced hospitalization

Q: “What really is the value proposition? If 75% to 80% of livers are still within the region, is it really a benefit from spending \$100,000 versus doing it the traditional way with cold storage?”

A: “So, now you can pump a liver, a liver that historically you would not use based on - **I will freely admit most surgeons are using mostly subjective criteria** for what is a good organ or not....and so, they attribute mostly right, but **somewhat wrongly subjectively the benefit from all this to the device. The biggest cheerleader for this is a busy surgeon who doesn’t want to come in at 2AM,** would rather come at 8 in the morning and do a transplant, perceives that utilization of these devices reduces the hospitalization - **of course, when you go say that to the hospital system, they love that...the main and perhaps only value proposition was scheduling convenience.”** -Transplant hepatologist and director of the liver program at a Midwest academic center; one of

The off-label usage is simply unsustainable – centers are on to the game that surgeons are playing and cracking down on usage. An influential, longtime OPO executive who serves on UNOS committees called it “bizarre that they would take on such a cost for convenience...you can’t have outrageous organ acquisition charges...for the surgeon’s convenience.” He doubted the legality of surgeon’s simply fabricating a clinical indication – “they can pump anything and say, look, I was wondering about the functionality of the lung, and I need to check it for a couple of hours...I don’t think anybody would say that out loud.”

“Outrageous organ acquisition charges” simply for surgeon convenience; surgeons fabricating clinical excuses as a cover
“The convenience of pumping...is even more **bizarre that they would take on such a cost just for convenience**...the organ procurement area, **I mean, legality** - you would think that there would be some overseeing body that says you have to charge x amount or you have to cover x percent. You **can’t have outrageous organ acquisition charges** to make up for your organ procurement poor performance or for **the surgeon’s convenience**. But there just isn’t. You could say you need to pump. You wanted to see how it does. You could always say that. I mean, sure, I don’t know if they will put that in the chart, but they’ll pump. **They can pump anything and say, look, I was wondering about the functionality of the lungs, and I need to check it out for a couple of hours**. And these surgeons want their convenience, so **they would say that in a heartbeat and not blink**. It’s hard to prove that the surgeon didn’t want to see the liver function improve or, lung O2 improve, or heart function improve, and that it was really around OR convenience. I doubt anybody would say that out loud....they wouldn’t put in a patient’s chart that they’re pumping something to free up an OR.” – Longtime executive at major OPO (organ procurement organization), and on UNOS committees

The head of a major OPO stated that TransMedics is “preying” on surgeons seeking to improve their personal lifestyles, who then concoct clinical excuses to manipulate and “bully” naive transplant administrators – “he’s probably preying on the transplant surgeons...I think it’s going to the surgeon and then using the surgeon to bully the administration...they hide behind the excuse that ‘well it’s going to improve the outcomes’...the administrators, they don’t know what going on half the time...they’re probably getting bullied by the surgeons.”

“Preying” on surgeons who then “bully” administrators and “hide behind excuses” to use device for convenience

A: “If I’m TransMedics, I’m hitting up the surgeons because they’re going to surgery in the middle of the night. These poor surgeons are stuck, like okay, do I either continue to not sleep like I have in the last decade? Or do I have this solution where I can have somewhat of a lifestyle. **It sounds to me like he’s probably preying on the transplant surgeons.** That’s what I think. I don’t think it’s the administration. **I think it’s going to the surgeon and then using the surgeon to bully the administration; “we’re going to do this.”**

Q: “And to use their device?”

A: “Yeah, and to use their device and **they hide behind the excuse that, “Well, it’s going to improve the outcomes,** and we’re going to be able to do more of it, and we’re going to have less turnover, etc., etc., etc. **In my experience, the administrators, they don’t know what’s going on half the time. So, I think they’re probably getting bullied by the surgeons.”** -CEO of a major OPO (Organ Procurement Organization), longtime industry executive

However, hospitals and transplant administrators are increasingly realizing that surgeons are disingenuous when they say they need “to see liver function improve, lung O2 to improve, or heart function to improve” – and are beginning to severely limit or eliminate TransMedics usage entirely, as we shall cover in more detail. UCSF, a high-volume user who told us they intend to shut down most TransMedics usage in early 2025, clued into the game its surgeons were playing across all three programs - heart, lung, and liver: “its not like anything that required the device but they would use it out of convenience, which is a very expensive backup plan...even when it wasn’t clinically indicated”; “we’ve since remedied that.” The “surgical director is very mindful of cost...and questioned if we’re using it too liberally, like on organs we would have taken otherwise before TransMedics was available...the cost has been a concern.”

Surgeons using device when it “wasn’t clinically indicated”; administrators have cracked down

“Our heart transplant group was using it because they might not have a procurement surgeon available to go get a heart, and this could just be a standard heart. **It’s not like anything that required the device but they would use it out of convenience, which is a very expensive backup plan.** We’ve since remedied that...we were **using it even when it wasn’t clinically indicated.** For lung transplant, our surgical director is very mindful of cost. [Redacted] will only use it when absolutely necessary.” – UCSF hospital administrator

“Using it too liberally” as “surgeons are preferring it” because it’s “more convenient”

Q: “Why did the surgical director and/or you make the decision to try to move away from TransMedics? Is it primarily cost or is it you don’t see much clinical benefit from it?”

A: “I think the surgeon, Dr. [redacted]. **questioned if we’re using it too liberally**, like on organs we would have taken otherwise before TransMedics was even available to us, but now the **surgeons are preferring it because it might be more convenient.** If you have this option at your fingertips...they’re making that call and they’re using it. I think he thinks it’s being used a little bit too liberally, and the cost has been a concern.” - UCSF hospital administrator

Our research interviews lead us to believe that TransMedics highest-volume users – such as Houston Methodist, Mayo Arizona, Henry Ford, UC San Diego, University of Utah, and others – are particularly notorious for off-label usage and that most of their utilization is not for clinical reasons. UCSD and Utah told us so directly. A Utah surgeon described the device as sort of a drug for surgeons’ lifestyles: “you don’t have to work all night...daytime is wonderful...it’s so much easier for me...the bitching and moaning from the OR is so much better.” A UCSD surgeon echoed the bliss of scheduling transplants when you want them, even if it extends the time an organ is outside of a body: “it’s the genie out of the bottle once you get a taste...transplant surgeons make our money from 11 at night to 5 in the morning...but it’s not fun...I can do it in the morning.”

High volume Utah surgeon admits device usage is for convenience

Q: “It looks like you use their NOP service for each of these?”

A: “Yes, the liver just shows up at your door at 7 am, and **you don’t have to work all night... this paradigm shift of someone else doing the donor** and us doing the transplant during the **daytime is wonderful. It’s so much easier for me. The bitching and moaning from the OR is so much better.**” – Transplant surgeon at a high volume academic center in Utah

UCSD states primary reason is convenience and lifestyle

“...it’s **the genie out of the bottle once you get a taste** - transplant surgeons, we make our money from 11 at night to 5 in the morning. But it’s not fun. We don’t like it—flying all over the country and being up all night is exhausting, especially when you get into your 50’s. Doing a case that is scheduled and is going to go well, **I can do it in the morning**, it’s hard to put a price tag on that. It really changes the dynamics of transplant.” - Transplant surgeon

Surgeons have become creative in finding excuses to jam payors for \$100,000 to \$400,000 per TransMedics case, even coming up with self-serving studies that purport to show that errors are higher if they have to operate at night. They were more forthcoming during our interviews, however – you can “consciously or subconsciously...find a reason to put it on the machine so that you don’t have to do it in the middle of the night”; and that it’s “another good question” whether the device even has any clinical benefit for organs. The surgeon flatly stated that the main rationale is stacking and scheduling transplants as they only have one operating room – “we’ve had three livers on OCS at the same time”; “you can do that case at your leisure and you can stack cases”; “even take a break for dinner in between.”

“Super user”: use case is scheduling and stacking transplants

Q: “How much the use case for OCS is for OR scheduling and convenience?”

A: “What has become **more and more normal**, and **even more dramatic than what I predicted** - once surgeons and teams and anesthesiologists get a taste of a liver that is placed on machine perfusion and **you can do that case at your leisure and you can stack cases** - we’ve had situations where **we’ve had three livers on OCS at the same time** and you can just kind of do them one after the other after the other and **even take a break for dinner in between, that has really changed the way we approach it. Whether it’s consciously or subconsciously, you can find a reason to put it on the machine so that you don’t have to do it in the middle of the night**...I would say **the biggest thing is that you can schedule cases when you want them**, and if you have a busy program, it’s easier for everybody if you can do that case when you want to do it. And **that has become an increasing indication for putting a liver on the device.**” - Transplant surgeon

Unclear if the device even has a clinical benefit in assessing marginal organs

“We just can use a lot of organs that we were afraid to use before; whether TransMedics makes all the difference or whether it just gives us courage is another good question. Logistically, we’re structurally a program that **can only operate in one operating room at a time.** So, I can now do a lot more transplants because **I can just stagger them** and do them when I want and accept organs whenever it comes up, and I don’t have to worry about the logistics of it.” - Transplant surgeon

Mayo Arizona and Houston Methodist came up repeatedly in our interviews, with surgeons alleging that they use dubious practices to spike their TransMedics device usage. We quoted an OPO executive who alleged that Methodist uses the pump even when the donor and recipient are in the same hospital. UCSF shared similar conduct at Mayo, where both retrieval and transplant were at their center but “the surgeon decided to put that organ on TransMedics because they were doing another case” – “when I shared that with one of our surgeons, he was like oh, I could see how that would be cool.” A leading liver transplant surgeon, perhaps the most influential in the world, described what he stated were conversations with Ghobrial at Methodist, which suggested off-label usage as driving their growth and TransMedics usage: “Ghobrial has definitely grown his program, more than doubled its size...the guy in Mayo Clinic in Scottsdale has more than doubled theirs...this is potential abuse of technology...a very expensive abuse...you’re really doing a disservice because you’re increasing the cost without incremental increase in outcomes.”

Mayo Arizona allegedly using device when donor and recipient are both in the same hospital

“I heard that there was a donor at Mayo in the hospital where the transplant program is. And the surgeon decided to put that organ on TransMedics because they were doing another case. So, that kind of convenience. It’s not like it was coming from a far distance or anything clinically was required for this donor. But it was a convenience thing. When I shared that with **one of our surgeons, he was like oh, I could see how that would be cool.”** - Executive at a pre-eminent West Coast academic transplant center, one of the highest volume in the world

Highest volume users have doubled transplant volumes through “very expensive abuse” with no improvement in outcomes

“Mark Ghobrial has definitely grown his program, more than doubled its size. The guy in Mayo Clinic at Scottsdale has more than doubled theirs. Part of it is that they’re willing to take organs that others haven’t been willing to - older donors, all that stuff.... I remember Congress jumping up and down about the Cost Report because they were saying it’s being abused. Well, this is potential abuse of technology, a very expensive abuse...if you took it to the extreme and talk about an \$8 billion industry, you’re really doing a disservice because you’re increasing the cost without incremental increase in outcomes.” –One of the top liver transplant surgeons in the world, based at a leading Midwest academic center

Source: Scorpion Capital consultation calls with experts

The information that Ghobrial at Houston Methodist allegedly shared with this surgeon is troubling – that ~70% of their TransMedics cases are simply “to push the operation off until the morning...it is really unjustified to use this...to make life more convenient for the transplant team.” We highlight that the surgeon who relayed this color is perhaps the world’s top liver transplant KOL, who stated that Methodist is using TransMedics to “basically make transplants a semi-elective operation”; “I frankly find that unjustifiable.” He estimated that Methodist and Mayo Arizona are using ~400 TransMedics pumps – a significant percentage of the company’s revenue particularly when NOP transport charges are included.

Methodist and Mayo use of TransMedics is for convenience and “really unjustified”

Q: “So, you’re saying most of the use now is livers where it doesn’t need to be used?”

A: “I know this because I talked to Mark Ghobrial about this. Seventy percent of their transplants use TransMedics. And when I asked him, I asked are these are all DCD’s? And he goes, no. **He goes, in their program, only about 30% to 40% are DCD or older donors, so the rest of them are being used to basically make transplants a semi-elective operation.** In other words, instead of operating at midnight or 1 or 2 in the morning operating all night long, they’re **using these devices to push the operation off until the morning.** **To me, that’s a really unjustified use** of resources to make the operation more convenient. **It is really unjustified to use this technology to make life more convenient for the transplant team** because I didn’t train that way; I don’t.”

Q: “I’ve heard from calls that the main use case is that people don’t want to operate at night. For the transplant center, it’s just logistically easier to push it to the next day. They’re leaving these livers on these pumps now for 10, 15, 20 hours, 30 hours sometimes.”

A: “I agree with that. **For me, I frankly find that unjustifiable.** I can see once in a while, you’ve got a transplant scheduled, and then they call you up, and you have a great organ for another patient, but you only have one operating room.”

Q: “If Ghobrial said 70% of their liver transplants use TransMedics, and for most of them, it’s just because they pump it overnight because it’s more convenient?”

A: “Yes, that’s what it sounds like to me. This is what our discussion was. They’re doing it to make the operation semi-elective.”

Q: “How many cases is Ghobrial doing as a high-volume user?”

A: “I think they’re doing 200 transplants a year. That’s a lot. That’s like 125, 150 - you do the math, but 70% of those use cases are for convenience.”

Q: “So, single-handedly alone, he’s probably driving \$15 million in revenue for the company.”

A: “Oh yeah, possibly. **Look at who the big users are. The same thing with Mayo. Mayo went from 100 transplants and they’re doing 200-plus a year also.**” –One of the top liver transplant surgeons in the world, based at a leading Midwest academic center

Surgeons and OPO's indicate that off-label misuse for convenience is now taking a more extreme turn, leading to injured and lost organs: "back to base" use where centers retrieve the organ in an ice cooler, and then put it on the pump back at the hospital for convenience and scheduling. Such use is not indicated, approved, nor was it the protocol in the pivotal trial, given the shock to the organ from different environments and three temperature changes – donor (warm body) to hypothermic (cold, ice) to a normothermic (warm) TransMedics pump. A leading KOL stated "there are two different practices now for TransMedics" and that "back to base" is now a third of device usage: "smaller programs using it...to delay the OR time until the morning...bring them in and then put them on the machine...the ratio is probably 2:1..." – two via NOP and one via back to base.

"Back to base" usage where organs retrieved on ice and then put on the pump

"There are two different practices now for TransMedics. There's the one that they do, which they bring the machine to the donor hospital. They procure the organ. They put the organ on the machine at the donor hospital, and then they transport the whole device with the organ to your center. And then, there's a practice which is logistically easier, it's certainly a lot cheaper, which is **you procure the organ the old-fashioned way, put it into an ice bucket/chest. You transport it back to your institution, then you put the organ on the machine, and then you do the pumping at your own hospital.**" – One of the top liver transplant surgeons in the world, based at a leading Midwest academic center

2:1 ratio of organs procured via NOP vs back to base

Q: "Of the people out there - I'm sure you have tons of colleagues, other liver transplant surgeons you talk to, is that how they predominantly use it for liver now? They bring it with cold storage, and they pump it at their facility."

A: "It depends on the institution. The programs that use NOP for livers tend to be larger. They tend to use them where the donor is a little bit more unstable or the transportation distance is a little bit longer. **Smaller programs using it for other reasons, logistics to delay the OR time until the morning or something, tend to be good organs. They bring them in, and then put them on the machine. I would say that the ratio is probably 2:1: 2 procured the NOP way and 1 the back-to-base way...technically, I guess it is off-label because of how the original trials were done.** Having said that, that's not true for the hypothermic. The hypothermic ones, which are the organ recovery system that's being reviewed by the FDA and then there's a new one that's coming out from Bridge to Life. That one also is back-to-base. But those are both hypothermic because they're at 8 degrees Centigrade." – One of the top liver transplant surgeons in the world, based at a leading Midwest academic center

The CEO of an influential OPO in the Midwest echoed that centers are increasingly transporting organs on ice and then storing them on the device at the hospital for scheduling convenience. He estimated that 10% of livers in his region are now lost to damage through this practice, whether via physiologic injury from ice to pump or because organs are left on the pump for excessive periods: “that could get TransMedics into some trouble”; “they’re under increased scrutiny by the federal government...we have some frustrations with them because transplant centers will use it off-label”; “we have a surgeon here locally [who] recovers the liver and sends it to transplant program...and then that program puts the liver on the pump...well, they transplanted the liver late...they actually lost the liver on the pump.”

Centers increasingly using the device off-label for storage at the hospital after initial recovery/transport on ice

A: “It’s an ultra-expensive device. **They’re under increased scrutiny by the federal government. We have some frustrations with them because transplant centers will use it off-label**, meaning they don’t want to pay for TransMedics for the recovery and the flight. They bought crazy, like brand-new, \$13 million aircraft, and then people are forced to use it. So, what they’ve done is we’ve had this happen a few times, where **we have a local recovery. We send them the liver, and then they put it on the TransMedics device**. But they don’t want to pay for the recovery and the flight.”

Q: “Explain that off-label dynamic to me.”

A: “They should send the TransMedics recovery surgeon, and they should travel with the device, and they should do the recovery in the OR, obviously, and then place the organ on the device in the OR **as opposed to having it recovered locally and then sent to the transplant program, and then they use the device**. The benefit really of the pump is that it allows the surgeons to sleep and transplant during normal business hours. They can put it on the pump, and then they can transplant at 7 am instead of 2 am...it’s more of a convenience factor. I’m not a huge fan of TransMedics....I’ll give you an actual example. A transplant center accepted a liver from one of our donors. And they wanted to use a TransMedics device but they didn’t want to pay for their recovery team and logistics program to go get it because it’s so expensive. And we didn’t know this. So, they asked, will you have someone local recover it for us? So, **we have a surgeon here locally. He shows up and recovers the liver, and sends it to that transplant program. And then, that program puts the liver on the pump and then transplanted the liver later. Well, they transplanted the liver late. They actually lost the liver on the pump**. What we’ve seen - **it’s happened three times in the last few months** is transplant centers don’t want to pay the excessive cost to use their logistics, so **they’ll have the organ recovered locally and then sent to them. So, I think that could get TransMedics into some trouble**...the behavior of some of the transplant programs because of the TransMedics cost. .” – CEO of a major OPO (Organ Procurement Organization), longtime industry executive

The CEO of the Midwest OPO stated that “it’s happened three times in the last few months” that livers were lost to damage when centers left them on the pump too long after bring them back to the hospital on ice. He stated the livers were lost at three different transplant centers, indicating the growth in the practice as centers rebel against extortionate NOP recovery/transport charges, with some apparently able to extricate themselves from the otherwise forced bundling of the device and logistics service – “the liver is supposed to go on the pump in the operating room right after recovery...it’s not supposed to be shipped and then go on the pump four or five hours later...that’s been our experience lately...they’re like, we’re going to put it on and then...transplant in the morning and then the lactic acid is too high,” which kills the liver.

“Three different transplant centers” lost livers recently from off-label usage; 10% of livers in region damaged/lost to the practice

Q: “What’s the off-label aspect here?”

A: **“The liver is supposed to go on the pump in the operating room right after recovery. It’s not supposed to be shipped and then go on the pump four or five hours later.”**

Q: “You’re saying what centers are doing is they’re just using it at the recipient location just to pump it overnight to be able to wait until the next day. That’s the use case?”

A: “Correct. That’s what we’ve seen. **That’s been our experience recently, and it’s happened a few times to us, and we’ve lost three livers because of it.**”

Q: “Are they just keeping the liver too long on it?”

A: “Yeah, they’re like, okay, **we’re going to put it on, and then we’re going to transplant in the morning and then the lactic acid is too high.** In the old system; you would have just transplanted, and it probably would have been fine.”

Q: “And you’ve seen it three times in the last few months where you lost three livers. What percentage of total liver transplants is that in the last few months? How prevalent is this?”

A: **“For us, it’s probably 10%.** I mean, we’ve just seen it recently the last few times. **Three different transplant programs.”**

Q: “But doesn’t TransMedics tightly control the device and say you have to use their perfusionist? So, how did the center get the device?”

A: “I’m not sure. I’m not sure how. I thought they did as well. I thought they kept it at the recipient center, and then they did it that way. So, maybe TransMedics is doing this, you know, maybe they’re using it that way for these centers...And I don’t know if it’s because they increased the cost, and then it’s put the added pressure on the transplant program, and they’re trying to cut corners.” – CEO of a major OPO (Organ Procurement Organization), longtime industry executive

Off-label misuse of TransMedics device is, we think, centered on two aspects. The first, utilization for surgeon lifestyle and convenience with no clinical indication or benefit, exploits hospitals and payors who incur six-figure sums per case so surgeons can work hours of their own choosing. The second type of misuse is just as prevalent but far more insidious: placing dubious, high-risk organs on the device in violation of specific criteria on the FDA label – by our estimate, almost every organ placed on the device violates one or more of these criteria and is therefore off-label use. We focus on the liver version of OCS as it is the core of the company's revenue and growth, as the heart and lung pumps are basically a failure. Per the label, the device is only allowed for livers that meet 3 critical parameters of viability, plus a fourth not explicitly mentioned in the label but which was key in the pivotal trial and FDA approval: 1) the DCD donor is ≤55 years old; 2) ≤30 minutes of warm ischemic time; 3) ≤15% macrosteatosis; and 4) the time the liver is on the device. We explain the significance of each in turn.

FDA PMA Approval Order Statement, Summary of Safety and Effectiveness Data – Sept 28, 2021

II. INDICATIONS FOR USE

The TransMedics® Organ Care System (OCST™) Liver is a portable extracorporeal liver perfusion and monitoring system indicated for preservation and monitoring of hemodynamics and metabolic function which allows for ex vivo assessment of liver allografts from donors after brain death (DBD) or liver allografts from donors after circulatory death (DCD) ≤55 years old and with ≤30 mins of warm ischemic time, macrosteatosis ≤15%, in a near-physiologic, normothermic and functioning state intended for a potential transplant recipient.

Most organs placed on the TransMedics pump, we believe based upon research interviews, fail to meet the labeled criteria of organ viability – rendering the entire DCD use case a scam. Most DCD livers were historically never used for transplant for exactly this reason. The bull case for TransMedics is that its pump suddenly enables the use of DCD organs and creates a new category of transplant, expanding the pool with previously unusable, high-risk organs. The reality is that its pump has simply fueled the usage of DCD organs that reputable centers still won't touch - from extremely old or diseased donors – by providing a “clinical” pretense” that perfusion magically makes them usable. As example, an ex-TransMedics employee made therapeutic claims that are quackery: “just as if they were in a human body...allows perseveration for...an unlimited amount of time...the organ is actually in better condition than it was...it actually has a therapeutic impact...absolutely right, it's in better condition...no limitation, even regulatory...on how long the organ can be on the machine....”

TransMedics employee claims the device improves the organ after transplant

“The organs on the OCS device are **just as if they were in a human body**, which means an organ can stay on a device for as long as needed. There's no more time limitation with 10 hours and whatever, and there are no distance limitations. The TransMedics OCS device not only allows the preservation of the organ for much longer, **for pretty much an unlimited amount of time**...when you place organs on an OCS device; you replicate a healthy human body, and **the organ is actually in better condition than it was when it was removed from the donor's body. It actually has a therapeutic impact.**” – Ex senior employee at Transmedics

...also claims organ is in “better condition” than before it went on the device

Q: “Is there any clinical change or degradation that you guys observe in a liver when it goes to like 10-12 hours or 40 hours?”

A: “That's absolutely the key. The organ is being perfused with the donor's blood and it's replicating the donor's body. There is no change because of the same temperature conditions, the same pressure, and it's the donor's blood so there are no clinical effects. There's an FDA certification for one of our devices at this point. The device is certified as therapeutic to the organ while it is on the device, which means that **when the organ is on the device, it's actually recovering**. As I mentioned to you before, it is recovering from trauma and any damage. **Absolutely right; it's in better condition than it was before it went on the device. There is no limitation, even regulatory limitation, on how long the organ can be on the machine.**” – Ex senior employee at Transmedics

The three labeled criteria for use, which most DCD's livers fail to meet, are 1) the DCD donor is ≤55 years old; 2) ≤30 minutes of warm ischemic time; 3) ≤15% macrosteatosis. The age indication is self-explanatory. The older the donor, the more compromised and high-risk the liver. The average donor age in the pivotal PROTECT trial for OCS Liver was 45.9 years, well below the labeled age, as the trial cherry-picked the best donors to manipulate the outcomes on the pump. However, one high-volume TransMedics center after another told us that most of their device use is for liver donors well over 55 – into their 60's and older. One such “super user” alleged that Houston Methodist, who we think is one of top two users, is “in the business of transplanting people with 70-year-old livers.”

Donor age - OCS Liver PROTECT pivotal trial, JAMA Surgery paper, March 2022

Table. Donor and Recipient Demographic Characteristics and Risk Factors for Transplanted Organs and Recipients

Characteristic or risk factor	Donors or recipients in primary analysis per-protocol population, No. (%)		Donors or recipients in as-treated population, No. (%)	
	OCS Liver	ICS	OCS Liver	ICS
Donor				
No.	151	142	153	146
Age, y				
Mean (SD)	45.9 (14.9)	46.9 (15.3)	45.8 (14.9)	47.0 (15.2)
Median (range)	47.5 (10.9-83.7)	45.8 (13.0-80.6)	47.3 (10.9-83.7)	46.4 (13.0-80.6)

The second indication on the label - ≤30 minutes of warm ischemic time – is a critical concept in donors after circulatory death (DCD), and refers to the period during which the organ is deprived of blood supply, typically when life support is removed, while still at body temperature and prior to being placed on the pump. The duration is of paramount importance, as prolonging it causes cellular injury and higher complications such as liver failure or ischemic cholangiopathy. One of the world's top liver transplant surgeons - who is sharply critical of and refuses to use TransMedics - told us that “two out of three DCD's never die in time” – meaning that the warm ischemic period exceeds 30 minutes, a key reason why DCD organs were rarely used prior to TransMedics: “if that period of time exceeds 30 minutes...we just walk away from the donor.” He stated that the organ “deteriorates very quickly, almost exponentially” at 30 minutes...if you try to get an organ out and use it for 50 minutes, it'll never work...it doesn't matter what you put into it.”

Most DCD's have excessive warm ischemic time; organs deteriorate rapidly after 30 minutes and unrecoverable

Q: “As far as somebody like Mark Ghobrial and these heavy TransMedics users, how much off-label use is there? The label says less than 55 years old. How much of the use is more than 55 years old? It says less than 30 minutes of warm ischemic time. By the way, what does that mean—30 minutes of warm ischemic time?”

A: “This is an interesting area because this relates to DCD's, donation after cardiac death. There's a period of time when after the life support is removed and before the heart stops; that period is called “donor warm ischemia time.” From the time of withdrawal from life support to the heart-stopping, it's variable and depends on a number of things. **If that period of time exceeds 30 minutes, generally, we just walk away from the donor, at least for that organ. What happens is two out of three DCD's never die in time. So, that 30 minutes relates to that period of time we call “donor warm ischemia time” in DCD's. What we're seeing is the expansion of the pump's use - certainly, older than 55, for sure, and pushing the 30 minutes to longer, we don't know. I've heard people doing it, but I also don't think it's going to be pushed much more than 35 to 40 minutes because the organ deteriorates very quickly, almost exponentially, once you get past a certain point, and that point being 30 minutes. If you try to get an organ out and use it for 50 minutes, it'll never work. It doesn't matter what you put into it.”** - One of the top liver transplant surgeons in the world, based at a leading Midwest academic center

The blunt statement by perhaps the world's top liver transplant surgeon and KOL - that organs deteriorate “exponentially” after 30 minutes of warm ischemic time and cannot be saved – indicates why the label precludes such livers on the pump. However, he noted that “what we’re seeing is the expansion of the pump’s use...and pushing the 30 minutes to longer...we don’t know...I’ve heard people doing it, but I also don’t think it’s going to be pushed much more than 35 to 40 minutes because the organ deteriorates very rapidly”; “I can’t imagine people pushing it for much more than 30 minutes...they would be remiss...the organs they procure are damaged or don’t work, which would be a terrible stain on their reputation...so I don’t think it happens to any great extent.”

Would be a “terrible stain on their reputation” to push warm ischemic times past 30 minutes

Q: “What percentage of uses do you think is over 55? Because that’s technically off-label.

A: “In all these criteria, age and whatever, I would say at least 25%. It may be even higher than that, but it’s not 50%. Between 25% and 35%.”

Q: “The bulk of that is driven by patients being more than 55 years old or more than 30 minutes of warm ischemic time.

A: “Yes, probably age. Probably not warm ischemia time. **I can’t imagine people pushing it for much more than 30 minutes...They would be remiss - lowering the threshold of quality to the point where the organs that they procure are damaged or don’t work, which would be a terrible stain on their reputation. So, I don’t think it happens to any great extent, pushing the boundaries outside the age.** And the age thing, I think, is it works at 50, it works at 55, it works at 57, it works at 60. But I don’t think they’re going to be going up into the 65, 70-year-olds. **The problem is that the national registry, the UNOS database, doesn’t collect information about the perfusion machine or the perfusion time or some of the parameters that we think are very important, like donor warm ischemia time.**” - One of the top liver transplant surgeons in the world, based at a leading Midwest academic center

The surgeon's speculation – that very few device users would be reckless enough to push the warm ischemic time beyond 30 minutes, and for only another 5-10 minutes at most even if they do so, given rapid organ deterioration – is naïve and merely the tip of the iceberg. The concentrated group of “super users” who drive TransMedics revenue regularly, we were told, push the period far beyond 30 minutes, to 2 hours or more. To his credit, the KOL noted he is in the dark: “the problem is that the national registry, the UNOS database, doesn’t collect information about...the perfusion time or some of the parameters we think are very important, like donor warm ischemia time.” We quote another surgeon below, one of the TransMedics’ highest-volume users, who stated that “more than 50% [of donor livers] are over 30 minutes” and that “we’ve waited two hours...we wait 45 minutes” – a disturbing admission given his explanation for why the label requires <30 minutes: “the rationale is you had a significant increase in the rate of complications...where the liver just doesn’t work.”

Pushing warm ischemic time to 2 hours despite acknowledging it causes injury

Q: “You said extended warm times. That’s that stage in between circulatory death and procuring it? I think there was a 30-minute warm ischemic time. How far are you able to successfully push that?”

A: “From the time of extubation, **we’ve waited two hours**. And from what we call—some people, there are lots of different ideas about when to start warm time, so when is the agonal phase? It used to be we would measure, at least in the United States, we would measure from the time of withdrawal of life support as starting the clock. So, from that point, we’ve gone two hours. And from the time of what we call agonal where we define agonal here is a systolic blood pressure of 80. **We wait 45 minutes.**”

Q: “When it was approved, there was language saying less than 30 minutes of warm ischemic time. What does that mean? What was the rationale for that initially before people had experience with it?”

A: “The rationale is you had a **significant increase in the rate of complications**, what we call ischemic cholangiopathy, and also primary nonfunction, where the **liver just doesn’t work.**”

Q: “What percentage would you say are now more than 30 minutes?”

A: “If you start the clock at the withdrawal of life support, which some people still do, then I would say **more than 50% are over 30 minutes**. If you start the clock at a systolic blood pressure of 80, that were probably over 30 minutes. I would say it is like maybe 20%.”

-Transplant surgeon at a high volume West Coast academic center

Closely related to warm ischemic time is the time the organ is on the pump. Although time on pump is not explicitly indicated on the label, it was a key measure in the pivotal trial and FDA approval. The OCS Liver PROTECT trial doesn't appear to specify the average pump duration, but surgeons indicated it was short – around 6 hours, with some up to 8-10 – given damage the longer a liver is perfused outside a body. Our review of the clinical literature indicates significant organ damage with extended warm perfusion periods >10 hours, such as hepatocellular damage, reperfusion injury, and histological changes. Machine perfusion vendors like TransMedics would jeopardize their existence by studying such damage, but we note emerging studies like one just published in Dec 2024 measuring biomarker concentration in perfusate samples from a pump – showing “significant accumulation of markers” for injury and pathological processes; “a self-perpetuating cycle of inflammation and additional cell damage”; “which may limit the extent of benefit of this technology....”

Dec 2024 study on liver damage from warm perfusion machines

Dynamics of Ischemia/Reperfusion Injury Markers During Normothermic Liver Machine Perfusion

studied. Normothermic machine perfusion (NMP) tries to rebalance cellular metabolism by maintaining the organ at physiological temperature and also supplying oxygen and nutrients. Although these conditions may allow viability assessment before using these organs for liver transplantation (LT), a comprehensive mechanistic assessment of NMP is still lacking. Moreover, this is an essential step toward identifying biomarkers to assess and predict liver function and viability after LT.

“a comprehensive mechanistic assessment of NMP is still lacking”

A ex-employee at TransMedics boasted to us that livers retrieved via their NOP program are often on the pump for 40 hours, with the average being “at least 10-11 hours”; while doctors are “getting their beauty sleep, our clinical specialist is sitting right next to the machine with the organ...so, it’s kind of weird.” He added that few centers use the heart pump and NOP because it’s simply too slow – “they’re used to four hours, and if it’s over eight hours, they start to get a little edgy.” A leading liver transplant KOL indicated users are pumping past 20 hours and that anything over 12-15 hours is “pushing it” and that the pivotal trial times on pump were half that or less. He questioned the transplant outcomes with long pump times – “and that’s a problem...it’s very hard to get the data...because they’re not collecting it.”

Lives on pump up to 40 hours

Q: “So, you’re not getting as much adoption in heart because they want it fast, within a certain amount of time and your value proposition is lower - what is their window that they want the heart within typically?”

A: “Well, they’re used to four hours, and if it’s over eight hours, they start to get a little edgy.”

Q: “What’s your average for liver, from harvesting to transplant?”

A: **“For liver, some of our missions last up to 40 hours - probably like 28 or 30 hours on OCS.** Because the doctors are so comfortable with the machine now, they are not in a rush. While they’re sleeping and getting their beauty sleep, our clinical specialist is sitting right next to the machine with the organ. So, it’s kind of weird. **So, on some of these missions, our clinical specialists are gone up to 40 hours long because there’s no rush for procurement.”**

Q: “What’s the average amount of time between harvesting and implantation with your service now, on average?”

A: **“I would say that it’s at least 10-11 hours for livers now.”** – Ex-employee at Transmedics

Extended times on pump are reckless and were not studied in the trial

“Once you get past about 12 to 15 hours, **you’re pushing it.** Those machine devices are **in the pivotal trials, you look at that, and they were in the 6 to 8 to 10-hour range.** So, if you get past the 50% increase, you’re up to 15 hours; I think you might say that’s **excessive.** And I’ve seen people push it to in the 20-hour range. I don’t know what the outcomes are, so it’ll be up to somebody to do a deep dive analysis about the outcomes. **And that’s a problem; it’s very hard to get the data out of the current system because they’re not collecting it.”** – One of the top liver transplant surgeons in the world, based at a leading Midwest academic center

The third labeled criteria for OCS Liver is ≤15% macrosteatosis, which refers to a low level of fat accumulation, as even mild levels of liver fat ~5% pose risks, while levels >20-30% lead to substantially higher levels of transplant failure. One study found 2-3-fold higher odds of death in patients receiving macrosteatotic livers versus those with non-steatotic grafts. Livers with 30% macrosteatosis are “frequently discarded as they are associated with an increased risk of mortality and graft loss. The clinical literature is unequivocal, per another study: “steatosis in the donor liver has been identified as a predictor of graft failure...so these livers are often considered...marginal....”

Clinical literature on steatotic liver outcomes – Transplantation journal, 2020 and 2023 papers

Minimizing Risks of Liver Transplantation With Steatotic Donor Livers by Preferred Recipient Matching

Donor livers with ≥30% macrosteatosis (steatotic livers) represent a possible expansion to the donor pool, but are frequently discarded as they are associated with an increased risk of mortality and graft loss. We hypothesized that there are certain recipient phenotypes that

Impact of Donor Liver Macrovesicular Steatosis on Deceased Donor Yield and Posttransplant Outcome

Macrovesicular steatosis in the donor liver has been identified as a predictor of graft failure, and so these livers are often considered extended criteria or marginal livers.² Previous studies have demonstrated an association

Predictably, the third area where high-volume TransMedics users blatantly and openly disregard the FDA label for the pump is the off-label misuse of highly fatty/steatotic livers, on top of abuse related to donor age and warm ischemic time – pushing the fat percentage to levels that lead reputable surgeons to reject the organ. An OCS “super user” stated “a lot of centers are using it...the livers are fatty, metabolic syndrome, type 2 diabetes...the livers get replaced with fat...if it stays greater than 20% fat...then the liver doesn’t tolerate the transplant very well.” Shockingly, he then stated he uses livers with ~50% steatosis, and made claims about the pump that are textbook quackery: “you can push it up to about 50%...the problem is that the fat doesn’t transplant well, but if you can re-perfuse it and let the organ calm down on the TransMedics pump...then all that fat washes out over the next couple of weeks.”

High volume user putting livers with ~50% steatosis on pump despite stating >20% is worrisome

A: “The other thing that I know we’ve done, where the livers were fatty.”

Q: “For the steatosis, isn’t it not supposed to be greater than 15%, and you were able to push it beyond that?”

A: “If it’s greater than 20%, we would get worried in the past. You would be hard-pressed to transplant it. So, now I think **you can push it up to about 50%. The problem is that the fat doesn’t transplant well, but if you can re-perfuse it and let the organ calm down on the TransMedics pump...then all that fat washes out over the next couple of weeks.**”

Q: “So, you use it for livers with 50% steatosis?”

A: **“I don’t know if they were exactly 50%, but yes, up to that.”** – Transplant surgeon and surgical director at a high-volume academic center in Utah

Increasingly using for fatty livers in addition to those with diabetes and metabolic syndrome

“The other area I think that we’ll start using it, and **a lot of centers are using it...the livers are fatty. Fatty liver disease, metabolic syndrome, type 2 diabetes** is what’s happening in the American population: obesity. The **livers get replaced with fat, but if it stays greater than 20% fat, then the liver doesn’t tolerate the transplant very well.** But the TransMedics relieves that.” – Transplant surgeon and surgical director at a high-volume academic center in Utah

Every top TransMedics center told us their volumes are driven by marginal organs that most other centers reject – geriatric donors, excessive warm ischemic time, and fatty. Organs are scarce with ~10,000 patients on the national waiting list for a liver. By definition, the surgeons that TransMedics depends on are prolific only because they traffic in rejects – under the pretense that the pump magically restores them, the quackery at the heart of TransMedics business model. A head of a major liver transplant program in Pennsylvania is typical: “we’re using it for organs who pretty much challenged...if you’ve had a lot of people say no...because that donor is not ‘standard’....very old, over 70 or 65...some problems...like a fatty liver.” The surgeon stated that the livers they transplant are “almost always” this marginal to the point you have to be “really, really sick to get a normal liver offer here,” as though needing a transplant is not sick enough – “one was over 70, fatty liver or abnormal lab test, and a lot of people pass...enzymes are a little abnormal...most of them.”

Using livers others won’t touch, “very old” donors or “fatty liver”

- A: **“We’re using it for donors who are pretty much challenged**...if there’s a good donor that comes up, a young, healthy donor who’s brain dead, you’re going to take that donor; you’re pretty much not going to need to use anything TransMedics or anything else. **If you’ve had a lot of people say no to an organ because that donor is not “standard” – they may be very old, over 70 or 65. They may have some problems with their liver like a fatty liver.** That’s called DCD, donation after cardiac death. And those livers get many more complications. You have to have a really high score; **you have to be really, really sick to get a normal liver offer here.”**
- Q: “So, you’re **using the device for less-than-ideal donors.**”
- A: **“Almost always.** Most of them are DCD; some of them are brain dead, and some of them are DBD. Like the last couple that were brain-dead donors, **one was over 70, fatty liver or abnormal lab test, and a lot of people pass.**”
- Q: “Of these livers where you use TransMedics, the ones that are more difficult donors, what’s the typical profile of those donors age-wise roughly? Or is it all ages?”
- A: **“It’s upper 60’s because these days 65 isn’t old.** people are pretty obese; they have a lot of diabetes, so more and more of these donors have a little bit of a fatty liver and their **enzymes are a little abnormal.**”
- Q: “Is it fair to say that most of the livers that you’re putting on the TransMedics device are donors that are over 55?”
- A: **“Most of them.”**– Transplant surgeon and transplant program director at a major academic center in Pennsylvania

The previous surgeon – who just recently transplanted a liver from a donor over 70, after putting it on the pump – is a hair-raising example of the type of off-label misuse that the TransMedics device has normalized. “Super user” surgeons openly admit to using geriatric livers: “up to 65 now is our limit... and who knows, maybe we’ll increase that even further...most people that are using machine perfusion now are using it for older donors.” Another stated 30% of his donors are in the 60’s, and that other TransMedics users are even more aggressive, like Houston Methodist which appears to be in the top two by device volume: “we’re not in the business of transplanting people with 70-year old livers, like other places do...I know Methodist does a lot of older donors.” UCSD stated over half exceed 55, and a senior employee of OrganOx, a normothermic perfusion pump competitor, indicated that “the large majority” of TransMedics donors are over 55: “yes, 100%...the donors are becoming increasingly older.”

Using for 65 year old donors, most device usage is for “older donors”

“Up to 65 now is our limit. And who knows, maybe we’ll increase that even further as we start doing more and more that are in that age. I think **most people that are using machine preservation now are using it for older donors**, especially older DCD donors.” – Transplant surgeon and transplant program director at a leading Northeast academic center

High volume user: 30% of donors in their 60’s; some high volume centers transplanting livers from 70 year olds

A: “There’s a peak probably in the 30s, that’s the MVA group, and then there’s another peak in the 60’s.”

Q: “If you had to ballpark what percentage of the OCS liver donors are over 55, what’s a very rough guess?”

A: **“30%...the older peak, we’re not in the business of transplanting people with 70-year-old livers, like other places do. I know Methodist does a lot of older donors.”** – Transplant hepatologist and director of the liver program at a Midwest academic center

Competitor: Large majority” of donor livers placed on TransMedics are over the age indicated in the label

Q: “What are some other ways where usage is off-label? The label says it’s for a liver from a donor that’s less than 55 years old.”

A: **“Yes., 100%. The donors are becoming increasingly older.** I would not be surprised if a **large majority** of folks are above that age range.” – Senior medical employee at OrganOx

More than half of donors are over 55

“DCDs over the age of 55, I would say, are probably **more than 50%.**” - Transplant surgeon at a high volume West Coast academic center

The bull case for TransMedics is that a normothermic pump purportedly enables the use of DCD organs, which are of dubious quality and therefore unused prior to the pump – “historically, DCD livers were utilized, in a gray area, maybe 8% of the time...so very, very low utilization.” While TransMedics claims that its pump has grown transplant volumes by making DCD organs usable, the claim is false: DCD’s have spiked because OPO’s and transplant centers are under pressure to show volume for compliance reasons, with the prospect of smaller OPO’s and centers being shut down or losing their certification – and they have resorted to using organs they would never touch before, not because a pump magically heals them, but because it gives them a pretense. We quote a high volume user: “that’s where and why we started to use them because there’s a lot of pressure to transplant our patients...our cut-off was like 30 years old...now we’re going up to 60-65...so it’s increased that dramatically.”

TransMedics usage due to pressure on OPO’s to grow using questionable “gray area” DCD organs

“There’s been a mandate that an OPO’s DCD volume should be between 45% and 50% of total organ donors. **Historically, DCD livers were utilized, in a gray area, maybe 8% of the time. So, very, very low utilization. And so, how do you grow?** If the OPO is going to continue to increase DCD but you can’t really grow 80% but **what the TransMedics does it allows you to use those organs; it resuscitates them. That’s where and why we started to use them because there’s a lot of pressure to transplant our patients.**” – Transplant surgeon and surgical director at a high volume academic center in Utah

Using for 60-65 year-old donors when cut-off was 30, or 40 “if pristine”

“The donors are older because **we weren’t accepting DCD livers that were - I think our cut-off was like 30 years old**, maybe we considered 40 if they were pristine and, say, a gunshot wound to the head. Now, **we’re going up to 60-65. So, it’s increased that dramatically.**” – Transplant surgeon and surgical director in Utah

How does one know that the entire premise of TransMedics – using a pump to make historically unusable DCD livers usable and open a new category of transplants – is disingenuous? Because its highest volume users told us so. One surgeon after the other stated that they use the pump only for the highest-risk, marginal organs – DCD, by definition - while bluntly stating it does nothing to suddenly make those organs less marginal - a striking display of contradiction and intellectual dishonesty: “we are only using TransMedics for DCD organs...that are very poor quality and, the thing is, if you put them on a machine, you’re not going to make a poor-quality organ into a good quality organ...if the organ is poor quality...because the donor was obese or was scarred from drinking, you’re not going to reverse those changes...that’s a chronic injury...so it’s not as beneficial in my mind for those organs.”

Putting “very poor quality” organs on the device, while admitting it doesn’t improve them

*“We are only using TransMedics for DCD organs. Donors from cardiac death as opposed to donors from brain death. We’re not really convinced that the donors from brain death, the organs get a lot of benefit from being on a TransMedics device because **often the ones that you’d like to put on are those that are very poor quality and, the thing is, if you put them on a machine, you’re not going to make a poor-quality organ into a good-quality organ.** So, if the organ is poor quality because it’s fatty because the donor was obese or was scarred from drinking, **you’re not going to reverse those changes by putting it on a machine...** with a marginal organ that’s, let’s say, fatty or scarred, that’s a chronic injury over many, many years. You’re not going to reverse that by putting it on a pump. So, it’s not as beneficial in my mind for those organs.” – Transplant surgeon and transplant program director at a leading Northeast academic center*

A top TransMedics user at a high-volume West Coast academic center chuckled and demurred when we asked if the pump has any benefit on marginal organs, and then proceeded to share how they're nonetheless "pushing the limits – using organs that no one else would have otherwise used and had not been used before" – such as geriatric donors, organs left on the device "for more than 24 hours," and "DCD's that have really extended warm times," reckless practices that damage the organ and which reputable centers refuse to engage in. Similar to every high-volume TransMedics user we interviewed, the center's utilization of the device is driven entirely by organs that are rejects: "we've done a lot of livers that other people were not willing to consider, and that's how we've been able to power our program forward."

One of highest volume users: 24 hour pump time; extended warm ischemic time; organs others won't touch

Q: "You said people are still starting to push the limits of the device. I've heard that from some people. What did you mean when you said that?"

A: "That we push the limits? I was referring to that we were the first center to **leave it on the device for more than 24 hours. We've done DCD's that have really extended warm time;** older DCDs has kind have been our go-to now. **We've done a lot of livers that other people were not willing to consider, and that's how we've been able to power our program forward. That's what I mean by pushing the limits—using organs that no one else would have otherwise ever used and had not been used before."** -
Transplant surgeon at a high volume academic center

A former procurement surgeon who worked with TransMedics told us he resigned because he was forced to engage in various types of inappropriate and unsafe conduct. We redact the duration of his tenure, which was significant: “I was actually one of the top three surgeons...I probably did 100 to 150 livers.” He corroborated the vast scale of the off-label and over-utilization scheme, as he personally observed such organs being retrieved and put on the pump, and stated the pump was unnecessary for most of the organs placed on it – “it has become very, very expensive...centers are using them on so many organs that don’t need to go on the pump...I’ve seen very good organs that went on the pump.” He indicated that some centers were happy to jam Medicare for a \$90,000 consumable, even when the device was unnecessary, just to use the NOP organ recovery service – a practice our research is prevalent among smaller centers lacking staff and who we believe to be paying for the device as a kickback in exchange for organs steered their way by the NOP program.

Ex-procurement surgeon alleged device used for “organs that don’t need to go on the pump”

A: “The price keeps getting higher and higher to the point that doesn’t make a lot of sense for a lot of people, especially for a transplant program that has 20 or 30 or 40 livers. It has become very, very expensive. However Medicare is going to pay for the cost, and **I see some transplant centers are using them on many organs that don’t need to go on the pump.**”

Q: “You’re saying there are a lot of livers that go on the pump, the TransMedics pump, that don’t need to go on the pump?”

A: “Yeah, some centers are using them as a recovery service **as long as somebody else is going to pay this chunk of money.**”

Q: “So, they’re using it as a recovery service even though they don’t really need the—”

A: “Yes, I’ve seen organs in young patients - **I’ve seen very good organs that went on the pump.**”

Q: “So, young patient, very good liver, it’s fine for cold storage, then hospitals are using it just because they want to use the NOP?”

A: “Just because they want to use the pump or the service for TransMedics.” – Transplant procurement surgeon previously working with TransMedics

He stated the final straw that led to his resignation was when he was “requested” to place every organ on the pump and to never use ice, this overruling his clinical/professional judgement as the procurement surgeon – when ice is basically free and sufficient for most organs: “I left one day when I had been requested to only work with the TransMedics device and not use anything else...a very big red flag...working only with their pump and preventing surgeons from putting livers on ice...to use it on every organ....no organ on ice...the majority of organs would be fine on ice...the majority of organs in the US don’t need to go on a pump.”

Ex-surgeon says he resigned as forced to over-utilize the pump; device not needed for “majority of the organs in US”

A: **“I left one day when I had been requested to only work with the TransMedics device and not use anything else, so for me, this was a very big red flag...working only with their pump or preventing surgeons from putting livers on ice - to use it on every organ.”**

Q: “You’re saying that they wanted people that would put every organ on a pump. They didn’t want people that would say, ice is okay.”

A: **“Every organ on a pump, no organ on ice.”**

Q: “What percentage of organs would have been fine for ice?”

A: **“The majority of organs would be fine...the majority of the organs in the US don’t need to go on a pump.”**

Q: “You said TransMedics is hiring people that will only use the device and that won’t use ice.”

A: **“Yes, who will not use ice.”** – Transplant procurement surgeon previously working with TransMedics

The surgeon stated that off-label use for organs above the indicated donor age was rampant – “40-50% at least...they will use the device off-label.” He also indicated a fourth type of off-label misuse – “they used it all the time for abnormal anatomy, all the time.” He indicated that a third of livers placed on the pump had abnormal anatomy – with half of donors too old and a third with such anatomy, it appears on-label use of the device is rare. The FDA label contraindicates the device for abnormal anatomy, such as “livers with accessory arterial blood supply requiring back table anastomosis” – which refers to reconstructing vascular structures on a sterile table prior to implantation. The surgeon stated that the OCS Liver device “is approved for standard anatomy” only – when “the liver has abnormal anatomy, like excessive right hepatic artery...this is off-label.”

FDA PMA Approval Order Statement, Summary of Safety and Effectiveness Data – Sept 28, 2021

III. CONTRAINDICATIONS

The OCS Liver should not be used for:

- Livers with moderate or severe traumatic injury
- Livers with active bleeding (e.g., hematomas)
- Split livers
- Livers with accessory arterial blood supply requiring back table anastomosis.

The FDA label contraindicates the device for abnormal anatomy, such as “livers with accessory arterial blood supply requiring back table anastomosis”

“40-50% at least” of livers beyond labeled age; also used off-label for abnormal anatomy; 1/3 of livers abnormal

Q: “Of the livers that you were procuring, how many of those people were older than 55?”

A: **“I would say 40-50% at least...they will use the device off-label.”**

Q: “What are the different ways that they use it off-label?”

A: **“They used it all the time for abnormal anatomy, all the time if it’s an abnormal anatomy. It’s not something rare...let’s say I’m going to get a liver on a device. So, the liver is going to go on the device either on-label or off-label. For example, if I’m going to see this liver and the liver has an abnormal anatomy, like excessive right hepatic artery or excessive left hepatic artery, this is off-label. They will use it off-label, like abnormal arterial supply of the liver or nonstandard arterial supply of the liver. I’m aware that the machine is approved for standard anatomy, but they will use it off-label.”**

Q: “What percentage of the time do you think the use was on-label versus off-label?”

A: “Oh, it was quite a common thing to find abnormal anatomy. It was a common thing...**like a third of the time, it was common.**” - Transplant procurement surgeon previously working with TransMedics

The surgeon indicated that off-label misuse was promoted and directed by TransMedics in various, overt ways – by definition, as 1) their retrieval surgeons are allegedly, as ex-employees told us, directed to place every organ on the pump, 2) only TransMedics device technicians are allowed to operate the device. The surgeon stated that “they will tell you that this is off-label; it’s not something hidden.” We asked who would tell him it’s off-label: “everybody there...they were like, ‘This is off-label’...it’s abnormal anatomy, using it off-label...yeah, yeah, everybody knows...it’s not something that they’re trying to hide...you can’t...it’s documented on the liver anatomy pages where the OPO has to sign...they will use it for back table; they will do that.”

Off-label use is allegedly openly known and promoted at TransMedics

Q: “Did you ever say this is off-label or would you have gotten fired if you said it?”

A: **“They will tell you that this is off-label; it’s not something hidden.”**

Q: “Who will tell you it’s off-label?”

A: **“Everybody there. They were like, ‘This is off-label.’ It’s abnormal anatomy, using it off-label.”**

Q: “Did the TransMedics staff say you were using it off-label or the surgeon?”

A: **“Yeah-yeah, everybody knows.”**

Q: “Wow”

A: “Yeah, everybody knows...it’s **not something that they’re trying to hide. You can’t. It’s documented on the liver anatomy page where the OPO has to sign.**”

Q: “I see. Contraindications: “The OCS liver should not be used for livers with moderate or severe traumatic injury, livers with active bleeding, e.g., hematomas, split livers, donor livers with accessory arterial blood supply requiring back-table anastomosis.” Is that what you’re talking about?”

A: “Yeah, **they will use it for back table; they will do that.**”

Q: “That’s what you mean by the arterial abnormality?”

A: “Yeah, they will use it for abnormal artery.”

Q: “And that’s what you’re saying is about maybe a third of the cases, just very roughly.”

A: “Yeah, and I’m aware also that Duke actually used it for split livers as well. I mean, the liver is going to go on the pump as a whole, and then they will take it out and split it afterwards.”

Q: **“So, there’s a lot of off-label usage of this thing.”**

A: **“Yeah, it’s not uncommon.”** – Transplant procurement surgeon previously working with TransMedics

Part 7. The NOP organ procurement service is “a ticking time bomb” staffed with imported H1B surgeons unlicensed to practice medicine in the US, from high-risk areas like India, Pakistan, and the Middle East; resulting in butchered and lost organs.

An ex-TransMedics executive expressed shock at the incompetence of its retrieval surgeons, sharing a recent anecdote of a heart that was rejected after it was severed without enough aorta for transplant. The executive stated “it’s really hard to get a procurement surgeon,” so they’ve hired surgeons “that nobody wants to hire...these are people who need jobs”; “they weren’t good enough to stay employed at a transplant hospital” and do NOP recoveries as a last resort – “it’s an order of magnitude of pain and more to work for TransMedics.”

Hiring surgeons no else will hire; recently didn’t remove a heart properly and was rejected

“When I was at another scientific meeting recently, that was what a lot of people were talking about. **It’s really hard to get a procurement surgeon.** One surgeon that I was talking with said they actually got an organ with not enough aorta to transplant into the patient. So, there **wasn’t enough aorta for the poor patient to get the heart, but they rejected the organ.** I’ve heard repeatedly that the **people that they’ve hired to do the procurements are the surgeons that nobody wants to hire to be a transplant surgeon.** These are not the crème de la crème that would be working at Mayo Clinic. These are people who need jobs.” – Former executive in a key reimbursement role

TransMedics surgeons not skilled enough to stay employed at a transplant hospital

A: “They **weren’t good enough to stay employed at a transplant hospital.** That’s pretty much the long and short of it. If they were, they would be, I mean, it’s an **order of magnitude of pain and more to work for TransMedics** - if you could be a transplant surgeon at a transplant center, you would because then you are making millions as opposed to hundreds.”

Q: “What would one do to get bounced out of one of those jobs?”

A: “Probably you’re not a very good surgeon - a couple of deaths. Lots of different things could happen. **I met two of their transplant surgeons where I heard repeatedly about their lack of skill from others** but that they were very nice people. It didn’t seem like it was a personality thing; it was much more of a they couldn’t cut it.” – Former executive in a key reimbursement role

The executive provided names of particular NOP surgeons known for repeatedly making critical mistakes, suggesting they end up at TransMedics after running from something – “you’re not a very good surgeon...a couple of deaths...two of their transplant surgeons...I heard repeatedly about their lack of skill from others.” The executive provided the name of the surgeon who allegedly severed the aorta improperly: “I heard that from a reputable source...she got an unusable heart...it didn’t have enough aorta to sew into the recipient...nobody in their right mind would hire her” – “I could tell you seven surgeons who should be excluded from surgery for life because they’re very, very brutal...they have terrible outcome...they kill more patients than they save...and they still go on to jump around the country at various hospital...it happens a lot.”

Surgeons “should be excluded from surgery for life”; “they kill more patients than they save”

“I heard that one from a very reputable source, that she got an unusable heart because it didn’t have enough aorta to sew into the recipient. That’s not good. **I could tell you seven surgeons who would be, should be excluded from surgery for life because they’re very, very brutal - I mean, they have terrible outcomes. They kill more patients than they save, and they still go on to jump around the country at various hospitals. It happens a lot.** She was one of the procurement team surgeons, and I heard specifically about her from this friend who was telling me about the aorta, and he’s like, and they hired your favorite person – and I couldn’t believe it. **Nobody in their right mind would hire her.** There are deaths that you expect, and there are deaths that just seem to not be what you would expect.” -Former executive in a key reimbursement role

The ex-executive's surgeon grim assessment is shared by its highest-volume users – making their usage even more troubling, consistent with a perverse incentive whereby organs are steered their way as kickbacks. A surgeon and “super user” stated that 10% of organs retrieved experience significant mistakes or are unusable because of surgeon error - “they’ve brought on more surgeons that are little bit more questionable skillset...not always the best skilled people...organs have been lost due to inadequate skill of the surgeon...things that are significant.” Organs must be retrieved rapidly as they deteriorate after 30 minutes of warm ischemic time after circulatory death – but TransMedics surgeons are incompetent to the point it “takes four hours from the time of death”; “the most common thing...is it’s taken too long to get the liver from the donor to the pump...ischemic time in between has caused damage....”

Hiring inferior surgeons who “inadequate skills” lead to loss of organs; 10% of organs lost due to mistakes

“As they’ve expanded, they’ve brought on more surgeons that are a little bit more questionable skillset...they’re building out their program with not always the best-skilled people...mistakes get made when things take too long. I’m sure organs - I know that organs have been lost due to inadequate skills of the surgeon...things that are significant and not just transplant surgeon nit-pickiness, I would say maybe 10% of cases have something significant that should have been done that wasn’t done, or an organ that could have been used probably wasn’t used because the surgeon wasn’t skilled enough - 10% maybe.” - Transplant surgeon at a high-volume academic center

Taking four hours to retrieve a liver; not capable of retrieving livers properly

Q: “Ten percent is pretty significant.”

A: “It is, yeah. The most common thing that’s happened is it’s taken too long to get the liver from the donor to on pump, and I think that ischemic time in between has caused enough damage that the liver, even though it got put on the device, was not good. If it takes four hours from the time of death and the liver gets on the pump, you lost a lot of the benefits of the pump.”

Q: “Is that because of the surgeon’s skill, or just because they couldn’t get their crew out there on time?”

A: “No-no, surgeon skill. Surgeon skill. Yeah, the donor dies, and they just couldn’t get them or weren’t capable of getting the liver out, getting back tabled, and getting it on the pump on time. Sometimes, there might be a vascular reconstruction...perhaps an artery wasn’t handled well, or something wasn’t tied properly, and then you get some bleeding after the implantation and things like that that are of consequence...” Transplant surgeon at a high-volume academic center

A transplant administrator at Massachusetts General corroborated widespread, catastrophic surgical errors, providing recent examples, such as one where a TransMedics surgeon allegedly sawed another center's liver in half while retrieving a heart: "maybe their team is...putting the heart on the cannula...but then they damage the lungs for another center...that happens more often than you think...this just happened – I won't name the center but someone from there reached out to me and said, 'Has this ever happened to you guys?'...they literally took the sternal saw and sawed straight through the liver for another center's recipient...yeah, I saw the picture...like, I could do it better than this."

Allegedly sawing livers in two and damaging other organs like lungs; device failures

Q: "When you say never events shouldn't have occurred. You said adverse events, so you don't lose the organ. What are you referring to? Device failures or something else?"

A: "Something else and **device failures**. Device failure-wise, it just goes along with what I had mentioned before, how any time that there's even a **device failure, any time they could potentially even put the organ at risk or the fact that a recipient will be put under anesthesia waiting for the organ and a device malfunction occurs, and you lose the organ?** That's a never event. Or you're going out to—maybe their team is going out, and they're putting the heart on the cannula and bringing it through, but **then they damage the lungs for another center, that happens more often than you think. I've heard of—this just happened—I won't name of the center but someone from there reached out to me and said, "Has this ever happened to you guys?" And they literally took the sternal saw, and sawed straight through the liver for another center's recipient.**"

Q: "They sawed the liver in two?"

A: "**Yeah. I saw the pictures. This is like, I could do it better than this.** They were procuring for the heart. You see what I'm saying? It's not device malfunction, but it's the experience and expertise of the team and the surgeon who's procuring it." – Veteran transplant administrator/executive at Massachusetts General Hospital

The Massachusetts General executive stated that they would never hire a surgeon who ever worked there; and noted that TransMedics quickly hired a surgeon they fired due to “quality issues” - “that’s consistent with what I hear from my colleagues...as we’re all recruiting for more surgeons, any time that some at TransMedics applies, I just automatically say no...I had a surgeon we had to let go because of quality issues. TransMedics hired him right up....I also have concerns related to...what is their training? What is their ongoing continuing medical education?”

Centers automatically reject any surgeons who worked for TransMedics

Q: “Are there other anecdotes that you’ve heard about, like this liver being sawed into two? Such as lack of surgeon skill and losing the organ?”

A: **“That’s consistent with what I hear from my other colleagues. As we’re all recruiting for more surgeons, any time that someone at TransMedics applies, I automatically just say no. I had a surgeon we had to let go because of quality issues. TransMedics hired him right up.** And so, I also have concerns related to the quality of - and not just their hiring practices but **what is their training? What is their ongoing, like continuing medical education?** How do they stay abreast of some of the changes that are occurring down the line around new techniques?” – Veteran transplant administrator/executive at Massachusetts General Hospital

Ex-employees who traveled and worked with TransMedics NOP surgeons in the operating room during organ recoveries provided particularly devastating color about the level of incompetence and mistakes. A former OCS Specialist with extensive prior experience in operating rooms and proper protocols - the device technicians who transfer, manage, and transport organs on pumps - resigned due to a litany of disturbing practices: “The way they did things – all of it definitely was not for me...I wasn’t super comfortable with things, so I thought it was time to go.” The ex-employee stated NOP surgeons were thus viewed with wariness or alarm by hospital staff, and that some were so inexperienced that they had to FaceTime for help: “to send someone brand-stinking new that has a lot of questions and has to FaceTime someone in another state on a reconstruction feel a little not good to me.” When we noted that TransMedics seems to have alienated the field, the ex-technician stated “that is pretty accurate...a lot of people, if they know TransMedics is coming to a surgery, they’re like ‘Oh no’...they’re very disappointed...they don’t want to work with the people.”

Hospital staff dismayed when TransMedics NOP surgeons show up; OCS specialist resigned due to discomfort

Q: “Are you able to speak to why you quit?”

A: “[Chuckles] A couple of things. **The way they did things - all of it definitely was not for me, and I wasn’t super comfortable with things**, so I thought it was time to go.”

Q: “I had a surgeon the other day use the terms “extortion,” “mafia,” and “racket.” It seems like they’ve really pissed off the entire field.”

A: “I think all of that is pretty accurate. **A lot of people, if they know TransMedics is coming to a surgery, they’re like, “Oh no.” They’re very disappointed by it. They don’t want to work with the people.**” - Former TransMedics OCS specialist

Surgeons allegedly incompetent and FaceTiming for instruction on how prep a liver

Q: “You’re saying that transplant centers are paying tens of thousands for the service and they’re misrepresenting it because they’re sending incompetent surgeons?.

A: “Yeah, I feel like if you’re doing this, you want someone who their whole career has been procuring livers or transplanting livers—you know what I mean? **And to send someone brand-stinking new that has a lot of questions and has to FaceTime someone in another state for questions on a reconstruction feels a little not good to me.**” - Former TransMedics OCS specialist

The ex-OCS Specialist stated that TransMedics surgeons were in over their heads and incapable of operating autonomously, requiring extensive handholding; that they were accompanied by staff who “have never been inside an operating room before”; and that it was a frequent source of tension with OPO and hospital staff – “it was a recurring thing the OPO was getting mad about having to help certain TransMedics surgeons”; “with certain docs it’s a habitual thing where they’re like ‘I need someone to help me now,’ and it’s not...for five minutes...it’s for an hour...they don’t have the skills...but I can’t officially say that”; “surgeons are supposed to be able to do things independently and they were not able to....”

TransMedics surgeons habitually need handholding and “don’t have the skills”

A: “It was a recurring thing that the OPO was getting mad about having to help certain TransMedics surgeons like, “Every time we’re here, you’re asking for help,” and TransMedics keeps saying, oh no, they don’t need But it’s a recurring thing that they keep asking for help.”

Q: “What would TransMedics get help on from the OPO? What did the OPO have to keep helping their surgeons with?”

A: “Like an assist in surgery.”

Q: “Because the surgeon just wasn’t competent? It seems like something one person could do.”

A: “Yeah, this is what I’m talking about when I say that their surgeons could benefit from more experience. With certain docs, it’s a habitual thing where they’re like, “I need someone to help me now,” and it’s not for a second. It’s not for five minutes. It’s for an hour....they don’t have the skills. But I can’t officially say that.” - Former TransMedics OCS specialist

NOP surgeons and staff are allegedly incompetent and some had never been inside an OR before

Q: “You mentioned people would realize TransMedics is coming, and they would be disappointed – the receiving transplant center?”

A: “Yeah, a lot of the time - I still know people who are working at other hospitals or work for the OPO or something like that. Other surgeons who work for other hospitals on other teams that have to work alongside the surgeons at TransMedics, they would get a little grumpy knowing that they had to work with TransMedics because they didn’t like how they did certain things. Staff didn’t like the way that TransMedics staff or even the surgeons sometimes would come in and interact with the hospital that they were visiting. So, that was an issue. Because a lot of the time, the people who are coming in with TransMedics staff have never been in an operating room before so they don’t really know how to act. So, that was an issue. And then, there were certain times when the OPO had issues with needing to help certain doctors with things that they didn’t feel like they should be responsible for helping them because they were told by TransMedics that the surgeons are supposed to be able to do things independently and they were not able to do things independently.” - Former TransMedics OCS specialist

The ex-OCS Specialist stated that TransMedics surgeons jeopardize and lose organs due to an alleged inability to excise them within the 30 minute limit for warm ischemic time – the period after circulatory death and before the organ is placed on pump – after which the they exponentially degrade. Surgeons routinely took “a couple of hours” and “some livers were refused on case I was on because they took too long from incision to getting the liver out...they were like, this liver has been out too long, and I don’t want it now...I kept having really weird things happen, like I kept going out with brand-new surgeons and they were taking a really long time, so I was just paranoid about it.”

Surgeons incompetent with long ischemic times; don’t know how to pull out livers; pawning damaged livers to other centers

A: “Their surgeons could benefit from more experience. Surgeons have so much training in residency and fellowship. Their surgeons did not get that experience. They would benefit from it, when you’re talking about ischemia time for an organ and reconstruction.”

Q: “What would happen? They were just too slow and fumbling and not able to take it out within the proper ischemic time.?”

A: “Yes. **Some livers were refused on cases I was on because they took too long from incision to getting the liver out or they took too long between cross-clamp and reconstruction. They spent too long on the reconstruction, so they were like, this liver has been out too long, and I don’t want it now.”**

Q: **“And then they would just pawn it off to another center?”**

A: **“Basically, yeah.”**

Q: “How long would these ischemic times go because they were not properly experienced?”

A: “It depends if there’s reconstruction or not. **Like a couple of hours.”** - Former TransMedics OCS specialist

OCS Specialist didn’t trust TransMedics to tell the truth to recipient centers about ischemic times and organ quality/damage

Q: “When they would then call around and tell people we have an organ, were they honest that it took 45 minutes or two hours or whatever to recover a liver? Or were they just saying we have an organ, and they didn’t want it, and they wouldn’t tell you more?”

A: **“I never really 100% trusted that they were, so I always made sure to get exact times so they could make their own decision based on the numbers.”**

Q: “So, you were worried enough that TransMedics was not telling the transplant center everything they need to know about the organ while pawning it off, so you made sure that you did that.”

A: “I just wanted everyone to have the right information.”

Q: “Was there a time when you decided to overcompensate and do that where they should have said something and didn’t?”

A: **“I kept having really weird things happen, like I kept going out with brand-new surgeons and they were taking a really long time, so I was just paranoid about it.”** - Former TransMedics OCS specialist

In addition to OCS Specialists to who worked with TransMedics NOP surgeons in the operating room during organ recoveries, we spoke with organ procurement technicians at other hospitals who were also frequently in OR's. Their appraisals of TransMedics surgeons were just as scathing as those of ex-employees. We quote a highly experienced retrieval technician, who does runs for a leading academic transplant center and has numerous colleagues inside TransMedics: "I work with TransMedics closely in operating rooms and [I'm] hearing feedback from surgeons at the University of [redacted]"; "it was shocking to find out that surgeons are not licensed...some are problematic because they've been through a disciplinary system...and can't get jobs...often when you hear that you have a career procurement surgeon, you know that there's probably something shady in their past, and that's just the way It is."

Organ procurement technician has extensive interaction with TransMedics teams in operating rooms

"I work with TransMedics closely in operating rooms. The connections I have to TransMedics are just working with them in the operating room, hearing feedback from surgeons at the University of [redacted] and then my connection to [names redacted] and people that work at TransMedics. I'm on the procurement side. I travel with the surgeons to get the organs and then bring them back." – Organ procurement technician at a high-volume Northwest academic transplant center

"Shocking to find out that surgeons are not licensed"; troubling and "shady" backgrounds

"The procurement world is different; it's not regulated like the regular medical community is. I've been doing this for four years and it was shocking to find out that surgeons are not licensed...some are problematic because they've been through a disciplinary system like in this country and can't get jobs elsewhere. Often, when you hear that you have a career procurement surgeon, you know that there's probably something shady in their past, and that's just the way it is." – Organ procurement technician at a high volume Northwest academic transplant center

The organ procurement technician stated that TransMedics surgeons are “just a nightmare to work with and the general consensus across the board – I work with surgeons all over the country....it’s not a sustainable system...a lot of people don’t think they’re going to be around in a few years....the feedback I’ve heard from the University of [redacted]...that they don’t trust when the organs come back that they are being handled by somebody who’s qualified, and they’re hesitant to use the service for that reason”; “I could put you in touch with a lot of people that are just going to say the same thing, ‘I hate TransMedics’...the OPO’s, hospitals...it’s the same thing over and over again, that TransMedics, we don’t see them being here in a year...just the general consensus that oh, we have to deal with them again?”

TransMedics surgeons viewed as a “nightmare to work with”; centers don’t trust organs are handled by qualified surgeons

Q: “It sounds like you would know a lot about the TransMedics NOP team. What are your overall thoughts? You said feedback from surgeons - what did you observe personally? What have you heard from other people at University of [redacted]?”

A: “They’re just a nightmare to work with and the general consensus across the board—I work with surgeons all over the country, not just the [redacted]. It’s not a sustainable system. A lot of people don’t think they’re going to be around in a few years. As far as the feedback I’ve heard from the University of [redacted], the discussions are that they don’t trust when the organs come back that they are being handled by somebody who’s qualified, and they’re hesitant to use their services for that reason.” – Organ procurement technician at a high-volume Northwest academic transplant center

Widely despised in the field; “don’t see them being around in a year”

Q: “Are there any other folks, who have observed some of this behavior on the part of TransMedics, whose name you could provide?”

A: “The person I know would be [redacted]. I could put you in touch with a lot of people that are just going to say the same thing, “I hate TransMedics.”

Q: “What sort of people? At other hospitals?”

A: “The OPOs, hospitals. I had recently had a conversation with the surgeon from our local OPO because I fly with them often for Paragonix. It’s the same thing over and over again, that TransMedics, we don’t see them being here in a year. They’re just so disruptive to the operating room. Just the general consensus that oh, we have to deal with them again?”

Q: “What other hospitals or surgeons are saying the same thing?”

A: “Often we’ll walk in, and there are teams from Texas, and teams from here and there, and they’re like TransMedics - and the gloom and doom of having to deal with them.” – Organ procurement technician at a high-volume Northwest academic transplant center

The organ procurement technician stated TransMedics surgeons are manifestly incompetent, “struggling” in front of other teams in the operating room and that “it’s rare to see one that does it smoothly”: “each surgeon, the less skilled they are, the more you watch them struggle...sometimes the whole room – you’ve got a room of 10 people that are like, ‘What the hell is happening over there?’” The technician stated that the struggles are worse with hearts and lungs, not just livers: “when it’s a heart and lung situation...there are more gray areas as far as what the surgeons are doing....I can tell you the surgeon I worked with that went there...I see him struggle with the TransMedics device...there are other cardiothoracic surgeons with TransMedics that nobody likes to work with...they compromise the safety of the rest of the team...we cringe when we see who walks in the door when it’s TransMedics...I watched him struggle to get these organs onto the pump.”

NOP surgeons are inept and struggling; surgical staff wonder “what the hell is happening”; “rare” to see a smooth retrieval
“So, they’re sitting there on the back table with either warm or cold, going through all of these multiple steps to get them onto the pump or the rig is what they call it. Each surgeon, the less skilled they are, the more you watch them struggle. And the more that time extends where they’re trying to sew little sutures and put little pieces in and do little things. Sometimes, the whole room—you’ve got a room of 10 people that are like, “What the hell is happening over there?” And they just struggle. And it’s rare to see one that does it very smoothly.” – Organ procurement technician at a high-volume Northwest academic transplant center

TransMedics surgeons allegedly “struggle”; “compromise the safety”; “we cringe” when TransMedics surgeons walk in
“When it’s a heart and lung situation - I think there are more gray areas as far as what the surgeons are doing. The cardiothoracic stuff, I can tell you the surgeon I worked with that went there...I see him struggle with the TransMedics devices. I also can tell you that there are other cardiothoracic surgeons with TransMedics that nobody likes to work with. They compromise the safety of the rest of the team by just kind of doing things that would benefit only them or the organ that they’re there to procure. We cringe when we see who walks in the door when it’s TransMedics. I know for a fact that the surgeon that I worked - I watched him struggle to get these organs onto the pump...” – Organ procurement technician at a high volume Northwest academic transplant center

The organ procurement technician confirmed the color that NOP surgeons appear to lack the skills to procure organs within the 20-30 minute limit for warm ischemic time, beyond which organs rapidly deteriorate and become unusable: “they’re still struggling with the heart...20 minutes from cross-clamp...I often don’t even see them finish...and I’m like, ‘Whoa, that’s just crazy...it’s just crazy”; “it extends the warm ischemia time”; “I often leave, and I’m like, ‘There’s no way they’re going to be able to use that organ.’”

NOP surgeon incompetence leads to unusable organs, that TransMedics keeps trying to pawn to one center after

Q: “Have you seen an organ be jeopardized or not be usable because of this incompetence? It sounds like they don’t really care.

They’ll just pawn it off to one of their high-volume TransMedics users where they’re just looking for volume for the transplant center.”

A: “**I often leave, and I’m like, ‘There’s no way they’re going to be able to use that organ’...I feel like they just keep going and going and going, keep trying and trying and trying...it’s going to be in an operating room with a recipient **hours and hours later when the heart function is off.****” – Organ procurement technician at a high-volume Northwest academic transplant center

Can’t get any organ procured in time – whether heart, liver or lungs – “just crazy”

“Typically, the heart is the first thing out the door and it’s on its way. **They’re still struggling with the heart.** The goal with the heart when you’re not doing TransMedics is to have it packaged and out the door within **20 minutes from cross-clamp.** The goal with lungs is to be out door within 50 minutes of cross-clamp. What happens when TransMedics is in the room is we’re out the door packaged with our lungs and gone before they have the heart onto the rig. So, you’re taking a typically 20-minute packaged time, and we’re walking out the door. I often don’t even see them finish because I have to leave before they can get the heart onto the device...and I’m like, **“Whoa, that’s just crazy. The heart’s still in the room and the lungs. The kidneys and the liver are already out of the patient. It’s just crazy.”** – Organ procurement technician at a high-volume Northwest academic transplant center

NOP surgeons are inept and struggling; surgical staff wonder “what the hell is happening”; “rare” to see a smooth retrieval

“Yeah, they’re on the back table for a long time, so when the organ comes out, and there’s no blood flow,, **they bring them onto the back table, and they have to do 35 different things before they can get it on the pump and then start to troubleshoot.** They don’t send them help and typically, a surgeon would have at least one to two other people to accomplish what they’re expected to do just with one. So, there’s nobody else scrubbed in—that just doesn’t happen for those kinds of procedures outside of procurement. So, **it extends the warm ischemia time on the back table for the organ, which is not ideal.**” – Organ procurement technician at a high-volume Northwest academic transplant center

The organ procurement technician alleged conduct by TransMedics surgeons, during heart procurements - arresting the heart prematurely - that jeopardizes the retrieval of every other organ. The practice appears to be driven by surgeon ineptitude as well as a strategic decision, as the TransMedics pump seems to require collecting blood from the donor, which is used to then perfuse the organ – “if you’re trying to use the TransMedics heart device and you’re drawing blood, if you do it too quickly, you will stop the heart and in that moment, every other organ is being compromised”; “they will arrest the heart before the rest of the team is ready...that’s happened multiple times in operating rooms where I was...it’s usually a particular surgeon, in order for them to collect the blood that they put onto their pump....they put the other teams in a tough bind...it’s not ideal for the organs that they’re trying to procure...it consistently pisses off all the team...I don’t know if it’s bad surgical skill...or if it’s strategic.”

Pump allegedly leads NOP surgeons to arrest the heart early to collect blood, jeopardizing other organs

Q: “You said other surgeons don’t like to work with them and they cringe. You said they compromise safety and do things that only benefit them. Can you elaborate on what you mean there?”

A: “They **will arrest the heart before the rest of the team is ready**. That’s happened multiple times in operating rooms where I was. And then, **it’s usually a particular surgeon. In order for them to collect the blood that they put onto their pump, they put the other teams in a tough bind because it’ll arrest early** and it’ll start draining the blood, and then everybody else is kind of sitting around, which is not ideal. And sometimes, that doesn’t always go smoothly. They don’t always get the amount of blood they need, or they need more time to get the blood they need, and it kind of **puts everybody else in a pinch because it’s not ideal for the organs that they’re trying to procure**. It consistently pisses off all the teams. For some reason, there are certain TransMedics surgeons that will do that before everybody is in place and ready to go on a regular basis. I don’t know if it’s **bad surgical skill, which is a possibility. Or if it’s strategic. It just consistently happens.**” – Organ procurement technician at a high volume Northwest academic transplant center

Compromising other organs by arresting heart early

“If you’re trying to use the TransMedics heart device and you’re drawing blood, if you do it too quickly, you will stop the heart, and in that moment, every other organ is being compromised. Nothing is receiving oxygen anymore. I’ve not heard that they do that, but it doesn’t surprise me.” – Executive at a leading West Coast OPO, longtime industry veteran

The CEO of a major OPO called the use of unlicensed “foreign-trained physicians” a “ticking time bomb...they’re cutting corners...wait until somebody finds out.” The executive bemoaned that procurement teams working for reputable transplant centers would “get pissed” if a fellow was sent to a retrieval versus an attending – despite fellows being MD’s with “5 years of surgical residencies and 2 years of fellowship” – and that “now it’s regressed so far” with TransMedics surgeons who are not “licensed to practice in the United States.”

NOP surgeons “couldn’t hack it” elsewhere; “ticking time bomb”

*“What they do is they get these **foreign-trained physicians to do recoveries**. They haven’t gone through the process of being licensed to practice in the United States, and that’s exactly what it is. If you’re a recovery surgeon, you’re either in the twilight of your career and you’re just like the hell with patients, or **you’re somebody who couldn’t hack it**. You can do organ recovery as what’s called an organ recovery specialist. They might have their MD from wherever, but they haven’t gone through the process and have done the fellowship here. It’s a ticking time bomb. **It’s a ticking time bomb. They’re cutting corners. Wait until somebody finds out.**”* – CEO of a major OPO (Organ Procurement Organization), longtime industry executive

“Regressed so far” with quality of TransMedics surgeons vs use of fellows and attending surgeons previously

Q: “This person said somebody cut out a heart a few weeks ago, a TransMedics transplant surgeon, so they didn’t take enough of the aorta to be able to connect the heart, so it was unusable. And that they’re not skilled.”

*A: “No, because they’ve **never been on the other side of it in the United States, like putting it in**. When I was bedside and we flew all over the country and did this, we would fly with the surgeons, but we took an attending physician and a fellow, and I went, and we did the recoveries, and we’d fly back with it. And we **used to get pissed when they would send fellows. You people have done five years of surgical residencies and then two years of fellowship**, and they’re on their third year of fellowship because they want to do bowel. And we’d be like, “No, you’re not an attending.” And **now, it’s regressed so far where it’s like pull so-and-so and train them to do it because they’re just a “skilled” technician and have them go do it.**”* – CEO of a major OPO (Organ Procurement Organization), longtime industry executive

While a handful of transplant centers are aware that TransMedics appears to use surgeons who are unlicensed to practice medicine in the US, and therefore avoid its device – “that’s part of why we don’t use it...I don’t trust their surgeons...you don’t have top-tier surgeons working for TransMedics” – most TransMedics customers appear clueless and told us they just assume they’re licensed. A surgeon who is one of their highest-volume device users stated that “if you’re procuring an organ in an OR in the continental US, there’s no way you’re not licensed to practice medicine and surgery.” Another surgeon, also a “super user,” similarly assumed they meet stringent licensure requirements like residency/fellowship in the US, stating his local OPO wouldn’t use a surgeon who hadn’t: “you have to be credentialed by the hospital as a transplant surgeon...no one’s operating...without an MD, general surgeon residency...and a fellowship.”

Transplant surgeon won’t use NOP – “I don’t trust their surgeons”

Q: “A former employee indicated that the surgeons are not that skilled - they’re importing guys from Pakistan, India, the Caribbean.”

A: “You are correct, which is the piece that I said earlier about the trust issue...**“that’s part of why we don’t use it. It’s not just the cost, but it’s part of why we don’t use it. I don’t trust their surgeons...You don’t have your top-tier surgeons working for TransMedics.”** – Cardiac transplant surgeon at a major Midwest academic center

High volume TransMedics user ignorant and thinks NOP surgeons are licensed to practice surgery

A: “The surgeons that may be for procuring organs - by the way, just to be clear, **if you’re procuring an organ in an OR in the continental US, there’s no way you’re not licensed to practice medicine and surgery.”**

Q: “Don’t you have to be licensed to practice surgery to harvest an organ?”

A: “Yeah, to procure an organ...**you’re supposed to be licensed, yes, I’m pretty sure.”** - Transplant hepatologist and director of the liver program at a Midwest academic center

High volume user unaware that TransMedics surgeons are licensed; thinks requirements will become more stringent

“So, I think what you’re going to see soon is you **have to graduate from a transplant fellowship**; you have to have the certification, and you have to have a certain amount of case numbers. I think **that’s what you’re going to start seeing**....It’s just been discussed, **at our OPO, you have to be credentialed as a transplant surgeon. So, no one’s operating as a transplant surgeon without an MD, general surgeon residency and/or thoracic residency, and a fellowship.**” – Transplant surgeon and surgical director at a high-volume academic center in the western US

When we queried customers about TransMedics alleged use of unlicensed surgeons, particularly high-volume users, they were stunned and/or embarrassed: “Oh really? I didn’t know that.” The OPO CEO appears prescient - “it’s a ticking time bomb...wait til somebody finds out” – judging by the reactions we observed. A transplant physician in a leadership role at Massachusetts General appeared shaken and then admitted that “their surgical procurement is the black box that I’ve never known anything about...here I am thinking a national organ program could be surgeons from accredited programs who are hired guns for the company...but that’s such a simplistic assumption.”

High volume user unaware that TransMedics surgeons are licensed; thinks requirements will become more stringent

Q: “The TransMedics people, a lot of their surgeons are not licensed in the US. They’re importing them from Pakistan, India, and stuff like that. “

A: **“Oh really? I didn’t know that.”** – Transplant surgeon and surgical director at a high-volume academic center in the western US

Transplant centers would be “shocked” to find out TransMedics surgeons are unlicensed and “probably put a stop to it”

Q: **“Would a medical director ever hire a procurement surgeon that didn’t have a California medical license?”**

A: **“No.** We have a new transplant center that might open up in Reno, and the person they hired from the New York team, he’s only credentialed to do kidneys. He’s not credentialed to do abdomen and other organs and **UCSF and Stanford said until he gets credentialed, he’s not coming out on cases.**”

Q: “Yet, those same hospitals are forced to use a TransMedics NOP with a non-credentialed surgeon.”

A: **“If they knew that, they’re either accepting that risk or they’re not aware of it...yeah, they’d be shocked. They would probably put a stop to it. I would think they’d have to** because I don’t know if they absorb the risk of a poor recovery or surgical mistakes. The OPO wouldn’t.” – C-level officer of one of the largest OPO’s (organ procurement organizations) in the country

MGH appears to have no idea NOP surgeons are not licensed

“Their surgical procurement is the black box that I’ve never known anything about. Here I am thinking a national organ program could be surgeons from accredited centers who are hired guns for the company. But that’s such a simplistic assumption.” – Transplant hepatologist Massachusetts General Hospital

A former procurement surgeon who worked with TransMedics told us he resigned because he was forced to engage in inappropriate and unsafe conduct. We redact the duration of his tenure, which was significant: “I was actually one of the top three surgeons...I probably did 100 to 150 livers.” He stated “they hired somebody from England” who “brought quite a large number” of surgeons who “were never trained in the United States...they’re not licensed to practice medicine” - referring to Magdy Attia, “Chief of OCS Liver Transplant program at TransMedics.” He alleged that TransMedics improperly credentialed them as procurement technicians in collusion with the local OPO – New England Donor Services, run by a controversial CEO and the subject of various investigative pieces, and who we heard is close to TransMedics’ CEO: “they managed to give them an authorization to recover organs through their local OPO in Boston...it’s amazing because those surgeons are not licensed by any licensing state or any licensing entity in the US but they still managed to work because they gave them the authorization....”

Ex-procurement surgeon procured 100-150 livers for TransMedics

“I did work with them for a good [redacted timeframe]. I was actually one of the top three surgeons who performed liver procurements for them. I probably did 100 to 150 livers.” – Transplant procurement surgeon previously working with TransMedics

Surgeons not licensed to practice in US, allegedly certified improperly through the local OPO

“They hired somebody from England. He brought quite a large number of surgeons from England to recover...however, they were never trained in the United States...they’re not licensed to practice medicine, but they managed to give them an authorization to recover organs through their local OPO in Boston. They have a very good relationship with them. It’s amazing because those surgeons are not licensed by any licensing state or any licensing entity in the US but they still managed to work because they gave them authorization through the OPO in Boston...” – Transplant procurement surgeon previously working with TransMedics

The ex-TransMedics surgeon alleged that the credentialing/licensing scheme involved the granting of “ACIN numbers” through the OPO, and that he was told so directly by senior staff at TransMedics: “when [redacted] asked, he said all of them have the ACIN through...the New England OPO...how did he do that? I don’t know.” The Association of American Organ Procurement Associations (AOPO) provides ACIN numbers to credential organ retrieval technicians for identification and tracking, per guidelines excerpted below and its ACIN Portal. He stated the conduct was highly unusual and that no other OPO would do this - “I’m not aware of any other people who do that...”

Allegedly obtained ACIN numbers thorough New England OPO; highly unusual

Q: “How were they able to get the ACIN - you said they have some special relationship with the Massachusetts OPO?”

A: **“When [redacted] asked, he said all of them have the ACIN through the OPO, through the New England OPO. How did he do that? I don’t know.”**

Q: “Is that appropriate? Would the other OPO you work with ever do something like that?”

A: **“I don’t think [redacted] himself would ever do that.”**

Q: “How common is that in the OPO space to give an ACIN number to somebody who’s not—?”

A: “I’m not aware of any other people who do that.”

Q: “That sounds like a huge red flag.”

A: **“I’m not aware of anybody else who can do that.”** – Transplant procurement surgeon previously working with TransMedics

AOPO Standards and Interpretative Guidelines, Manual, Jan 2020

CL 6.2	The OPO must have protocols in place for quick verification of the credentialing and/or qualifications and training of recovery personnel prior to any recovery. Documentation of the verification must remain on file and confirm that verification was done before recovery.	Assess the OPO’s methodology for maintaining the credentialing, and/or qualifications and training documentation. The OPO must conduct an annual review of licensure and must use AOPO’s ACIN database or some other documentation method, for surgeons outside the OPO’s service area.
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The ex-TransMedics surgeon laid out the gravity of the deception that would be required to improperly obtain ACIN credentials through an OPO, and pointed to Magdy Attia as an architect and/or participant of the alleged scheme. Attia appears to be a washed-out Egyptian transplant surgeon imported via the UK and appears to run the liver transplant program at TransMedics and oversee its surgeons. His name came up repeatedly during interviews, with multiple surgeons painting him as a henchman-type figure badgering and pressuring them to accept dubious organs they resisted – with one surgeon alleging that he lied to their center and hid that a liver was defective. The ex-TransMedics surgeon stated that Attia was unlicensed to practice medicine in the US, and that an OPO may not grant an ACIN without extensive verification, which TransMedics surgeons would fail: “all of them request a medical license to issue an ACIN for you...the OPO in New England...did not request the license...the OPO’s obligation is to make sure that this person is certified, licensed, and has the minimum training...if they’re not licensed to practice medicine in the US....but no, their surgeons practice like the top person on the liver side at TransMedics” – that is, Attia.

TransMedics allegedly obtained ACIN numbers for their surgeons by circumventing OPO licensing/credentialing procedures

A: “When [redacted – major organ procurement organization] hired me, **they needed my license, my credentialing, the number of cases that I did, all of that. I actually contract with multiple OPO’s and all of them request a medical license to issue an ACIN for you. Apparently, the OPO in New England, they did not request their license**...UNOS has the contract from the government to give authorization for transplant surgeons to recover organs. But surgeons don’t apply directly to UNOS. We apply to the OPO, and **the OPO’s obligation is to make sure that this person is certified, licensed, and has the minimum training** and this and that. To my understanding, **if they’re not licensed to practice medicine in the United States - this is a form of practicing medicine. But no, they practice, like the top person on the liver side at TransMedics.**”

Q: “Who’s that person?”

A: **“Magdy Attia. He used to work in England. I don’t think he’s licensed to practice medicine in the United States.”**

Q: “What is his role at TransMedics?”

A: “He is the chief of the liver program at TransMedics.”– Transplant procurement surgeon previously working with TransMedics

The ex-TransMedics surgeon echoed others who called the use of unlicensed surgeons a “ticking time bomb” – “it’s their biggest risk...having people that don’t have licenses...I still don’t know how they did it.” He stated that TransMedics NOP employs or contracts “10-15 at least or even 20 full-time surgeons” and that “maybe half, maybe a bit less” are licensed – “do their surgeons have a license? No, they don’t.” When we asked how long it takes to get licensed, he detailed the years of study, exams, fellowships and so forth he underwent for licensing, and stated it is unrealistic for TransMedics foreign-trained imports to do the same: “it’s a lot of work...it’s not something definitely you’re going to bring a full-blown surgeon at the age of 50 and ask him to do.”

“Biggest risk” is unlicensed surgeons ; ex-surgeon surprised they pulled it off

“I mean, **their biggest risk**, I think is actually having people that don’t have licenses. **I still don’t know how they did it.**” -Transplant procurement surgeon previously working with TransMedics

Allegedly half of NOP surgeons are unlicensed; takes years to satisfy licensure requirements

Q: “How many surgeons do you think they have now?”

A: **“I’m aware of 10-15 at least or even 20 full-time surgeons.”**

Q: “And how many of those are licensed?”

A: “Maybe half, **maybe a bit less than that**, maybe half [...] **do their surgeons have a license? No, they don’t.**”

Q: “Why can’t they get US-licensed physicians?”

A: “That’s a lot of work. You need to do USMLE exams, which is four big exams. Like I did four exams. And then you need to get at least one year training. Some states will ask for two or three to be licensed in the US. You cannot get licensed without doing step exams which take at least two years. **It’s a lot of work. It’s not something definitely you’re going to bring a full-blown surgeon at the age of 50 and ask him to do.**” – Transplant procurement surgeon previously working with TransMedics

Our checks on TransMedics procurement surgeons confirm the concerns raised by ex-employees, surgeons, OPO executives, and hospital staff – that a reasonable person may view them as generally unemployable by reputable transplant centers; that they are imported from high-risk foreign jurisdictions like India, Pakistan, and the Middle East; appear to be unlicensed to practice medicine in the US; and/or are associated with institutions abroad implicated in organ trafficking scandals. One TransMedics surgeon appears to be associated with one of the most gruesome medical device frauds in history, led by the infamous Paolo Macchiarini, who was charged with manslaughter for patient deaths and sentenced to prison. Another surgeon appears to be currently associated with a suspicious transplant tourism middleman in India. We begin with Magdy Attia, the “Chief of OCS Liver Transplant program at TransMedics” per his LinkedIn bio. An Egyptian import by way of the UK, an ex-TransMedics procurement surgeon alleged that he is unlicensed to practice medicine in the US.

Magdy Attia, shown last month as a speaker at the Egyptian Liver Transplantation Society



Attia's alleged lack of a license to practice medicine in the US is troubling, given that he appears to manage TransMedics procurement surgeons. The ex-TransMedics procurement surgeon pointed to Attia as not only recruiting many of these unlicensed surgeons, but as an architect or a participant of an alleged scheme to improperly obtain ACIN credentials for these surgeons via their local OPO, as detailed earlier: “they hired somebody from England” who “brought quite a large number” of surgeons who “were never trained in the United States...they’re not licensed to practice medicine.” For example, **the “Associate Director of Cardiothoracic Surgical Services @ TransMedics” appears to have attended medical school in Jamaica, and his LinkedIn bio states he is also presently a surgeon at a facility in Trinidad/Tobago called Premier Heart care.**

LinkedIn bio for Stacey Brann, excerpts; Premier Heart Care, based in Trinidad/Tobago - website



Stacey B. · 3rd

Associate Director of Cardiothoracic Surgical Services@ TransMedics

Experience



Associate Director of Cardiothoracic Surgical Services

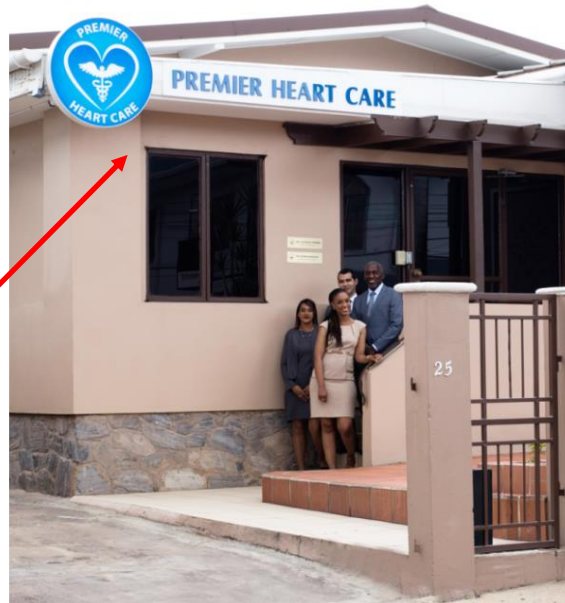
TransMedics, Inc. · Full-time
Nov 2021 - Present · 3 yrs 2 mos
Andover, Massachusetts, United States

♥ Surgery



Consultant Cardiovascular Surgeon

Premier Heart Care Limited
Jan 2021 - Present · 4 yrs
Trinidad and Tobago



Source: Scorpion Capital consultation calls with experts; <https://www.linkedin.com/in/staceyhbrannfrcs/>; <http://www.premierheartcarett.com/our-consultants-dr-stacey-herbert-brann/>; <http://www.premierheartcarett.com/our-practice/>

LinkedIn bios for other TransMedics surgeons lead us to think that they are generally imported from India, Pakistan, Iraq, China, Eastern Europe, and so forth; have short tenures at the company of a year or two; and/or may be near the end of their careers. One even lists Delhi, India as his current location.

LinkedIn bios for TransMedics procurement surgeons - excerpts



Mazin Al Salihi · 3rd
Principal Cardiothoracic Procurement Surgeon in Transmedics Inc



Farhan Zafar · 3rd
Vice President of Cardiothoracic Clinical Development at TransMedics, Inc.



James Zhan · 3rd
MD, PhD.



Anand Jothidasan · 3rd
Principal Cardiothoracic Donor Procurement Surgeon



Anant Dinesh · 3rd
Transplant and Hepatobiliary-Pancreatic Surgeon



Marian Porubsky · 3rd
Procurement Surgeon, TransMedics



Raffaele Girlanda MD · 3rd



Donald Dafoe · 3rd



ANURAG KAW · 3rd
Principal Cardiothoracic Procurement Surgeon, TransMedics



Luzielio Filho · (He,

Source: <https://www.linkedin.com/in/donald-dafoe-932778310/>; <https://www.linkedin.com/in/marian-porubsky-6270011a6/>; <https://www.linkedin.com/in/anand-jothidasan-30656820a/>; <https://www.linkedin.com/in/james-zhan-b2019361/>; <https://www.linkedin.com/in/raffaele-girlanda-md-642ab4282/>; <https://www.linkedin.com/in/farhan-zafar-6014b088/>; <https://www.linkedin.com/in/anant-dinesh-729123181/>; <https://www.linkedin.com/in/anurag-kaw-97882b14/recent-activity/comments/>; <https://www.linkedin.com/in/mazin-al-salihi-2738178b/>

As examples, we detail the backgrounds of three surgeons who appear to work with or at TransMedics, which we suspect are representative, to illustrate the executive's opinion of why they end up at TransMedics. One cardiothoracic surgeon, who lists TransMedics on his LinkedIn bio, was associated with Stanford per the same bio. His tenure appears to have ended abruptly around the time he ended up at TransMedics. Online chatter leads us to believe his affiliation with Stanford may have ended due to inflammatory statements on Twitter/X: "she's a cunt," referring to CNN reporter Christiane Amanpour. We count 10 tweets from his account in 2024 alone where he calls various individuals "cunt." Stanford issued a terse statement on Twitter/X stating he is not affiliated with the institution.

Social media posts on Twitter/X

The image displays a collection of social media posts and a formal statement. At the top left is a tweet from Stanford Medicine (@StanfordMed) stating that Dr. Alan Malki is not affiliated with the institution. To the right and below are several tweets from Alan Malki, M.D. (@_amalki) dated throughout 2024, each containing the word "cunt" in a red box. The tweets include: "Cunt" (10/27/24), "You really are a cunt" (11/12/24), "Cunt" (6/7/24), "You're a cunt" (5/20/24), and "Vote this cunt of a racist out" (3/27/24). A larger tweet from Malki (4/14/24) says "She's a cunt. There it needed to be said" and is also boxed. A reply to @FFT1776 (9/24/24) also contains the word "cunt" in a red box. Another tweet (9/2/24) says "Cunt doesn't do him justice" with "Cunt" boxed.

Stanford Medicine @StanfordMed

We have received inquiries regarding statements made by Dr. Alan Malki on Twitter/X. Dr. Malki is not employed by or affiliated with Stanford Medicine.

Alan Malki, M.D. @_amalki · 10/27/24
Cunt

Alan Malki, M.D. @_amalki · 11/12/24
You really are a cunt

Alan Malki, M.D. @_amalki · 6/7/24
Cunt

Alan Malki, M.D. @_amalki · 5/20/24
You're a cunt x.com/AmericafirstZO...

Alan Malki, M.D. @_amalki · 3/27/24
Vote this cunt of a racist out

Alan Malki, M.D. @_amalki · 4/14/24
She's a cunt. There it needed to be said

Alan Malki, M.D. @_amalki · 9/24/24
Replying to @FFT1776
Cunt

Alan Malki, M.D. @_amalki · 9/2/24
Cunt doesn't do him justice

The surgeon regularly makes incendiary statements on X. We include a small sample below.

LinkedIn bio, and social media profile/posts on Twitter/X



Alan Malki · 3rd
Cardiac Surgeon.



Cardiothoracic Surgeon

TransMedics, Inc. · Part-time

Oct 2023 - Present · 1 yr 3 mos

Los Angeles, California, United States · Hybrid



- Alan Malki, M.D.** @_amalki · 2/5/24
Islam is fast growing by violence and intimidation. A so- called religion that deprives women its most basic rights where rape, honor killings and genital mutilation is a common occurrence that is encouraged. It's a **disease** not a religion.
- Alan Malki, M.D.** @_amalki · 1/8/24
Oh yes. **Islam** is never the problem. They love everybody.
- Alan Malki, M.D.** @_amalki · 3/20/24
No o it started with the birth of **Islam**.
- Alan Malki, M.D.** @_amalki · 2/3/24
I am bamboozled by this statement. All we see from **Islam** is destruction, hatred and misery.
- Alan Malki, M.D.** @_amalki · 11/19/23
We understand **Islam** very very well. That is precisely the problem. We know who you are and we know what you want.
- Alan Malki, M.D.** @_amalki · 1/31/24
Actually it started when **Islam** raped, murdered and pillaged its way from the Arabian peninsula into the fertile crescent
- Alan Malki, M.D.** @_amalki · 8/6/24
Replying to @GadSaad
Yes **Islam** is civilized

However, he is a sideshow relative to the far more suspect backgrounds of other TransMedics surgeons. Saverio LaFrancesca, a “Principal Cardiothoracic Surgeon,” was previously the President and Chief Medical Officer of Biostage (NASDAQ: BSTG), a micro-cap pump-and-dump engaged in quackery and which was involved with the notorious surgeon Paolo Macchiarini. BSTG was previously named Harvard Apparatus Regenerative Technology, and developed a “bioengineered trachea implant” – the initial surgical use of which “was led by Professor Paolo Macchiarini” per the BSTG S-1. A Harvard Apparatus press release noted La Francesca’s promotion to EVP, after which he became President, per his LinkedIn bio.

[Press release announcing Saverio La Francesco promotion and Harvard Apparatus name change to Biostage, 2015 and 2016 respectively](#)

Harvard Apparatus Regenerative Technology Promotes Chief Medical Officer Saverio La Francesca, M.D. to Executive Vice President

-Dr. La Francesca leads scientific and clinical development including HART’s second-generation bioengineered organ implant products-

Harvard Apparatus Regenerative Technology Changes Name to Biostage, Inc. (NASDAQ: BSTG)

Source: Biostage S-1 <https://www.sec.gov/Archives/edgar/data/1563665/000155837022010152/tmb-20220621xs1.htm>;
<https://www.businesswire.com/news/home/20150812005293/en/Harvard-Apparatus-Regenerative-Technology-Promotes-Chief-Medical-Officer-Saverio-La-Francesca-M.D.-to-Executive-Vice-President>; <https://www.prnewswire.com/news-releases/harvard-apparatus-regenerative-technology-changes-name-to-biostage-inc-nasdaq-bstg-300243888.html>

Before we detail Paolo Macchiarini's offenses, we note that La Francesca appears to play a meaningful role at TransMedics, and is listed as an author on its papers. His LinkedIn bio states he is "the company's leading surgeon in California" and leads "approximately 100 surgical interventions each year." The same bio spills extensive amounts of ink on his various roles at Biostage/Harvard Apparatus Regenerative Technology such as CMO, EVP, and then President and CMO.

Saverio La Francesca LinkedIn bio; TransMedics



Saverio La Francesca, MD · 3rd
 Cardiothoracic Surgeon, Innovator, Entrepreneur | Driving the
 Evolution of Organ Transplantation and Medical Logistics with
 Advanced Air Mobility Technologies
 Los Angeles, California, United States · [Contact info](#)

 TransMedics, Inc.

(509)

Performance of the National OCS Program (NOP) in US Heart and Lung Transplants

W. Hassanein,¹ F. Zafar,² A. Hassanein,¹ M. Al Salihi,³ S. Reddy,⁴
 N. Dang,⁵ K. Ihnken,⁶ A. Kaw,⁷ **S. LaFrancesca,⁸** M. Omara,⁹ L. Tsonis,⁷
 Y. Huang,¹⁰ P. Lezberg,² A. Jothidasan,¹¹ L. Damme,² X. Zhou,¹
 F. Gensini,¹² S. Brann,¹³ and T. Khayal.¹ ¹TransMedics, Inc., Andover,



Biostage, Inc.

3 yrs 7 mos

Holliston, Massachusetts, United States

President & Chief Medical Officer

Full-time

Mar 2017 - Oct 2017 · 8 mos

Biostage (NASDAQ: BSTG) develops bioengineered organ implants to treat congenital defects, cancers, and life-threatening conditions of the esophagus, bronchus, and trachea (biostage.com).

- Led all facets of the company, clinical enterprises, and technology advancement across multiple indications in complex regulatory environments.
- Invented Cellframe™ technology and flagship implant product Cellspan™ (bioartificial esophagus) from ideation to world-first status and winning two NI Engineering Impact awards.

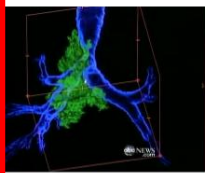
Harvard Apparatus Regenerative Technology, prior to its name change to Biostage, bet the farm on Paolo Macchiarini – despite attempts to quickly distance itself after his conduct came to light. We located various materials on Internet Archive’s Wayback Machine, such as company presentations, that showcase the trachea implants that Macchiarini would utilize. The IR page and presentations appear to have had no shortage of photos of Macchiarini.

Harvard Apparatus Regenerative Technology – Internet Archive Wayback Machine screen captures

The Patient was “condemned to die” – Prof. Macchiarini



The patient and Prof. Macchiarini before the surgery



CT scan used to make the scaffold. The 6cm tumor is in green



The scaffold prior to seeding with bone marrow stem cells



Seeding the cells in the InBreath bioreactor



The seeded graft after 2 days of culture



Mr. Beyene after the surgery

HARVARD
BIOSCIENCE

Tools to Advance Life Science Research and Regenerative Medicine

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16

HARVARD APPARATUS Regenerated Organs for Transplant
REGENERATED TECHNOLOGY

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April 30, 2013 Harvard Apparatus Regenerative Technology's "InBreath" Scaffold and Bioreactor Used in First-of-its-Kind Transplant of a Regenerated Trachea Into Two-Year-Old

June 27, 2012 InBreath Bioreactor used in World's First Successful Laryngotracheal Transplant

Jan 9, 2012 Harvard Bioscience, Inc. Announces Participation in the Regenerative Medicine Insight Track at the Biotech Showcase

July 7, 2011 Harvard Bioscience's Bioreactor Grows a Synthetic Tissue-Engineered Trachea Used in World's First Successful Human Transplantation

Nov 29, 2011 Harvard Bioscience's "InBreath" Bioreactor Used in World's Second Successful Synthetic Trachea Transplant

Nov 29, 2011 Harvard Bioscience's InBreath Bioreactor Highlighted in Article in The Lancet Summarizing the World's First Synthetic Trachea Transplant

The trachea implants that Macchiarini employed turned out to be one of the most gruesome cases of medical device quackery and fraud in history. Seven of his eight patients died. Biostage was sued. Macchiarini's patients who had their tracheas removed died "horrible, painful deaths" – with the one who sued Biostage being left to "cough up pieces of her own flesh." Macchiarini was convicted of various crimes and sentenced to prison, becoming internationally infamous and the subject of a Netflix series as well as an opera.

Netflix series; press coverage on Macchiarini's fall and lawsuits against Biostage



1. Episode 1

Celebrity "supersurgeon" Paolo Macchiarini is a pioneer in regenerative medicine. At the height of his success, he meets journalist Benita Alexander.



2. Episode 2

Paolo's stories grow in grandeur, but Benita doesn't question him. A shocking slip during one of his lectures ignites suspicion among his colleagues.

Holliston's Biostage faces wrongful death suit over organ implant surgery

As Surgeon Paolo Macchiarini Faces Charges, Biostage Pushes Ahead With New Tech



Disgraced surgeon Paolo Macchiarini, whose crimes inspired an opera, headed to prison

Dr Con Man: the rise and fall of a celebrity scientist who fooled almost everyone

When we did a name search for another TransMedics surgeon, his bio came up on the page of a suspect transplant tourism site in India. Shortly afterward, the site was taken off-line and the contact numbers no longer work. However, his profile is still visible on a cached version at Internet Archive's Wayback Machine. The surgeon is an import from India who appears to have arrived in the US a year ago. We note that he previously worked at Apollo Hospitals, which was implicated in organ trafficking schemes in 2016 and 2023. His LinkedIn Bio shows only two years of total surgical experience – in India – before joining TransMedics.

TransMedics surgeon – LinkedIn bio; Apollo Hospitals organ trafficking allegations – press coverage



Yashavanth kumar · 3rd
HPB and Liver Transplant surgeon, Multi organ retrieval surgeon

Experience



Surgeon
TransMedics, Inc. · Full-time
Oct 2023 - Present · 1 yr 3 mos
Andover, Massachusetts, United States · On-site



Medical Consultant
Apollo Hospitals · Full-time
Aug 2021 - Sep 2023 · 2 yrs 2 mos
Mysore, Karnataka, India · On-site

♥ Organ retrieval , Hepatobiliary Surgery and +1 skill



surgeon
safdarjung hospital
Jul 2023

Revealed: Global private hospital group embroiled in 'cash for kidneys' racket

Desperate young people from Myanmar selling their organs to rich patients of India's prestigious **Apollo Hospitals chain**

Surgeon at 'cash for kidneys' hospital linked to new organ trafficking claims

Delhi police arrest doctor over alleged links to a gang enticing patients from Bangladesh into an illegal kidney transplant racket

According to the Times of India and multiple local media outlets, Dr Vijaya Rajakumari, a visiting consultant at the Indraprastha **Apollo Hospital** in southeast Delhi and Yatharth Hospital in Noida, was arrested along with six others accused of running an organ donation network between Bangladesh and India.

Apollo Hospitals seems to be a recidivist when it comes to organ trafficking scandals – which seems par for the course for hospitals in major cities in India. In 2016, five individuals at Apollo were arrested. In the 2023 scandal, The Telegraph implicated Sandeep Guleria, one of India’s most prominent transplant surgeons and a Padma Shri awardee, one the country’s highest honors, indicating the level of rot that pervades India’s transplant field – “Dr. Sandeep Guleria – who trained in the UK...was named by recipients and agents as the surgeon who conducted the transplants.”

Apollo Hospitals - 2016 and 2023 organ trafficking scandals, respectively

Kidney Racket Busted In Delhi's Apollo Hospital, 5 Persons Arrested

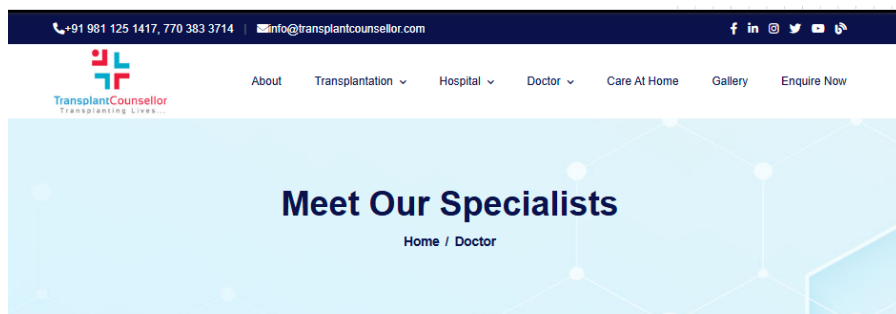


One of India’s leading medics, Dr Sandeep Guleria – who trained in the UK and received one of the nation’s highest honours, the Padma Shri, for his services to medicine – was named by patients and agents as the surgeon who conducted the transplants.



We recognize that Apollo Hospitals is a large Indian chain and make no allegation that this TransMedics surgeon was involved in any scheme. Nonetheless, we are troubled to see his profile on an Indian site called Transplantcounsellor.com, run by a transplant middleman/broker with an aggressive, promotional social media presence – who states in his bio he is focused on transplant tourism. Similar middlemen are at the center of recent trafficking scandals at Apollo and elsewhere. India is a leading destination for foreigners seeking illegal organ transplants, particularly from Persian Gulf states like Dubai.

Transplantcounsellor.com cached page shows TransMedics surgeon bio; broker's Instagram profile

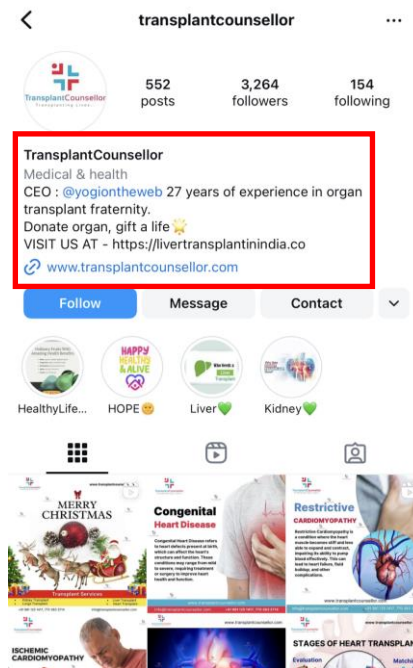


Dr Yashavanth Kumar V
Liver

About Dr Yashavanth Kumar V

Dr Yashavanth Kumar V is a highly skilled and experienced HPB and Liver Transplant Surgeon who has gained national and international experience from institutes of repute in the field of HPB and Liver Transplantation and Liver Surgery. He has been trained at King's College Hospital, London, one of the world's busiest centres for HPB surgery and Liver Transplantation in the world and had the opportunity to work under pioneers in the field of Liver Transplantation at St James University Hospital, Leeds which is among the top three Liver transplant centres in UK. He has worked as a Lead Retrieval Surgeon for National Organ Retrieval Services, UK. Dr. Yashavanth has undergone a Multi-organ retrieval training course (2020) from NORS, UK while participating in several research and publications.

Qualification



The individual behind Transplantcounsellor.com runs various accounts online, such as one for liver transplants, as well on Twitter/X, LinkedIn, and perhaps other platforms. His LinkedIn bio indicates his parent entity is focused on transplant tourism – and mentions living donor transplants, which we understand to be illegal in India unless from a relative. Living donor transplants are, of course, at the center of cash-for-organs schemes. We note also that his LinkedIn bio shows that he worked at Indraprastha Apollo Hospitals New Delhi – one of the Apollo hospital locations implicated in organ trafficking – as well as endorsements from colleagues there.

Transplant Counsellor CEO – LinkedIn bio

About

I am working as a LIVER TRANSPLANT COORDINATOR CUM COUNSELLOR in Hospitals with TRANSPLANT COUNSELLOR Unit of M2Y Medi-Aid Solutionsa Medical Tourism Company. My job is to maintain the database of all the medical facilities and their patients as per hospital policies. Also handling



Founding Director & CEO

Transplant Counsellor (A unit of M2Y Medi Aid Solutions) · Self-employed
Jan 2009 – Present · 16 yrs
New Delhi Area, India · Hybrid

Complete coordination and facilitation to the patients. Bridge between the patients and surgical team. providing the transportation and accomodation for outstation patients. Financial counselling. Pre transplant evaluation for cadaver or living related liver transplantation.



TC_logo_png.png

Skills

Healthcare Management

Endorsed by Dr. Osman El Bakry (MB.Bch, MSc, MD, MBA, PhD) and 4 others who are highly skilled at this

Endorsed by 2 colleagues at Indraprastha Apollo Hospitals New DELHI

Apollo doctor held for Bangladesh-India kidney transplant ‘racket’

Last week, Delhi Police arrested a 50-year-old doctor from Delhi-based Indraprastha Apollo Hospital for her alleged involvement in an organ transplantation racket across Bangladesh and India.

Part 8. The NOP service, we believe, traffics in and pushes unsafe, damaged organs rejected by reputable centers, as an unplaced organ means loss of device and aircraft fees; with TransMedics allegedly concealing adverse organ information.

We believe that a key part of TransMedics modus operand is preventing its procurement surgeons from communicating with recipient surgeons and centers, in order to conceal or overrule their concerns about defective or unusable organs when they arrive at the donor site – any organ not accepted means the loss of \$90,000 in device revenue plus potentially another \$25,000 to \$300,000 in dubious air transport charges. An ex-procurement surgeon who worked with TransMedics – who was “one of the top three surgeons” and “probably did 100 to 150 livers – stated he resigned because TransMedics was “not respecting the professional judgement of the surgeon and pushing for stuff that might not be reasonable or safe from the perspective of the transplant surgeon.” When we asked who was pushing to retrieve and put unsafe organs on the device, he stated it was the headquarters, which we interpret as the executive team given color from other ex-employees about their alleged active participation on internal case chats: “on every case, at least two or three people watching this case from TransMedics...they don’t want you to give professional judgments...they want you to do the job, that’s it.”

Allegedly ignored their procurement surgeons’ judgment that a liver is not usable

A: “The fact that I didn’t want to really work with TransMedics anymore is due to their **not respecting the professional judgment of the surgeon and pushing for stuff that might not be reasonable or safe from the perspective of a transplant surgeon.**”

Q: “Who was pushing in a way it was not safe?”

A: **“On every case, at least two to three people watching this case from TransMedics.”**

Q: “What are they pushing that you think that doesn’t respect the judgment of the transplant surgeon?”

A: **“They don’t want you to give professional judgments. They want you to do the job, that’s it.”** – Transplant procurement surgeon previously working with TransMedics

The surgeon's color was alarming - that TransMedics NOP is allegedly seeking to retrieve and sling any organ onto the device no matter how unsafe to the receiving patient. We asked the surgeon what the company required that made him uncomfortable: "Some organs, I don't think they're safe for transplant. I think they might not work, or they might not work long enough...this is my professional judgment...maybe I've done 800 of these transplants or more than a thousand...I trained for years, so when a liver doesn't work, it's a disaster." When he told TransMedics an organ was unsuitable, they would disingenuously say "the machine will do the trick...the transplant centers...make the judgement" – a disingenuous ploy as "the transplant program who's accepting the liver is not here...if I don't tell them my professional judgment, they don't know....it was simply, just put the liver on the pump."

Pressure to send any and all livers to centers

Q: "What there would make you uncomfortable? What would they ask you to do that a surgeon may not want to do?"

A: "Some organs, **I don't think they're safe for transplant. I think they might not work**, or they might not work long enough, or they're not very good for a transplant, and this is my professional judgment that every transplant surgeon - maybe I've done 800 of these transplants or more than a thousand..., we know for sure and we hear from the transplant centers which organ can work. **I trained for years, so when a liver doesn't work, it's a disaster."**

Q: "Give me some examples or some anecdotes. You would, as a surgeon say, I don't think this liver is going to work. And the three administrators or whatever at TransMedics, what would they say? They would say, shut up and do your job and get the liver?"

A: "They would say that we don't make those decisions. Those decisions are going to be for the transplant program, but the transplant program who's accepting this liver is not here. They are not here. **If I don't tell them my professional judgment, they don't know.** They say the machine will do the trick. It may work on the machine; it may not work on the machine. The transplant centers are the ones who make the judgment. Our job is to provide the procurement service – **it was simply, just put the liver on the pump."** – Transplant procurement surgeon previously working with TransMedics

He described control exercised by TransMedics to shut off the retrieval surgeon's ability to communicate his/her professional judgement about the organs they are sent to retrieve, and stated that all communication was routed through a case coordinator: “my judgement – everything is going through TransMedics; nothing is doing directly most of the time....” If he felt an organ was unsuitable for transplant, he indicated he was not allowed to relay his concerns to the surgeon waiting for it: “no, all the communication goes through the TransMedics people.”

Surgeons allegedly prevented from communicating with receiving centers to relay their concerns

Q: “Would they let you communicate that to the transplant center? Or they wouldn't let you communicate your judgment?”

A: **“Very much the latter. My judgment - everything is going to go through TransMedics; nothing is going directly most of the time.”**

Q: “When you said I don't think it's a good organ and they would say let the transplant center decide. Would they let you tell the transplant center your judgment?”

A: **“No, all the communication goes through the TransMedics people.”** – Transplant procurement surgeon previously working with TransMedics

The surgeon stated he was kept completely in the dark, and that in cases where he concluded the organ was defective or otherwise not appropriate for transplant, that he doubted that TransMedics communicated his professional judgement as the procuring surgeon to the recipient center – “I don’t think so. In our field, if a receiving surgeon hears that the recovery surgeon said I don’t think the organ should go, it’s very unlikely it will go...very unlikely.” He added that he wasn’t even told if the organs he procured were transplanted or not – particularly the ones he thought were unsuitable – and that no information was ever shared about outcomes after transplant, a complete circumvention of a typical organ procurement process: “I don’t even know if that organ ended up transplanted...I don’t even know if it did well afterwards or not...we’re not in communication with transplant centers...they don’t even tell us...we don’t know...we simply don’t know.”

Allegedly keeping their own procurement surgeons in the dark; refuse to even tell them if a liver worked or was even transplanted

Q: “So, there were times when you told the TransMedics person your professional judgment as a surgeon that the organ didn’t make sense, and you have no idea if they communicated that or not.”

A: “**I don’t think so. In our field, if a receiving surgeon hears that the recovery surgeon said I don’t think it he organ should go, it’s very unlikely he will go.** Very unlikely...I don’t even know if that organ ended up transplanted or not. I don’t even know if it did well afterwards or not. We’re not in communication with the transplant centers... **they don’t even tell us if this organ got transplanted or not or if this organ ended up transplanted and is doing well or not. We don’t know. We simply don’t know.**”

Q: “So, TransMedics didn’t share any information with the transplant surgeons about outcomes?”

A: “No.”– Transplant procurement surgeon previously working with TransMedics

Our calls lead us to conclude that it is TransMedics NOP service that controls whether an organ is deemed suitable for recovery and transplant – not the recipient center’s surgeon, who is not in the operating room for retrieval - thereby giving it the ability to withhold adverse information. A senior medical employee at OrganOx, a competitor with a near-identical normothermic pump, stated he has spoken with numerous centers, who indicated that TransMedics makes the call on whether an organ is usable and whether to put in on the pump, as part of the soup-to-nuts NOP service: “TransMedics is taking care of literally everything...actually I’m unaware if the decision is even made by the transplanting surgeon because once TransMedics takes the organ from the donor...they can make the call right then and there...if you have a lousy organ...and they’re already worried...and it doesn’t look good...they should discard it right then, but they don’t...what I have heard is that once the organ goes into the hands of TransMedics...the receiving team has very little input on anything that happens....”

TransMedics allegedly makes and controls the decision on whether an organ is accepted

A: “TransMedics is taking care of literally everything up until that point...**actually, I’m unaware if the decision to discard the organ is even made by the transplanting surgeon because once TransMedics takes the organ from the donor, I believe they can make the call right then and there.** So, if you have a lousy organ coming out and the transaminase are in the thousands or something, and they’re already worried about it and it doesn’t look good, **they should discard it right then, but they don’t** because they’re not footing the bill...what I have heard is that **once the organ goes into the hands of TransMedics, once the donation is made, the receiving team has very little input on anything that happens.”**

Q: “Where have you heard that TransMedics has all of the control over the organ, and they’re the ones to make the decision? I think they’re very hesitant about admitting that.”

A: **“I heard that directly from healthcare workers while I was on sites, from hospitals.”** – Senior medical employee at OrganOx

The potential for abuse is self-evident, given that TransMedics naturally has a perverse incentive to never reject an organ no matter how dubious. The OrganOx employee in a clinical role noted complaints from hospital sites he has visited, with centers dismayed at their lack of control: “...autonomy is huge...I get the sense that’s being stripped away...that entire process is controlled by TransMedics...little autonomy that the surgeon has until the organ actually reaches the recipient...that’s the pattern that we’ve seen.” A transplant cardiologist at Vanderbilt – one of the busiest centers in the world, which has cut off most TransMedics usage – lamented their experience with the NOP service and dependence on surgeons they didn’t trust: “just sending you cell phone videos of...what the heart looks like...it makes it very challenging if there’s some injury...during...dissecting out that heart.”

“Entire process” of organ procurement and decision-making is “controlled by TransMedics”; stripped control from centers
“Autonomy, I think, is big with healthcare providers, especially when it’s something as impactful as an organ transplant; I think autonomy is huge. I get the sense that that’s being stripped away from them in the current distribution model...In our experience and in my experience, we’ve seen that sites prefer to use a device which they have more control over. If a site were to elect to use the OCS system, everything from collecting the organ from the donor all the way up to bringing the OCS to the recipient, that entire process is controlled by TransMedics. So, there’s little autonomy that the surgeon has until the organ actually reaches the recipient. And that’s something that I believe surgeons don’t really like. I think they like to have that level of autonomy and see the organ themselves and have the clinical adjustment there...that’s the pattern that we’ve seen.” – Senior medical employee at OrganOx

Receiving centers receiving haphazard organ info via “cell phone videos”

A: “The other thing that’s really driven us away from this is they were essentially mandating, requiring you to use their surgeons and their procurement teams for these hearts. I trust my surgeons. My surgeons trust our surgeons when they go and visualize something and say something is funny but it’ll be fine, etc. It’s harder to do this when a team that you don’t work with on a regular basis is just sending you cell phone videos of this is what the heart looks like, etc..”

Q: “That’s how it works with their team? They’re sending cell phone videos to the team on the other end to see if they like it?”

A: “It’s like phone calls and cell phone videos and that’s it, yeah. Because none of our team members are there, so that’s the only way if there are any concerns whatsoever. It makes it very challenging if there’s some injury that happened during the donor chest opening and dissecting out that heart.” -Transplant cardiologist at one of the top centers in the US/world

Ex-employees who traveled and worked with TransMedics NOP surgeons in the operating room during organ recoveries provided smoking gun information that leads us to conclude that TransMedics pawns rejected and damaged organs to other centers – such as those that took longer than 30-minutes for their surgeons to retrieve, past the limit for warm ischemia time after which the organ rapidly deteriorates – and then allegedly may withhold or misrepresent the info from/to receiving centers. A former OCS Specialist, who resigned due to a litany of disturbing practices - the device technicians who transfer, manage, and transport organs on pumps – stated organs would experience “a couple of hours” of warm ischemia due to the surgeons’ inability to remove them within the defined period, and replied “basically, yeah” when we asked if they would then just pawn them off to another center.

Surgeons incompetent with long ischemic times; don’t know how to pull out livers; pawning damaged livers to other centers

A: “Their surgeons could benefit from more experience. Surgeons have so much training in residency and fellowship. Their surgeons did not get that experience. They would benefit from it, when you’re talking about ischemia time for an organ and reconstruction.”

Q: “What would happen? They were just too slow and fumbling and not able to take it out within the proper ischemic time.?”

A: “Yes. **Some livers were refused on cases I was on because they took too long from incision to getting the liver out or they took too long between cross-clamp and reconstruction. They spent too long on the reconstruction, so they were like, this liver has been out too long, and I don’t want it now.**”

Q: **“And then they would just pawn it off to another center?”**

A: **“Basically, yeah.”**

Q: “How long would these ischemic times go because they were not properly experienced?”

A: “It depends if there’s reconstruction or not. **Like a couple of hours.**” - Former TransMedics OCS specialist

The ex-OCS Specialist expressed concern as to whether TransMedics lied to receiving centers when organs had experienced long ischemic times – perhaps the single most important criteria in assessing the viability of a DCD organ. We asked if the company was honest in conveying the duration of ischemic time: “I never really 100% trusted they are, so I always made sure to get the exact times so the centers could make their own decision based on the numbers.” We then asked if the ex-technician had to overcompensate and convey the info independently because TransMedics did not: “I just wanted everyone to have the right information”; “I kept having really weird things happen...so I was just paranoid about it.”

OCS Specialist didn't trust TransMedics to tell the truth to recipient centers about ischemic times and organ quality/damage

Q: “When they would then call around and tell people we have an organ, were they honest that it took 45 minutes or two hours or whatever to recover a liver? Or were they just saying we have an organ, and they didn't want it, and they wouldn't tell you more?”

A: “**I never really 100% trusted that they were, so I always made sure to get exact times so the centers could make their own decision based on the numbers.**”

Q: “So, you were worried enough that TransMedics was not telling the transplant center everything they need to know about the organ while pawning it off, so you made sure that you did that.”

A: “I just wanted everyone to have the right information.”

Q: “Was there a time when you decided to overcompensate and do that where they should have said something and didn't?”

A: “**I kept having really weird things happen, like I kept going out with brand-new surgeons and they were taking a really long time, so I was just paranoid about it.**” - Former TransMedics OCS specialist

The ex-OCS Specialist stated that **organs were damaged and often refused by the first center on the match list**, because of long ischemic times due to surgeon incompetence, **or were diseased – and that TransMedics would then start shopping them aggressively to other centers**. The ex-employee expressed **“shock” that anyone took these organs**: “I’ve had a lot of organs that were refused by the initial center...’We actually don’t want it’...that usually boiled down to withdrawal time...or the pathology results”; “when those more conservative centers...refuse it...it’s supposed to go to who’s next on the list...and that’s when those phone calls in the background were happening”; “I’ve seen **organs that I was very surprised that anybody wanted**...I’d look at some of these histories”; “they were drinking a fifth of liquor a day, and you’d be like, **‘Wow, I can’t believe that anybody wants this liver...stuff where you’re just shocked...I was shocked to learn this is what was happening or an overdose was being transplanted...**”

Organs rejected due to long warm ischemic times or pathology but still allegedly steered to aggressive TransMedics users

A: **“I’ve had a lot of organs that were refused by the initial center that called in the request for service. This happened a lot where they were like, “We actually don’t want it.” But that usually boiled down to withdrawal time as a DCD - or the pathology results.**

Most of the time, they were refused because of the withdrawal time - the time limit exceeded the policy for that transplant center.”

Q: *“But you’re saying in those cases, TransMedics would find somebody to take the organ anyway?”*

A: **“So, when those more conservative centers are the ones who initially were supposed to get the organ, and they refuse it, it’s supposed to go to who’s next on the list. And that’s when those phone calls in the background were happening.**

Q: *“Of the aggressive transplant centers, did you ever see them turn down an organ or they would literally take any organ?”*

A: **“I’ve seen organs that I was very surprised that anybody wanted.”**

Q: *“What happened there about the organ? You and the other team could have probably been like, “Wow. What the hell just happened?” What made you surprised that somebody took it?”*

A: *“It was a combination of things. But it usually boiled down to what the actual history was. And so, I was sometimes surprised that an alcoholic liver was going to be transplanted into somebody else...I’d look at some of these histories, and they’d say that they were **drinking a fifth of liquor a day, and you’d be like, “Wow. I can’t believe that anybody wants this liver.” It was like stuff like that where you’re just kind of shocked, or at least I was shocked to learn this is what was happening or an overdose was being transplanted. Those were always surprising to me...**” - Former TransMedics OCS specialist*

The ex-OCS Specialist stated every single organ was ultimately accepted and that “I never had a turn down,” no matter what the pump indicated for lactase, bile, or other biomarkers of viability – unsurprising, as TransMedics allegedly goes down its list of its aggressive customers until it finds one rapacious enough – perhaps Mayo Arizona, Houston Methodist, or Henry Ford – to transplant what more responsible centers won’t touch. The color is consistent with what we heard from surgeons who are high-volume users of the device and NOP service – that they have never seen TransMedics say “thumbs down” on an organ they went to retrieve. The specialist stated “they always wanted people to get people to accept the organ right away, and that is what it boiled down to...get it on the machine, and everything else was questionable...I never had a turn-down...I had thought that maybe people wouldn’t want my liver because it wasn’t putting much bile out, but they were like, no, we’re actually happy with this.”

OCS specialist never saw an organ turned down, even defective ones; surprised the organs were accepted

“They always wanted to get people to accept the organ right away, and that is what it boiled down to. Get it on the machine, and everything else was questionable where they’re like, it kind of looks okay, but I want to see how it does. That happened a lot where they were like, “It looks fine, but I want to see how it performs.” And they could turn it down based on that. So, they wanted to check the lactase and if it was putting bile out. But I never had a turn-down. Everybody ended up being like, okay, yeah, this is good enough, “Okay, we’ll take it.” There were a couple of times where I had thought that maybe they wouldn’t want my liver because it wasn’t putting much bile out, but they were like, no, we’re actually happy with this.” - Former TransMedics OCS specialist

We suspect another reason why organs retrieved by TransMedics seem to find a taker, no matter how diseased the patient or damaged the organ, is that the retrieval surgeon and OCS Specialist monitoring the organ on the pump are allegedly directed by case coordinators and prevented from communicating directly with the receiving center. An ex-procurement surgeon we quoted earlier stated **“all communication goes through the TransMedics people.”** The ex-OCS Specialist stated the same, and noted that these case managers are simply device technicians who were moved to support. As we detail in the next section, the technicians tend to be **inexperienced and incompetent, not to mention unlicensed to practice medicine, and allegedly receive only one week of training before being handed organs to manage:** “I always had to report everything to whoever my clinical lead was that day...everything had to be like, take a picture or video and **send it to whoever was running the case remotely**”; “they did my job...they were me until they got promoted or asked to start doing some support.”

“Clinical leads” in control of procurements are untrained OCS Specialists who got promoted

Q: “Who was monitoring and making adjustments to the organ? Was that you, or was that the surgeon, the recipient surgeon, or did you have full ownership of monitoring the health of the organ?”

A: “It was me, but I **always had to report everything to whoever my clinical lead was that day**. So, it was like I was actually physically adjusting things, but everything had to be like, **take a picture or a video and send it to whoever was running the case remotely.**”

Q: “And who were these clinical leads? These were people based in Boston? Who were these clinical leads, and what do they do?”

A: **“They did my job. They were me until they got promoted or asked to start doing some support.”** - Former TransMedics OCS specialist

In addition to OCS Specialists who worked side-by-side with TransMedics NOP surgeons during organ recoveries, we spoke with organ procurement technicians at other hospitals who were also frequently in OR's – who stated that they witnessed TransMedics aggressively trying to pawn unusable organs. An experienced retrieval technician, who does runs for a leading academic transplant center and has numerous colleagues inside TransMedics, echoed that their surgeons lack the skills to procure organs within the 20-30 minute limit for warm ischemic time, beyond which organs deteriorate – “it's rare to see one that does it smoothly; “they're still struggling with the heart...20 minutes from cross-clamp...and I'm like, ‘Whoa, that's just crazy...it's just crazy’; “it extends the warm ischemia time.” Yet, the technician witnessed persistent efforts to push such suspect organs – “I often leave, and I'm like, ‘There's no way they're going to be able to use that organ....they just keep going and going and going, keep trying and trying and trying...hours and hours later when the heart function is off....”

NOP surgeon incompetence leads to unusable organs, that TransMedics keeps trying to pawn to one center after

Q: “Have you seen an organ be jeopardized or not be usable because of this incompetence? It sounds like they don't really care.

They'll just pawn it off to one of their high-volume TransMedics users where they're just looking for volume for the transplant center.”

A: **“I often leave, and I'm like, ‘There's no way they're going to be able to use that organ’...I feel like they just keep going and going and going, keep trying and trying and trying ...it's going to be in an operating room with a recipient hours and hours later when the heart function is off.”** – Organ procurement technician at a high-volume Northwest academic transplant center

Can't get any organ procured in time – whether heart, liver or lungs – “just crazy”

“Typically, the heart is the first thing out the door and it's on its way. **They're still struggling with the heart.** The goal with the heart when you're not doing TransMedics is to have it packaged and out the door within **20 minutes from cross-clamp** [...] and I'm like, **“Whoa, that's just crazy. The heart's still in the room and the lungs. The kidneys and the liver are already out of the patient. It's just crazy** [...]”

Yeah, they're on the back table for a long time, so when the organ comes out, and there's no blood flow,, **they bring them onto the back table, and they have to do 35 different things before they can get it on the pump and then start to troubleshoot** [...] So, **it extends the warm ischemia time on the back table for the organ, which is not ideal.**” – Organ procurement technician at a high-volume Northwest academic transplant center

Part 9. Organs on TransMedics devices are managed by inexperienced, high-risk technicians called OCS Specialists, who allegedly receive only a week of training prior to engaging in the practice of medicine, putting organs and recipients in jeopardy.

As part of its National OCS Program (NOP), TransMedics pumps are operated by technicians called OCS Specialists who are patently incapable and unqualified to do so – with allegedly such little training, support, or relevant experience that a reasonable person may call it gross negligence and/or malfeasance – and with high turnover as they appear to quit from the mistakes, stress, and chaos from insufficient training. Transplant centers appear to be in the dark and may be alarmed at their liability for allowing them into their OR's, as the technicians appear to clearly exercise medical/clinical judgment and manage, monitor, and medicate organs for 24 to 40+ hours. Previously, TransMedics trained hospitals on the device but now they must use its technicians - as part of the scheme alleged by Congressman Gosar and others - a bait and switch after FDA approval when it “began to change the entirety of its business model” and ceased training centers.

Congressman Paul Gosar letter to TransMedics CEO, Feb 21, 2024

Once FDA approval was achieved, TransMedics began to change the entirety of its business model. Transplant centers that participated in the trial and previously purchased the equipment were then mandated by TransMedics to utilize the OCS no less than three to five times per month, or centers were forced to return the OCS machine without reimbursement of their upfront purchase capital. TransMedics informed these initial centers that if volumes were not maintained, they would not offer more cassettes for purchase. Almost immediately, the cost of the one-time, disposable cassette utilized to encompass the organ during transportation and perfusion increased from the initial \$7,000 to greater than \$60,000 per disposable cassette. Additionally, transplant centers would no longer receive training for their medical teams to utilize the device at their discretion. Instead, TransMedics created its own team of individuals as the sole source for any initiation of the OCS device and labeled this the National OCS Program, or NOP. Transplant centers could no longer purchase the medical device, rather lease the device and request the necessary TransMedics personnel for any OCS heart, liver, or lung organ recovery. Costs for

TransMedics claims that its technicians must operate the device for quality control – a disingenuous ruse, given color from two ex-OCS Specialists, as well as other hospital organ recovery staff interacting with them in operating rooms. We begin with an experienced retrieval technician, who does runs for a leading academic transplant center and works side-by-side with TransMedics OCS Specialists and surgeons inside OR's, and who has colleagues at the company. The color was devastating – and entirely consistent with what former OCS Specialists admitted to us themselves: “...just how unqualified the people are that are running their devices...those should be nurses...they're not even people from the operating room, which puts everybody in a pinch....these people are coming in and they don't know...the other teams...have to help them...they're clearly not prepared...one of my co-workers who is not qualified in any way, shape, or form...she's still working for TransMedics...she's not qualified to be pulling or pushing medications.”

Device technicians are unqualified and “unprepared”; require help; “not qualified to be pulling or pushing medications”

Q: “You had a lot of interesting anecdotes – any others?”

A: “Just **how unqualified the people are that are running their devices**. Those should be nurses running those devices and they're **not always nurses**. A lot of times, they're not even people from the operating room, which puts everybody else in a pinch that is used to working in operating rooms. There's a culture and way that you do things in an operating room, and these people are coming in, and they don't know, and then that takes away from the other teams because they **have to help them, and they have to explain to them that you can't do this, you can't do that**. They're clearly not prepared to come in. One of my co-workers who is not **qualified in any way, shape, or form to work for TransMedics, and she's still working for TransMedics. She's not qualified to be pulling or pushing medications.**” – Organ procurement technician at a high volume Northwest academic transplant center

The procurement technician emphasized the importance of the role as it is essentially the practice of medicine with the organ going into a living recipient, and expressed shock at the level of incompetence: “it just blows my mind that they’re hiring people who are not qualified...I see them struggle...I talk to the people that run their devices...I’m standing next to them when they’re stressing and sweating and freaking out and not knowing what to do and calling support lines and trying to pull things up on iPads...I’ve stepped in several times to help them...when they can’t pull it together”; “they’re scrambling and can’t do it...I’ve never been in the OR where TransMedics was able to run their perfusion of the organ and device...these poor people are just stressed beyond words and have had a week of training”; “they don’t know how to maintain a sterile field...it’s just a complete nightmare every time.”

“Blows my minds” that technicians are so unqualified; struggling, scrambling, and stressed while trying to operate the device
“I understand it’s not a human, but **it’s an organ that’s going to go into a live human. It just blows my mind that they’re hiring people who are not qualified to be doing the stuff that they’re doing. I see them struggle. I talk to the people that run their devices. I’m standing next to them when they’re stressing and sweating and freaking out and not knowing what to do and calling support lines and trying to pull things up on iPads. I’ve stepped in several times to help them do things that I’m qualified to do that they just are completely—when they can’t pull it together.** They often will recruit somebody in the room to run their fluids when they initially cross-clamp and perfuse the organs because they have so much stuff to do, and **they’re scrambling and can’t do it . I’ve never been in the OR where TransMedics was able to run their perfusion of the organ and device at the same time.** They have to recruit somebody every time for a heart. And they’re completely—**they need almost two other people to be able to run that.** I sit in the back with them and talk to **these poor people who are just stressed beyond words and have had a week of training.**” – Organ procurement technician at a high-volume Northwest academic transplant center

“Majority” of technicians have allegedly never been “inside an operating room” – “complete nightmare every time”
“The **majority of the people they’re hiring have never been inside of an operating room.** So, that puts the rest of the teams, when they come in, and they **don’t know how to maintain a sterile field,** they can’t perform all the functions that their team needs to do to keep the whole case rolling and moving, **they recruit us to help them** do certain—like the perfusion part of it so that they can run the pump because they have to collect - **it’s just a complete nightmare every time.**” – Organ procurement technician at a high-volume Northwest academic transplant center

A former OCS Specialist indicated the importance of the role: “putting [the organ] on the machine and taking care of it – transporting and monitoring it,” including reacting to the data the machine spits out, managing fluids/solutions, and so forth – all after “they do a week-long training at the headquarters to get certified, and then you do a couple of cases, and then you’re on your own.” We asked if the minimal training caused worry: “yeah, I’m pretty stressed out with this job, that’s why I don’t work there anymore.” When we asked if technicians were engaging in the practice of medicine after one week: “yeah, it’s terrifying...because nobody ever answered that question of if something happens, then what...I definitely would have liked more training.” We then asked if issues occurred from the lack of training: “Oh yeah, I was terrified...you’re by yourself....it’s not like you can just go ask somebody to help you...then you’re on your phone calling for help.”

OCS Specialists state only one week of training before operating devices and managing organs

A: “And then, once the organ is out, **putting it on the machine and taking care of it - transporting it and monitoring it.**”

Q: “Whatever data the machine puts out, you’re making sure whatever fluids, solutions it needs - that was all part of what you were doing?”

A: “Yes.”

Q: “How long was the training? How long does it take to get trained on this device?”

A: “They do a **week-long training at the headquarters to get certified, and then you do a couple of cases, and then you’re on your own.**” - Former TransMedics OCS specialist

OCS Specialists quit because “stressed out”; “terrifying” situation; feel untrained; on their own and “calling for help”

Q: “So, you’re responsible for monitoring everything about the health of the organ and then the doctor assumes that whatever you’re saying is - it seems like an intimidating situation. Did you ever worry - you mentioned you wished they’d given you more training?”

A: **“Yeah, I’m pretty stressed out with this job; that’s why I don’t work there anymore.”**

Q: “It’s like you’re basically being **asked to engage in the practice of medicine after a week’s worth of training.**”

A: **“Yeah, it’s terrifying. Because nobody ever answered that question of if something happens, then what? And they just kind of were like moving on to the next thing....I definitely would have liked more training.”**

Q: “Were there any issues that happened because they didn’t train you properly?”

A: **“Oh yeah, I was terrified, and you’re by yourself. It’s not like you can just go ask somebody to help you.** You’re the only person that knows how to run that machine. So, then you’re **on your phone calling for help.**” - Former TransMedics OCS specialist

This former OCS Specialist, unlike others, came with extensive experience inside operating rooms, and alleged that TransMedics forced technicians into reckless, unsafe practices that contaminated the sterile field, even after they raised the risks. The technician stated that the consumable is packaged in corrugated cardboard, which TransMedics allegedly would not allow technicians to open prior to an organ being accepted – placing cost of goods above safety: “I asked when I first started...’Oh, this is going to be an issue, bringing this stuff in...and they’re like, ‘It’s totally fine. Just do it.’”; “going into OR’s, there are certain procedures...you can’t bring that past a red line in the OR”; “you cannot do that, and I would never do that...but they wanted you to pass the red line”; “I’ve heard a lot of complaints about their people...touching things they shouldn’t...ruining a sterile field...if you keep upsetting the staff and the hospitals, they’re going to be like...we don’t want these visitors anymore.”

Allegedly forcing device specialist to take non-sterile equipment into OR’s

A: “You would have questions about, like, **“Why are we doing things this way?”** **Going in OR’s, there are certain procedures you have to follow.** You’re not supposed to bring certain stuff in, like hardware, for example. All their boxes are in these corrugated cardboard boxes. That’s an issue literally anywhere you go is that you can’t bring that stuff behind a red line. **And I had asked when I first started. I was like, “Oh, this is going to be an issue, bringing this stuff in?” And they’re like, “It’s totally fine. Just do it.”**

Q: “What was the issue? The machine is not sterile or there’s cardboard--?”

A: “The package in the box that the module insert is in is corrugated cardboard, and you **can’t bring that past a red line at the OR.** You’re supposed to take this out because you can’t bring the box past that line technically. **But they were like, “Oh, it’s totally fine. You can do it.” I would never take it into an operating room because you cannot do that, and I would never do that. But they wanted you to pass the red line.”** - Former TransMedics OCS specialist

Both NOP surgeons and device technicians allegedly “upsetting” hospitals by constantly jeopardizing the sterile field inside OR’s

A: “In an OR, people are high-strung about what you cannot touch. Certain things you do not do. **I’ve heard a lot of complaints about their people coming in and doing that, touching things they shouldn’t, being too close to the sterile field, ruining a sterile field - if you keep upsetting the staff and the hospitals, they’re going to be like, we don’t want these visitors anymore.”**

Q: “And who’s doing this? The surgeon that TransMedics is using for procurement? Somebody who’s tagging along? Who’s doing that?”

A: **“Sometimes it’s the surgeon, sometimes it’s the OCS specialist.”** - Former TransMedics OCS specialist

The OCS Specialist detailed similar disarray upon arriving at recipient centers, left to wander the halls for 24 hours or more with an organ, unable to sleep, with receiving transplant surgeons simply disinterested in checking on the organ until surgery. The specialist only recalled one time when “someone actually met and showed me where to go...it was like, figure it out yourself...I would wander through until I found the OR...a long time, like hours...overnight.” The technician called out practices at UC San Diego in a particular, one of TransMedics highest-volume users and who we believe to be using the device off-label for scheduling and for dubious organs other centers won’t touch, to boost their volumes: “with San Diego...it was like whatever times you showed up...it would be like the next morning...I had to wait for another liver to do...I got there at night and I was waiting until noon or 2pm the next day”; surgeons “would show up right before they would take the patient back...they just don’t put eyes on the organ, and we’re not sending pictures to the receiving surgeon after it’s on the machine.”

OCS specialists “wander” around with no procedure/protocols at receiving centers; surgeons disinterested in monitoring organ

Q: “When you would arrive at a transplant center, would you meet the surgeon? Who would show up? What would the procedure be?”

A: “[Laughs] I would show up and I think it happened once where someone actually met me and showed me where to go and checked me in and everything. But other than that, it was like figure it out for yourself. But I would get dropped off at the recipient hospital, and I would wander through until I found the OR. Sometimes somebody would be kind enough to show me the way, and then I would find the desk. Sometimes, they didn’t want to see it until they wanted me in the OR.”

Q: “How long did they keep you waiting sometimes?”

A: “A long time, like hours. Like overnight...they would want you to wait overnight, and then they would do the first case in the morning, so you would just wait all night long, and then they would do a 6 am or 7 am start. And then the same with San Diego. I think I always waited there. It was always like whatever time you showed up, and then it would be like the next morning. Another time I had to wait for another liver to go, and then I went after it. I got there at night and I was waiting until noon or 2 pm the next day before I went in.

Q: “And nobody’s coming to even look and check out the organ and talk to you. You’re just on your own?”

A: “There were certain places that were very good about coming to look at it. But other docs were less careful about coming and seeing it—they would show up right before they would take the patient back...they just don’t put eyes on it, and we’re not sending pictures to the receiving surgeons after it’s on the machine.” - Former TransMedics OCS specialist

A second OCS Specialist provided similar color, and elaborated on their improper use as surgical assistants despite no such training or credentials, as well as their engaging in the practice of medicine: “a lot of the OCS specialists struggled...a lot of these people were coming in and they’ve never navigated an OR...so much stuff they don’t know about an OR...it is stressful because you only get that one week of training...it kind of blows your mind...you do it once and then you’re kicked out of the nest.” The ex-specialist noted particular stress when roped in by TransMedics retrieval surgeons: “it is stressful because they’re wanting you to call out times...you have a certain amount of time to get the amount of blood out that you need to go on the OCS...they’d be like, ‘can you run the cardioplegia for me?’” – that is, arresting the heart by administering a specialized solution to minimize ischemic damage. The specialist detailed tasks that strike us as the practice of medicine – after a week of training: “all these drugs you have to put on...you have to put on the blood...and sample it.”

OCS specialists struggled; no OR experience; kicked out after one week of training

“A lot of the OCS specialists struggled...a lot of these people were coming in and they’ve never navigated an OR. There’s so much stuff they don’t know about an OR. I think a lot of them struggled with that, so navigating that and then navigating the job, too....It is stressful, because you only get that one week of training, and then you get maybe one case where you’re getting precepted. You get one run out with somebody to do it with a second set of eyes. It kind of blows your mind, They’re like, “Okay, you do it once, and then you’re kicked out of the nest.” – Former OCS Specialist who left recently

OCS specialists allegedly forced to practice medicine; arresting the heart; overwhelmed with medical tasks

“You are by yourself. It’s like an OCS specialist and then a surgeon, and it is stressful because they’re wanting you to call out times, making sure you’re keeping track of time because you have under a certain amount of time to get the amount of blood out that you need to go on the OCS and then you also have to put in the cardioplegia. What I was seeing was if I was there, they’d be like, “Can you run the cardioplegia for me?” The cardioplegia has to go in. You have to get a certain amount of blood out, and then the surgeon is working on getting the heart out, and then while they’re working on getting the heart out, the OCS specialist has to prime the machine, so there are all these drugs you have to put on. You have to put on the blood and get that pumping and you have to sample it to make sure it’s where you want it to be. So, they’re doing all that before the heart comes out.” – Former OCS Specialist who left recently

The second ex-OCS Specialist noted the predictable, rampant turnover and the company's inability to hold on to technicians – with the CEO allegedly lashing out when asked during a town hall: “there is this huge turnover...they go through people in one or two years, sometimes less...they're like, if they leave, they leave...the CEO...he got really kind of nasty...Well, if you don't like it, then leave’...there people quit in a month here in [major city redacted].” The technician stated the company didn't even have a protocol for needlesticks, as an indication of the overall level of recklessness: “every hospital I've ever worked at always has protocols...it's mind blowing...it's a common thing....people aren't reporting that it's happening...they told, you can go to the ER if you want...it's wild.”

Rapid and “huge turnover” in OCS specialists; CEO allegedly told them to quit during town hall question; “nasty”

A: **“There is this huge turnover. They go through people in one or two years, sometimes less** than that. They just keep bringing people in. It costs so much to train people, and then they're like, if they leave, they leave. And even during one of the town halls or some meeting that **the CEO was like, “Well, if you don't—” he got really kind of nasty. He's like, “Well, if you don't like it, then leave,”** when somebody was asking questions about improving things, and that was just their answer to it, which I thought was weird that you're not worried about turnover in a job like this.

Q: “Were there any particular areas where you thought there was huge turnover? Was it people in your role, the OCS specialist?”

A: “Yeah, people in my role specifically were the ones who they can't keep track of. I don't know if you knew, but **three people quit in a month here in [major city redacted] ..** There was someone who had a needlestick, so they were trying to get a hold of HR. This was the only person available, so they finally got a hold of this person who was **totally clueless, which blows my mind.** Every hospital I've ever worked at always has protocols...it's mind-blowing to me. And it's a common thing, and **people aren't reporting it that it's happening** but it's still happening. They told her, you can go to the ER if you want. That was their only response...it's wild.” – Former OCS Specialist who left recently

Part 10. Customer rage: TransMedics business practices have antagonized the small world of transplant, past the point of no return – a short-tempered CEO; allegedly threatening centers who state that organs have been held hostage from ill patients.

TransMedics practices have broadly antagonized the entire transplant field, past the point of no return and consistent with a company in the midst of a customer exodus and death spiral. The level of rage, venom, and expletives – towards its CEO and management team, in particular - is unlike anything we have ever heard. Across dozens of interviews with surgeons, transplant center administrators, and ex-executives/employees, the sentiment was universal. Notably, its highest-volume users exhibited similar animus and signaled their intent to eliminate or sharply reduce use of the TransMedics device as soon as possible. We begin with a prominent KOL and Director of a leading academic transplant center – whose colorful language was representative. The KOL is well-published with a national reputation, participated in TransMedics trials, and knows the CEO well: “their company, from a corporate culture point of view, is dishonest...their claims are exaggerated”; “I don’t like Waleed or a lot of their upper management...he’s doing the fake it til you make it thing...what a fuck-face he is...he’s so disingenuous.”

CEO is “so disingenuous” – disliked by hospitals

“I don’t like Waleed or a lot of their upper management...he’s doing the fake it ‘til you make it thing. He doesn’t acknowledge facts. He doesn’t acknowledge the fact that reimbursement is an issue. He doesn’t acknowledge that there are other companies that are out there with technology. He always badmouths every other company when, in fact, other companies have got some really great ideas to the point, by the way—and this goes to show you what a fuck-face he is...he’s so disingenuous...” – Prominent surgeon, Director of a leading academic transplant center

“Dishonest” and “claims are exaggerated”

“[W]e’ve been involved in some of their trials...their company, from a corporate culture point of view, is dishonest, to be blunt, and I think their claims are exaggerated.” – Prominent surgeon, Director of a leading academic transplant center

The surgeon commented on the alienation and backlash in the transplant community, and noted the CEO's allegedly reactive personality – a recurring theme of interviews which described “screaming” episodes: “the alienation...some of us really hate Waleed because we just think he's dishonest”; “he gets angry when people say, no, we're not getting reimbursed...it pisses him off...they always get angry at the meetings when people get up...they tried to force it down everybody's throat...this is ridiculous that you are forcing us to use your hired surgeon and aviation company...this hard sell, push down your throat approach to the surgical community....there are a lot of people they pissed off.”

“The alienation” - CEO and his team “get angry” when hospitals indicate the product is uneconomic to use

“The alienation...some of us really hate Waleed because we just think he's dishonest with the way he talks...and he gets angry when people say, no, we're not getting reimbursed, and it pisses him off and his people off. They always get angry at the meetings when people get up and say, yeah, but we actually did a financial analysis, and Duke University did a beautiful analysis - they presented it at the meetings where they showed how much of a hit they took with TransMedics liver use at their center. They presented this at one TransMedics user meeting, and then they've also presented at a symposium at the ATC.” – Prominent surgeon, Director of a leading academic transplant center

Alienated the field by forcing NOP use; “ridiculous” for organs nearby

“TransMedics has been charging a lot of money for not only the device but also their NOP program, their aviation company - all the stuff that they've rolled in and they tried to force it down everybody's throat that to use their liver device, you had to do everything. It was a price-fixed dinner; you couldn't do a la carte...the push from all of us in the surgical community saying this is ridiculous that you are forcing us to use your hired surgeon and aviation company when we would prefer to use our own surgeon...if you're going to be getting an organ that's from a DCD donor that's only 150 miles away and you can drive there with your liver team, why would you need to use their aviation company and their surgeon when you could provide those services yourself and improve your contribution margin because these technologies do really whittle away at your bottom line for the hospital.” – Prominent surgeon, Director of a leading academic transplant center

“Push down your throat” – pushing people to use any competitor, like LifePort

“Cleveland Clinic and the guys at the University of Chicago, two centers I think are really good, both are on board with the LifePort study and, in addition, because of the fact that TransMedics acted as this hard sell, push down your throat approach to the surgical community saying this is our way or the highway, there are a lot of people they pissed off. So, I think even if LifePort trials showed the exact same results as TransMedics, I think there's actually be people who would say I'm using LifePort just because it's not TransMedics.” – Prominent surgeon, Director of a leading academic transplant center

We quote two other surgeon KOL's, both prominent and widely respected in the field – one in a leadership role in the transplant program at Vanderbilt, a pioneering center and one of highest-volume in the world; the other is considered the top liver transplant surgeon in the world, at an academic center in the upper Midwest, to whom we spoke after other KOL's suggested him as a key expert. His distaste was palpable: “dealing with Waleed...it's all about the money...I'm just hoping that with these other devices coming into the market....” The Vanderbilt surgeon played a role in TransMedics pivotal PROTECT trial and seethed at the bait-and-switch after FDA approval when the pricing spiked: “I don't like the TransMedics people...to be honest with you, a lot of people are very pissed off...they are acting too greedy and predatory...I really support that letter from the Congressman...a lot of the people in different transplant centers are very pissed off...I can tell you, I hate the CEO.”

People are very pissed off”; bait and switch

A: **“I don't like the Transmedics people that much** - I was a [redacted] when they did the initial trial, the PROTECT trial. And, actually, when they were doing the trial, the way that they were advertising the product is like, well, once we get the trial, once we get approved by the FDA, we are going to sell or we are going to lease the pump and then you only have to buy all the stuff that you have to do for each case. So, that's the way that they were advertising. But once it got approved by the FDA, again, I don't know what happened at the upper level of the administration of the company, but they completely changed.”

Q: “They basically did a bait-and-switch with the investigators.

A: “Yes, they did. That's what they did. And that's the reason that, to be honest with you, **a lot of people are very pissed off with the company...a lot of people are very pissed off because they are acting too greedy and predatory. I really support that letter from that Congressman**. Again, a lot of people in different transplant centers are very pissed off with the company.... **I can tell you, I hate the CEO.**” - Transplant surgeon at Vanderbilt

“All about the money”; eager for any alternative

“Dealing with Waleed on contracts and talking to them also about using their device for other things for research purposes and stuff like that, **it's all about the money...I'm just hoping that with these other devices coming into market**, it's going to drive the cost down, and TransMedics is going to have to come up with some other pricing.”– One of the top liver transplant surgeons in the world, based at a leading Midwest academic center

Former employees provided various anecdotes to illustrate the CEO's questionable reputation and behavior. An ex-executive, with a long tenure and extensive contacts at transplant centers around the country, state that “when I joined, my phone just kept ringing, saying are you crazy? That man is insane.” The executive stated that “he is not well liked” due to his “style with the community of physicians,” mentioning an illustrative cardiovascular surgeon – “he was just like, I will do anything that I can to not use this device.” Other anecdotes included an alleged email that insulted physicians at a key annual conference: “the public shaming was the topic of conversation...it was just the buzz of the meeting...a cloud as was cast...over the enthusiasm of using OCS.”

Allegedly “not well-liked” and viewed as “insane” by the transplant community

“He is not well-liked. His style with the community of physicians. My contacts are mainly with the heart transplant centers, including the administrators all the way up to the C Suite in lots of cases. **When I joined, my phone just kept ringing, saying are you crazy? That man is insane.** The first anecdote came from a cardiovascular surgeon who told me that he was invited to be part of the scientific advisory. He was kicked off and after that, **he was just like, I will do anything that I can to not use this device.”** – Former executive in a key reimbursement role

An alleged incendiary email from CEO – “public shaming” at a key meeting alienated users of the device

“The International Society of Heart-Lung Transplant. It’s the big scientific meeting. So, this was in April of 2022. And he sent it out the day before ISHLT begins, and literally, the world of heart transplant and lung transplant attends this meeting, saying that they had 24 hours to submit their reviews or it was going in without their review and basically giving them a public spanking. The public shaming was the topic of conversation. That’s all anybody could talk about. It was just the buzz of the meeting. A cloud was cast, as you can imagine, over the enthusiasm of using OCS for DCD.” – Former executive in a key reimbursement role

The executive commented on the CEO's allegedly short temper as a source of alienation among transplant surgeons, sharing another anecdote where a surgeon accidentally sat in an assigned seat at Hassanein's table during a conference: "Waleed literally stood up and said, 'Who are you and why do you think you can sit at my table?' It was very upsetting and insulting" – nothing that the surgeon he berated "does more than 25 heart transplants year...he is never, ever going to use an OCS." A former longtime employee in senior medical safety/clinical safety roles stated "Waleed is very short-tempered...he screamed at people a lot and he could be pretty unprofessional...there was a running joke that whenever a customer is asking too many questions...he'd be like, 'Oh my God, such and such idiots.'"

Allegedly berated and insulted KOL who accidentally at his dinner table at a conference

*"The International Society of Heart-Lung Transplant. It's the biannual meeting. There was a table of honor right in the middle, and that was Waleed's table. So, of course, the surgeon from MGH was seated at the head table. My surgeon contact was invited to the dinner. And he and this MGH surgeon, the two of them are talking, walking together to sit for dinner. The other surgeon has no idea that he's got assigned seating; it's not obvious. So, he just sits down. **Waleed literally stood up and said, "Who are you, and why do you think you can sit at my table?" It was very upsetting and insulting,** and you better believe even though he does more than 25 heart transplants a year, he is never, ever going to use an OCS." – Former executive in a key reimbursement role*

CEO is "very short-tempered" and "screamed at people a lot"; lacks credibility in the field

A: **"Waleed is very short-tempered. He screamed at people a lot and he could be pretty unprofessional** when he was getting impatient or the stuff wasn't going his way. There was a running joke that whenever a customer is asking too many questions, the moment they would leave, he'd be like, **"Oh my God, such and such idiots."**

Q: "He was calling customers idiots if they asked too many questions?"

A: "Yes, or challenging questions. He did not like those. Waleed did not have a name in the world of medicine, so he didn't even finish his training and residency. **He does not have academic credibility,** unlike other perfusion companies are founded by well-known names, like celebrity surgeons. He was not very savvy. He definitely **lacked knowledge, and people would question his academic knowledge and he was getting pretty frustrated.**" - Former longtime employee in senior roles including medical safety and clinical affairs

An administrator at a pre-eminent, high-volume center detailed a pattern of “ridiculous” air transport charges with no invoice transparency, which were not part of any contract – involving multiple aircraft sent for a single recovery, allegedly resulting in charges of several hundred thousand dollars. When the center delayed payment, the executive alleged that members of TransMedics management team – COO Tamer Khayal and OCS liver head Magdy Attia – attempted to pressure them for payment by holding a heart hostage from an ICU transplant patient – “I hate flying their team all over the fucking country. I hate it. I hate paying for private jets...I don’t know why we have to pay for all these ridiculous transportation invoices...they just started flying their teams and then sending us these invoices...there’s very little transparency...we just get a piece of paper...it’s just a number...they could have made it up.”

“I hate flying their team all over the fucking country”; sending extortionate invoices with no transparency

“Now, I don’t even bother. I would never even attempt to negotiate anything with them because they won’t negotiate. The transportation is a major problem for us. Part of the reason why our PO took so long to execute is because they just created their own transportation company and started flying - I hate flying their team all over the fucking country. I hate it. I hate paying for private jets to fly their surgeons all over. I don’t know why we have to pay for all these ridiculous transportation invoices. And I hate when, now that they’ve got their own transportation company, they just started flying their teams and then sending us invoices. We can’t just pay a random—we can’t pay for flights from a company that we’re not contracted with. They have all these outstanding flight invoices because we don’t have a contract with...there’s very little transparency. Contracting with them for the transportation felt ridiculous because they couldn’t tell us what the rates were. Basically, we just get a piece of paper, an invoice that says you owe us \$20,000 for this flight going from here to here. It doesn’t say anything about what kind of aircraft it was, how long was it in the air—all this stuff. It’s just a number. They could have just made it up. I find that to be very frustrating.” - Administrator at a pre-eminent academic transplant center

The administrator stated that TransMedics then denied a heart to a status one patient, when payment for suspect invoices was not forthcoming. Given the severity of the allegation, we confirmed the details with a second source, an executive at an OPO who was privy to the situation. The administrator voiced outrage: “I cannot stand working with them...it feels like I’m talking to a used car salesman...we had a recent situation, where we had some A/R...they declined a heart transplant for one of our patients in the ICU...they refused to...get the heart for us...I could not believe it...I had to spend eight hours...trying to beg and plead with them to release the hold because we didn’t want to miss an opportunity to get an organ for one of our patients...I was groveling...this person is in the ICU...they’re going to die...they’re status one...the sickest patient in need of an organ, and TransMedics made that call...it was their COO, Tamer Khayal, who is not a clinician...a used car salesman...[Magdy Attia] was part of the people threatening us.”

COO allegedly held a heart hostage

A: “Me, personally, I cannot stand working with them. It feels like I’m talking to a used car salesman. I don’t talk to their founders as much as I used. We had a recent situation, where we had some A/R...they declined a heart transplant for one of our patients in the ICU because of this credit hold. They refused to go out and get the heart for us. I could not believe it. They eventually agreed that the majority of the invoices were liver, so they released the heart and lung programs from the hold, but they were going to hold the liver program until we paid all of our invoices. I had to spend probably eight hours of my week on the phone with their rep, trying to beg and plead with them to release the hold because we didn’t want to miss an opportunity to get an organ for one of our patients. It was very frustrating. I felt like I was groveling. The whole thing was a terrible experience.”

Q: “So, they put a patient’s life at risk, essentially.”

A: “Yes. These heart patients, this isn’t like a kidney transplant. This person is in the ICU. They’re going to die. They’re status one. They’re getting this organ offer because they’re the sickest patient in need of one, and TransMedics made that call. It was their COO, Tamer Khayal, who is not a clinician, who is the person that I find the most challenging to work with, like a used car salesman.”

Q: “There’s this other Egyptian guy named Magdy Attia who interacts with surgeons, and I’ve heard that surgeons sometimes have said that they don’t want to take an organ, and he’s badgering them to take it, telling them it’s okay. Have you heard that?”

A: “Yeah. He was part of this whole week where I was scrambling - he was part of the people threatening us.” – Transplant administrator

Another high-volume academic center provided similar anecdotes suggestive of badgering and extortion: “typically, this is not how companies behave with somebody who helped them get FDA approval...but that’s Waleed’s method...he and Tamer Khayal, his COO...that’s how they roll...they play hardball...they’ve really alienated a lot of people and a lot of my friends have said, ‘No way, I’m not ever using those guys because of how they do business...they’re using OrganOx and doing just fine.’” He also singled out Magdy Attia, the head of the OCS Liver Program despite being unlicensed to practice medicine in the US, as threatening him over suspect invoices, as well as to accept defective organs: “I get badgered very regularly by their medical director...I get badgered constantly...he just talks in circles....”

CEO/COO “play hardball” and alienated surgeons who now use other perfusion machines

Q: “I’m a little surprised about how aggressive they’re being with you. You’re listed as an author on their paper, and the company’s playing hardball.”

A: “I do agree.”

Q: **“Typically, this is not how companies behave with somebody who helped them get FDA approval - one of the investigators, essentially. It’s very unusual. We’re a very high-volume user but that’s Waleed’s method, he and Tamar Khayal, his COO....that’s how they roll. They play hardball. They’ve really alienated a lot of people and a lot of my friends have said, ‘No way. I’m not ever using those guys because of the way they do business.’ And my friends, of course...they’re using OrganOX, and they’re doing fine, and they’re happy with their volume.”** - Transplant surgeon at a high-volume academic center

Allegedly “badgered very regularly” by Magdy Attia, who leads the TransMedics liver program

“I get badgered very regularly by their medical director - their director of surgeons. My impression is that he’s getting badgered by Waleed and Tamer to get our institution in line because he seems uncomfortable when he’s talking to me. But I get badgered constantly about it. Can you check on the contract? What’s going on? And then, when I call him to say, why are you sending a team from [redacted city] when you have a team in [redacted city]? That makes no sense. And you can see the logic is so flawed; he just talks in circles.” - Transplant surgeon at a high-volume academic center

The surgeon stated that the “hardball” reminded him of a protection racket in a Mafia movie, speaking of TransMedics Chief Commercial Officer Tamer Khayal: “I would call him Waleed’s enforcer...he’s someone you wouldn’t be surprised to find out is knee-capping people...he’s a very intense guy...Tamer’s just going to tell you...’this is how it’s going to be’...and you’re going to like it, and if you cause any trouble, I’m going to keep you in line...those movies where the guys go around and collect the protection money...that’s what he reminds me of – the guy who’s going around to collect the protection money.”

CCO reminds high-volume user of a protection racket

Q: “You said they make it more expensive if you don’t use their air transport?”

A: “Right.”

Q: “I was shocked to hear some examples of their sending two planes – that they don’t really provide any visibility on the invoice. And I heard a lot about bullying by the COO, Tamer Khayal. And you mentioned Magdy Attia. Have you had interactions with Tamer Khayal?”

A: “Yes.”

Q: “And what was the interaction with him like?”

A: “[Chuckles] I would call him Waleed’s enforcer. He’s someone you wouldn’t be surprised to find out is knee-capping people.”

Q: “What is he doing or saying?”

A: “He’s a very intense guy. Tamer’s just going to tell you, ‘Take it or leave it, this is how it’s going to be, you’ll pay our fees.’ He’s not trying to work with you. He’s not trying to schmooze you. He’s trying to just tell you you’re going to get what you get, and you’re going to like it, and if you cause any trouble, I’m going to keep you in line - that kind of thing. You see those movies where the guys go around and collect their protection money. That’s what he reminds me of - the guy who’s going around to collect the protection money.” - Transplant surgeon at a high-volume academic center

The surgeon noted badgering and car salesman-type pressure by Magdy Attia to accept suspect organs that he thought may fail, due to excessive time lingering on the pump or other reasons – as TransMedics would lose the device revenue if the organ is rejected. Attia is an Egyptian import by way of the UK who appears to be unlicensed to practice medicine in the US, yet seems to do exactly that with customers referring to him as the “medical director.” His LinkedIn bio states his official role is “Chief of OCS Liver Transplant program at TransMedics,” in addition to apparently overseeing its procurement surgeons. The surgeon described the squeeze: “they might push you and say, well, I think this liver is fine, you should use it... they are pushy...they’re a pushy company...they will say ‘I think this liver is going to work...I think it’s usable...you should use it...it’s their medical director, Attia...they lose the money...so they try really hard not to lose organs.’”

Pressuring surgeons to use organs even if surgeons reject them

- A: **“If you decide you don’t want to use a liver, like it’s been on the device a while and you’ve got it back at your center, they might push you and say, well, I think the liver is fine, you should use it...they are pushy. They’re a pushy company.”**
- Q: “So, they’re pushing you to use a liver that, in your judgment as a surgeon, your initial reaction is, I don’t want to use it?”
- A: “Yeah, I would say they’re encouraging it for sure. But ultimately, you get to say no but yeah, **they will say, “I think this liver is going to work. I think it’s going to be fine. I think it’s usable. You should use it.”**
- Q: “The surgeons don’t seem like they would be that commercial. Who is pushing it?”
- A: **“It’s the medical director, Attia, who might say this liver is going to be fine. I think you should use it. Of course, the typical qualifying statement, but it’s up to you. It’s up to you.”**
- Q: “And if you don’t use it, is there a cost? Who absorbs the cost if the liver is not used? They went up there, they put it on the device with a disposable. They lose money, or do you still get charged for it?”
- A: “They lose money. You have to pay for the transportation and everything, but **they lose the money. So, they try really hard not to lose organs...**” - Transplant surgeon at a high-volume academic center

The surgeon stated that although he is experienced and confident enough to assess organs, that smaller centers – typically TransMedics highest users – are more vulnerable to the badgering to accept bad organs, and that he receives calls frequently from novice users: “I could definitely see that happening...I get a lot of phone calls and texts from people around the country saying...do you think I should use it...what should I do...I get a lot of those texts and calls...[from] new users and people with less experience using marginal organs that they know we’re using...a good word to describe the company is top-down aggressive...there’s no question....”

Allegedly pressuring smaller centers into using questionable organs

Q: “When they’re badgering people, particularly smaller centers where they have a less confident surgeon or administrator - what percentage of the time do you think they’re badgering people into using an organ they don’t want to use?”

A: “That’s a good question. I have enough experience that I’m confident, but I would imagine somebody who’s a new user, yeah, I can see that happening. I couldn’t give you a percentage, though, but **I could definitely see that happening**. Like I said, **I get a lot of phone calls from people around the country** saying, here’s what’s going on. Do you think I should use it? Or do you think it’s going to be okay, or what should I do? **I get a lot of those texts and calls.**”

Q: “From new users or just people that use the device?”

A: “Yeah. **New users and people with less experience using marginal organs that they know we’re using**. A lot of calls from friends and acquaintances around the country.”

Q: “A former employee that I talked to said that their procuring surgeons would sometimes, in their judgment, say we should not use this liver, and they’re basically told it’s perfectly good; go ahead.”

A: “A good word to describe the company is top-down aggressive. **There’s no question they’re aggressive.**” - Transplant surgeon at a high-volume academic center

Chief of OCS Liver program is allegedly unlicensed to practice medicine but engaging in the practice of medicine

Q: “Is Attia the person who’s calling centers and doctors and saying you have to use this liver even if they don’t want to?”

A: **“He is the person that would call, yes.”**

Q: “And he’s engaging in a clinical conversation?”

A: **“Yes.”**

Q: “I heard he’s not even licensed to practice medicine in the U.S.”

A: “That’s probably true. Yeah, he’s from the UK. **He definitely oversteps at times**. But again, I think that’s par for the course with the company. I think that is **their personality from the top down.**” - Transplant surgeon at a high-volume academic center

Other surgeons provided alarming examples suggestive of coercion and concealment of adverse information about organs - by senior TransMedics management, notably Magdy Attia, the “Medical Director” and Chief of the Liver Program – who, we repeat, appears to be unlicensed to practice medicine in the US. The Chief of transplant at a center in a major metropolitan area – a high-volume TransMedics user, although not for much longer – described “procurement issues” by the retrieval surgeons overseen by Attia: “there has been pressure to use grafts when we don’t like the way they’re performing on the pump, but they lose out if we don’t use the liver...there is pressure to use some of these livers.” The surgeon singled out Attia: “he has to guide transplant surgeons because we’re new at using this platform...what does it mean that the lactate went from .8 to 1? Is that concerning?...we’ll call Magdy and say ‘What do you think about this?’...he’ll say maybe you should watch the liver for another couple of hours....I think it’s going to be ok to use it...that’s his job to be on call at all hours...he’s that point person.”

Pressure to use organs not performing on the pump; Attia appears to be engaging in practice of medicine

A: “There have been some procurement issues. There has been some pressure; **there has been pressure to use grafts when we don’t like the way they’re performing on the pump, but they lose out if we don’t use the liver. We lose out if we don’t use it - so there is a pressure to use some of these livers.**”

Q: “I’ve heard that from other people where there’s a liver and it’s a marginal with an issue and you don’t want to use it. And the transplant surgeon’s getting badgered. One of the surgeons called out Magdy Attia, who I don’t even think is licensed to practice in the US. Walk me through some of that.”

A: “Magdy is the company spokesperson. **He has to guide transplant surgeons because we’re new at using this platform. What does it mean that the lactate went from .8 to 1? Is that concerning? In those situations, we’ll call Magdy and say, “What do you think about this?” And he’ll say maybe you should watch the liver for another couple of hours and then decide or no, we see that, I think it’s going to be okay to use it. That’s his job to be at all hours of the day and night, he’s that point person.**” – Chief of transplant surgery at a major transplant center; high-volume TransMedics user

The surgeon provided a recent example of a necrotic liver where “part of the right lobe of the liver was just completely dead” – which the surgeon then realized, after reaching out to colleagues around the US, is a prevalent issue. The surgeon indicated that the TransMedics retrieval surgeon was aware of the damage but withheld it – although, as indicated previously, TransMedics allegedly prevents them from communicating directly with receiving centers, which leads us to think that he/she told headquarters who then concealed it: “situations where the liver is on pump...the numbers may be favorable but there’s something we don’t like about the liver...there was a liver a month ago where the right lobe felt very firm, like it was an infarction, that part of it had died...one of my colleagues was already at a point of no return in the recipient operation...he had already started before realizing that this was happening...so we ended up having to use that liver, and sure enough, post-transplant, there was an area of infarction on the right lobe...we don’t know if it happened because of the pump...we don’t know if it happened because of the procurement...but we have this less than ideal outcome for our patient where part of the right lobe of the liver was just completely dead.”

Necrotic liver due to pump or TransMedics procurement issue; transplanted anyway

A: “Examples where it hasn’t gone so great are situations where the liver is on pump. **The numbers may be favorable but there’s something we don’t like about the liver. There was a liver a month ago where the right lobe felt very firm, like it was infarction, that part of it had died.”**

Q: “It felt very firm?”

A: **“It felt very firm, yes. At this point, unfortunately, it wasn’t my case but one of my colleagues was already at a point of no return in the recipient operation. He had already started before realizing that this was happening in the graft. And so, we ended up having to use that liver and, sure enough, post-transplant, there was an area of infarction on the right lobe.** Now, we don’t know if it happened because of the pump. We don’t know if it happened because of the procurement. But **we have this less than ideal outcome for our patient where part of the right lobe of the liver just was completely dead.** We did let TransMedics know. After the transplant, we communicated our findings and our concerns and we let them know about it.” – Chief of transplant surgery at a major transplant center; high-volume TransMedics users

The surgeon stated that TransMedics was fully aware that part of the liver was dead, and that they recently encountered another necrotic liver, just weeks after the other one – and that their “Medical Director” Magdy Attia falsely told them they were the only center that had encountered the issue. For the avoidance of doubt, we asked the surgeon if TransMedics didn’t tell them about the first liver’s damage: “Until after we felt it, exactly...we reported this to TransMedics and we were right...our concerns were correct...the procuring surgeon could feel it but it was not reported...but that’s just the first part of the story...fast forward three weeks later, another colleague, we had another liver on pump for a different patients...this time, because of what happened...we inspected the graft more thoroughly...and again, this time, the right lobe of the liver fell off...so we called Magdy...the response was ‘Wow, yours is the only center that is having these issues,’ which is an outright lie...there were other centers that definitely had concern about a graft and how it felt before implantation.”

Failed to report liver was necroscopic; TransMedics allegedly lied to the center to cover up a prevalent issue

Q: “So, when you got the liver, you didn’t even know until you felt it and **they didn’t tell you?**”

A: **“Until after we felt it, exactly. So, there was an issue and then the liver, the pump - we reported this to TransMedics and we were right. Our concerns were correct.”**

Q: “Was this the kind of thing that should have been obvious to the procuring surgeon just by feeling it?”

A: **“The procuring surgeon could feel it, but it was not reported.** It could have been a procurement issue. But that’s just the first part of the story. **Fast forward three weeks later, another colleague, we had another liver on pump for a different patient. And this time because of what happened the previous time, we inspected the graft more thoroughly before we started the operation. And again, this time, the right lobe of the liver felt off.** But the numbers, the perfusion numbers were fine. So, we called Magdy and we called TransMedics and we said, “The numbers are okay. It looks okay. But that right lobe, we’re nervous because of what happened with our previous patient, and **the right lobe doesn’t feel right.” And then the response was “Wow, yours is the only center that is having these issues,” which is an outright lie. There were other centers that definitely had concerns about a graft and how it felt before implantation.** – Chief of transplant surgery at a major transplant center; high-volume TransMedics user

The surgeon's color implies a widespread issue with livers becoming necrotic on the device - a known risk and potentially fatal flaw with normothermic perfusion pumps – and suggests a conspiracy to cover it up: “their advice to us was, ‘Why don’t you wait a couple of hours’...and the right lobe still didn’t feel much better...so we declined the liver...it left a bad taste in our mouth...the trust was fractured at that point...Magdy was the one who told us about that...then we called other colleagues around the nation...and they said, ‘Oh, we’ve definitely seen this before...sometimes it’s a procurement issues, sometimes it’s pressure on the pump’...so, it’s just a fucking lie...it’s a lie that we’re the only center that’s having these problem but that was the communication to us....”

Failed to report liver was necroscopic; TransMedics allegedly lied to the center to cover up a prevalent issue

A: “So, then their advice to us was, “Why don’t you wait a couple of hours,” which we did. And the right lobe still didn’t feel much better. So, we declined that liver. Fortunate for TransMedics, [transplant center redacted], our colleagues down the road, decided to use that liver for one of their patients. So, that liver did get transplanted. I don’t know what the outcome was for their patient but it left a bad taste in our mouth because we are trying to do what’s right by our patient and the feedback makes us feel like it’s something that we’re doing wrong in our assessment....the trust was fractured at that point.”

Q: “Who was saying to you you’re the only center having this issue and lying? Was that Magdy Attia?”

A: “Magdy was the one who told us about that. So, then we called other colleagues around the nation. We called [redacted]; we called [redacted] - what do you guys think? And they said, ‘Oh, we’ve definitely seen this before. Sometimes it’s a procurement issue; sometimes it’s pressure on the pump. We’ve definitely seen that before.’ So, it’s just a fucking lie. It’s a lie that we’re the only center that’s having these problems but that was the communication to us. We don’t know what’s going on with you guys, but everyone else is fine using our pump. And this situation, we were like, “You know what? We’re not going to put our patient at risk. We’ll eat the cost, fine.” – Chief of transplant surgery at a major transplant center; high-volume TransMedics user

The surgeon indicated that the two recent livers with necrotic tissue – and the fact that other centers are encountering the same issues – calls for the publication of data about device problems and outcomes, which TransMedics has allegedly refused: “members of the transplant community, myself included, have called for publication of the livers that are not used or the complications related to livers that have been on the OCS pump, and they declined...TransMedics declined to publish that data...that is not being very honest...it’s their responsibility....they should publish it and be forthcoming.”

TransMedics allegedly refusing to publish data on necrotic livers

“I can’t speak to their outcomes overall. But I do know that there are **members of the transplant community, myself included, have called for publication of the livers that are not used or the complications related to livers that have been on OCS pump, and they declined.** TransMedics declined to publish that data. Like that is not being very honest...it’s their responsibility to vet their surgeons and they should publish it and be forthcoming.” – Chief of transplant surgery at a major transplant center; high volume TransMedics user

In addition to surgeons and transplant center administrators, executives at leading OPO's - organ procurement organizations – provided similar examples of hardball conduct suggestive of extortion, and exhibited the same animus toward TransMedics. The CEO of a major OPO stated “their CEO is pretty arrogant...he's an arrogant prick....I mean, look at what they did...they bought \$12-13 million aircraft...it's just price gouging...it's unbelievable...it's not what transplant is meant to be...people don't like him.” A second executive, at one of the largest OPO's, stated they ceased working with the company after repeated cases of aggressive, improper behavior and described a volatile CEO: “we bailed for these reasons...the business practices were starting to come to light...if you ask any questions, they are very aggressive and confrontational...it is frankly putting people's lives in jeopardy...it's crazy...[Waleed] just launched on me...threatening that if we didn't work with them...it was just very aggressive.”

CEO viewed as “an arrogant price” due to “price gouging”; “people don't like him”

*“Their CEO is pretty arrogant. He's an arrogant prick. I mean, look at what they did. They bought \$12-13 million aircraft. It's beautiful, but guess what? It works just as well as a \$2 million aircraft or lease the damn thing and have a reasonable logistics expense, but it's just **price gouging. It's unbelievable. It's not what transplant is meant to be.** It's not meant to be this crazy. I don't care. Make your margin, crush it, but don't withhold it and make it almost impossible to use your technology that's going to help people just because **you're a prick. People don't like him.**”* – CEO of a major OPO (Organ Procurement Organization), longtime industry executive

Large OPO ceased working TransMedics; “putting people's lives in jeopardy”

Q: “Some surgeons have used words like racket, mafia, and extortion to describe TransMedics when I spoke to them. I'm just quoting verbatim. It's a complex space and I get the sense that they are engaging in some sketchy behavior through OPO's.”

A: “We were a customer and **we bailed for these reasons...** some of the **business practices were starting to come to light that we didn't agree with...**I've been on a few calls with them. **If you ask any questions, they are very aggressive and confrontational...**what they charge for it and what they expect you to do to be able to have their blessing to use **it is frankly putting people's lives in jeopardy. It's crazy. Personally, we did not agree with some of their business practices...the CEO was on the call... he just launched on me, threatening that if we didn't want to work with them, then that's fine. They could take their device elsewhere. I really don't know how else to explain it. It was just very aggressive. They don't like receiving questions or perceiving that they're being questioned.”** - Executive at a leading OPO, longtime industry veteran

The OPO executive detailed particularly disturbing behavior where TransMedics allegedly threatens to leave organs such as kidneys in a donor body unless paid double the Medicare rate for extraction – thus preventing patients from getting life-saving organs - or unless the OPO agrees to “back and forth favors” such as improperly credentialing TransMedics’ unlicensed, imported surgeons – “if they’re sending their liver team...they just blanketly refuse to do it...they tell them if you want us to recover your kidneys, it’s going to be \$3,000 – extortion....just to tell you about their business practices, it’s all back and forth favors...they threaten to leave the kidneys in the body and risk two people not being able to come off dialysis because they want to charge these crazy fees.”

Refusing to procure kidneys, standard practice; allegedly trying to extort another \$3,000, “extortion”

*“The other thing they do - in the transplant community, often there are donors where seven organs get procured, but most of the time it’s livers and kidneys. Historically, the liver team goes in and procures the liver for their patient and they’re there already. It takes an extra five minutes to recover the kidneys for transplant. I’ve been here for [redacted] years. **This is just standard professional surgical courtesy.** If you’re on the liver team, you’re going to spend a few hours dissecting the liver, procuring it for transplant, and by that point, the kidneys are almost detached from the body. **You spend an extra five minutes - TransMedics, if they’re sending their liver team, they won’t do it. They blanketly just refuse to do it. For most OPOs out there, they tell them that if you want us to recover your kidneys, it’s going to be \$3000 - extortion.**”* – Executive at a leading OPO, longtime industry veteran

“Back and forth favors”; “threaten to leave kidneys in the body and risk two people coming off dialysis”

A: “We’re a non-profit. The kidney side of the operation is totally paid for by Medicare. So, for them to charge us \$3000, we would just pass it to Medicare. So, we just refused that. **Just to tell you about their business practices, it’s all back and forth favors. Because we happened to credential one of their surgeons, we were able to negotiate that they recover the kidneys for us without these additional fees. But I know for a fact that they’re charging the OPO in Sacramento or these other OPOs when they’re in a bind because it’s only the TransMedics surgeon present; they threaten to leave the kidneys in the body and risk two people not being able to come off dialysis because they want to charge these crazy fees.**”

Q: “You said standard operating procedure is a transplant surgeon goes to get the liver, and at the end, it’s an extra five minutes and \$3000 to get the kidneys. So, just out of professional courtesy, every procurement team does that. Their surgeon won’t do that, so they’re forcing people to send a whole other team?”

A: “It’s standard courtesy, and Medicare tells us that what we can pay for that surgical service, \$625 per kidney, so \$1250 for the two. **TransMedics will look you in the eye and say, we’re going to leave them in the body unless you pay us \$3,000.**” – Executive at a leading OPO, longtime industry veteran

The OPO executive alleged that the CEO's sister Amira Hassanein, who runs the failed lung program, is as "confrontational" as other executives in badgering and coercing surgeons to accept and transplant organs they have already rejected as dangerous to their patient: "Waleed's sister is in charge of the lung program... we had some lungs...where the accepting center declined...the bronch was not good...the lungs were contused...these calls are 100% happening...I saw her on the line with a transplant surgeon...insisting that these lungs would have a good outcome...I was thinking, 'How do you even know that?'...well beyond her scope to insist, badger a surgeon who made a decision that transplanting this organ could put her patient in jeopardy and to sit here for 15 minutes working to actively convince them to make a different clinical decision...I've never seen anything like that...confrontational, speaking over people...speaking as if she's their boss, their attending surgeon, insisting that the organ will have a good outcome...'you don't understand the machine, you don't understand the machine...."

CEO's sister allegedly badgering people to transplant defective lungs

Q: "One surgeon told me that there are organs sometimes that he doesn't want to take, and he then gets badgered by Magdy Attia pressuring him to take them. Have you seen them try to force people to take marginal organs?"

A: "Waleed's sister is in charge of the lung program....she is the one that I have witnessed do this. We had some lungs in that situation where the accepting center declined; the bronch was not good. The lungs were contused when they were recovered and I was able to witness some of those conversations myself....these calls are 100% happening...I saw her on the line with a transplant surgeon who's trying to weigh the clinical outcome for her recipient. And Amina would be on the line with her, just insisting that these lungs would have a good outcome after they were transplanted. I was thinking, 'How do you even know that?' This is confusing. I'm also not a surgeon, but it just seemed well beyond her scope to insist, badger a surgeon who made a decision that transplanting this organ could put her patient in jeopardy and to sit here for 15 minutes working to actively convince them to make a different clinical decision, I've never seen anything like that...confrontational, speaking over people, really speaking as if she's their boss, their attending surgeon, insisting that organ will have a good outcome and that they're jeopardizing the patient's safety if they don't transplant it, and it's all because "you don't understand the machine. You don't understand the machine." – Executive at a leading OPO, longtime industry veteran

Part 11. An accelerating customer exodus, the fastest such stampede we have ever encountered. Large, pre-eminent transplant centers have eliminated or sharply reduced usage; its remaining high-volume customers indicate they are in the process of doing so.

Our research indicates that TransMedics is in the middle of an accelerating customer exodus. One transplant center after another indicated that 1) using TransMedics wipes out their margin on transplant cases; 2) that despite the company's claims to the contrary, Medicare provides only partial reimbursement to centers and that private payors offer none, forcing centers to eat the cost; 3) that alternatives are radically cheaper, whether NRP which is perhaps a mere 3 to 5% of the cost of an OCS case, or a tsunami of new entrants with cheaper, alternative perfusion/storage devices; and 4) that they plan to imminently sharply reduce or entirely eliminate their usage – within the next few months or quarters – or have already done so. We begin with a surgeon who highlighted a recent Duke University study that showed their contribution margin dropped by an astounding 60% per transplant case when they used TransMedics. The surgeon indicated “Waleed and his people got so angry at the Duke guy for bringing it up” and that Duke had to sharply reduce their usage “because they were taking such a bath.”

Duke study showed 60% reduction in contribution margin per case when using TransMedics

*“I talked to the Duke guys - the way that they analyzed this, what they looked at, what their pre-OCS usage was of DCD organs, but **then they looked at, especially when they started using the NOP program which also involved using the surgeon and the aviation part of it, and they saw - because I've seen slides where they actually analyze what their contribution margins were** for cases that used standardized storage, OCS at the simple level, with just the technician, and OCS through the NOP program. If I remember correctly, say, the direct contribution margin for static cold storage was just rated at a dollar. The **contribution margin was something in the 40-cent range for the cases you use TransMedics on.**”* – Prominent surgeon, Director of leading academic transplant center

TransMedics angry at cost analysis; Duke had to reduce device usage

*“The presentation at that meeting - **Waleed and his people got so angry at the Duke guy for bringing it up. I know for a fact that the Duke guys were on target to do a huge percentage of their liver transplants using TransMedics in a given year but they dropped their volume by 20% because they were taking such a bath** based on their Medicare reimbursement that they needed to make sure that they stayed profitable and not start actively losing money, which, again, at a center like mine that has a worse payor mix than Duke, is even more at risk for. So, the centers that were really big enthusiastic users, **like any new technology, there's this explosion in people using whatever the technology or the drug is, and then the reality of the situation sets in, whether it's the finances or the fact that it's being overused or beyond indication** from the FDA and what we call “clinical creep.” It was approved for one indication, and suddenly, doctors are using it for a whole bunch of indications.”* – Prominent surgeon, Director of leading academic transplant center

The surgeon stated that TransMedics largest users are beginning to curtail usage – we heard from others that Duke and Massachusetts General were ~100% of the usage during the trial, and that both are moving on. MGH told us directly they are interested in eliminating usage entirely. The surgeon who is privy to the Duke economic analysis stated that it was the talk of the meeting when presented, given already prevalent concerns about the device's cost: "I heard Waleed's head almost exploded"; "the centers that were the really big enthusiastic users...then the reality of the situation set in." We quote two other surgeons below, both of whom also direct their liver programs – and both are high-volume users, indicating TransMedics jeopardy. One stated his margin per case drops by more than 90%, from \$100-120K to \$10K with the device; the other stated "you are eating up a very, very substantial amount of the margin, no doubt about it."

Duke analysis was the talk of the conference

*"Duke just got up at the user's meeting and said we're taking a bath on this because Waleed's people had gotten up at that meeting—my administrator was actually in the room at the time, and she was calling me frantically because we were in the midst of our big value analysis committee discussion of this and she said, **"You gotta hear the shit that's going on."** And she told me. And then subsequently, I spoke to the Duke guys. I said **I heard about what happened at that meeting. I heard Waleed's head almost exploded.** And they said, yeah, they don't like us very much anymore." – Prominent surgeon, Director of leading academic transplant center*

Margin drops 90% or more if using TransMedics

*"Whether I use TransMedics or whether I put it in a cooler of ice, **I still get \$300,000. But when I put it in the cooler of ice and bring the organ, my margin on the case is, let's say, \$100,000-120,000 or something. When I do it, and I then use the OCS device, my margin on the case is \$10,000 because I still have to pay for using the TransMedics device** out of that global contract. The only way I get reimbursed is that I can put the utilization of that on my Medicare Cost Report...but for liver, it's less, and at our center, for example, Medicare patients make up roughly about 30% of our liver transplant patients. So, 30% of the cost will get reimbursed to us on the Medicare Cost Report." – Transplant surgeon and transplant program director at a leading Northeast academic center*

Eats up "very substantial amount of the margin"

Q: "Several surgeons said it wiped out the margin by 35% or 50%. Can you comment on that?"

A: "All of this is very true and it very much eats up your margin...**you are eating up a very, very substantial amount of the margin, no doubt about it.**" -Transplant hepatologist and director of the liver program at a Midwest academic center

A transplant surgeon at Vanderbilt, who plays a leadership role at their liver center - one of the busiest in the world – did a deep dive to explain reimbursement dynamics and why TransMedics nukes his margin by half per case, given the center’s Medicare patient percentage, a key factor in the formula that determines reimbursement. He stated that, contrary to TransMedics protestations, that Medicare only covers half the device cost and that private insurance covers none: “because of the cost, people are going to start shifting gears....the transplant center has to suck the cost.”

Unsustainable economics; almost no private reimbursement

“Because of the cost, people are going to start shifting gears, and they’re going to be changing to a different pump because **it’s not sustainable for a transplant center,** and I understand that TransMedics, **they’re going to tell you, this is going to be covered by Medicare and by private insurance and yadda, yadda.** The reality is that this is going to drive the healthcare cost - you already know that the healthcare in America is destroyed - so **this is going to kill the healthcare system.** The reality is that right now, Medicare, when you use the pump, you add this cost into your acquisition cost for Medicare, but **you get paid about 50-60% of the cost.** That is with Medicare. And then you have commercial insurance, which is a completely different animal. Commercial insurance right now, unless you renegotiate the contract, commercial insurance, they don’t care whether or not you use the pump. **You don’t get paid by commercial insurance. So, the transplant center has to suck the cost.”** – Transplant surgeon at Vanderbilt

Profit per case drops by half; insufficient Medicare reimbursement and almost none for commercial

Q: “What are the economics, with the organ acquisition cost, for your center and profit if you use TransMedics?”

A: “Last year, our profit on a liver transplant on average was about \$170,000. If you are using TransMedics and it is a Medicare patient, you get paid something. **Maybe instead of \$170, it goes down to \$90,000 or 100,000.”**

Q: “And 90 with TransMedics, so you lose more than half the profit or about half the profit.”

A: “Yeah, of course.”

Q: “But the CEO of TransMedics runs around telling everybody, no, everything is a pass-through to Medicare.”

A: “Again, Medicare doesn’t pay you 100%. In these cases, you prefer to have Medicare versus commercial insurance because the commercial insurance, **they don’t care. They don’t pay you anything. They pay you zero** if you use TransMedics unless you renegotiate a contract. But even with Medicare, this is something that I had spoken about with my administration. She was telling me that when **we put all the organ acquisition costs with Medicare, we get paid; I think she said between 50 to 60% of the cost. So, if that cost is 100,000, you get reimbursed 50,000. So, you have to absorb the other \$50,000.”**

Q: “And how about with private insurance?”

A: “We have to absorb 100,000 because they don’t reimburse you any money.” -Transplant surgeon at Vanderbilt

The Vanderbilt surgeon's sharp comments are representative of the tsunami of anger and new entrants causing TransMedics customers to flee. He stated his colleagues on the heart side loathe TransMedics, which we confirmed by interviewing a transplant cardiologist there: "Vanderbilt is the biggest heart transplant program not in the country but in the world. The volume is gigantic. They do zero TransMedics...they are so upset...they hate the company." A surgeon and head of the program at a major Pennsylvania academic center echoed the comments, stating his hospital has cracked down and rationed usage, and that he would get fired if he uses the device too much: "we started using TransMedics a year ago...we had to talk about using it a reasonable amount because of the cost...the price of the plane, I mean...let's be honest...the question you have to ask yourself is hard...I know I can't use it too much, or I'm not going to be making any money, and then I'm not going to have a job."

Vanderbilt heart program refuses to use TransMedics

"Vanderbilt is the biggest heart transplant program not in the country but in the world. The volume is gigantic. They do zero TransMedics. Zero because they are so upset. They were extremely upset. So, they don't use TransMedics at all. They hate it. They hate the company. They hate is so much. They really hate it." -Transplant surgeon at Vanderbilt

Had to ration usage because of the cost

"We do about mid-80s to mid-90s, so just under 100 liver transplants a year...we started using TransMedics about a year ago...We had to talk about using it at a reasonable volume because of the cost and we have a budgeted percentage of transplants we're allowed to use it in... Yeah, their technician has to put it on the pump and has to run the pump. They keep the pump, and they run the pump...The price of the plane, I mean...let's be honest. And then the cost of the disposables and the charge, it's a lot. The question you have to ask is hard." – Transplant surgeon and transplant program director at a major academic center in Pennsylvania

Surgeon stated he'll get fired if using the device too much

"I know I can't use it too much, or I'm not going to be making any money, and then I'm not going to have a job...if a hospital thinks they're losing money by providing a service, are they really going to go out and provide a service? So, that's why I said I had to have an actual financial talk with my head of finance and describe the margins and how they're doing." – Transplant surgeon and transplant program director at a major academic center in Pennsylvania

A second Vanderbilt source – a transplant cardiologist – echoed the color from his liver transplant colleague for heart/lung, stating that they are the highest volume transplant program in the world. He stated their OCS usage fell from 60-70 cases from 2020 to 2023 to four cases in 2024, and that he sees the same trend at leading heart transplant programs around the country, due to the superiority of NRP and the fact that newer coolers provide the same benefits at a fraction of the cost: “the trend has been going down...three years ago I thought we were going to be using it a lot more and it dropped into the low single digits in this timespan...we’re involved in all these different cooler trials...and we’re getting the same distance benefits from those coolers as well.”

Vanderbilt heart usage of OCS has tanked

A: “Over the last four-ish years or so, we’ve done about 125 heart or multi-organ transplants, including heart per year. We’re shooting to around 140, I’m told, by the end of 2024. We perform the most heart transplants globally. Roughly, because I have the numbers in hand, between January 2020 and January 2023, and this is publicly available from data we’ve published [...] the **use of OCS was roughly about a third of those hearts**. 21 out of 122—that’s like 1/6 of the DCD hearts during that three-year time frame used OCS. Out of 120 cases and that was just for DCD. We used it probably another 30 or 40 times amongst another 260 brain-dead hearts during that same time frame. So, probably brings **our total OCS volume between 2020 and 2023 to roughly about 60-70 or so.”**

Q: “How is that trending?”

A: “**The trend has been going down. For example, thus far this year, we’ve used it four times**. Last year, OCS overall for the entirety of January to December 2023, we used it a total of 12 times over all DCD cases.”-Transplant cardiologist at Vanderbilt

Vanderbilt has ceased usage; high-volume users like UCSF are next with competition from NRP and coolers

Q: “Are there programs that you know of **where the utilization of OCS has just really gone down dramatically? They stopped using it completely**, like high-volume ones?”

A: “**That was essentially us. Three years ago, I thought we were going to be using it a lot more and it dropped down into the low single digits in this time span**. The more we talked to other high volume centers like UCSF, etc., there’s just lots and lots of utilization of it. We’ve all gotten good at NRP. We all have large surgical teams to send our surgeons and perfusionists to procure organs via NRP, and **we’re involved in all these different cooler trials regardless of what cooler it may be, and we’re getting the same distance benefits from those coolers as well**. But it really cuts down the need for OCS in terms of what situations are we really using it for. We’re not using them for marginal hearts anymore, really. We’re not really using them for distance. It really comes down to what are those few scenarios per year where we generally don’t have the surgical bandwidth when we have a potential offer for one of our listed patients.” -Transplant cardiologist at Vanderbilt

The list of major transplant centers which have reduced, eliminated, or are on a path to terminate TransMedics usage is long and growing quickly. We document others beyond the ones already quoted, with the following on this page alone: Massachusetts General, University of Cincinnati, University of Virginia, and University of Washington. MGH played a key role in the pivotal PROTECT trial, but two sources - a transplant physician and a hospital executive, both in leadership roles – expressed growing dismay and alarm, respectively, at the company’s practices. The MGH executive: “We do use TransMedics quite a bit, and I do have thoughts regarding the future.” An OPO executive close to Cincinnati and Virginia stated they’re both “getting away from” Transmedics and moving to another device. An ex-employee indicated that UW is “trying to block all TransMedics business because they’re so upset...they were making an effort to not use it all.”

MGH signaled their intent to reduce usage of TransMedics

“I am not solely loyal to TransMedics. **We do use TransMedics quite a bit, and I do have thoughts regarding the future**” – Veteran transplant administrator/executive at Massachusetts General Hospital

Major centers moving to competing devices

“I know **University of Cincinnati was using TransMedics, and they’re getting away from it** because it’s just so darn expensive, and they’re moving towards a Paragonix-type device. It’s called “Liver Guard.” It’s what they use a lot down in Tampa General. We’ve got a relationship with them, so we partner with our logistics company called Mobile Network 24, and then they house the Paragonix devices, and we’ll pop a liver on one of those and then send it to the transplant program; it buys us some time. But I think Cincinnati is going that route and getting away from the TransMedics device. **University of Virginia is another one.**” – Executive at a major OPO

University of Washington allegedly trying to kill off TransMedics usage; questionable outcomes

“One of my contacts works here locally at UW and she was in one of the meetings about - on their transplant board or whatever - in a meeting, they were going into how they were **trying to block all TransMedics business because they’re upset about the cost and everything**, and they’ve gone way over-budget and all this stuff, so **they were making an effort to not use it at all**, especially because they were saying Washington wants to do NRP now instead of using TransMedics. **I’ve heard a lot of complaints about TransMedics just from local people.** Someone told me recently at the hospital that they were **worried about the outcomes - they were worried about the outcomes from a couple of hearts that they’d gotten from TransMedics.** There was a poor outcome on two different hearts that they had used TransMedics for. They were saying that something had gone wrong.” - Former TransMedics OCS specialist

A CA-based transplant center – one of the largest in the country by volume of liver transplants and a heavy OCS user – illustrates the disillusionment with TransMedics rapidly unfolding in the transplant field in recent months. A surgeon who directs the program expects to terminate their usage in the next 6-12 months: “Yes, it will go to zero.” The surgeon is actively shopping for new, competing devices at a quarter of the cost, such LifePort or XVIVO; and noted that centers which have already made the switch now “don’t use TransMedics ever, very rarely...they use their own pump...Wash U in St. Louis, you should talk to [redacted]...they don’t use TransMedics”; “I’m very interested in their platform.” The surgeon warned that “if I were an investor, I would be very wary of investing in TransMedics...I don’t the field is going to move in their direction....we are going to get other options...there is going to be competition.”

Shopping for alternatives and TransMedics likely to “go to zero” shortly

A: “As a surgeon, I am still going to use TransMedics now. But **I am definitely shopping. I’m shopping for other options.**”

Q: “And when do you think it’ll impact your use of TransMedics, particularly for liver?”

A: “Once we get a pump here at [center redacted] that we like, we’re looking—so, XVIVO, they also have a hypothermic pump like LifePort. They are not FDA-approved in the United States. They are widely used in Europe, so they’ve been incredibly successful in Europe and **I’m very interested in their platform** because they have excellent outcomes on their pump and they have a back-to-base model, which would allow the transplant center to buy their pump, use their own procurement team, and pump livers at home, and it’s a **lot cheaper, like \$20,000 versus \$80,000. I would like to buy a pump in the next year, either LifePort or XVIVO and have that option for my team so that we don’t have to use TransMedics.**”

Q: “Once you do that, does your volume of TransMedics go close to zero?”

A: **“Yes, it will go to zero.”**

Q: “And do you think this is a realistic possibility in the next 6-12 months?”

A: **“I think it is. I think the centers that have LifePort, for example, they don’t use TransMedics ever, very rarely. They use their own pump. Or Wash U in St Louis, you should talk to [surgeon name redacted]. They don’t use TransMedics; they use their own pump... if I were an investor, I would be very wary of investing in TransMedics.** I don’t think the field is going to move in their direction. I think it’s still going to be judicious utilization of the pump because of the cost associated with the pump, and I think **we are going to get other options. There is going to be competition.**” – Chief of transplant surgery at a major transplant center; high-volume TransMedics user

The surgeon laid out a competitive and regulatory avalanche, such as a cap on the device cost: “competitors are coming down the pipe pretty quickly...they’re not going to have the monopoly anymore....there’s going to be regulatory change...we really can’t justify the cost...there’s going to have to be some kind of cap...again, as a consumer, I am shopping.” The surgeon noted irksome calls and pressure from TransMedics to use the device even when not clinically indicated – “I don’t see any advantage in using their pump anymore” – and chafed at the lack of control after being forced to use and pay up for TransMedics device technicians and procurement, versus newer entrants that give the surgeon full control and autonomy: “then we can control the procurement...what goes on the pump, and we can decide...the field is going to move towards companies that are willing to sell their pump to the actual transplant center.”

No advantage in using the pump

“I think the biggest risks are that, one, there are competitors coming down the pipe pretty quickly. So, they’re not going to have the monopoly anymore. I think that there’s going to be regulatory change. I think pretty soon the US government is going to or insurance companies are going to say yes, you guys are increasing the total number of transplants for this country but **we really can’t justify the cost. So, there’s going to have to be some kind of cap** or something. I think that it could be that some of these negative stories also make the company more vulnerable...**again, as a consumer, I am shopping...they will reach out to us and they’ll say, we notice that you haven’t been using our pump a lot recently. What’s going on? I do find it off-putting.** Like, I will use your pump when I decide to use your pump. **I don’t see any advantage in using their pump anymore.** They want us to use their pump as much as possible. They’ll call and say, we noticed there was a drop-off. You had a great January, February - and now you’re not really using our pump in March. What’s going on?” – Chief of transplant surgery at a major transplant center; high-volume TransMedics user

Competitors coming online; allow centers to control the device; field moving in that direction

“The FDA just approved LifePort, which is a hypothermic pump. Sometime this year, and it’s been around a lot longer than TransMedics. But they have been really slow at getting their FDA approval. So, now they’re ready to roll and sell LifePort...we are interested in purchasing LifePort. Another company is called XVIVO, another hypothermic platform. And the benefit of those pumps is that **you can sell the actual pump to a transplant center, and then we can control the procurement, and we can control what goes on pump, and we can decide. So, I think that the field is going to move towards companies that are willing to sell their pump to the actual transplant center.**” – Chief of transplant surgery at a major transplant center; high-volume TransMedics user

Massachusetts General indicated their usage has already dropped off and that they expect to terminate all usage within a few months, indicating a wave as the field is rapidly shifting away from TransMedics due to “clinicians who have been very, very concerned about their poor business practices and misalignment of incentives”: “I would switch right away”; “for liver, probably in the next six months...I think we’ll probably stop using it...it’s gotten to that point....there’s going to be no real utility in using OCS in comparison to other cheaper, more widely-available...it’s a no-brainer.”

MGH expects to cease OCS liver use in next few months

A: **“Livers have been used less because one of our leading surgeons left**, and that’s one of the reasons why utilization has gone down.”

Q: “I’m hearing from major centers that they’re trying to move to OrganOX, Paragonix, and NRP as fast as possible because they’ve alienated people. It sounds like Paragonix wasn’t able to produce enough devices. I talked to a major OPO today, and they’re like, “We just got our device a few days ago.” It sounds like there’s a full-court press to just switch out TransMedics as fast as humanly possible.”

A: “I think you’re really good at your job in picking up on that. **That’s what I believe will happen.**”

Q: “Do you see a wave coming?”

A: **“Yeah, that’s what’s going to happen**. It’s not new technology, I mean, it’s been around for a while. **Like NRP, I think it’s going to be a huge game-changer**. And you see some of this initial data that’s coming out even from Europe, and you’re like, **“Why are we not doing this?” NRP is going to probably be standard. There’s going to be no real utility in using an OCS in comparison to other cheaper, more widely available, and more preferable from a business-to-business standpoint—it’s a no-brainer.**”

Q: “When do you think your volumes will take the next leg down?”

A: **“For livers, probably in the next six months.”**

Q: “In the next six months, do you think your liver utilization will take a hit?”

A: “Oh yeah.”

Q: “By how much do you think it’ll go down?”

A: **“I think we’ll probably stop using it. It’s gotten to that point...the idea that their device is the premier and one and only best device that is on the market and that can be used to save human life is not true simply because the market has shifted...the clinicians who have been very, very concerned about the poor business practices and misalignment of incentives with TransMedics, that there are now going to be other technologies** that are coming up that actually...**it’s very promising that there’s going to be a different option**, not only from the actual product standpoint but just from a business standpoint, **I would switch right away.**” – Veteran transplant administrator at Massachusetts General Hospital

UCSF indicated the same intention as MGH – that they expect to eliminate TransMedics usage starting in early 2025 and replied affirmatively when we asked if they expect their usage to “drop off a cliff”: “that’s our plan, yeah”; “we’re trying to reduce our utilization”; “we will shortly be up and running with OrganOx...which would negate the need for TransMedics altogether”; “our surgical director is not a fan...he’s actively working on strategies to minimize our reliance on it”; “we really don’t love doing business with them...they’re gouging us...as soon as we can detangle ourselves from them, the better.” UCSF also indicated they plan to shift aggressively to NRP in addition to OrganOx, with an NRP procurement service starting in Jan 2025 for both liver and heart.

Liver is highest volume and seeking to use other options “whenever possible”

*“The liver program is our highest volume. We, also on the liver side, are **interested in doing more NRP and using other devices as alternatives whenever possible...we don’t want to have to rely so heavily on TransMedics**...I’m struggling from an administrative standpoint. These costs are really making it challenging for us.” – UCSF administrator*

UCSF eliminating TransMedics by early 2025 through combination of OrganOx and NRP

Q: “Is there a timeline for when you guys think you’ll reduce your usage? You used them 70 times in about the last 12 months - walk me through what the usage has been the last couple of years on liver and where you think it will go.”

A: “I was just doing this analysis recently because **we’re trying to reduce our utilization**. We’re trying to identify—our OPO and then our neighboring OPO both recently finally got OrganOX machines, and their staff is being trained right now. Our surgeons are scheduled for training. So, **we will shortly be up and running with OrganOX...which would negate the need for TransMedics altogether.**”

Q: “When do you think your NRP usage will eliminate your DCD use of TransMedics?”

A: “There’s a procurement company that we’ve contracted with for thoracic, and we’re trying to get them to do liver NRP for us. I would suspect that **by January 2025, they could be doing NRP for us.**”

Q: “**So, theoretically, your usage of TransMedics could fall off the cliff in 2025?**”

A: “**That’s our plan, yeah. I hope...**for liver transplant, we use it obviously a lot more than heart or lung. I know **our surgical director is not a fan of TransMedics and he’s actively working on strategies to minimize our reliance on it.** I think the surgeons really like it because of the convenience of being able to send someone out to get the organ for you. But I think everybody’s totally on board with adopting something new, and I can speak for the surgical director; he and I are on the same page. **We really don’t love doing business with them and it really does feel like they’re gouging us. And as soon as we can detangle ourselves from them, the better.**” – UCSF administrator

An University of Washington source stated that their use of OCS devices has dropped 90% in the last year, and that NRP and other devices like Paragonix and XVIVO will wipe out TransMedics within a year: “I could put you in touch with a lot of people that are just going to say the same thing....‘I hate TransMedics,’ the OPO’s, the hospitals...it’s the same thing over and over again...we just don’t see them being here in a year”; “I mean, from Hawaii to Texas, I’ve heard people say that it’s just not sustainable”; “NRP is about to get approved...it’s going to completely change how much business TransMedics can do in this area...NRP is the new big thing for sure”; and beyond declining usage for liver and heart, “our lung surgeon won’t use TransMedics...he never has, despite the push...”

Cost is “unsustainable” for UW; moving to NRP

Q: “You heard some scuttlebutt about how UW is thinking about moving away from them and that there was a meeting about this?”

A: “It’s cost. It’s just not sustainable. There have been meetings where they are coming down on everybody really hard to be more responsible when it comes to cost. That’s completely separate from the talk of them not being comfortable with being taken off of the pump by qualified people. But the talk of cost for TransMedics is, I mean, from Hawaii to Texas, I’ve heard people say that it’s just not sustainable, that they have to find different ways because they just can’t sustain the cost...but it’s also being disrupted a lot with Paragonix devices, with the XVIVO, and with NRP. In this area, NRP is about to get approved, and it’s going to completely change how much business TransMedics can do in this area. I mean, the University of Washington is 100% ready to go with NRP. We’ve been traveling to other states to do NRP to avoid using TransMedics, and it’s about to get approved in the state of Washington, so we’re going to start doing a lot more of—like the hearts that we can’t do right now because their DCDs and TransMedics is going—but that’s going to change with NRP. NRP is the new big thing, for sure.” – Organ procurement technician at a high-volume Northwest academic transplant center

UW reducing use; already down 90%; “we don’t see them being here in a year”

“Our lung surgeon won’t use TransMedics; he never has, despite the push from TransMedics to get him to do it. Our heart surgeons only use them for DCDs...I know that the push is less and less and I see less and less because I’m involved in—I’ll know if a heart is going to TransMedics or if it’s going to my team. I could probably go back and look this year how many they’ve done. It’s probably less—it’s definitely less than 10, maybe less than 5 this year, which is a significant decrease from last year. It’s probably 90% down from last year...I could put you in touch with a lot of people that are just going to say the same thing, ‘I hate TransMedics....’I had recently had a conversation with the surgeon from our local OPO because I fly with them often for Paragonix. It’s the same thing over and over again, just TransMedics, we don’t see them being here in a year.” – Organ procurement technician at a high volume Northwest academic transplant center

Part 12. TransMedics device has no value proposition. Even its largest customers admit it has no clinical benefit to organs, using it off-label strictly for “surgeon lifestyle” and scheduling convenience; or, we believe, in exchange for organs steered improperly.

The TMDX bull case is that the OCS pump is a multi-organ platform for lung, heart, liver, with kidney in development, that revolutionizes transport, preservation, and monitoring. In reality, it is a one-trick pony in liver, with zero chance in kidney, the most widely transplanted organ by orders of magnitude; OCS Lung, run by the CEO's sister, is a colossal failure; and OCS Heart is currently in freefall. That leaves only OCS Liver, a gimmick used not for any clinical benefit but for a) off-label use for scheduling; b) by questionable centers whom we believe receive kickbacks and organs in exchange for taking them on the device. We cover each organ in turn – starting with the failure of OCS in Europe, the canary in the coal mine – proof that it exists in the US solely due to a Medicare loophole. The head of a large transplant center: “You should look at the European market...you know what market share TransMedics has in Europe? Zero because they understand that they don't have a shot at competing...there are so many competitors and options...they cost 20% of TransMedics.”

No market opportunity for TransMedics in Europe

*“To do your due diligence, you should look at the European market. The Europeans have been using pumps a lot longer than the Americans and they use a variety of different pumps at their disposal: normothermic, hypothermic. **And you know what market share TransMedics has in Europe? Zero because they understand that they don't have a shot at competing in the European market** because there are so many other competitors and options for a transplant center to use that are much cheaper. **They cost 20% of TransMedics.**”* – Transplant surgeon and division head at a major transplant center; high-volume TransMedics user

One leading center after another indicated that the OCS platform is a commodity with no value proposition, detailing why it failed in heart and lung, and why liver is next. One of the world's top liver transplant surgeons, perhaps the most influential KOL in the field, sharply criticized its use and indicated it has zero clinical benefit: "the problem with the market is that they've expanded use which doesn't need to be expanded...it's ridiculous...you're replacing a very cheap technology that costs a few thousand dollars with one that costs \$100,000"; "patient and graft survival are no different between cold storage and machine perfusion." A transplant physician at Massachusetts General called it **a commodity: "I don't think there's any magic to TransMedics...it's got cute bells and whistles...let's face it: there's no secret sauce here"; and that it will be replaced shortly with devices that are "...a whole lot less costly, then you do that way...there's no question...I think there will be a low threshold to jump over."**

Zero value proposition; no difference in outcomes; no TAM

A: "The problem with the marketing is that they've **expanded use which doesn't need to be expanded**. In other words, they're talking about an \$8 billion industry. That means that every organ would be pumped on the machine at \$100,000; **it's ridiculous. You're replacing a very cheap technology that costs a few thousand dollars with ne that costs \$100,000.**"

Q: "You're saying the device really doesn't add value beyond the traditional method of cold storage."

A: "No, not for every organ because the pristine, what we call blue ribbon organs, they do perfectly fine. One thing that you can take away from all of these studies also is that the **patient and graft survival are no different between the cold storage and machine perfusion...there was already a trend towards improved outcomes before the machines came out.**" - One of the top liver transplant surgeons in the world, based at a leading academic center

MGH calls perfusion pumps a commodity

"I don't think there's any magic to TransMedics per se. It's got cute bells and whistles; it travels. **Let's face it: there's no secret sauce in this.** Perfusion is in many ways perfusion and, in this case, normothermic perfusion is normothermic perfusion. It would be **very hard to ascribe a special quality to one modality necessarily over another.** If you can get that with one device comparably to another and logistically it's less challenging and certainly a whole lot less costly, then you go that way, **there's no question. So, I think there will be a low threshold to jump over.**" – Transplant hepatologist at Massachusetts General Hospital

A transplant surgery KOL at the University of Chicago trashed the OCS pump for liver, heart, and kidney - every TransMedics product - stating they are a massive program that transplants 100 livers, 60 hearts, and 15-20 lungs per year, but see no value in liver; lung use dropped from 50% share to nothing; and that they still have a few units left over from 4 years ago when TransMedics forced them to buy 10 consumables – “for liver, we don’t use it all...we use OrganOX”; “we were using them for about 50% of our lung transplants...the last time we did a lung on TransMedics was close to 9 to 12 months ago”; and that for heart “the new director isn’t planning to use it in many case...we expect to drop off.”

One of the largest transplant programs basically uses no TransMedics

“We are a multi-organ transplant program at the University of Chicago. **We do about 100 liver transplants a year, 200 kidney transplants, and about 60 hearts. We probably will do about 15 or 20 lungs this year.** I do primarily liver. But I’m the [redacted] of the transplant institute, so I work closely with all the other programs...we transplant 60-65 hearts a year. We just started doing the DCDs about two and a half years ago now. And we’ve been **using the NOP DCD program that TransMedics has for about 4-5 cases a year. The new director isn’t planning to use it in many cases. So, we expect to drop off. And then for liver, we don’t use it at all.**” – Leading liver transplant surgeon/KOL

Lung use went from 50% to zero; none in 4-6 months

“We were using them for lungs. **Our director has already said he’s probably not going to use TransMedics to any great extent. So, we would anticipate a drop-off in TransMedics use in lungs.** About five years ago, we went to TransMedics and **we were using them for about 50% of our lung transplants** that year. Our practice is primarily using Paragonix for the preservation of lungs. So, he doesn’t pump the lungs that he’s transplanting now...” – Leading liver transplant surgeon/KOL

Still have lung devices unused from 4 years ago; last lung case on TransMedics was almost a year ago

“The lungs, **we have units still left over from four years ago. We haven’t done a lung transplant on TransMedics, probably in about four to six months**....originally, when we contracted with them, you had to buy 10 units. And so, you basically stored them in your storage area...**the last time we did a lung on TransMedics was close to 9 to 12 months ago...I haven’t tried to replenish our stock.** I think we may still have units left over from the past.” - Leading liver transplant surgeon/KOL

No use in liver, use OrganOX instead due to cost

“For livers, we don’t use it at all. We use OrganOX. When TransMedics approached us for the combination heart-lung-liver, we viewed the contract and the pluses and minuses, and we eventually ended up going with OrganOX for liver...it was primarily a decision that was based on cost. – Leading liver transplant surgeon/KOL

We quote three other surgeons as well as an executive at one of the largest OPO's in the US, who provided equally damning color on the lack of any value proposition and TransMedics dismal future – we note two of the below are among its highest-volume users, suggesting its precarious position. One stated “our liver transplant surgeons don't use TransMedics...a lot of people don't think you need it on the liver...cold storage works very, very well...ischemic times can be 8 hours...our liver guys are not interested in using it at all.” A high-volume liver user stated “our heart surgeons...do not care much for the machine...they're going to continue to lose cardiac volume.” An OPO executive confirmed that “there's not a ton of utilization of their heard device out here on the West Coast.” And one of its top liver users stated cold storage is just fine – “you don't need the Cadillac to transport the organ...you can get by with a less expensive version.”

Cold storage works fine for liver

“Our liver transplant surgeons don't use TransMedics. I think a lot of people don't think you need it on the liver. Cold storage works very, very well. Ischemic times can be 8 hours. They've turned that operation into a quasi-elective case...Our liver guys are not interested at all in using it.” – Cardiac transplant surgeon at a major Midwest academic center

Ice and cheap machines are sufficient for liver

“TransMedics and OrganOX are examples of normothermic machine perfusion, which is the most expensive way to preserve an organ. Less effective in preservation but still pretty good is something called hypothermic machine perfusion, for example, Bridge to Life or LifePort. And that costs significantly less. It provides better preservation than just putting it in a cooler of ice. But for many cases, you don't need the Cadillac to transport the organ. You can get by with a less expensive version.” – Transplant surgeon and transplant program director at a leading Northeast academic center

Minimal heart use left on West Coast

“There's not a ton of utilization of their heart device out here on the West Coast...” – Executive at a leading OPO, longtime industry veteran

Cardiac surgeons have no interest in TransMedics pump,

“I do think and I hear the same from our heart surgeons. They do not care much for the machine and prefer other modalities for their hearts...they're going to continue to lose cardiac volume.” -Transplant surgeon at a high-volume academic center

Both heart/lung and liver transplant surgeons undermined the fundamental premise of the pump: that it spits out data on biomarkers like lactate levels that purportedly help surgeons assess the viability of – and thereby expand the use of – marginal DCD organs. Surgeons provided devastating color that there is no correlation between how organs look on the pump and how they turn out in the recipient – that organs that look great may fail, and that organs which look questionable on the pump have performed fine. A liver transplant surgeon who runs a major program stated no one knows how to use the machine data, indicating cases where “the data from the machine has been good, but we have a liver that it not good in the end...so we don’t have perfect data to tell us whether a liver is going to be good or not.” He noted the confusion exists despite almost a decade of study and use since the clinical trials started – “there’s still not enough data that can tell us when something’s good...so you get a lot of surgeon cross-talk....’What should I do? Should I give it more time? Should I dump it?’....we’re still trying to sort that out.”

No correlation between organ performance on pump and after transplant

- A: “They have a specialist that’s with the machine and the organ at all times and you’re checking labs hourly. And so it requires a lot of manpower to manage. I would say that we’ve had a few cases where all of the **data from the machine has been good, but we still have a liver that is not good in the end. And so we don’t have perfect data to tell us whether a liver is going to be a good liver or not..**”
- Q: “I talked to a cardiac transplant surgeon. He also said that there’s no correlation between how well a heart pumps, how it looks once you put it on the pump versus what ends up happening when you implant it. So, they’ve seen cases where it didn’t look good at all, and it ended up being great, and cases where it looked great, and it was a disaster. Is that what you’re observing?”
- A: “Yeah, I would say that is true and it’s such a novel device even though it’s been out and we’ve had a trial and everything. There’s still **not enough data that can tell us when something’s good or not.** And so, you get a lot of surgeon cross-talk, people talking to one another, high-volume users saying I’ve got this scenario on the device. Would you use that? Or what would you do? What should I do? Should I give it more time? Should I dump it? **I think we’re still trying to sort that out...**” - Transplant surgeon at a high-volume West Coast academic center

Based on surgeon color, we expect papers on the lack of correlation between OCS pump data versus actual organ viability and outcomes – which would only accelerate the customer exodus. A transplant cardiologist at Vanderbilt – the world’s busiest heart transplant institution – indicted the same lack of predictive benefit for assessing hearts as for livers: “we haven’t seen anecdotally a correlation between how well it looks on the rig and how it does in the recipient”; “it was sold initially that...it’s a beating heart to see how it performs over time during transport...we’ve really noticed that is that there doesn’t seem to be a strong correlation between how the heart is performing on the OCS machine and how it does in the recipient after implantation....we’ve had some great-looking hearts clearing lactate, and these patients just crashed and burned horribly....so, while it is a selling point...we haven’t found the trend to be always accurate....”

Organ performance on pump has no correlation with organ viability

*“It was sold initially that unlike most of our procurement methods, there’s a physiological perfusion...it’s a beating heart to see how it performs over time during transport. **All of these indirect indicators show that perhaps, even though it looks marginal initially on our echocardiograms or in the donor’s chest, that this year will probably do okay if you gave it a little bit of time after we implant it in the recipient and get them through those early first couple of days or so. But that’s one potential selling point.***

*This is something we speak about with our surgeons all the time – one of the things we’ve really noticed is that there **doesn’t seem to be a strong correlation between how the heart is performing on the OCS machines and how it does in the recipient after implantation.** We’ve had some that look really crappy by the time they showed up to the recipient operating. But at this point, we’re already in the chest, so we have to go through with it, and we’ve been pleasantly surprised at how well that heart did afterwards. On the flip side, **we’ve had some great-looking hearts on the machine with clearing lactate, and these patients just crashed and burned horribly afterwards. So, while it is a selling point...we haven’t found the trend to be always accurate .”** -Transplant cardiologist at one of the top centers in the US/world*

No predictive value or value proposition even for marginal DCD hearts

Q: “When you said you don’t use it for marginal hearts anymore; that’s just because it doesn’t help you predictively?”

A: “In part because **we haven’t seen anecdotally a correlation between how well it looks on the rig versus how it does in the recipient.** And then, two, there’s just been more and more observational research data coming out and in our experience that even if the heart looks marginal based on other clinical characteristics of the donor...we can reasonably predict how well that heart’s going to do in the coming 72 hours.” - Transplant cardiologist at one of the top centers in the US/world

TransMedics lack of a kidney pump is a critical tell that undermines the OCS bull case as a multi-organ platform, as kidneys are by far the largest transplant market with 25K transplants per year in the US, versus only 10K livers, 4k hearts, and 3K lungs, per OPTN data. It purports to have a kidney product in development - for 2029 launch, amusingly - but showed nothing but a conceptual vaporware slide during its recent investor day. TransMedics has – and will never have – any opportunity in kidney as 1) only 40% of kidneys are pumped due to cost, despite the pumps being around for decades; 2) LifePort has 85% market share with its kidney pump; and 3) their pump is a third of TransMedics OCS cost at \$30K.

Vaporware TransMedics slide from Dec 2024 Investor Day vs. LifePort Kidney Transporter

TMDX OCS Kidney Timeline - Gear 3

OCS Kidney: **Launch 2029**

OCS Kidney FDA Trials:
Initiating 2027

OCS Kidney in
Development **in 2025**



Indicating the commodity nature of TransMedics OCS device, transplant surgeons indicated that kidney “portable perfusion pumps...have been around for literally 40 years” and are “a good example of technology that’s similar to TransMedics...but despite the fact that kidney transporting technology has been around for 40 years, less than 40% of kidneys are pumped...for financial reasons...TransMedics and XVIVO have no device even in trials...kidney is really easy to talk about....there is no other company out there other than LifePort.”

Kidney pumps used for 40 years but less than 40% pumped “for financial reasons”

“For kidney, there’s actually been pumping technologies, portable perfusion devices that have **been around for literally 40 years**, and that’s a good example of a technology that’s similar to TransMedics. But **despite the fact that kidney transporting technology has been around for 40 years, less than 40% of kidneys in the United States per year are pumped for financial reasons**. All of these companies - XVIVO, LifePort - **had to segue to liver transplantation**...because those are the fields that there are tens of thousands of transplants as opposed to just a few thousand hearts and lungs per year.” – Prominent surgeon, Director of leading West Coast academic transplant center

LifePort has 85% share of kidney pump market

“Right now, the only people in that space, which is **the largest, with their 20,000-25,000 kidneys a year, are the egg-shaped devices that have been around for four decades**. The company that’s emerged as being the largest in terms of the market share is the one we use, when we have a kidney rigged from an older donor, somebody, say in their 60s or 70s, is the **LifePort kidney device**. Bridge to Life may have something, but right now, LifePort, of the kidneys that are pumped, which in the U.S. is probably about **30% of the kidneys total in the U.S. are pumped, 85% of that is under LifePort. To date, TransMedics and XVIVO have no device even in trials for kidney that I know of**. They’ve got just prototypes in the lab. Kidney is really easy to talk about. There is no other company out there other than LifePort. When we pump a kidney, it’s an extra \$30,000 per case.” – Prominent surgeon, Director of leading West Coast academic transplant center

We next turn to OCS Lung, which TransMedics began commercializing in late 2018 after receiving PMA approval, its first FDA-approved product and the one it has marketed the longest – making its failure an inconvenient fact for the bull case, as it remains a negligible percent of total sales. A former employee in senior medical/clinical affairs roles stated the product serves no purpose versus storing lungs on ice: “the worst-performing product is definitely the lung...that organ is preserved on ice just as well, if not better, because of the complexity of the physiology and anatomy of the lung...it’s pretty expensive...the majority of the cases can be safely preserved on ice...this is a method that was working pretty well.”

“Product doesn’t work” for lung

A: “I think **the worst-performing product is definitely the lung. I think internally, that organ is preserved on ice just as well, if not better, just because of the complexity of the physiology and anatomy of the lung.**”

Q: “You’re saying that the device doesn’t add any value for the lung?”

A: “I don’t think it does, yeah. I think there are very isolated cases, maybe like a long-distance retrieval that lasts 11 hours, 12 hours, something from Hawaii to New York and such. But anything within six hours, I don’t think they really need that device for the lung.”

Q: “The entire U.S. is less than six hours flight time.”

A: “Exactly, that’s what I’m saying.” - Former employee in senior roles including medical safety and clinical affairs

“Majority of cases” can be “safely preserved on ice”

Q: “Does the device add any value versus the status quo in cold storage? It’s almost \$100,000 for the device.”

A: “It’s pretty expensive. **I do believe that the majority of the cases can be safely preserved on ice. This is a method that was working pretty well.** There’s a competitor who actually preserves organs on ice and then connects them to the circuit for assessment purposes only, not for the transfer.” - Former employee in senior roles including medical safety and clinical affairs

A prominent cardiothoracic transplant surgeon who leads a large academic program stated the lung product is as an utter failure as **“no one’s using it anymore in the US”**; that the **“really enthusiastic centers that were heavily involved in the trials...have stopped using it”**; that **“I don’t see them having any future in lung”**; and that a variety of cheaper alternative devices, already available or in trials, are not only superior but **“just much, much cheaper.”** He stated that extensive research and clinical trials since the OCS Lung approval have shown that cold storage around 10 degrees Celsius – slightly above traditional static cold storage – provides superior benefits over TransMedics warm perfusion pump, undermining the entire premise of its – **“I do not see them having any market at all in lung anywhere in the US or, for that matter, even worldwide compared to a much cheaper technology like 10-degrees Celsius storage.”**

“No one” using lung product anymore; total failure

Q: “Does their device add anything in lung versus cold storage or other methods to justify the cost?”

A: “The short answer is no. The longer answer is that despite that being the first approved device for TransMedics of an organ type, almost **no one’s using it anymore in the US**. It’s being used maybe 100-120 times per year, and **most of the centers that I know, including the really enthusiastic centers that were heavily involved in the trials and the trial design, have stopped using it, and I don’t see them having any future in lung**. And the reason for that is not just because of XVIVO’s collaboration with Lung Bioengineering, but more importantly, two years ago, the University of Toronto published their 10-degree Celsius storage story, and so there’s a spinoff company called Traferox that’s Canadian-based in conjunction with the University of Toronto that is currently doing a multi-center, large, randomized trial in the U.S. for transporting lungs at 10 degrees instead of a regular ice chest to give it a little bit like Paragonix. The short answer to your question is that **I do not see them having any market at all in lung anywhere in the United States or, for that matter, even worldwide compared to a much cheaper technology like 10-degrees Celsius storage.**” – Prominent surgeon, director of leading academic transplant center

Radically cheaper to use cold storage for lungs

“It’s just much cheaper. It’s much, much cheaper...the costs of the technology versus the reimbursement are so prohibitively expensive that if you can do something that only costs \$5000 or \$8000, which is probably where Traferox could come out, maybe even if you charged as much ridiculous charges as Paragonix does of \$14,000 or \$15,000 like they’re doing with their stupid device. The fact is it’s still much, much cheaper than \$85,000, and the logistics are also more complicated. ” – Prominent surgeon, director of leading academic transplant center

The cardiothoracic surgeon indicated that the only reason OCS Lung even got any initial traction was due to a promotion and then plummeted after a bait-and-switch – “this is again where the company’s dishonest”; “I personally have not seen a pair of lungs rigged on a TransMedics device in over a year”; and indicated a fatal blow to its chances not just from 10-degree Celsius, but by “enthusiasm for XVIVO’s device with Lung Bioengineering” and Paragonix which has also taken TransMedics share: “I just wanted to make sure you know that Paragonix does have a growing business in lung,” versus TMDX which has flatlined.

Initial traction only due to special offer; killed off by XVIVO and cold storage

“They initially got traction in lung. I think their peak year - and this is again where the company’s dishonest. They won’t tell you, but I can tell you what they did. In 2022, post-FDA approval, to gin up interest, they had a deal, which they did with liver, too, I believe. They would not charge the transplant center if the organ was rejected while it was on the device. You only paid for lungs that were used. And then when 2022 ended, in calendar year 2023, you paid whether you took the organ or not. And a lot of transplant centers got really pissed with that. In fact, that’s the reason why there was a lot of enthusiasm in 2022, which I believe was their peak year for lung usage in the U.S. It plummeted for two reasons. One, you now had to pay for an organ whether you used it or not. Two, enthusiasm for XVIVO’s device with Lung Bioengineering. Number three, which really put the nail on the coffin, was the 10-degree Celsius storage story.” – Prominent surgeon, director of leading academic transplant center

Allegedly trying to hide failure in lung; “have not see a pair of lungs rigged” in “over a year”

“My understanding is that, and you can get this directly from the company if they don’t smoke-and-mirror you with total sales and revenue but the number of disposable kits per organ sold year-to-year or quarter-to-quarter...they know exactly how many lung kits they’re selling. My guess would be without having actual proof, but based on everybody that I know in the field, my guess would be that you’d see that their peak number for lung usage was in 2022, it plummeted in 2023 and 2024. I personally have not seen a pair of lungs rigged on a TransMedics device in over a year.” – Prominent surgeon, director of leading academic transplant center

Lung business has disappeared, Paragonix has taken share in lung

A: “Just be aware that TransMedics’ competition is with 10 degrees, with XVIVO through Lung Bioengineering as its collaborator, and then Paragonix is eroding some of the market for TransMedics because I’ve seen actually more lungs put on the Paragonix device; even if I think the Paragonix device is a waste of money for lung, they are definitely putting lungs on that device that in 2022 would have been put on the TransMedics device. Just for completeness, I just wanted to make sure you know that Paragonix does have a growing business in lung.”

Q: “But the main takeaway is their lung business is gone.”

A: “At least in the U.S., lung is only going to be 100 cases a year at best, is my guess.” – Prominent surgeon, director of leading academic transplant center

After kidney and lung, we now discuss the heart product. Lung failed as cold storage is provably superior to warm perfusion – TransMedics entire premise was fatally flawed. Heart usage is in freefall for similar reasons: the pump is inferior to NRP in assessing heart function pre-transplant. We could find no cardiac surgeon still enthusiastic about TransMedics heart pump: “the problem, compared to what you can do with NRP or other technologies, is that the heart on the TransMedics device is beating but it’s empty...it’s not doing any work...it’s not supporting circulation...it’s not generating a pressure...you’re guessing...on the device based on the lactic acid dropping...that it will function well enough that your patient doesn’t end up on ECMO or an Impella.” The surgeon stated that “30-40% of heart programs in the US” are now using NRP – “you can assess the heart function completely...you’re doing it in three to four minutes...you can assess all the normal physiologic parameters: is the heart beating again in the patient.”

Device is fatally flawed for heart as no pressure; NRP better

“The initial trials that TransMedics did were in conjunction with the guys from Cambridge in the UK, but they showed that if you got the heart out fast enough and put it on their rig, the TransMedics OCS heart system, add the lactic acid bloc steadily, you could use that heart safely. **The problem, compared to what you can do with NRP or other technologies, is that the heart on the TransMedics device is beating, but it’s empty. It’s not doing any work. It’s not supporting circulation. It’s not generating a pressure. So, you’re guessing at that part, which is beating on the device based on the lactic acid level dropping, that it will function well enough that your recipient patient doesn’t end up on ECMO or an Impella or on something else.**” – Prominent surgeon, director of leading academic transplant center

30-40% of heart programs already moved to NRP; can assess the heart better with NRP

“My center is doing the same protocol that probably **30-40% of heart programs in the U.S. are doing now is we’re using NRP**, normothermic regional perfusion, which was first really described in a big way at Vanderbilt University and then UC San Diego is that **you can assess the heart function completely** because you’re putting the donor who’s arrested, you’re opening their chest and cannulating them and going on bypass rapidly, but **you’re doing it in three to four minutes**. And you reanimated the heart, and then after 45 minutes of resting the patient on your heart-lung machine, you then wean off bypass, and now you’ve basically taken a DCD donor and **converted them to a DBD donor, and you can assess all the normal physiologic parameters: is the heart beating again in the patient.**” – Prominent surgeon, director of leading academic transplant center

Cardiothoracic transplant surgeons indicated that NRP has rapidly eroded TransMedics usage in DCD hearts in the past 12 to 24 months. TransMedics had 100% of the DCD heart market a mere 2 to 3 years ago. That share dropped to 60-70% vs. 30% NRP over the last year or two, but NRP has now flipped to pole position with 50-60% share at TransMedics expense, with the share loss rapidly accelerating at OPO's and centers roll out NRP capabilities. Surgeons noted various new entrants in heart pumps – such as the XVIVO Heart Box, a “miniaturized version of the TransMedics system,” but noted that they expect NRP to decimate both Transmedics and XVIVO's heart pumps, with XVIVO taking half of TransMedics remaining share: “both XVIVO and TransMedics are going to have serious competition from NRP...I think at least 50% of heart centers will feel very comfortable doing NRP...they're going to split that remaining market right down the middle.”

NRP rapidly eroding TransMedics market share in DCD hearts

“TransMedics’ device first started getting used in the US, there were a total of less than 10 DCD hearts used. And from 2019, in less than five years, last year, in 2023, there were over 600 DCD hearts done, of which probably about 60-70% were on TransMedics and 30% were NRP. This year, it’s projected based on what everybody’s talked about at the ISHLT team meeting that was in Prague recently. Everybody’s estimating that we’ll probably do about 800 DCD hearts in the United States, and my guess is that the market is shifting from three years ago. All the DCD hearts that were done were with TransMedics; probably 30% to 40% of them will be done with NRP, and 60% will be done with TransMedics.” – Prominent surgeon, director of leading academic transplant center

XVIVO heart system will take half of TransMedics remaining heart volume; but both being decimated by NRP

“In addition to NRP you should be aware that XVIVO has finished their Australian trials for their XVIVO Heart Box, which is a miniaturized version of the TransMedics system; it’s even smaller than TransMedics’, and they are right now in a multi-center phase 3 trial in the US. Both XVIVO and TransMedics are going to have serious competition from NRP because I think at least 50% of the heart centers will feel very comfortable doing NRP, but TransMedics, which had two or three years ago, 100% of the DCD heart market. When the XVIVO Heart Box is FDA-approved, probably two years from now, they’re going to split that remaining market right down the middle, probably with TransMedics to XVIVO, but both of those companies will be competing against NRP. If 50 go to NRP and 50 go to machine perfusion, that 50 machine perfusion will probably split 25-25 cases between XVIVO and TransMedics.” - Prominent surgeon, director of leading academic transplant center

A cardiac surgeon at a major Midwest center – who was on the FDA panel that voted to approve TransMedics device – now barely uses the heart pump, in ~6 out of 60 total transplants per year. He stated the pump is controversial and still “hotly debated” years after FDA approval, with no proven benefit for lung or heart, and that it may in fact injure the heart – “there are concerns with pumping a heart for a long period...a pump is not a benign thing...if you go back and read the comments on the FDA panel, there’s data from TransMedics studies that suggest myocardial edema...whether the data will ultimately support this is still being hotly debated at transplant meetings...there still remains skepticism about pumping organs...and increasing the ischemic time...the data that surrounds OCS, does the patient really benefit, and is there truly no harm done to that organ?”

Surgeon on FDA panel that approved heart device barely uses it anymore

“I am a cardiac surgeon. Our program does about 60 heart transplants a year. We do not do lung transplants. We do Impeller liver transplants; I’m not involved in those. **We use TransMedics in 10-12% of our cases a year. For disclosure, I was on the FDA panel that voted to approve TransMedics, so I’m very familiar with that product.** We began to use it about two years ago.” - Cardiac transplant surgeon at a major Midwest academic center

Likely harm and no proven benefit from TransMedics pump for heart or lung

“Lung is marginal, whether pumping the lung is a good thing. And there are **concerns with pumping a heart for a long period of time. A pump is not a benign thing. If you go back to read the comments on the FDA panel, there’s data from TransMedics’ studies that suggest myocardial edema** and longer pump times and abnormally higher pressures are not necessarily good for the heart...I don’t know of anybody that uses OCS that keeps the heart or whatever on the pump longer than they need to. So, nobody is going to keep a liver on a pump for 40 hours...**Whether the data will ultimately support this is still being hotly debated at transplant meetings...there still remains skepticism about pumping organs longer than you need to and increasing the ischemic time, particularly for hearts and lungs...the data that surrounds OCS, does that patient really benefit, and is there truly no harm done to that organ?**” -Cardiac transplant surgeon at a major Midwest academic center

A transplant cardiologist at a leading Ivy League center detailed the same concerns as Vanderbilt regarding an inability to properly assess a heart's viability while on pump and its lack of predictive value – the entire point of the device, in contrast to NRP which he stated is the only way for a robust evaluation: “there are limitations in terms of the actual assessment of the heart on that device...you watch it beat in the box...there are physiological reasons why it's not the most robust test of organ quality...lactate trends, not necessarily a marker that your organ is good or bad...there's no way to assess it...we assess functionality...against resistance and load...with a TransMedics device, there is no resistance...if you resuscitate with NRP...you can actually have a clear assessment...that's missing on the TransMedics device: true functional assessment of this organ...if it looks good on the machine, will it look good once you put it in...there's no alternative other than NRP, really.”

No way to assess a heart on the device; NRP is only method

“At the end of the day, the device is a preservation machine. There are **limitations in terms of the actual assessment of the heart on that device**, that is not perfect. You watch it beat in the box. There are physiological reasons why it's **not the most robust test of organ quality and then lactate trends, not necessarily a marker that your organ is good or bad...there's no way to assess it**...the way your heart works and how we assess functionality is based on the ability to pump against **resistance and load**. And because you are in a solo, free-standing system with a TransMedics device, there is no resistance that it's pumping against. If you resuscitate with NRP, which is normothermic regional perfusion, within the body of the donor, the resistance of that donor, blood vessels, etc., is put on the heart, and you can **actually have a clear assessment of its functionality. That's missing on the TransMedics' device: true functional assessment of the organ. If it looks good on the machine, will it look good once you put it in and restart it and it's not as strong?... There's no alternative other than NRP, really.**” -Transplant cardiologist at a major Ivy League academic center

Vanderbilt stated that cheap, improved cooler systems they've simply repurposed internally are safer for hearts than warm perfusion pumps – devastating color for the TransMedics bull case, given Vanderbilt's gigantic transplant volume and access to every worthwhile device trial: “For longer distance of marginal grafts, we use other cooler systems that are really coming around, which we've found have less re-perfusion injuries...post-transplant.” A transplant cardiologist there stated that coolers are sufficient even for long-haul retrievals, indicating bluntly that TransMedics has no value proposition vs. coolers and NRP: “for both DBD and DCD, we've been trying to prioritize using these coolers that keep the heart closer to 8-10 degrees or so”; “even for long distance using 8 or 10-degree coolers is the preference...we've traveled as far as Alaska to procure hearts and haven't had any issue.”

Cooler systems for long distance use have less injuries to organ

“We use them predominantly for DCD because for DBD, it's pretty rare for us to resort to using OCS. **For longer distance or marginal grafts, we use other cooler systems that are really coming around, which we've found have less re-perfusion injuries and things like that post-transplant.** So, our preference these days for any of these DCD hearts is to **really prioritize NRP** unless we're absolutely in a pinch and don't have the staffing bandwidth to procure an organ. **For both DBD and DCD, and NRP, we've been trying to prioritize using these coolers that keep the heart closer to 8-10 degrees or so.** There are a couple of commercially available ones, for example, Paragonix. One we use mostly is one that's **repurposed from an internal cooler** that our lung surgeons use to keep the heart around 10 degrees. And then we're actively involved in a clinical trial with XVIVO, which is another carrier that keeps the heart at roughly 10 degrees Celsius or so...**even for long distance using 8 or 10-degree coolers is the preference. We've traveled as far as Alaska to procure hearts and haven't had any issues.**” -Transplant cardiologist at one of the top centers in the US/world

Vanderbilt indicated that TransMedics “absolutely” has no value proposition, with an exodus from the device among the top centers and those who participated in its trials: “they enrolled some of the highest-volume transplant centers in their clinical trials and these were the main ones using them until the last couple of years...there’s been a steady decrease in utilization of TransMedics”; “so many centers are actively involved in research, whether it’s Paragonix or XVIVO...or these other competing coolers that have really come out over the last couple of years...there’s less of an incentive to use the OCS for long-distance hearts”; “its usage steadily has been declining amongst the high-volume transplant centers, and I don’t know how many of the low-volume ones are using them.”

No value proposition; usage has tanked in heart among high volume centers

Q: “I would expect that 80% to 90% of the volume has to be the high-volume major centers of excellence, no?”

A: **“They enrolled some of the highest-volume transplant centers in their clinical trial and these were the main ones that were using them until the last couple of years** where amongst our group of folks that we talk to across the country who are other high-volume transplant centers as well, there’s been a **steady decrease in utilization of TransMedics.”**

Q: “So, you talked to other transplant centers, heart transplant centers, and they don’t use it for the same reasons you do?”

A: **“It’s gotten to the point that the folks that we speak with typically resort to it if there are issues with not having bandwidth,** etc. But because so many centers are actively involved in research, whether it’s Paragonix or XVIVO, whatever it may be, or these other competing coolers that have really come out over the last couple of years, there’s **less of an incentive to use the OCS for long-distance hearts** because we can just use these instead. So, it’s really coming to do I have a certain bandwidth?”

Q: “So, you can just use what instead?”

A: “Coolers, these things that keep the heart at a different temperature than the typical preservation method.”

Q: “So, you’re saying really has **no value proposition.**”

A: **“Absolutely. Its usage steadily has been declining amongst the high-volume heart transplant centers, and I don’t know how many of the low-volume ones are using them.”** -Transplant cardiologist at one of the top centers in the US/world

The cardiologist indicated that their remaining use of the heart device is solely as a last resort when their surgeons are tied up and they send an NOP retrieval team, which then bundles the device with the procurement service; and that, at best, their value proposition is for small centers without enough surgeons. However, he stated that “NRP is going to rapidly skyrocket in utilization,” and that a trickle-down effect from larger centers will lead the smaller centers - TransMedics last bastion of usage - to wind down as well: “NRP really is taking off...as more and more big academic centers...start spreading their surgeon graduates...around to other centers...we’re going to see smaller programs that don’t have an NRP program to start building their own”; “my suspicion, honestly, is more utilization at those low to mid-range programs who don’t have surgeon bandwidth.”

“NRP is going to rapidly skyrocket in utilization”

Q: “How much traction is NRP getting? Because if it’s taking off, then that’s going to be bad for TransMedics.”

A: “NRP really is taking off, especially for DCD organ.”

Q: “For DCD, it’s all going to be about NRP, basically. You’re seeing that take off in all the big academic centers?”

A: “At the big academic centers, for sure. And I think as more and more big academic centers do these and they start spreading their surgeon graduates or their faculty surgeons around to other centers as well because that’s normal. There’s some degree of turnover at academic centers. We’re going to see smaller programs that don’t have an NRP program to start building their own ones as well. I do expect that lung DCD is going to be a much bigger deal in the next five years and two, NRP is going to rapidly skyrocket in utilization..” -Transplant cardiologist at one of the top centers in the US/world

No value proposition; just for some smaller programs who need surgeons to retrieve a heart

Q: “So you haven’t seen much of a value proposition. You’re just using them as an on-call procurement service when you’re in a pinch?”

A: “Essentially, yeah. I think the programs that may have the potential and the biggest benefit in utilizing OCS currently are these smaller programs trying to build up their transplant volume who don’t have the upfront institutional capital or protocols in terms of NRP procurement, etc., so they can just hand off the procurement to another team [...] my suspicion, honestly, is more utilization at those low to mid-range programs who don’t have surgeon bandwidth, who don’t have to worry about trying to put in the upfront organization cost that are required to build up an NRP program because that’s still relatively new in the game. It’s been around for less time than the OCS system, so it’s harder to gain traction because you have to build your surgeon’s experience and your team’s experience with it, which takes more money in the beginning.” -Transplant cardiologist at one of the top centers in the US/world

Cardiac surgeons indicated that TransMedics NOP procurement/transport service has no value proposition for heart, leaving aside the device – they have no trust in third-party retrieval surgeons; and they demand hearts back faster than the NOP service can fly them: “I don’t know anybody in cardiac transplant as opposed to liver that are using a TransMedics-hired surgeon...the NOP offering, the soup-to-nuts thing, is useless for cardiac because I don’t know many cardiac surgeons that would ever trust anybody except their own surgeons.” A second surgeon stated “Do I trust somebody that I don’t know personally to go cut out an organ...and haven’t screwed it up? That’s a big trust factor that’s hard to overcome.” A TransMedics ex-logistics employee confirmed that NOP is dead-on-arrival: “the heart community is less comfortable with us...a refueling stop only adds 45 minutes...but they are still so sensitive about it....”

Zero value proposition for NOP in heart

*“Liver surgeons are more likely to use the NOP program because they’re okay with having someone else recover the liver. Whereas in cardiac transplantation, if you guess wrong or something technically goes wrong, you’re going to have a dead recipient on your hands back home. **I don’t know anybody in cardiac transplant as opposed to liver that are using a TransMedics-hired surgeon. The NOP market, the soup-to-nuts offering, is useless for cardiac because I don’t know many cardiac surgeons that would ever trust anybody except their own surgeons to do the procedure.**” – Prominent surgeon, Director of leading academic transplant center*

Lack of trust in third-party procurement surgeons is an insurmountable barrier in heart

*“A procurement service is counter to the way hearts and lungs have been harvested because they are infrequent events compared to routine heart operations - so there’s a big trust issue. **Do I trust somebody that I don’t know personally to go cut out an organ from a donor, bring it back to me, and I trust that they’ve done it correctly and haven’t screwed it up? That’s a big trust factor that’s a hard to overcome.**” - Cardiac transplant surgeon at a major Midwest academic center*

TransMedics NOP unable to meet heart surgeon requirements

*“**The heart community is less comfortable with us.** They rush to the point that if you are flying a plane that needs to stop to refuel, they will say no, get us a bigger plane that can fly farther. A refueling stop only adds 45 minutes to a 10-hour journey but **they are still so sensitive about it.** They don’t want to have even that extra hour added. That tells you that from a logistics point of view, we are still under that pressure, and **this is the biggest constraint, to make sure that our logistics network is able to support it.**” – Ex-logistics employee at Transmedics*

UCSF, Massachusetts General, and other top centers indicated the TransMedics heart pump serves no clinical purpose and is detrimental to the heart, confirming the devastating color from Vanderbilt. UCSF stated their heart transplant group used it as a last resort if a surgeon was too busy for a retrieval run, and that some use for convenience, but they are eliminating usage in heart just as they intend for liver – and that their lung usage is down to nothing: “it’s not like anything required the device but they would use it out of convenience, which is a very expensive backup plan...we were using it even it wasn’t clinically indicated”; “the heart program, our surgical director is really aggressive and wants to be doing NRP for every donor, so I think his goal would be to discontinue TransMedics”; we’re probably the fourth-largest program in the country right now, doing over 100 lungs...we’ve used it five times in the last year.”

UCSF surgeons only use occasionally for convenience, not for clinical reasons

*“Our heart transplant group was using it because they might not have a procurement surgeon available to go get a heart. **It’s not like anything required the device but they would use TransMedics out of convenience, which is a very expensive backup plan** if you don’t have a surgeon. We’ve since remedied that. We no longer have any issues with covering the procurement schedule, but for a while, we were **using it even when it wasn’t clinically indicated**. For lung transplant, our surgical director is very mindful of cost. [Redacted] will only use it when absolutely necessary.” – UCSF administrator*

UCSF discontinuing use for heart; almost no use in lung despite big program

A: **“The heart program, our surgical director is really aggressive and wants to be doing NRP for every donor, so I think his goal would be to discontinue TransMedics as well.”**

Q: “Heart and lung it just sounds like they were just really using it as a backup option at some point when they were in a pinch. It doesn’t really seem like something that is really part of their protocol. Is that fair to say?”

A: “I’d say for sure with lung that is true. I wouldn’t say lung is using it—heart was using it more as a backup for procurement at times. I have to say our lung program has grown a lot recently. **We are probably the fourth-largest program in the country right now, doing over 100 lungs. We’ve used it five times in the last year.** So, we’ve been able to manage to grow with the new allocation system without relying on TransMedics. For heart, I know it partly was a procurement convenience thing.” - UCSF administrator

Massachusetts General indicated the lung product has no use case and that cold storage is superior – universal feedback across our customer calls; and that the device is dangerous to hearts: “hearts tend to get boggy and fluid overloaded and congested on the pump”; “for lungs...we don’t see a use for OCS...we use a different device that’s static cold storage...the lungs are very resilient...we have a better technology...cold storage manned or shipped unmanned to the final destination...and the lungs are fine.”

MGH won’t use for lungs; no value proposition versus cold storage

A: **“For lungs, it’s more of the fact that we don’t see a use for OCS. We use a different device that’s static cold storage and the lungs are much more resilient than the heart, and so, we’ve actually done a lot more research around that. We actually prefer using that method versus using the OCS and it would take a whole different team to go out there. It’s called Lung Guard.”**

Q: “What’s the cost of that, just so I know the comparison to OCS? Their lung business has gone nowhere, so obviously, nobody’s really using it for lung.”

A: “With lung, it’s a little bit of everything. One is that the amount of centers that have lung transplants is much less than any other organ. Two, the lung distribution continues to expand everything way beyond what we were able to do. Three, the procuring teams for certain lungs, it’s very, very limited. So, even though there might be an excess of lungs, the team available to get those lungs to the right recipient is much less, so you really have a supply and demand issue. And then, I would just almost say four. **It’s also because the lungs are very resilient; they can be tuned up much easier than a heart or a liver, and we have a better technology that can put a lung that’s more resilient on it in static cold storage manned or shipped unmanned to the final destination of the recipient hospital. And the lungs are fine. So, there’s a little bit more of a different approach for lungs than there are for hearts and livers.**” – Veteran transplant administrator at Massachusetts General Hospital

Pump adversely affects hearts

“Our team goes out and procures the heart, and it’s usually used in conjunction for DCD hearts. And the duration on pump is much shorter. For heart transplant, there is some functionality that’s assessed on the pump, but the duration on the pump is much shorter because the **hearts tend to get boggy and fluid overloaded and congested on the pump...you don’t want to keep the heart too long on pump.**” – Veteran transplant administrator at Massachusetts General Hospital

Part 13. Competitive tsunami. TransMedics is being rapidly replaced by superior devices at a fraction of the cost, which began rolling out only recently. Customer interviews indicate delivery of OrganOx, Paragonix, and other devices at a weekly clip.

It is rare to see a medical device company in the middle of such a rapidly unfolding tsunami of new entrants and competitive disruption. The typical transition from monopoly device to a crowded space involves devices that are cheaper or superior – in this case they are both, dealing what we expect to be a fatal blow. The new entrants are each a fraction of the price – \$20-30K versus \$80-90K, with NRP at a mere \$3-10K per case. Thus, every surgeon, transplant administrator, OPO executive, and ex-employee we interviewed painted a dire situation with 1) FDA approved competitors rapidly rolling out and gaining adoption; 2) other devices on the cusp of FDA approval with surgeons enthusiastic about the trial data and potential; and 3) the accelerating rollout of a superior technique called NRP. A surgeon who leads a large center stated that “if I were Waleed’s people, I’d be selling stock like mad” – “I think they know damn well” that they’re in trouble. Another surgeon: “they’re sort of riding a wave right now...the stock is way over-priced and that thing is going to fall because once these devices come in and NRP catches on, people will see....”

Management should be “selling stock like mad” due to competition

Q: “I noticed the management and the board are selling stock hand over fist.”

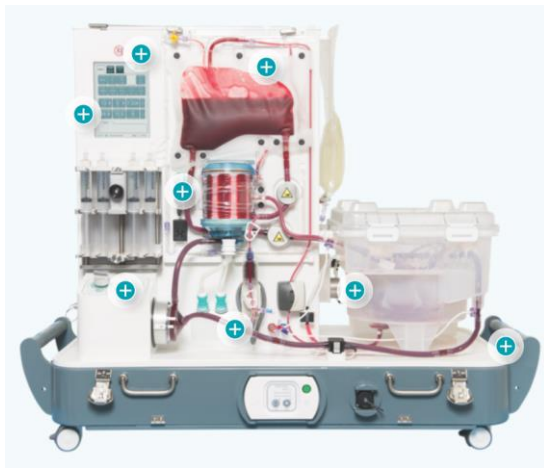
A: “**I think that they know damn well** – Waleed said they’re making a device that’s going to look a lot like XVIVO’s heart box. They know - even though they poo-pooed it - and **everything I know about LifePort and the research published out of Columbia shows that it is superior to TransMedics for liver**. I have no reason to believe that when Martine Rothblatt does for liver what they’re doing in lung right now with their back-to-base system, that they’re not going to take a big chunk away. **If I were Waleed’s people, I’d be selling stock like mad.**” – Prominent surgeon, director of a leading transplant center

Stock will crash with new entrants which are already coming online

“My own take is that they’re sort of riding a wave right now. The stock is way over-priced, and **that thing is going to fall because once these devices come in and NRP catches on, people will see** when the stock has gone from 7 to what in a span of two years. It’s way overpriced.. **they’ve started coming online. The XVIVO device is FDA-approved**. The Bridge to Life device is submitted for FDA approval and is going to get FDA approval soon. The LifePort device is approved and is getting some traction. NRP is certainly getting traction across the country.” -Transplant surgeon and transplant program director at a leading Northeast academic center

The surgeons provided a laundry list of newer devices and technologies: 1) OrganOx, normothermic pump basically identical to TransMedics OCS and less than a third of the cost, rapidly being installed at centers now after delays in producing enough units to satisfy overwhelming demand from transplant centers and OPO's; 2) Paragonix, a hypothermic system which is similarly gaining share at TransMedics expense; 3) LifePort Liver Transporter, a hypothermic perfusion machine and an ominous threat given LifePort's 90% share in kidney pumps; 4) VitaSmart's hypothermic system, with an FDA submission expected this quarter; 5) XVIVO's combo device with hypothermic, normothermic, or other perfusion modes; 6) United Therapeutics' Lung Bio-Engineering (NASDAQ: UTHR); and 7) NRP, which is currently spreading like wildfire and renders TransMedics obsolete.

OrganOx



Paragonix



Lifeport



XVIVO



Surgeons indicate that the new devices are the buzz at transplant conferences, with the field eager to defect to alternatives given widespread alienation with TransMedics business practices – and that the company is basically toast. We quote three surgeons below, all at large centers, two of whom are high-volume TransMedics devices, though not for much longer. Surgeon #1: “there was certainly hallway conversations at ATC...we’re glad there’s competition...almost everybody was saying that....I’ll leave it to you whether this would devastate their earnings...this is not going to stay exclusive much longer.” Surgeon #2: “if there is not much of an advantage of having this incredibly expensive setup vs. a \$30K different technology...they’re going to have a very hard time selling it.” Surgeon #3 stated the devices in the FDA queue are “guaranteed” to get approval: “the cost is going to go down...you are going to have other competitors...people are to going to start shifting gears...to a different pump because it’s not sustainable.”

Buzz at transplant conferences about new entrants and competition

“This was certainly hallway conversations at ATC. The discussion that people had in the hallways was, first of all, we’re glad there’s going to be competition, so TransMedics can’t be the only game in town. Almost everybody was saying that. Again, I’ll leave it to you whether this would devastate their earnings reports or not...this is not going to stay exclusive much longer.” – Cardiothoracic transplant surgeon, director of a leading academic transplant center

Will be game with if new devices are comparable as so much cheaper

“If there is not much of an advantage of having this incredibly expensive setup versus a \$30k different technology, then I think they’re going to have a very hard time selling it.” – Transplant hepatologist and director of the liver program at a Midwest academic center

“Guaranteed” FDA approval for new entrants and market shift to different pumps

“There’s going to be a point where the cost is going to go down. Right now, TransMedics doesn’t have big competition. The only competition is called OrganOX. That is the same type of machine perfusion. There are other new companies that are going through the FDA to get approved. Those are different types of pump like hyperthermic machine perfusion. So, I can guarantee you that these companies, they are going to get approved by the FDA. I think that in the next year or two years at the most, you are going to have other competitors with TransMedics. Because of the cost, people are going to start shifting gears, and they’re going to be changing to a different pump because it’s not sustainable for a transplant center....” – Transplant surgeon in a leadership role at Vanderbilt

In addition to being cheaper, surgeons stated the new devices have superior technology and trial data, for example: “everything I know about LifePort and the research published out of Columbia shows that it is superior to TransMedics for liver”; about United Therapeutics, “I have no reason to believe that when Martine Rothblatt does for liver what they’re doing in lung....that they’re not going to take a big chunk away.” Another surgeon, one of its top users with ~100 liver cases a year, is itching to switch: “I definitely think there are threats on the horizon”; “OrganOx...let you keep control...and it’s a third of the cost...that is definitely appealing”; “XVIVO and HOPE are going to be appealing to people...Paragonix”; “TransMedics got big market share, but the question is, what’s going to happen?” A third surgeon: “they’re going to see significant market erosion, especially with liver being their major source of income...liver is absolutely going to have competition from XVIVO and LifePort 100%...and VitaSmart.”

One trick pony with competitive threats on the horizon

“I definitely think there are threats on the horizon. I think OrganOX had some stumbles with getting enough machines to roll out. They offer a similar device, but they let you keep control over it, and it’s a third of the cost. So, that is definitely appealing. I think that the XVIVO and the HOPE are going to be appealing to people. Paragonix, which is cold storage at different temperatures. So, I think that TransMedics certainly got big market share push, but the question is, what’s going to happen...with other things coming online?... If you’re only looking at the whole ex vivo perfusion market...this field is going to have a lot of competitors. TransMedics is not going to be alone” - Transplant surgeon at a high-volume West Coast academic center

“Going to see significant market erosion”

“They’re going to see significant market erosion, especially with liver being their major source of income. Liver is absolutely going to have competition from XVIVO and from LifePort 100%. And VitaSmart, which was just presented last week...administrators are going to let the surgeons use whatever’s less expensive..” – Transplant surgeon, director of a leading academic transplant center

LifePort, XVIVO, other entrants

“The only companies in the U.S. approved for lung are XVIVO and TransMedics. For heart, TransMedics is the only device approved, but they’re competing against NRP, which is what we use. For liver, the only FDA-approved device is the TransMedics OCS. There’s also LifePort, which is in trials and has got very promising results. And XVIVO also has a liver perfusion device. And they’re all in various stages of phase 2, phase 3 trials.” – Prominent surgeon, Director of leading West Coast academic transplant center

Some of the newer systems are “back-to-base” which poses a particular threat to TransMedics, which is portable – “the data is not quite there to prove that a fully transportable device is superior, and logistically it’s easier and cheaper to Vita Smart rig the device after it’s been transported.” Non-portable pumps - located at hospitals, local OPO’s, or United Therapeutics dedicated perfusion centers – are less expensive and also provide more organ assessment and therapeutic capabilities, having no size constraints. TransMedics portability is irrelevant, as its primary use case is not clinical but scheduling and convenience – in other words, it is already mostly used off-label as a back-to-base device while OR’s wait for surgeons to wake up or stack cases. Moreover, our research indicated that a rapidly increasing use case is transporting organs to the center on ice and then putting them on TransMedics - precisely back-to-base usage.

Back to base is superior, cheaper, no benefit to fully transportable perfusion pump

*“Simultaneously and, in fact, ahead of TransMedics, XVIVO was working on this. They had a different concept. Instead of doing the fully transportable system that TransMedics has, their concept is what’s called “back to base.” And there are **several companies are out there that are like that, including VitaSmart who along with XVIVO feel that the data is not actually quite there to prove that that a fully transportable device is actually superior, and logistically, it’s easier and could potentially be cheaper to rig the device after it’s been transported to the transplant center or to the OPO.**” – Prominent surgeon, Director of leading West Coast academic transplant center*

United Therapeutics’ Lung Bio-Engineering has already rigged 600 lungs at perfusion centers

*“**Lung Bio-Engineering and United Therapeutics** have a completely different business model, which is they have their recovery centers, one in Silver Springs, Maryland and the other in Jacksonville, Florida, associated with Mayo. They are **rigging lungs, and they’re eventually going to do livers. [They] can physiologically evaluate a lot more than what TransMedics can do because they’re in a facility;** they’re not just on a little portable machine. And if the organs are considered acceptable, in the case of lung, based on oxygenation, lung compliance, and pulmonary pressures, then the organ is flushed cold again and then transported to the transplant center. Just to give you an example that TransMedics is not alone in this enterprise, there have been **over 600 lungs that have been rigged by the lung bioengineering folks in Maryland that have been transplanted from Toronto down to Duke all the way to Cleveland, New York at Columbia Presbyterian.** So, this is not some Star Wars concept; this is something that’s practically going on, and they, in fact, are going to go for FDA approval third quarter of this year, is my understanding from what they’ve said at the meeting.” – Prominent surgeon, Director of leading West Coast academic transplant center*

Our research indicates OrganOx penetration is increasing rapidly and taking TransMedics market share – both are nearly identical normothermic pumps - with centers installing devices and getting trained, starting aggressively about 6 months ago. We spoke with an OrganOx senior medical employee: “OrganOx is rapidly growing...they’re expanding quite quickly...they’re increasing their US presence quite a bit...things are going well.” One of the top liver transplant surgeons and KOL’s noted an unpublished paper he recently reviewed, which surveyed 43 transplant centers in the US – it indicated OrganOx already has more share than TransMedics: “I hadn’t heard this until I read it in the paper – more people are using OrganOx...in the United States than are using TransMedics...I would have thought it was the other way around.” The surgeon indicated his center went with OrganOx over TransMedics: “we elected not to do it for livers...OrganOx was about \$30,000, so it’s about half the price.”

OrganOx employee: “rapidly growing”

“OrganOX is rapidly growing...they’re expanding quite quickly...they’re increasing their US presence quite a bit. They have a new physical presence in New Jersey. Things are going well...it’s a virtual direct competitor to TransMedics...in layman’s terms, you can think of it as a machine that does essentially the same thing...these are the only two normothermic machine perfusion devices on the market...there are **a few others that are about to make an entry into the US market.**” – Senior medical employee at OrganOx

OrganOx already has more share than TransMedics

“I just reviewed this paper and it was a survey of all the transplant programs around the world. There were like 43 respondents in the United States, and what was interesting - and I hadn’t heard this until I read it in this paper - **more people using OrganOX for normothermic machine perfusion of livers in the United States than are using TransMedics**. And I would have thought it was the other way around. I would have thought TransMedics was higher than OrganOX.” – One of the top liver transplant surgeons in the world, based at a leading Midwest academic center

Perhaps the top liver transplant surgeon and KOL in the world declined to use TransMedics, uses OrganOx

“Although we signed onto it for livers, **we decided not to purchase it**. Initially, you had to buy half a million dollars worth of devices. They were charging \$60,000 each, and you had to buy 10 units. And so, basically, that was your investment into their program. **So, we elected not to do it for livers. OrganOX was about \$30,000, so it’s about half the price. That’s why we elected to stay with OrganOX.**” - One of the top liver transplant surgeons in the world, based at a leading Midwest academic center

Surgeons indicated enormous pent-up demand for OrganOx, which it was only recently able to start fulfilling. The head of a large academic center in Pennsylvania: “OrganOx is their main competitor...they’re coming out now...they have so many orders...it was like buying a Lexus when it first came out...you pay for it now and you’ll get it in 9 months....a third of the price....they literally don’t have enough machines...I’m not kidding...we have a contract out for one...we just don’t have it yet – and stated “it goes down significantly” when we asked how it will impact his TransMedics usage. Another surgeon – a top TransMedics customer who does ~100 livers a year with the device: “to be transparent, we actually just acquired an OrganOx for our institution.” A transplant hepatologist in a leadership role Massachusetts General noted good date for OrganOx’s approach and that it will “knock TransMedics off its high horse.”

OrganOx “coming out now” with huge pent up interest

“OrganOX is their main competitor. So, they’re coming out now. They have so many orders; it was like buying a Lexus when it first came out. You pay for it now, and you’ll get it in nine months. That’s **a third of the price**...OrganOX is now available to buy. ...It’s been out for a couple of years...it’s been probably six months to a year. **I think they literally don’t have that enough machines; I’m not kidding.”** – Transplant surgeon and transplant program director at a major academic center in Pennsylvania

TransMedics use “goes down significantly” after receiving OrganOx machine

A: “Yeah, it is FDA-approved. We have a contract out for one; we just don’t have it yet.”

Q: “Is that going to impact your TransMedics usage once you do the OrganOX?”

A: “Yes...**it goes down significantly.**” – Transplant surgeon and transplant program director at a major academic center in Pennsylvania

One of TransMedics largest customers just acquired OrganOx

“We have used OrganOX and **to be transparent, we actually just acquired an OrganOX for our own institution.**” -Transplant surgeon at a high-volume West Coast academic center

MGH: combo of back-to-base with OrganOx and NRP will make “TransMedics come down from its high horse”

“OrganOx’s has presented data around their approach...they have pretty **good statistics around the use of this strategy in livers**. Competitive forces are going to change what TransMedics does...**you’ve got two levers here that are going to knock TransMedics off its high horse and come down from its cost.**” – Transplant hepatologist in a leadership role at Harvard/Massachusetts General Hospital; key role in oversight of the liver transplant program

In addition to centers buying OrganOx directly, OPO's – of which there are 57 that run the US organ transplant system under UNOS – are now purchasing the machines themselves to set up back-to-base facilities for all transplant centers in their territories, thus making machine perfusion essentially free for every hospital they work with. The CFO of one of a Midwest OPO that's one of the largest in the US, stated they're on the waiting list for OrganOx: "whether it's because TransMedics is too expensive or OrganOx is better, there is a desire for our transplant center to do more OrganOx...that is a risk for the market share for TransMedics...liver is why OrganOx is taking off." A West Coast OPO and also one of the largest, stated OrganOx is about to eliminate TransMedics use in their territory, and that they tried to buy 3 machines: "if you see the liver use start to go down, I can tell...it will all be attributed right away to OrganOx...they can't produce the machines fast enough...as soon as deploy this program...we would have purchased three of them...they can only give us one at the moment."

OPO's keen to move to OrganOx

"Whether it's because TransMedics is too expensive or OrganOX is better, there is a desire for our transplant center to do more OrganOX when it becomes available. So, that is a risk for the market share for TransMedics...liver is why OrganOX is taking off...once we go live with it, we anticipate anywhere from 35 to 50 cases a year doing local, under 250 miles from [OPO headquarters city] cases using OrganOX. And that could grow if the transplant centers like it a lot because there are definitely more cases." – CFO of one of the largest OPO's (organ procurement organizations) in the country

Tsunami of OrganOx and NRP will lead to collapse of TransMedics usage "soon"

"If you see the liver use start to go down, I can tell you, at least for here in the Bay Area, it will all be attributed right away to OrganOX. They can't produce the machines fast enough. We would have purchased three of them, but they can only give us one at the moment. So, then as soon as we deploy this program, we are in conversations with the perfusion group to bring in NRP. And if we want to provide NRP on 100% of our DCD donors, which is the bulk of the OCS liver machine use, is on those DCDs." –

Executive at a leading West Coast OPO, longtime industry veteran

The West Coast OPO - which works with some of the highest-volume centers like UCSF, Stanford, and Sutter - stated they are simultaneously kicking off NRP and OrganOx, which they would use on “100% of our DCD donors, which is the bulk of the TransMedics liver machine use” – killing off their TransMedics use completely. Given the dire situation he painted for TransMedics, we asked when the window would close: “you’ll see it happen soon because of all these OPO’s...OrganOx has been having trouble producing the machines in these volume...they’re dropping them at an OPO, at this point, every couple of weeks with us being the second one”; “you want to know why TransMedics is going to blow up?...it’s really liver...most OPO’s I know hate working with them...to the point where’s we’ll all out there getting OrganOx devices...we’re going to do trainings at Stanford, CPMS, and UCSF...and then we’re live.”

Tsunami of OrganOx and NRP will lead to collapse of TransMedics usage “soon”

Q: “How long do you think the window is? It seems like it’s a fast-approaching wave, like a tsunami between OrganOX and NRP. It sounds to me like UCSF could be at risk, Washington could be at risk, Vanderbilt’s gone away. It seems like wildfire that people are just getting sick of them and are looking for alternatives.”

A: **“I think you’ll see it happen soon** because all of these OPOs, OrganOX has been having trouble producing the machines in these volumes, and it seems like they’re **dropping them at an OPO, at this point, every couple of weeks with us being the second one.**” – Executive at a West Coast OPO, longtime industry veteran

OPO’s aggressively moving to OrganOx; costs transplant centers nothing

“You want to know why TransMedics is going to blow up? You’ve picked up on their heart and lung products have gone nowhere. **It’s really liver. Most OPO’s I know hate working with them even though they’re not our vendor, to the point where we’re all out there getting OrganOX devices. We just got ours,** and the manufacturer is sending reps out here, and **we’re going to go do trainings at Stanford, CPMC, and UCSF. And then we’re live.**” – Executive at a West Coast OPO, longtime industry veteran

The West Coast OPO stated that OPO's are now eliminating the need for centers to use TransMedics, by simply offering OrganOx as a free, back-to-base service near the hospitals they work with – and named the largest OPO's in the country as quickly moving to the same model: “you can pick TransMedics and pay \$100,000-plus, plus you'll for my liver acquisition charge...or you could choose OrganOx normothermic perfusion and it costs you nothing, only the standard acquisition chare...that's it...that's all we charge....we're launching our program...I don't see why you, as UCSF or any of the transplant centers, would choose to use TransMedics...for any of the Bay Area donors....the group out of LA, the largest OPO in the country, they just received their machine...the OPO in Nevada has received their OrganOx machine...so all of these are going to be owned by the OPO and there is zero additional charge to the transplant center.’

OPO's aggressively moving to OrganOx; costs transplant centers nothing

“We are paying for that. And if you're a transplant center in Northern California, and you want normothermic perfusion, you can pick TransMedics and pay \$100,000-plus, plus you'll pay for my liver acquisition charge. Or you could choose OrganOX normothermic perfusion, and it costs you nothing, only the standard acquisition charge. That's it. That's all we charge. But we have three machine operators. We have purchased vehicles to move the machine around in. We're launching our program. I don't see why you, as UCSF or any of the transplant centers, would choose to use TransMedics over the OrganOX machine for any of the Bay Area donors. There's another OPO based out of Sacramento; it's a small group. They received their machine on the [redacted], and they'll be live on [redacted] The group out of LA, the largest OPO in the country, they just received their machine, and I believe they're arranging for their training sometime next month and that's going live. The OPO in Nevada has received their OrganOX machine. So, all of these are going to be owned by the OPO and there is zero additional charge to the transplant center.” – Executive at a West Coast OPO, longtime industry veteran

Paragonix, a hypothermic preservation system for the heart, lung, and liver that costs only \$10-\$20K, is also disrupting TransMedics usage. An organ retrieval specialist at a Northwest academic transplant center, who travels to operating rooms across the US, indicated rapid uptake of the devices in cases which previously used TransMedics: “you’re seeing more and more liver devices show up from Paragonix versus where it was TransMedics”; “liver is taking off like crazy because it’s enabling them to not have to use TransMedics for long hauls”; “typically that would have only been TransMedics a couple of years ago, but they’re changing the game”; “we’ve able to go further, and then we’re seeing other centers – East Coast, West Coast”; “we would never have seen lungs come from the East Coast with a Paragonix before...it’s rare that you go into an operating room now anywhere in the country and not see Paragonix...that was all TransMedics a year ago.”

Paragonix already taking TransMedics market share; only \$10-20K

“The Paragonix devices are changing the whole - I was in the operating room the other day – Mass General, was getting lungs from California, in a Paragonix. And typically, that would have only been TransMedics a couple of years ago. but they’re changing the game. Paragonix is just cold static storage but they maintain a temperature between 4 and 8 degrees, they’re not freezing and they’re maintaining them at these ideal temperatures. The University of Washington has taken the third liver from Hawaii, which was always coming back on TransMedics. We’ve done three livers at the University of Washington with Paragonix devices. We’re able to go further, and then we’re seeing other centers come—the East Coast, West Coast...the devices run between \$10,000 and \$20,000 a device, so significantly less.” – Organ procurement specialist at a Northwest academic transplant center

Paragonix liver “taking off like crazy”; for long-haul transport as well

“I mean, we would never have seen lungs come from the East Coast with a Paragonix device prior to this. That would all be TransMedics. It’s rare that you go into an operating room now anywhere in the country and not see a Paragonix device. And so, those longer hauls, like Hawaii livers, that was all TransMedics a year ago. It was all TransMedics. We did an Alaska lung that went to Duker, and that would have been TransMedics a couple of years ago - it started with heart. Lung was the second device, and liver is the third. And the liver is taking off like crazy because it’s enabling them not to have to use TransMedics for long hauls...it’s just getting busier and busier because of the ability to ship across the country. You’re seeing more and more liver devices show up from Paragonix versus where it was TransMedics.” – Organ procurement specialist at a Northwest academic transplant center

A leading transplant surgeon and KOL stated that OrganOx and Paragonix are merely the tip of the iceberg, with a second wave of devices in trials that will render TransMedics obsolete. The surgeon participated in TransMedics studies and interacts with management, offering an “inside baseball” view of the centers/investigators now conducting competing trials and stated that LifePort’s Liver Transporter is a critical threat, given its kidney device has a lock with 90% share, surgeon excitement for its liver data, and TransMedics alienation of surgeons: “the real competition based on everything I know of the Columbia Presbyterian data...is LifePort...centers I think are really good...are on board with the LifePort study...TransMedics did this hard sell, push down your throat approach to the surgical community...a lot of people they pissed off...people who would say I’m using LifePort because it’s not TransMedics; “we have no loyalty whatsoever to TransMedics...they are at real risk...LifePort is going to be very, very competitive and may exceed the interest level” for TransMedics.

LifePort is a key threat to TransMedics

“The real competition based on everything I know of the Columbia Presbyterian data, and I know James Guarrera, who is the lead investigator for that, is LifePort, the company I’ve already mentioned to you for kidney, has got their LifePort liver device, which is in a multi-center trial, and that device is transportable, so you can bring it to the donor OR. You can measure a whole bunch of physiologic parameters, not just lactic acid, like you can with the TransMedics device. And the guys at Cleveland Clinic and the guys at the University of Chicago, two centers I think are really good, both are on board with the LifePort study and, in addition, because of the fact that TransMedics did this hard sell, push down your throat approach to the surgical community saying this is our way or the highway, there are a lot of people they pissed off. So, I think even if LifePort trials showed the exact same results as TransMedics, I think there’s actually be people who would say I’m using LifePort just because it’s not TransMedics.” – Transplant surgeon, Director of a leading academic transplant center

LifePort will be “very, very competitive” and surgeons “have no loyalty whatsoever to TransMedics”

“If LifePort’s trial and subsequent FDA approval show that you can get more information or the same amount of information from LifePort than TransMedics provides, we have no brand whatsoever to TransMedics...they are at real risk.. my only point is that there’s going to be competition in liver.... knowing what I know about James Guarrera in Columbia, I think LifePort is going to be very, very competitive and may exceed the interest level for most liver surgeons in the U.S. once that device eventually gets FDA-approved.” – Transplant surgeon, Director of a leading academic transplant center

The KOL stated that top liver centers – “really big names in the field” – from the TransMedics trial have already defected to LifePort: “these are centers, some were in the TransMedics trial but they have clearly jumped ship to be involved with LifePort...some very, very good centers that are involved in the LifePort liver study...that’s the device that a lot of really big names in the field...think is going to be very, very competitive with TransMedics.” In addition, he pointed to VitaSmart’s hypothermic system which uses a temperature of 4 to 12 degrees Celsius to reduce an organ’s metabolic rate to 10% of normal and prevents re-perfusion injury common with TransMedics pump. It expects to file for PMA approval this quarter – the surgeon stated that recently presented data indicates no advantage for TransMedics vs. Vitasmart: “the decreases in EAD were fairly equivalent...which surprised a lot of people...it’s already approved in Europe..”

Top liver centers who were involved with TransMedics have defected to LifePort

“The LifePort study is a really good example. **These are centers; some were in the TransMedics trial, but they clearly jumped ship to be involved with LifePort** because that device has got real promise. And in that trial right now—and I know this for a fact because I know a lot of the investigators—you have Northwestern in Chicago, which is a major center in liver. You have the University of Chicago, which is huge. You have Columbia University, which is a big program. There’s Rochester. There’s the group at UVA in Charlottesville, and there’s the group in Utah. **So, there are some very, very good centers that are involved in the LifePort liver study...that’s the device that a lot of really big names in the field in liver really think is going to be very, very competitive with TransMedics**” – Transplant surgeon, Director of a leading academic transplant center

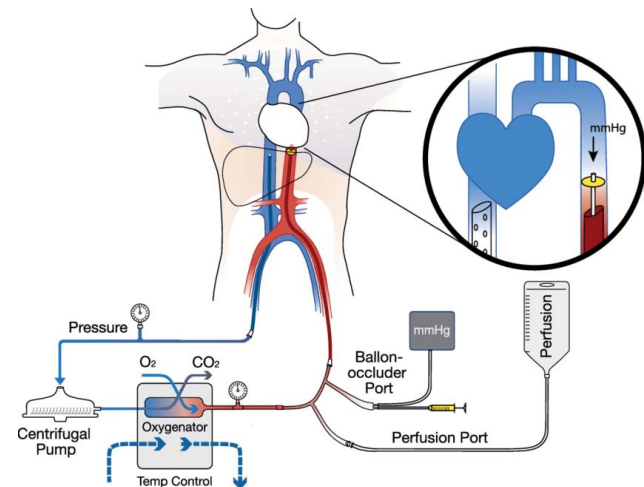
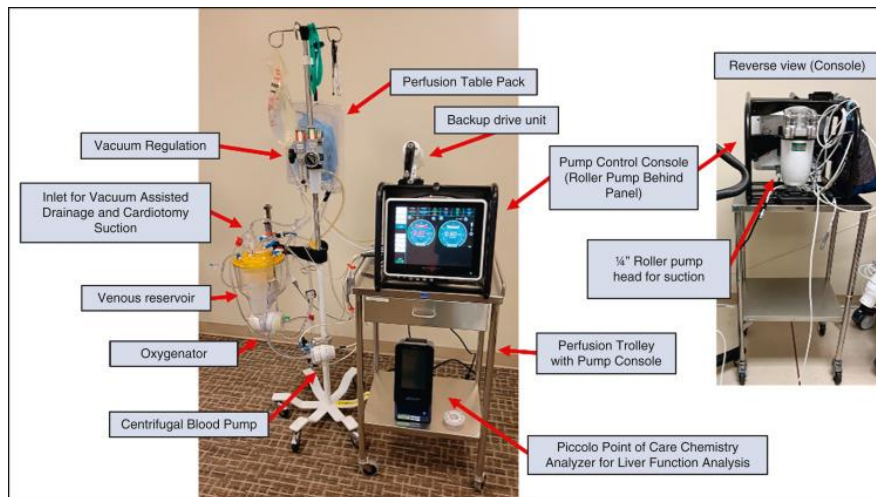
Surgeons in TMDX liver trial excited about recent new data for VitaSmart system

“The study that was the basis of the FDA submission for TransMedics for their liver approval was JAMA surgery in 2022, What’s interesting is that there was huge enthusiasm,, all the liver surgeons I work with in Region 5 in the United States, and a lot of those surgeons in those centers were in the trial. What’s interesting is the data that was presented at the ATC. The results that Dave Reich from Cleveland Clinic down presented at the meeting, their results were not all that different. If you look at their EAD rate for the stuff that was presented at the ATC, the experimental group, there was like 20% versus 40% in the standard criteria group, and for the TransMedics trial, that was the basis of the FDA submission, there was an 18% rate, which is almost the same as the 20% rate in the VitaSmart trial versus in the 30’s range for the ice chest storage. **The decreases in EAD were fairly equivalent, which surprised a lot of people when Dave presented the results last week. It’s already approved it in Europe. The data presented would argue that it’s actually not as big a difference as we thought.**” – Transplant surgeon, Director of a leading academic transplant center

Part 14. Disruption by NRP, spreading like wildfire. OPO's and transplant centers are standardizing around an inexpensive, alternative technique which has long been the standard of care in Europe, with TransMedics 2025 revenue at risk of collapse.

The recent explosion of normothermic regional perfusion (NRP) – an inexpensive way to retrieve DCD organs, the entire use case for TransMedics and driver of its growth – renders TransMedics essentially obsolete, given NRP is \$3,000 per case versus \$80-90K for OCS plus another \$40-100K or more for its NOP procurement service. Instead of removing an organ after circulatory death and resuscitating on a TransMedics, NRP is simpler and more logical – blood is pumped and recirculated while the organs are still in the body, using a \$1,000 ECMO pump (extracorporeal membrane oxygenation) commonly also known as a heart/lung machine. Circulating the donor's own oxygenated blood after cardiac arrest mitigates warm ischemia and enables organ assessment and preservation. NRP has been the standard of care in Europe for decades – with various counties mandating its use in DCD, by law – and explaining why TransMedics was dead-on-arrival abroad. The technique works for all organs.

Examples of cardiac and abdominal NRP circuits



Vanderbilt University – a transplant pioneer and perhaps the highest-volume center globally – also pioneered the use of NRP in the US, for both heart and liver. A transplant surgeon in a leadership role stated that NRP has now wiped out almost 100% of their use of TransMedics devices for both liver and heart – “it has been done in Europe for 20 years...over the last two years, Vanderbilt was the pioneer in using this technique” in the US. He stated they do 200 liver transplants a year, with TransMedics use now down to just 10 cases – “for each organ that you recover with TransMedics, it’s almost \$100,000. NRP is \$3,000, and with \$3,000 you can recover the heart, lung, liver, kidney, and pancreas...so \$3,000 versus \$300,000 if you want to recover all the organs. TransMedics cost is ridiculously high.”

NRP used in Europe for 20 years; Vanderbilt was the pioneer in US

*“NRP means—again, you have a brain-dead donor and what are called DCD. On the DCD, basically, when the donor expires, we put the donor on extracorporeal circulation, so we still can circulate blood throughout the body for one hour, one and a half hours, to help us preserve the liver and decreases the rate of having complications. That is a technique that initially—it has been **done in Europe for almost 20 years**, but here in the US, it never picked up for liver but it was for some philosophical or ethical reasons. But now, over the last two years, **Vanderbilt was the pioneer in using this technique here in the country**. So, right now, I think we are the biggest center doing NRP on DCD livers..” – Transplant surgeon at Vanderbilt*

Vanderbilt NRP usage has crushed use of TransMedics

A: “At Vanderbilt, each organ is a separate division. I am the [redacted] for liver transplant and hepatobiliary surgery. **We are on track to do 200 liver transplants this year.”**

Q: “And you use TransMedics for 10 of those a year?”

A: “**We are a big user of NRP or normothermic regional perfusion. Vanderbilt is like the pioneer center in the country with this technique. We do a lot of these NRP cases.** And we have a protocol on which cases to do the pump or the TransMedics machine. The main reason is cost. It’s cost-saving; that’s the main thing.” – Transplant surgeon at Vanderbilt

NRP is 3% of TransMedics device cost

“For each organ that you recover with TransMedics, it’s almost \$100,000. NRP is \$3,000, and with \$3,000, you can recover the heart, lung, liver, kidney, and pancreas. So \$3,000 versus \$300,000 if you want to recover all the organs. TransMedics cost is ridiculously high.” – Transplant surgeon at Vanderbilt

He stated that the only that the cost savings versus TransMedics “is insane...that’s the reason we do NRP so much...you can get one of these pumps for \$1,000...it’s very cheap.” He stated their local OPO is now standardized on NRP, eliminating the use of pumps like TransMedics: “our OPO is Tennessee Donor Service...they do NRP on every single DCD...we never use machine perfusion...our OPO has their own surgeon...the surgeon is recovering all of the organs via NRP.” The surgeon added that TransMedics badmouths NRP because they know it’s a serious threat: “TransMedics speaks a lot of garbage about NRP...they talk trash because it’s a big competitor and way cheaper.”

OPO’s trained to do NRP’s almost exclusively vs. TransMedics

A: “Our OPO is Tennessee Donor Service in our area. So, **they do NRP on every single DCD**. So, every time that we have a DCD, our OPO will always use NRP. **We never use machine perfusion**. But when we have to recover a DCD in a different area, in those cases, we use TransMedics. Those are the 10% cases that we do per year. They are done outside Vanderbilt.”

Q: “So, you trained your OPO in NRP, basically?”

A: “Yes, we did. Our OPO has their own surgeon, and the surgeon is recovering all of the organs via NRP. **So, our OPO, they do all NRPs.**” – Transplant surgeon at Vanderbilt

NRP is so inexpensive “it is insane”

A: “To do NRP, the main thing that you need is you need a perfusionist because you need to bring a little pump with you. It’s a portable pump; it’s very cheap. **You can get one of these pumps for \$1, 000, and this is disposable; it’s very cheap.** You have to go with this little pump with a perfusionist, so then you can put the donor on this extracorporeal circulation during the recovery of the organ.”

Q: “There’s literally a \$1000 device and a total cost of \$3000 versus \$100,000.”

A: “**It is insane. And that’s the reason that we do NRP that much.** That’s the main reason....TransMedics speaks a lot of garbage about NRP. **They talk trash because it’s a big competitor and way cheaper.** And more and more and more centers are using NRP now.” – Transplant surgeon at Vanderbilt

Our interviews indicate that OPO's and transplant centers large and small have now begin aggressively rolling out NRP. An Oct 24, 2024 paper just published in JAMA surveyed 55 of the 57 OPO's in the US to gauge their rate of adoption – 49 indicated they already have experience with the technique, indicating it is on the cusp of becoming the standard of care as it is in Europe. Transplant surgeons pointed to enthusiasm and buzz around other recent conference presentations and published papers, showing extraordinary cost savings versus TransMedics-types pumps as well as superior outcomes – such as another Oct 2024 paper in the American Journal of Transplantation showing an all-in cost labor and consumables of \$9,000 per organ with NRP vs. \$80,000 for machine perfusion.

Two Oct 2024 papers: rapid adoption and cost savings from NRP vs. machine perfusion

Original Investigation | Surgery

October 24, 2024

Normothermic Regional Perfusion Experience of Organ Procurement Organizations in the US

Our findings that virtually all OPOs expressed a desire for information and/or resources to develop and/or refine their NRP process should prompt action. Given the expansion of NRP across the US and our finding that the majority of OPOs have participated in NRP recoveries, it seems clear that this recovery technique will continue to be used as part of DCD donation. Recent reports show increased organ utilization from DCD donors with NRP compared with SRR in the US.^{6,16}

Source: <https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2825303>;
[https://www.amjtransplant.org/article/S1600-6135\(24\)00237-5/abstract](https://www.amjtransplant.org/article/S1600-6135(24)00237-5/abstract)

Association of procurement technique with organ yield and cost following donation after circulatory death

DCD-SRR with ex-situ NMP. In our institutional experience, the total average cost per DCD-NRP donor was \$9463.22, whereas prior reports suggest a per allograft cost of \$80 000 for DCD-SRR with ex-situ NMP⁴⁵⁻⁴⁷ However, it is important to consider the

Table 3
Normothermic regional perfusion cost projections.

	Seventy cases (Δ = -20%)	Eighty-eight cases FY24 forecasted	One hundred six cases (Δ = -20%)
Total indirect fixed cost per case (\$)	2542.87	2022.72	1679.25
Total average direct variable cost per case			
Completed run (\$)			
Labor	1550.00	1550.00	1550.00
Disposables	5250.00	5250.00	5250.00
Pharmacy	1281.00	1281.00	1281.00
Total	8081.00	8081.00	8081.00
Incomplete run (\$)			
Labor	1550.00	1550.00	1550.00
Disposables	5250.00	5250.00	5250.00
Pharmacy	-	-	-
Total	6800.00	6800.00	6800.00
Total average cost per case (\$)			
Completed run	10 623.87	10 103.72	9760.25
Incomplete run	9342.87	8822.72	8479.25
Averaged run	9983.37	9463.22	9119.75

Ww earlier quoted various surgeons who indicated that NRP is not only radically cheaper but that it enables better organ assessment and preservation as the organs are still inside the donor – “there’s no alternative other than NRP, really”: “the heart on the TransMedics device is beating but it’s empty...it’s not supporting circulation...it’s not generating a pressure...you’re guessing”; “there are limitations in terms of the actual assessment of the heart on that device...it’s not the most robust test of organ quality...there’s no way to assess it...with a TransMedics device, there is no resistance...if you resuscitate with NRP...you can actually have a clear assessment...that’s missing on the TransMedics device: true functional assessment of this organ...if it looks good on the machine, will it look good once you put it in.” Unsurprisingly, OPO’s and centers are flocking to NRP, either training their own staff or contracting with inexpensive, third-party NRP perfusion services which are proliferating – and marketing via Google ads.

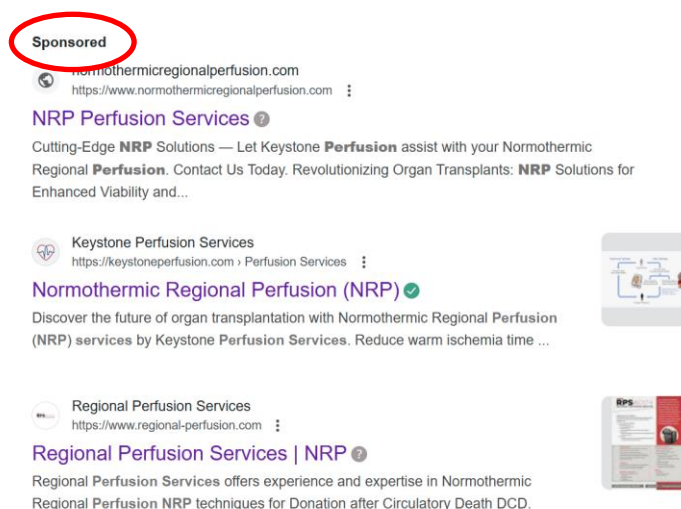
Google ads for third-party NRP services; example provider



CONTACT US

**Let us provide the
Normothermic Regional
Perfusion (NRP) Services that
you need.**

Our unique ability to provide immediate, and potent emergency NRP coverage, bolstered by our successful track record in staffing health systems around the country makes us an enviable NRP services partner.



Some previously large users of TransMedics like Vanderbilt have already eliminated its use after moving to NRP. Other large customers indicated they are on the cusp of adopting NRP. One of TransMedics largest users, with 100 livers per year on OCS, stated NRP is better and cheaper – a canary in the coal mine: “NRP for DCS’s...has the potential to really eat into the overall volume for TransMedics...personally, I feel quite confident that the outcomes are better with NRP than they are with OCS...NRP, I think, is better...and it’s obviously significantly cheaper...we are starting to see OPO’s offer their own NRP services.” Massachusetts General stated they’ve now wetted their beak: “we’ve done a few NRP’s...there is a very strong case for changing...our economic model to NRP...you’re going to start seeing real dents in the market share for TransMedics.”

NRP taking off; outcomes are better; OPO’s beginning to offer it

Q: “Where is that NRP trajectory in the liver community?”

A: “It’s a great question, at least when it comes to DCDs because really you only do NRP for DCDs, that has the **potential to really eat into the overall volume for TransMedics. Personally, I feel quite confident that the outcomes are better with NRP than they are with OCS...NRP, I think, is better. And it’s obviously significantly cheaper.** That said, it does require more manpower of some form or another, whether it’s on the OPO side or whether it’s on the transplant center side, to do the NRP...and **we are starting to see OPOs offer their own NRP services.** Las Vegas, for example, the OPO in Las Vegas. The OPO in San Francisco, CADN. I know that the OPO in Los Angeles, which is one of the biggest in the country, is starting to roll that out as well. They’re getting their own surgeons, so they’re going to start offering it. Arizona is offering it as well.” - Transplant surgeon at a high-volume West Coast academic center

MGH interested in moving to NRP

A: “I mentioned other perfusion devices but NRP is an even more salient comparator here....our NRPs have been restricted to very local sites...**we’ve done a handful; we’ve done a few NRPs.**”

Q: “Are you going to shift to NRP or are you kind of slower than some of these other centers?”

A: “It’s a great...**I think there is a very strong case for changing our business model or changing our economic model to NRP...I think we’re going to see NRP just gain in stature and in use, and I think you’re going to start seeing real dents in the market share for TransMedics**...they can’t sustain that kind of cost model, and it really seems haughty...to even think they can get away with the kinds of additional charges that they’re pinning on centers.” – Transplant hepatologist Massachusetts General Hospital

Consistent with the Oct 2024 paper that surveyed 55 OPO's on their NRP adoption, our interviews indicated OPO's are rapidly rolling it out to transplant centers in their respective geographies. The high-volume TransMedics user on the previous page noted the OPO's in Las Vegas, San Francisco, Arizona, and Los Angeles – the largest in the US – as among these. We quote two OPO executives below: “NRP is even cheaper...an OPO out of Seattle that covers parts of Alaska, Montana...some transplant centers have really figured it out beautifully...in Salt Lake City...they're got a small, tiny device and a great team, and they travel all over the place doing NRP, bringing all their organs that way.” A second executive: “we're just trying to iron out all the kinks before we offer it to more...we'd like to do more NRP cases...I know we're late to the game...we know of other OPO's and transplant centers that only use that now...I don't know if it's a Texas or Louisiana OPO....who is just interested in only using NRP now...I just thought that was a dramatic change.”

NRP taking off with OPO's and centers who have “really figured it out beautifully”

Q: “And I've also heard NRP is taking off, particularly in heart.”

A: “**Yes, NRP is even cheaper**...here in California—there's an OPO out of Seattle that covers parts of Alaska, Montana, you need a big team; you need perfusionists, a really specialized team on top of the transplant surgeons that will need to come in and run the NRP. So, sometimes, there are **some transplant centers that have really figured it out beautifully, like Intermountain Medical Center in Salt Lake City. They've got a small, tiny device and a great team, and they travel all over the place doing NRP, bringing all their organs in that way.**” – Executive at a West Coast OPO, longtime industry veteran

OPO's increasingly moving to NRP exclusively

“And then we started to do NRP...we're just learning it. We're actually a little late. We were very cautious. But one of our transplant centers, [high volume transplant center - redacted], has capacity and we've been using one of their facilities to do some cases. **We're just trying to iron out all the kinks before we offer it to more** than just [center name redacted]...**We'd like to do more NRP cases...I know we're late to the game, meaning that we know of other OPOs and transplant centers that only use that now...** I don't know if it's a Texas or Louisiana OPO; it's one of our friends in the south that my CEO met recently **who is just interested in only using NRP now...I just thought that was a dramatic change.**” – CFO of one of the largest OPO's (organ procurement organizations) in the country

A transplant surgeon who heads the program at a leading Northeast academic center stated they've already eliminated 1/3 and 1/2 of their TransMedics liver and heart use, respectively, by moving to NRP. His comments suggest the rest of the volume is at high risk, with liver being cut again shortly to half: “why should it cost \$70,000-80,000 for disposables?...the disposables are essentially plastic tubings and plastic containers...that’s more than the cost of a brand-new luxury car...paying for a set of things you throw in the garbage...it is, no question overpriced...there’s lots of precedence for this sort of stuff happening in the healthcare industry.” He stated the price cuts will be like the first Hepatitis C treatments like Harvoni when fell from \$120K to \$20-30K – “I think that’s what will happen here...all these other devices are coming in....no question NRP is picking up....last year [we] started utilizing NRP and ended up doing 50/50 for NRP versus OCS...so, that’s also going to cut into it”;

Overpriced and being disrupted by new entrants and NRP

“Why should it cost \$70,000-80,000 for disposables? The disposables are essentially plastic tubings and plastic containers and cartridges, and that’s more than the cost of a brand-new luxury car that you’re paying for a set of things that you throw in the garbage, essentially. It is, no question, overpriced...there’s lots of precedence for this sort of stuff happening in the healthcare industry. If you remember when the first medications came out for Hepatitis C, Harvoni. Harvoni was priced at \$120k per treatment when it first came out, and that’s because it was the only treatment that was available at that time. But **within a year, there were five other treatments that were available and the price went from that to about \$20-30k. I think that’s what will happen here.** You now have OrganOX; you have XVIVO, you have Bridge to Life, you have LifePort—so you have **all these other devices that are coming in**, and the price will go down because there are all of these competitive products on top of which there’s now NRP, and **there’s no question NRP is picking up.** As I said, our center has started using it. **This year, we’ll see probably two-thirds of our cases with OCS and one-third with NRP...I think what will probably happen is it’ll be 50/50: half the cases will be with NRP, and half will be with OCS. I know our heart program, for example, was exclusively using OCS for their DCD cases for two years, but then last year, started utilizing NRP and ended up doing 50/50 for NRP versus OCS. So, that’s also going to cut into it.”** – Transplant surgeon and transplant program director at a leading Northeast academic center

Part 15. TransMedics device is at risk of recall, plagued by what we think is a high failure rate, resulting in necrotic livers with dead tissue and other issues. Surgeons report losing a significant percentage of organs, and allege a cover-up by dishonest management.

Ex-employees, surgeons, and OPO's indicate that TransMedics OCS devices are plagued by failures leading to the loss of a significant percentage of organs; that livers in particular are prone to becoming necrotic, essentially rotting on the device, with dead tissue and parts of livers falling off; that the issues are prevalent enough that customers question why there hasn't been an FDA recall; and that TransMedics is allegedly engaged in a systematic cover-up by lying to physicians, failing to report device failures to the FDA as required, with the CEO allegedly pressuring employees to doctor safety reports. We begin with an ex-employee in medical safety roles, who stated that "it was a pretty complex device, so many malfunctions...kinks so the fluids and the gas were not able to flow properly...between 5-7% of the time, they lose the organ because of a failure." The employee indicated a cover-up: "yeah, that was definitely the case...I participated in those investigations heavily...when it came to reporting, everything was done to basically not report as much as possible."

Significant percentage of organs allegedly lost to device failure

"It was a pretty complex device, so many malfunctions....there were some kinks so the fluids and the gas were not able to flow properly. Some faulty filters...I think it's **between 5-7% of the time, they lose the organ because of a failure.**" - Former employee in medical safety and clinical affairs

Allegedly trying to "not report" safety incidents "as much as possible"

Q: "There are not many cases reported in MAUDE. Is the company just not reporting them to the FDA?"

A: "I think there's **a certain number of events that go unreported**. Their system is lacking a lot historically in terms of reporting."

Q: "Did you ever try to report something and you were told by the CEO or the VP that it doesn't need to be reported? Were they trying to underreport?"

A: "**Yeah, that was definitely the case...when some malfunction was recognized, everyone was blamed but the device**, so the users, the support staff, the engineers and such and such. There was a lot of finger-pointing and blaming...upper management was taking every failure very, very personally. The investigation was very thorough for every case of malfunction. That much I can tell you because **I participated in those investigations heavily. But when it came to reporting, everything was done to basically not report as much as possible.**" - Former employee in medical safety and clinical affairs

When we asked if the CEO was the one pushing to conceal device/safety events, the ex-employee stated “yeah, yeah, absolutely” and alleged “a lot of changes” to replace compliance-minded staff with more amenable ones – “I was asked to consistently phrase stuff differently, especially safety...I had meetings with Waleed on multiple occasions, when he said I don’t like what you wrote here...why don’t we just try to rephrase it?...it went through cycles and cycles of editing until he was satisfied...when I was writing my narrative, he’d be like, well, you need to write here that this death is not device-related...he was, like, you have to write it’s not related to the device...I’m like, I cannot write this....” The ex-employee indicated similar practices with respect to safety data submitted to the FDA prior to approval: “TransMedics data was always under scrutiny because of multiple violations...and warning letters...because the FDA was aware that they are not conducting themselves in the most honest manner.”

CEO allegedly pressuring staff to doctor and cover up device safety reports

Q: “Was it the CEO saying device malfunction or safety incidents don’t need to be reported?”

A: **“Yeah, yeah, absolutely.** He takes everything very personally, and so a lot of changes in the quality department – from people trying to reinforce industry standards as to those listening to him. **So, yes, he’s one of those crazy visionaries who cannot take criticism.**

That is absolutely a correct assessment.”

Q: “Were you ever stopped from reporting something by him?”

A: **“I was asked to consistently phrase stuff differently, especially safety. I had meetings with Waleed on multiple occasions when he said I don’t like what you wrote here** because it’s finger-pointing. Why don’t we just try to rephrase it? It went through **cycles and cycles of editing until he was satisfied; how does this read?** I was forced, as they say, to kind of brush it or make it more presentable. **When I was writing my narrative, he’d be like, well, you need to write here that this death is not device-related.** And I was, I can’t write it because I’m not sure. This went through independent adjudication with my clinical events committee. They had their doubts. They never said that this is not related 100%. **He was like, you have to write it’s not related to the device. I’m like, I cannot write this.** Things like this... **everything that went to the FDA to get the device approved..** In terms of the wording, the FDA - **TransMedics’ data was always under scrutiny specifically because of multiple violations** that they’d committed, and warning letters and such because the FDA was aware that they are **not conducting themselves in the most honest manner.”** - Former employee in medical safety and clinical affairs

The ex-employee further alleged that TransMedics has no safety function or even one person in such a role, and that the VP of Global Regulatory Affairs featured on its management team is just “an ornament because she worked at the FDA before...she was never in the office...she was not really involved with anything...she had never been to Boston...she was living in Washington, DC...she was not really engaged with any of us...we didn’t know where she was or what she was doing.” The ex-employee further alleged that TransMedics had only one person in a safety role but terminated the entire function: “I still talk to a few people at the company...they do not have a safety person...they haven’t hired anyone...it was a one-person operation...doing all safety for all trials and devices.” We conducted a brief LinkedIn search to check, which showed ~900 employees but none with a profile consistent with such a safety role; the search did indicate a handful of clinical and regulatory affairs employees, albeit without any detail suggesting that their roles encompassed safety.

VP of Regulatory Affairs is allegedly just “an ornament”

A: “Especially the vice president of regulatory, she was hired, but **she was more, so to speak, an ornament because she worked at the FDA before.** So, you know, to be able to say on their website that so-and-so used to be at the FDA. But **she was never in the office.** She may be close to 70 years old, but maybe they were just using her name. **She was not really involved with anything. She had never been in Boston. She was living in Washington, DC.”**

Q: “Was she on calls? Was she working? Was she interacting with you or no?”

A: **“Not really. And she was not really engaged with any of us.”**

Q: “Was she working with the CEO?”

A: “No idea. We didn’t know where she was or what she was doing... **I still talk to a few people at the company. They do not have a safety person. They haven’t hired anyone.** They haven’t replaced that position.”

Q: “You said they let go the entire safety team?”

A: **“It was one person. It was a one-person operation who was doing all safety for all trials and all devices.”**

Q: “They only had one safety person the entire time?”

A: “Yeah.” – Former employee in a regulatory affairs role

A medical director at OrganOx - which markets a nearly identical FDA-approved normothermic perfusion pump – indicated the device problems were even more widespread at 10-20% of transplant cases: “one recurrent theme that I seem to be hearing about...is the reliability of these machines...while an organ is on a machine, things can go very, very wrong...I’ve seen maybe...10-20% have some sort of issue...if we had to discard an organ due to a machine error, that gets reported full stop...it’s take a very loud voice to do that...I don’t know if TransMedics has that.” The ex-TransMedics safety employee stated some centers stopped using the device due to such issue: “I don’t think the patient outcomes were as good...they had actually one of the worst outcomes.” Another ex-employee, an OCS device operator, confirmed that centers ceased using the device due to poor outcomes: “I heard a lot of complaints...they were worried about the outcomes...they were saying that something had gone wrong.”

Centers allegedly stopped using the device due to device problems

Q: “Were there any centers that stopped using it because they had problems with the device?”

A: “Yeah, there were. **The University of Washington had some problems. I don’t think the patient outcomes were as good; they had actually one of the worst outcomes.**” - Former employee in medical safety and clinical affairs

Hospitals concerned about poor outcomes

“I’ve heard a lot of complaints about TransMedics. Someone told me recently at the hospital that they were **worried about the outcomes - the outcomes from a couple of hearts that they’d gotten from TransMedics.** There was a poor outcome on two different hearts that they had used TransMedics for. They were saying that something had gone wrong.” - Former TransMedics OCS specialist

Allegedly widespread device malfunctions and lack of reporting on errors

“One recurrent theme that I seem to be hearing about - his isn’t restricted to OrganOX but before then, when I worked at a different transplant company - **is the reliability of these machines.** There’s a lot of tubing and a lot of moving parts. Sometimes things go wrong, and there are malfunctions and defects. **While an organ is on a machine, things can go very, very wrong.** So, there’s a fair bit of that where there’s just a lot of QA in play here that’s really applicable to every organ care system...**I’ve seen maybe, let’s say in a total number of 10 transplants, maybe 10% to 20% have some sort of issue**...the allograft should be treated as a human being, that’s how I treated every single case. If we had to discard an organ due to a machine error, that gets reported full stop. **It takes a very loud voice in an organization to do that. I don’t know if TransMedics has that. I sure hope they do.**” – Medical director at OrganOx

One of the most serious issues – which came up repeatedly in our interviews – is livers becoming necrotic on the device, a known risk specific to normothermic perfusion machines like TransMedics but which our research indicates is far more widespread than known – with surgeons alleging a cover-up by the company, a refusal to publish related safety data, and raising the need for a device recall. For reasons we shall explain, liver tissue begins to die and renders the organ unusable, or leads to poor outcomes in the recipient. We were first alerted to the issue by the CEO of a large OPO, who stated TransMedics was trying to keep it “hush-hush” and that he knew a TransMedics procurement surgeon who refused to use the device because it was killing liver tissue - and resigned: “have you heard about the concerns around necrotic tissue on the liver when it goes on a TransMedics device?...when they put the liver on it, it’s almost upside down...concern as there were a couple of livers where this pressure was too high...the tissue was necrotic – it was dying...it was kept pretty hush-hush... I heard about it though a surgeon that used to do recoveries for them...he refused to do it...he’s like, ‘I’m not doing recovery for TransMedics anymore...he left.’”

Procurement surgeon allegedly resigned due to livers becoming necrotic on the device

A: “Have you heard about the concerns around the necrotic tissue on the liver when it goes on a TransMedics device? When they put the liver on it, it’s almost upside down, and there’s some concern as there were a couple of livers where the pressure was too high and with the warm ischemic - or the normothermic perfusion and the pressure, that some of the tissue was necrotic on the outside of the liver - it was dying. But I think it was kept pretty hush-hush, and they blamed the surgeon and said they didn’t put it on the device correctly. I heard about it through a surgeon that used to do recoveries for them, and then, ultimately, he refused to do it. He’s like, “I’m not doing recovery for TransMedics anymore,” because I guess when you’re doing recoveries for them, you have to sign an agreement that says you’ll never recover an organ in any other way than - as opposed to normothermic perfusion.”

Q: “So, you’re saying one of TransMedics’ transplant surgeons refused to do surgeries for them? He resigned because of this?”

A: “He left.” – CEO of a major OPO (Organ Procurement Organization), longtime industry executive

We then spoke to this former TransMedics procurement surgeon, who provided some of the most damaging color on the company's practices, as detailed in other chapters. He stated livers become necrotic as the machine's reservoir "scrunches" them into a small container: "it happens even more with TransMedics...it doesn't have enough blood supply...the reservoir....can be tight or the liver can be under pressure...a portion of the liver can get necrotic before or after transplant...if it happens after transplant, the outcome could be bad...it's reported only in the normothermic machine...I know for sure that happened...I don't know how many times....I know for sure that it happens."

TransMedics pump "scrunches" organs into a small area and results in necrotic livers

Q: "[Redacted – CEO of major OPO] mentioned the issue of liver necrosis where the liver is upside down. He said there's **an indentation and they lost some livers because of it**. He stated you didn't want to be involved with the device anymore because it had some issues. I don't know if that's accurate, What is the issue?"

A: "The reservoir that holds the liver, sometimes the liver can be big, and it's just scrunched into a small area, and with the other machine, it happens even more than with TransMedics; **a portion of this liver is necrotic** - it doesn't have enough blood supply and gets necrotic. The reservoir of the liver itself, where they put the liver in the machine, can be tight, or the liver can be under pressure. And **a portion of the liver can get necrotic before or after transplant. Before transplant, that's unfortunate, but you can't always tell from looking at the liver. If it happens after transplant, the outcome could be bad. It's reported only in the normothermic machines. I know for sure that happened. I don't know how many times**...I know for sure that it happens." – Transplant procurement surgeon previously working with TransMedics

While the clinical literature on normothermic perfusion-induced liver necrosis is lacking – and surgeons told us TransMedics is refusing to publish the data even though they’ve pushed - we note a handful of papers that reference “cradle signs” that appear on livers from being jammed into a pump. Images A and B (middle) both show the noticeable compressions marks immediately after perfusion. Image B (far right) shows the persistence of compression injury post-transplant, as the latticed indentations are visible on a CT scan of the recipient. The couple of papers on the topic attempt to downplay its significance, but are dated and prior to wider adoption of perfusion – in contrast to current color from the TransMedics ex-surgeon, and other surgeons whom we shortly detail, which indicated the seriousness of the issue.

Clinical literature on “cradle signs” from machine perfusion

Machine Perfusion—Leaving Its Mark on Liver Transplantation



The risk is exacerbated as livers are kept on the pump for 10-40 hours – far longer than the few hours in the pivotal trial – while surgeons “get their beauty sleep.” TransMedics recklessly appears to promote the device as capable of keeping livers going for days, with an ex-employee even making claims akin to quackery, suggesting organs are regenerating and “healing.” A medical director at OrganOx expressed alarm at the risk of extended perfusion times – similar to various surgeons we interviewed who have declined to use the device: “to put it mildly, a lot of different things are going to happen clinically to that organ...if you have blood doing through very, very small tubing...that is not healthy...these plastic tubes...red blood cells lyse and cause hemolysis...enzymes will leak out of the liver...all of these markers are essentially early markers of liver failure...increasing at a very high rate if the liver is on for an extended period”; “infection risk increases the longer this thing is not transplanted....”

Injury to livers accelerates rapidly after long duration on the pump

Q: “I’ve heard they’re keeping livers on this for a really long time. **What risk is there in keeping it for 10-11 hours versus 4 hours?**”

A: **“To put it mildly, a lot of different things are going to happen clinically to that organ. Number one is blood. If you have blood going through very, very small tubing, that is not healthy in the long term.** Our own blood vessels are slightly elastic in that they kind of move a little bit. These plastic tubes, which these PRVCs have to move through are more rigid. Because of that, blood, these **red blood cells lyse and cause hemolysis** and not carry enough oxygen to the organ itself. So, that’s a very real thing that happens with these organ care systems. That’s number one. Then, there is the performance of the liver itself. I talked about transaminase being high. These **enzymes will leak out of the liver** if it’s sick, and this is a very traumatic thing for a liver to go through. So, it will periodically increase its own transaminase levels and it will show you that it’s not performing well. Similarly, with another biomarker called lactate, that will also increase. **All of these are essentially early markers of liver failure...**transaminase and lactase are going to start **increasing at a very high rate if the liver is on for an extended period of time.**” – Medical director at OrganOx

Infection risk from extended perfusion time

“Infection risk increases the longer this thing is not transplanted. For example, if you’re going to transplant something from Nevada to Seattle or vice versa or a very long distance, that’s when these kinds of limitations come into play where transporting it becomes a bit of a logistical nightmare. **Purely from a surgical point of view, there are some concerns there...The liver itself is more sensitive to that than other organ systems.**” – Medical director at OrganOx

The Chief of transplant at a center in a major US city provided a recent example of a necrotic liver where “part of the right lobe of the liver was just completely dead” – which the surgeon then realized, after reaching out to colleagues around the country, is a prevalent issue. The surgeon indicated that the TransMedics retrieval surgeon was aware of the damage but withheld it – although, as indicated previously, surgeons are prevented from communicating with receiving centers, leading us to think that TransMedics was told but buried it: “situations where the liver is on pump...the numbers may be favorable but there’s something we don’t like about the liver...there was a liver a month ago where the right lobe felt very firm, like it was an infarction, that part of it had died...one of my colleagues was already at a point of no return in the recipient operation...he had already started before realizing that this was happening...so we ended up having to use that liver, and sure enough, post-transplant, there was an area of infarction on the right lobe...we don’t know if it happened because of the pump...we don’t know if it happened because of the procurement...but we have this less than ideal outcome for our patient where part of the right lobe of the liver was just completely dead.”

Necrotic liver due to pump or TransMedics procurement issue; transplanted anyway

A: “Examples where it hasn’t gone so great are situations where the liver is on pump. **The numbers may be favorable but there’s something we don’t like about the liver. There was a liver a month ago where the right lobe felt very firm, like it was infarction, that part of it had died.**”

Q: “It felt very firm?”

A: **“It felt very firm, yes. At this point, unfortunately, it wasn’t my case but one of my colleagues was already at a point of no return in the recipient operation. He had already started before realizing that this was happening in the graft. And so, we ended up having to use that liver and, sure enough, post-transplant, there was an area of infarction on the right lobe.** Now, we don’t know if it happened because of the pump. We don’t know if it happened because of the procurement. But **we have this less than ideal outcome for our patient where part of the right lobe of the liver just was completely dead.** We did let TransMedics know. After the transplant, we communicated our findings and our concerns and we let them know about it.” – Chief of transplant surgery at a major transplant center; high-volume TransMedics users

The surgeon stated that TransMedics was fully aware that part of the liver was dead, and that they recently encountered another necrotic liver, just weeks after the other one – and that their “Medical Director” Magdy Attia falsely told them they were the only center that had encountered the issue. For the avoidance of doubt, we asked the surgeon if TransMedics didn’t tell them about the first liver’s damage: “Until after we felt it, exactly...we reported this to TransMedics and we were right...our concerns were correct...the procuring surgeon could feel it but it was not reported...but that’s just the first part of the story...fast forward three weeks later, another colleague, we had another liver on pump for a different patients...this time, because of what happened...we inspected the graft more thoroughly...and again, this time, the right lobe of the liver fell off...so we called Magdy...the response was ‘Wow, yours is the only center that is having these issues,’ which is an outright lie...there were other centers that definitely had concern about a graft and how it felt before implantation.”

Failed to report liver was necroscopic; TransMedics allegedly lied to the center to cover up a prevalent issue

Q: “So, when you got the liver, you didn’t even know until you felt it and **they didn’t tell you?**”

A: **“Until after we felt it, exactly. So, there was an issue and then the liver, the pump - we reported this to TransMedics and we were right. Our concerns were correct.”**

Q: “Was this the kind of thing that should have been obvious to the procuring surgeon just by feeling it?”

A: **“The procuring surgeon could feel it, but it was not reported.** It could have been a procurement issue. But that’s just the first part of the story. **Fast forward three weeks later, another colleague, we had another liver on pump for a different patient. And this time because of what happened the previous time, we inspected the graft more thoroughly before we started the operation. And again, this time, the right lobe of the liver felt off.** But the numbers, the perfusion numbers were fine. So, we called Magdy and we called TransMedics and we said, “The numbers are okay. It looks okay. But that right lobe, we’re nervous because of what happened with our previous patient, and **the right lobe doesn’t feel right.” And then the response was “Wow, yours is the only center that is having these issues,” which is an outright lie. There were other centers that definitely had concerns about a graft and how it felt before implantation.** – Chief of transplant surgery at a major transplant center; high-volume TransMedics user

The surgeon's color suggests a widespread issue with livers becoming necrotic on the device - a known risk and potentially fatal flaw with normothermic perfusion pumps – as well as a conspiracy to cover it up: “their advice to us was, ‘Why don’t you wait a couple of hours’...and the right lobe still didn’t feel much better...so we declined the liver...it left a bad taste in our mouth...the trust was fractured at that point...Magdy was the one who told us about that...then we called other colleagues around the nation...and they said, ‘Oh, we’ve definitely seen this before...sometimes it’s a procurement issues, sometimes it’s pressure on the pump’...so, it’s just a fucking lie...it’s a lie that we’re the only center that’s having these problem but that was the communication to us....”

Failed to report liver was necroscopic; TransMedics allegedly lied to the center to cover up a prevalent issue

A: “So, then their advice to us was, “Why don’t you wait a couple of hours,” which we did. And the right lobe still didn’t feel much better. So, we declined that liver. Fortunate for TransMedics, [transplant center redacted], our colleagues down the road, decided to use that liver for one of their patients. So, that liver did get transplanted. I don’t know what the outcome was for their patient but it left a bad taste in our mouth because we are trying to do what’s right by our patient and the feedback makes us feel like it’s something that we’re doing wrong in our assessment....the trust was fractured at that point.”

Q: “Who was saying to you you’re the only center having this issue and lying? Was that Magdy Attia?”

A: “Magdy was the one who told us about that. So, then we called other colleagues around the nation. We called San Diego; we called Washington University in Seattle - what do you guys think? And they said, ‘Oh, we’ve definitely seen this before. Sometimes it’s a procurement issue; sometimes it’s pressure on the pump. We’ve definitely seen that before.’ So, it’s just a fucking lie. It’s a lie that we’re the only center that’s having these problems but that was the communication to us. We don’t know what’s going on with you guys, but everyone else is fine using our pump. And this situation, we were like, “You know what? We’re not going to put our patient at risk. We’ll eat the cost, fine.” – Chief of transplant surgery at a major transplant center; high-volume TransMedics user

The surgeon indicated that the two recent livers with necrotic tissue – and the fact that other centers are encountering the same issues – calls for the publication of data about device problems and outcomes, which TransMedics has allegedly refused: “members of the transplant community, myself included, have called for publication of the livers that are not used or the complications related to livers that have been on the OCS pump, and they declined...TransMedics declined to publish that data...that is not being very honest...it’s their responsibility....they should publish it and be forthcoming.”

TransMedics allegedly refusing to publish data on necrotic livers

*“I can’t speak to their outcomes overall. But I do know that there are **members of the transplant community, myself included, have called for publication of the livers that are not used or the complications related to livers that have been on OCS pump, and they declined.** TransMedics declined to publish that data. Like that is not being very honest...it’s their responsibility to vet their surgeons and they should publish it and be forthcoming.” – Chief of transplant surgery at a major Los Angeles transplant center; high volume TransMedics user*

A transplant administrator/executive at Massachusetts General indicated that device problems are widespread, called them “Never Events” of the most urgent nature, questioned why there hasn’t been a recall, and alleged persistent efforts by TransMedics to cover-up device problems and conceal them from the FDA, and blasted the company’s ethics. The executive stated that transplant medicine is “a small world” and he’s been “bombarded” by colleagues around the country asking if he’s seen similar malfunctions: “people are now saying, are you guys facing the same issues? Am I the only one who’s constantly getting these bombardments of issues left and right without even being notified of these recurring issues or device failures?” He noted other medical device recalls: “that got self-reported to the FDA...why is that not happening here?...why does it have to be a community of concerned transplant administrators...to allow these types of individuals, organizations, misaligned incentives...”

“Bombardments of issues” re device failures and surgical problems

Q: “So, you’re referring to never events being surgical, primarily or device failure or a little bit of both?”

A: “Both. The example I just shared with you, that right there, didn’t impact my recipient because it was going for someone else, but **I’m starting to get people, again, just based off the fact this is a small world, that people are now saying, are you guys facing the same issues? Am I the only one who’s constantly getting these bombardments of issues left and right without even being notified of these recurring issues or device failures?** When there was the recall of the HeartMate 2, that went to everybody. When there was a recall because of the duodenoscopes for Olympus because there was an infection that was being spread and because of the lack of scope processing, but Olympus knew about it. **That got self-reported to the FDA. Why is that not happening here? And why does it have to be a community of concerned transplant administrators who are just trying to do their job having to ultimately be vulnerable and say, ‘I don’t know what’s going on. I need some help, guys.’** And that’s probably what you’re hearing, and that’s probably what you’re seeing, and we failed to put the patient first, and shame on us for not doing enough. I’m not surprised that there are individuals who are leaving an organization that allows this, that **promotes this at the expense of the patient**. And, on top of that, the wishes of the decedent and maximizing utility of their gift...and I don’t mean to be prophesizing but I’m literally just saying this is not why we came into this field. And **to allow these types of individuals, organizations, misaligned incentives to take over and put us in a corner? It’s not fair practice, but it’s also not sound business practice.**” – Veteran transplant administrator at Massachusetts General Hospital

The MGH executive stated that the device problems are now “a running joke” in the transplant field – “are you going to recall these because of a known issue?” - and detailed various incidents where TransMedics was evasive: “among other people in the field, it’s a running joke...we had a national conference...it’s like, come talk to me about your TransMedics story...it’s laughable...when it comes down to equipment issues...a malfunction of a panel or blood getting into one of them...then months go by and they say, ‘well, we reviewed it internally...we’re going back and making some changes’...and I’m going, ‘Okay, so what about all the other devices that are currently out there? Are you going to recall those because of a known issue?...and then they say, ‘this is actually nothing....’”

Rampant device failures; covering it up

Q: “How often are you seeing these never issues: device malfunction or surgical errors? Or how often are you hearing about them from other people in the field?”

A: **“Among other people in the field, it’s a running joke. We had a national conference; it’s like, come talk about your TransMedics story. It’s laughable. When it comes down to equipment issues, like dealing directly with, let’s say, a malfunction of a panel or blood getting into one of them** or something happening to the device mid-flight, but there’s no backup, you can’t flush the organ—I kind of go, “What’s your backup?” And how is there not a backup to the backup as well? And **then months go by, and they say, “Well, we reviewed it internally. It was a malfunction of the panel. We’re going back and making some changes.” And then I’m going, “Okay, so what about all the other devices that are currently out there? Are you going to recall those because of a known issue?”** And then they say, “This is actually nothing. It’s outside of the norm of how many we’re doing. There’s a certain defect rate that is allowable, and this does not fall within that defect rate. We’re a trusted partner. We appreciate you.” That’s the extent of it.” – Veteran transplant administrator at Massachusetts General Hospital

The MGH executive indicated that there are not minor malfunctions but “adverse events that should never occur,” calling them “Never Events” in the “highly, highly regulated world of transplants” – such as a failure that could “put the organ at risk or when a recipient will be put under anesthesia waiting for the organ and a device malfunction occurs, and you lose the organ...” He indicated “certain never events” with TransMedics that required that he “be called within a few hours by our governing body and regulatory body, so would my chief compliance officer...either it’s a sentinel event or adverse event”; “when there are certain never events...under the watch of TransMedics...I’ve been struggling to find anyone who takes ownership of it...to the point where I have to request them to show proof that they self-report to the FDA.””

Never events like losing organs due to device failure or human error; failing to report; called by regulatory bodies

“Device failure-wise, it just goes along with what I had mentioned before, how any time that there’s even a **device failure, any time they could potentially even put the organ at risk or when a recipient will be put under anesthesia waiting for the organ and a device malfunction occurs, and you lose the organ?** That’s a never event [...] these are adverse events that should never occur, and, obviously, you’d want to have many different countermeasures, many different plan Bs and Cs that would back things up so that you would never, **number one, lose the organ; two, have a situation where either the organ is not procured because of human error or such, or there are delays that were not mitigated.** And so, there are certain never events that have occurred where in the highly, highly regulated world of transplants, **I would essentially be called within a few hours by our governing body and regulatory body, so would my chief compliance officer, and we’d be on a phone call directly addressing the issue, either it’s a sentinel event or an adverse event** and there would be clear processes related to reporting of the event, either self-reporting to the DPH, CMS, OPTN, or UNOS and PFC. And that would happen roughly within 24 hours. **When there are certain never events that have occurred under the watch of TransMedics...I’ve been struggling to find anyone who takes ownership of it, anyone who either have self-reports to the FDA or not, and to the point where I have to request them to show proof that they self-report to the FDA.”** – Veteran transplant administrator at Massachusetts General Hospital

The MGH executive stated “these are situations where patients are not able to get a life-saving heart because of an issue....because the transplant world is small, we all talk to each other...the fact that we know of three or four situations where that’s occurred with other transplant programs, and TransMedics has not even self-reported that...I mean, any type of recall...TransMedics has not done that and I’m a bit taken aback...their response related to the device malfunction....that didn’t seem right to me...when we started this way back when, and it was still awaiting the FDA, we were working with them hand-in-hand...there was a fundamental alignment...that is not the case anymore.”

Failing to report device malfunction issues known among transplant centers

“There have been situations where they have **even stated that they have not needed to report it based on the volume that they have, which to me doesn’t make any sense because these are never events. These are situations where patients are not able to get a life-saving heart because of an issue.** And, on top of that, because the transplant world is small, we all talk to each other. We all know what’s going on. And the fact that we know of **three or four situations where that’s occurred with other transplant programs**, and TransMedics has not even self-reported that. I mean, any type of recall...**TransMedics has not done that and I’m a bit taken aback.**” – Veteran transplant administrator at Massachusetts General Hospital

MGH uncomfortable with lack of FDA reporting for device malfunctions

Q: “It sounds like they’re actively trying to conceal these times when a surgeon messed up, or the device malfunctioned. Have you seen a pattern of concealment or where they’re actually faking or misreporting it? People are telling me that and I’m wondering what you’ve observed directly or heard about.”

A: “...that one individual feeling pressured or bullied to either redact the report and adjust the content of that, that is, I mean, unethical is not even what I would call it, but I would just call it that for now...what’s sad to say out loud is **I wouldn’t be surprised if it has happened to us or around certain cases that we’ve been involved with.** As I mentioned before, **their response related to self-reporting to the FDA related to the device malfunction didn’t fall within their margin of error; that didn’t seem right to me, and we had to go through some other routes to get that resolved or at least looked at.** As I mentioned, when we started this way back when, and it was still awaiting the FDA, we were working with them very hand-in-hand because it did seem like it was a very promising technology. There was a fundamental alignment of our philosophies of what we wanted to do together. That is not the case anymore.” – Veteran transplant administrator at Massachusetts General Hospital